Capability Matters

A CASEBOOK

Learning Engineering for Next-Generation Systems

Capability

Matters

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LEARNING DESIGN AND TECHNOLOGY

FIRST EDITION · REVISED MAY 2026

An ongoing artifact of the LENS Practice Flywheel — Identify, Activate, Prototype, Analyze, Transition.

DEDICATION

To everyone who believes in a dream and refuses to give up.

COLOPHON

Learning Engineering for Next-Generation Systems: Capability Matters
A Casebook.
First edition, 2026.
Edited by William Gray-Roncal, PhD and James Diamond, PhD.
Compiled for the LENS specialization within the Learning Design and Technology program at the Johns Hopkins University School of Education.
Set in Instrument Serif and DM Sans.
Printed via Lulu at 8 × 10 in, perfect-bound.

HOW THIS VOLUME WAS MADE

This casebook was assembled using AI tools as part of an iterative learning-engineering process — the same Practice Flywheel (Identify → Activate → Prototype → Analyze → Transition) that the volume's case studies argue for. The methodology treats AI as the dual entity the curriculum names it: a creative partner that accelerated drafting, cross-referencing, layout, and citation lookup, and an epistemic risk that had to be hand-checked against the record. Every case in the book was reviewed by the editors and hand-checked by students for accuracy: confirming that the sources cited exist and are findable, that the quoted attributions are fairly represented, and that the case studies are accurate accounts of the incidents and the investigations they draw on. Items where the source could not be confirmed are marked "Paraphrasing…" so the attribution is honest about what is the author's reconstruction and what is verbatim from the record.

The volume is not finished. It is the first iteration of a reference dataset that LENS will continue to develop. We will revise the book based on reader feedback, reviewer findings, and our own ongoing practice of learning engineering — adding new cases as the operational record grows, correcting our record where it can be improved, and updating the curriculum crosswalk as the program itself iterates. Corrections, new cases, and reader feedback are welcomed.

SOURCES, ATTRIBUTION & LEGAL NOTICES

All cases reference publicly reported incidents and published investigations. Quoted material is attributed to source investigations, court documents, and academic publications. Reflection questions are original to this volume. The audit record of citations checked and changes made is maintained alongside the source repository.

References to convictions, settlements, and findings reflect the public record at the time of writing. Where individuals are named, they are named only as participants in investigations, court proceedings, or public inquiries already in the public domain. The casebook makes no legal characterization beyond what those records establish.

Corrections, requests for clarification, and notice of any inaccuracy can be sent to LDT / LENS at the Johns Hopkins University School of Education. Substantive corrections will be incorporated in subsequent printings and noted in the casebook's public errata.

WELCOME

Learning engineering is a toolbox, not a single tool.

At its center it is a dialogue between two traditions — the engineering disciplines that design, build, and sustain complex systems, and the learning sciences and education that study how people come to know and do. Around that dialogue gather many other fields: cognitive and human-factors engineering, measurement, implementation science, design, data and computation, and the operational domains themselves. The work is plural by design — many disciplines at once — and it draws on domain knowledge, coding, theory, systems thinking, design, teaching, and analysis.

It welcomes everyone who brings a real tool to it. It is not, for that, a free-for-all: learning engineering stands in an intellectual tradition with its own methods, evidence standards, and lineage, and it holds the work accountable to them. No one carries every tool. This is a team sport — capability gets built at the seams between people who each see part of the system clearly and trust the others to see the rest.

So you don't need to know everything before you start. You need to know what you bring, stay honest about what you don't, and find the people who carry the tools you're missing. What you bring is yours to decide. The hundred cases that follow are an invitation to find your part in the work.

— THE EDITORS

INTRODUCTION

Capability is a system parameter.

Every complex system depends on what people can do inside it. That dependency is measurable, designable, and too important to leave to chance.

> **01.** Systems don't fail because the technology breaks. They fail because someone couldn't do what the system required.
>
> **02.** We engineer every parameter of a system except the people.
>
> **03.** Capability is not a soft problem. It is a systems engineering problem.
>
> **04.** Decision-grade evidence. Not opinions about training.
>
> **05.** Capability without agency is automation. The goal is operators who can perform, adapt, and lead — not comply.
>
> **06.** Sometimes the constraints make the goal unreachable. Saying so early is a result, not a failure.

The cases in this book are not, individually, hard to understand. A pilot recovered from a stall the wrong way. A nurse could not stop a doctor who skipped a step. A safety operator was watching television on her phone when a pedestrian crossed the road. A captain shot down a civilian airliner because the radar data and the operational framing pointed in opposite directions, and the framing won. The proximate cause is usually obvious. The first investigator on the scene almost always finds it within hours.

What is hard to understand — and what this book exists to make legible — is the system that produced the gap into which each incident fell. In every case the proximate actor was operating inside an architecture of training, procedure, authority, measurement, and incentive that had, for years, been quietly degrading. The pilot had never trained to recover a stall at cruise altitude. The nurse had no procedural path to intervene. The safety operator had been hired into a role nobody had figured out how to make performable. The radar operator had been trained to a tempo and a presumption that the system above him no longer shared with the system around him.

Each of these architectures was designed. Each was funded. Each was reviewed. Each carried, written somewhere in its specifications, the assumption that the people inside it would be able to do what the system required of them when the system required it. That assumption is what this book calls the **capability parameter**. It is a property of the entire sociotechnical system, not of any individual operator. Capability lives at the **interface** between what a system requires of its operators and the impact the system has to deliver — and when that interface is wrong, the gap may originate in human development, in system design, in organizational performance, or in the interaction among them. Distinguishing which is itself a measurement and engineering problem; the casebook calls it **gap attribution**, and treats it as the diagnostic move that the rest of the discipline depends on. When the interface fails — wherever the source — the consequences are paid in lives, in dollars, in trust, and in time the institution will never recover.

The premise of the Johns Hopkins School of Education's Learning Design & Technology program — and of the Learning Engineering for Next-Generation Systems specialization that has grown out of it — is that the capability parameter is engineerable. Not by accident. Not by declaration. By the same kind of evidence-grounded, methods-grounded, cross-domain discipline that built modern reliability engineering, modern epidemiology, and modern systems safety. The discipline does not yet exist at the scale the evidence demands. LDT and LENS are organized to help build it.

I · THE COST OF THE GAP

The Institute of Medicine's **To Err Is Human** (1999) estimated that preventable medical error caused between forty-four thousand and ninety-eight thousand deaths a year in the United States — already, at the lower bound, a top-ten cause of death. The report catalyzed a generation of patient-safety work: surgical checklists, central-line bundles, team training, computerized order entry. Some of those interventions are documented in Part II of this book as successes. And yet in 2016 Makary and Daniel returned to the question from the same institution that produces this casebook and concluded that medical error remains the third leading cause of death in the United States, killing more than two hundred fifty thousand people a year [1].

The interval between the IOM report and the Makary update is approximately the canonical gap implementation science has identified between research evidence and clinical practice: seventeen years [2]. The average. The median. Only about fourteen percent of research findings ever reach practice at all [3]. The system to surface them, vet them, deploy them, sustain them, and measure their effect at the scale at which they would matter has never been built as an institution. It has been built as a series of grants. The difference is the point.

This is not only a healthcare problem. The U.S. Navy lost seventeen sailors in two avoidable destroyer collisions in 2017 because in 2003 it had cut a sixteen-week classroom-and-simulator course down to a set of CD-ROMs [4]. The world's most expensive aircraft program is operating at half of its design readiness in 2026 because the platform was fielded faster than the maintainer pipeline could be built [5]. One hundred million dollars of educational infrastructure dissolved in fourteen months because the institution building it did not engineer the governance and trust that the technology presupposed [6]. A predictive grading algorithm downgraded a third of an entire nation's high-school students in a single afternoon, and was withdrawn within a week, because nobody inside the agency could be specifically held to account for what the measurement instrument was measuring [7].

None of these failures is a failure of effort. None is a failure of intelligence. Each is a failure of the capability infrastructure to match the system it was operating. In each case, the gap was diagnosable in advance. In most cases, it was actually diagnosed in advance, by someone — and the diagnosis did not produce the remediation. The cost of the unrepaired diagnosis is why **capability** matters.

II · WHAT AN ENGINEERABLE DISCIPLINE LOOKS LIKE

Crew Resource Management did not exist as a discipline in 1976. In March 1977 two 747s collided on a foggy runway at Tenerife and 583 people died. Within five years United Airlines had operationalized the first CRM curriculum [8] within twenty years the Commercial Aviation Safety Team had built the closed-loop hazard-identification system that lets the FAA know whether CRM is still working [9]. The result over the next two decades was an eighty-three percent reduction in U.S. commercial aviation fatality risk and a ninety-five percent reduction in fatalities per hundred million passengers [10]. CRM works not because it taught individual airmanship — pilots were already excellent — but because it redesigned the **system of interaction** in the cockpit: the authority gradient, the communication protocol, the standard brief and debrief. The discipline treated team coordination as an engineerable property of the system.

The same pattern shows up in every domain where the cost of failure has been high enough to fund the discipline. After Three Mile Island the U.S. nuclear industry stood up the Institute of Nuclear Power Operations within nine months — before the Kemeny Commission's report was even finalized — because every utility had recognized that an accident at any single plant would affect every operator's license to operate [11]. INPO and the National Academy for Nuclear Training, founded in 1985, have presided over more than four decades of zero significant releases at U.S. commercial reactors. The comparison with the surface Navy, which cut its training during the same era, is the cleanest available test of what happens when capability is engineered versus when it is deferred.

Healthcare has its own version of this arc, but only partially. Peter Pronovost's central-line checklist, paired with the nurses' authority to enforce it, drove bloodstream infections to near zero across 103 Michigan ICUs and saved an estimated fifteen hundred lives in eighteen months [12]. Atul Gawande's nineteen-item surgical safety checklist, paired with three required pauses in the operative timeline, halved the surgical mortality rate across eight hospitals in eight countries [13]. TeamSTEPPS, jointly developed by AHRQ and DoD, translated five decades of high-reliability research from aviation and the military into clinical practice in years rather than decades — because the implementation infrastructure was funded as part of the intervention rather than as an afterthought [14].

These are not stories about individual hospitals or individual ships. They are stories about the difference between a domain that has organized itself to engineer capability and one that has not. The pattern is consistent across all of them: a catastrophe makes the capability gap visible; an institution is built that treats the gap as the institution's responsibility; the institution funds the measurement architecture that lets it know whether the gap is closing; and twenty years later the death rate is half what it was. The intervention is always paired. A technical artifact — a checklist, a brief, a cord — combined with a cultural artifact — protected authority to use it, a no-blame protocol, a credible governance body. Neither alone moves the curve. Both together move it dramatically.

III · THE DISCIPLINE WE HAVE, THE DISCIPLINE WE NEED

The intellectual material to build this discipline already exists. The learning sciences have produced four decades of converging evidence on how skill, knowledge, and judgment develop and decay [15]. Cognitive engineering has produced rigorous methods for analyzing human-machine systems [16]. Human factors and ergonomics have produced an evidence base for interface and workflow design [17]. Implementation science has produced frameworks for taking effective interventions to scale [18].

Systems safety engineering has produced both diagnostic tools — Reason's swiss-cheese model, Rasmussen's migration-to-the-boundary, Leveson's STAMP — and control-theoretic methods for design [19]. High-reliability- organization theory has documented what the institutions that have solved the capability problem actually do [20]. The discipline is not missing its evidence base. It is missing its integration.

Learning engineering is the integration. The term, in its modern usage, traces to Bror Saxberg and was elaborated by Goodell, Kolodner, and the IEEE Industry Connections Industry Consortium on Learning Engineering (ICICLE) [21]. The IEEE Learning Engineering Body of Knowledge (LEBoK) and its associated process model are the most developed available specification of what the discipline does [22]. The premise is that the work of building, deploying, and sustaining capability at scale is an engineering activity: it requires explicit requirements, evidence-based design, instrumented measurement, and feedback-driven iteration. It is not exclusively a research activity. It is not exclusively a managerial activity. It is engineering, and it can be taught and practiced as engineering.

LENS — Learning Engineering for Next-Generation Systems — is a specialization within the Johns Hopkins School of Education's Learning Design & Technology program that operationalizes this premise for high-consequence operational domains. Its core competencies — capability analysis and requirements, evidence and analytics, human-AI teaming, knowledge transfer, bias and governance, computational and AI methods — are organized to produce graduates who can do the work the cases in this book required, and who can build the institutions that the success cases in Part II had to invent. The curriculum threading commitments — implementation science as a through-line, equity as a design commitment rather than an audit, decision-grade evidence as a deliverable rather than a report, communication and system-of-systems integration as engineerable properties of the system, the disciplined use of learning- technology aids (from XR to adaptive platforms) as instruments whose evaluation is part of the work, and **AI as both a creative partner and an epistemic risk** — leveraging AI's generative reach while guarding against offloading the thinking — are drawn directly from the patterns the cases reveal.

Three of those commitments deserve a moment of their own — they are the substrate the rest of the curriculum runs on. The first is **communication, translation, and integration within and across disparate systems**. A striking number of cases in this book are, at their proximate cause, translation or integration failures across boundaries — failures of language, convention, unit, role, agency, or discipline to carry meaning reliably from one part of a system to another, or failures of subsystems built to different assumptions to compose into a working whole. Mars Climate Orbiter was a metric-versus- imperial translation failure (Case 54). Tenerife was a translation failure at the takeoff-clearance boundary between two cockpits and a tower under fog and time pressure (Case 12). The Patriot at Dhahran failed because the manufacturer's assumption about run time and the operator's actual run time were on opposite sides of a documentation boundary that had not been engineered to be crossed (Case 19). Eagle Claw failed because the rotary-wing, fixed-wing, ground, and command components were assembled across service boundaries that had no shared operating practice (Case 46). 9/11 was a cross-agency translation failure measured in thousands of dead (Case 86). Boeing 737 MAX was, at one level, an integration failure: a single-string sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together (Case 2). The successful cases — AlphaFold (Case 98), MICrONS, CRM (Case 12 paired with Case 23), Keystone (Case 14), the paired-intervention examples in Chapter 8 — are, equivalently, cases of disciplined translation and integration: biology into computation, science into operational practice, technical reform into cultural reform, multiple subsystems into a working whole. Engineering capability across these boundaries — language, discipline, agency, subsystem-to-subsystem, system-of-systems — is itself a designable target. The LENS curriculum treats it as such.

The second is **technology aids and learning platforms**. The discipline has at its disposal a fast-evolving toolkit: extended- reality (XR) simulation environments — VR, AR, and mixed-reality systems used for procedural training, spatial-reasoning practice, and high-fidelity rehearsal under conditions that would be unsafe or impossible in the field — together with learning- management systems, adaptive learning platforms (the lineage from Cognitive Tutor and ASSISTments to current ITS work; Cases 42, 38), intelligent tutoring frameworks such as GIFT, learning- analytics infrastructures such as xAPI and the Total Learning Architecture (Case 40), game-based learning environments, LLM-augmented tutors and authoring tools, and high-fidelity simulators in aviation, surgery, defense, and process industries. These tools are not the discipline. They are the discipline's instruments — and the cases in this book show, repeatedly, that powerful instruments do not by themselves engineer capability (Cases 8, 21, 25, 38, 96, 99). The LENS curriculum teaches the practitioner to choose, configure, evaluate, and govern these tools with the same evidentiary discipline applied to any other intervention, and to recognize when the binding constraint is the surrounding architecture rather than the tool itself.

The third — and the design constraint on all the others — is **human agency**. The capability discipline can be misread as optimization: produce operators whose behavior matches a specification. That reading collapses the very property the cases reveal as load-bearing. Every successful institution in Part II — INPO operators trained to challenge their reactor team, Keystone ICU nurses authorized to halt a procedure (Case 14), the Nuclear Navy's questioning attitude (Case 28), the first-officer authority CRM built into the cockpit (Case 12) — engineered for capability **and** for the operator's capacity to exercise independent judgment, to adapt to conditions the system was not designed for, and to shape the surrounding architecture rather than only execute inside it. Capability without that capacity is automation. LENS treats the preservation and development of agency — the room the system leaves its operators to think, decide, and lead — as a design constraint on every intervention, including those in which AI is the partner. The paired risk runs the same direction: an AI collaborator can accelerate the flywheel or quietly offload the thinking, and which one it does is a design and governance decision, not a property of the tool.

IV · LENS AS AN ITERATIVE LEARNING-ENGINEERING PROCESS

Learning engineering is not a one-time act of design. Its defining feature, as articulated in the IEEE Industry Connections Industry Consortium on Learning Engineering's process model, is iteration: understand the operational context, specify capability requirements, prototype an intervention, instrument it, measure, learn, redesign. The work cycles. The cases in this book that produced sustained outcomes — CRM and CAST, INPO, Keystone ICU, the WHO Surgical Safety Checklist, TeamSTEPPS — all share that loop structure. They were not built once and left alone. They were built, measured, rebuilt, remeasured, and rebuilt again. The institutional capability to sustain the loop **is** what distinguishes them from interventions whose effects decayed.

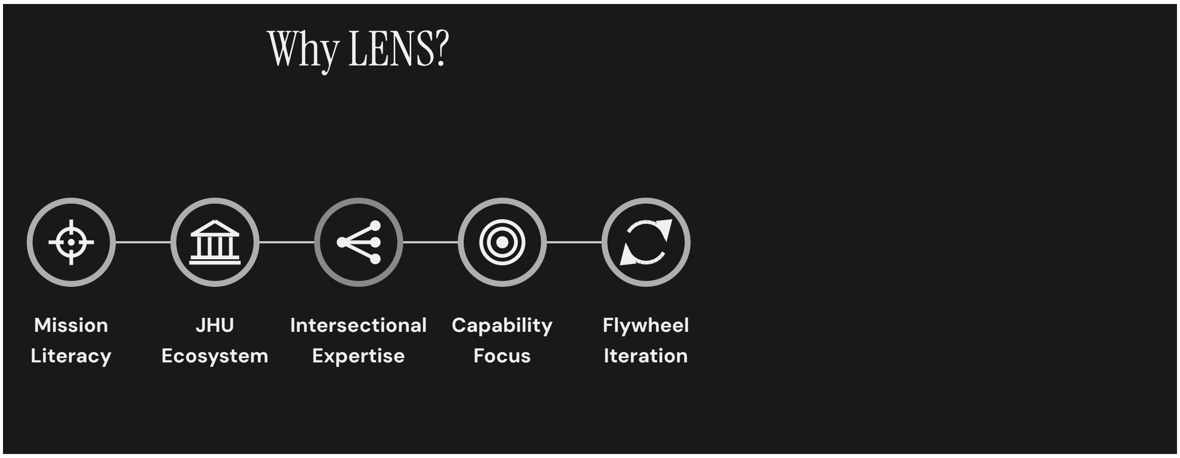
The loop also has a legitimate negative result, and the discipline has to be honest enough to reach it. Capability engineering always operates inside constraints — budget, schedule, regulation, institutional will, politics, ethics, public trust — and many of the binding ones are not technical. Part of the work is **managing those constraints and the risk they carry**: reading the constraint space accurately enough to recognize when external, non-technical factors have narrowed the solution space to the point where a particular capability goal is no longer realistic to pursue. Reaching that conclusion — and documenting **why** — is itself a valid outcome of a project, not a failure of one. It redirects effort toward goals that are actually reachable, surfaces the non-technical barriers for the stakeholders who can change them, and prevents the far more expensive failure of driving an infeasible target all the way to the operational record before it collapses there. A number of cases in this book are, in part, stories of institutions that could not say "not like this, not yet" in time.

LENS is organized as that loop made teachable. Required coursework carries students through the cycle: framing the system whose capability is at stake, eliciting requirements from operational analysis, building the evidence architecture that lets an institution know whether its interventions worked, examining the human and AI roles inside the designed system, and running the loop end-to-end on a real operational problem in an integrated capstone studio. Electives extend the loop into governance and risk, knowledge transfer and organizational learning, and computational methods. The program graduates a practitioner who has been around the loop enough times to know where it breaks. Specific course titles and sequencing evolve with the program — current titles and detailed descriptions live in the LDT/LENS catalogue; this casebook tags each case to the courses that most directly use it as a worked example so the mapping remains legible as the curriculum iterates.

The casebook is itself an iteration artifact. Each case is read, diagnosed, paired with a learning-engineering response, and tested in studio against students' own operational domains. The cases that return as success stories in Part II are the cases whose institutions completed the loop. The cases in Part I are the cases whose institutions did not — or did not loop fast enough. The pedagogical commitment of the program is that the practitioner who graduates can identify, instrument, and close the next such loop before the next case is written.

V · WHY LENS · THE FIVE PILLARS

A specialization that promises to teach a discipline still being built must show that its credibility rests on more than its ambitions. LENS stands on five pillars: **Mission Literacy**, **JHU Ecosystem**, **Intersectional Expertise**, **Capability Focus**, and **Flywheel Iteration**. Each is a real institutional asset, with a record. None of them, alone, would carry the argument; the pillars are load-bearing together.



WHY LENS — FIVE PILLARS

MISSION LITERACY

Real operational problems, real sponsors, real consequences — and the disciplined ability to read them. LENS is anchored in the conviction that the practitioner must be able to enter an operational setting, listen to the people doing the work, understand what the mission actually requires, and translate that understanding into design and measurement decisions that a serious reviewer would accept. Mission literacy is what makes the cases in this book teachable as cases and not as anecdotes.

JHU ECOSYSTEM

An institutional environment broad and deep enough to host the discipline. At Johns Hopkins this is the School of Education with its centers, doctoral programs, and faculty in learning sciences, educational measurement, and design-based research; the practice partners across the university (Armstrong Institute for Patient Safety and Quality at the School of Medicine; the Bloomberg School of Public Health's implementation-science tradition; Whiting School engineering and cognitive science; the Berman Institute's governance and ethics work); and the partnerships with sponsor and operational organizations that bring real problems through the door.

INTERSECTIONAL EXPERTISE

The distinguishing pillar — and the most demanding. The work sits at the intersection of the learning sciences, educational measurement, instructional design, design-based research, cognitive engineering, implementation science, the equity-and-governance disciplines, the relevant operational domain, and the engineering tradition that produced the artifact under study. No one of those disciplines is sufficient, and the work fails when the disciplines do not translate to one another. Intersectional expertise is what allows a single practitioner — or a single team — to hold the engineering, the learning, the measurement, the equity, and the operational context in the same conversation, and to make design decisions that survive scrutiny from any of them. Cultivating it is the central pedagogical task of LENS.

CAPABILITY FOCUS

The outcome — the central claim of the book. Capability is a designable, measurable, engineerable property of every complex system. The program produces it in graduates; the cases document where it was absent and where it was successfully engineered; the curriculum teaches how to recognize it, instrument it, and reproduce it. Capability focus is the discipline that keeps every artifact, every measurement, and every iteration accountable to what the system can actually do when the work is done.

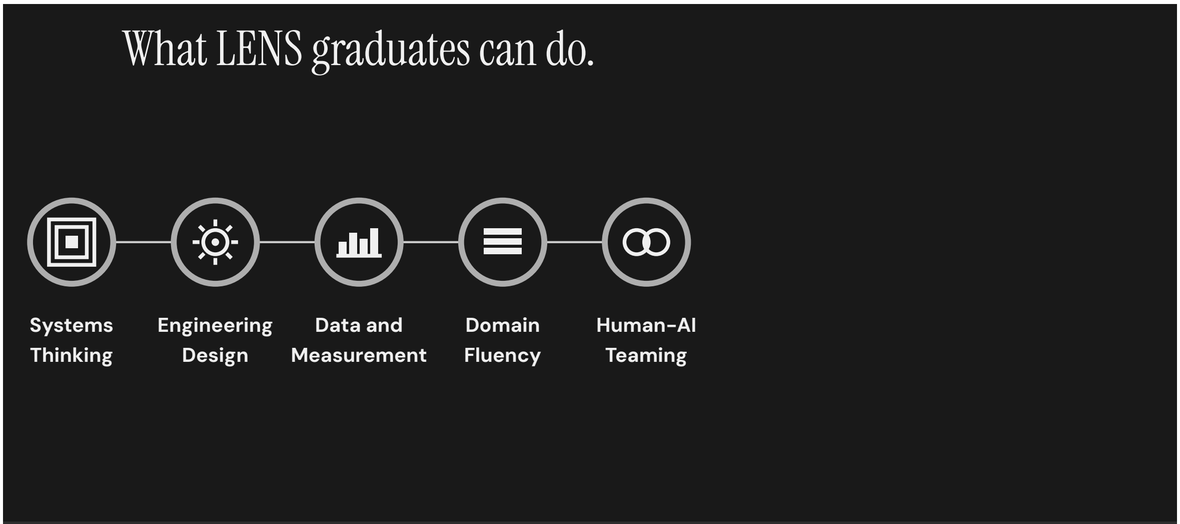
FLYWHEEL ITERATION

> The iterative method — the Practice Flywheel — that animates the other four pillars: **Identify → Activate → Prototype → Analyze → Transition**, the cycle through which mission literacy turns into capability and capability turns back into refined mission literacy. The flywheel is what prevents the discipline from being a series of unconnected projects and makes it instead a compounding institutional asset — each cohort, each case, each pilot shortening the implementation gap for the next. The casebook itself is a flywheel artifact: the work of prior cohorts produces the reference dataset for the next.

The pillars compound. Mission literacy reveals what is needed; the JHU ecosystem hosts the inquiry; intersectional expertise designs the intervention; capability focus keeps the loop closed on the outcome; the practice flywheel turns each iteration into the starting condition of the next.

WHAT LENS GRADUATES CAN DO · THE FIVE COMPETENCIES

The five pillars describe why LENS exists. The five concentration learning outcomes describe what its graduates can do. They map to the work the cases in this book required.



WHAT LENS GRADUATES CAN DO — FIVE COMPETENCIES

> 1 · SYSTEMS THINKING AND ANALYSIS
>
> Decompose system performance requirements into measurable human capability requirements; apply systems-engineering lifecycle models to scope, sequence, and evaluate capability interventions **within and across disparate systems**; analyze and communicate the interdependencies between human performance and system performance to predict operational impact at scale.

2 · LEARNING ENGINEERING DESIGN AND IMPLEMENTATION

Design capability-development interventions that integrate learning-sciences principles with measurable outcomes and system-design constraints, and that function at **speed and scale** in operational environments; construct and communicate implementation plans that address adoption, sustainment, and lifecycle integration across organizational and system boundaries.

3 · DATA, MEASUREMENT, AND EVALUATION

Design ethical instrumentation strategies that produce measurable evidence in regulated or high-consequence environments; construct **decision-grade evidence** artifacts — dashboards, reports, governance documentation — that link learning outcomes to operational impact; differentiate capability gaps from system-design and organizational- performance problems using mixed-methods **gap attribution**.

4 · CONTEXT AND DOMAIN FLUENCY

Analyze the regulatory, organizational, and cultural constraints that shape capability development in healthcare, defense, or education contexts; apply human-systems-integration frameworks to evaluate the measurable impact of capability approaches on operational environments; synthesize and communicate stakeholder requirements across disciplinary and institutional boundaries into coherent design specifications.

5 · HUMAN-AI TEAMING AND ADAPTIVE SYSTEMS

Design human-AI teaming configurations that **leverage AI as a creative partner** in capability development while preserving human agency and judgment; evaluate the measurable impact of AI-mediated learning systems on human performance — both the gains AI enables and the risks it introduces, including automation bias and cognitive offloading; communicate evidence-based recommendations for when AI augmentation improves versus degrades capability outcomes.

Three phrases recur across those five competencies and across the curriculum that produces them: **within and across disparate systems** (the scope at which capability is actually engineered), **speed and scale** (the operational tempo the discipline must match), and **gap attribution between learning, system design, and organizational performance problems** (the diagnostic that decides what kind of intervention will actually move the outcome). They are the working language of the program. They appear, repeatedly, in the cases that follow.

The remainder of this section names the three operational anchors the pillars are currently producing — the CIRCUIT–MERIT–COMPASS lineage, the Practice Flywheel in action across LE pilots, and the studio and applied-project model — as proof of work the pillars have already delivered.

The first operational anchor is the **CIRCUIT model, MERIT framework, and COMPASS content** — a more-than-decade record originating at the JHU Applied Physics Laboratory. CIRCUIT (Cohort-based Integrated Research Community for Undergraduate Innovation and Trailblazing) was built as the best operational solution to the IARPA MICrONS connectomics program's proofreading requirement, and was subsequently extended to additional sponsor problems. The **scientific and engineering impact** the trained cohorts produced — peer-reviewed contributions to the MICrONS reconstructions in **Nature**, the open-data infrastructure of BossDB and CAVE, the NeuVue proofreading framework — is the headline result; the student development was a deliberate and well-engineered byproduct of the mission solution, making doing well for the sponsor compound with doing good for the students. MERIT (Mentoring Exceptional Researchers to Innovate and Thrive) is the generalization of the CIRCUIT model into a replicable framework: holistic recruiting, the CIRCUIT eight pillars, the CCR competency framework, and the explicit "hidden curriculum" literature. **COMPASS** is the layer of targeted mentor interventions that operates MERIT inside a specific institutional setting — designed to help students uncover the **tacit and explicit knowledge gaps** that block independent research practice. COMPASS is not bound to a fixed workshop count; the COMPASS-Core curriculum is one instance, with an example set of workshops covering orientation and goal-setting, resilience and mindset, research-navigation, networking and faculty engagement, scientific reading and writing, presentations and figures, metacognition and advanced study, professional conduct and ethics, STEM identity, and long-term career and funding strategy. The CIRCUIT → MERIT → COMPASS lineage is documented in the peer-reviewed engineering-education literature [24]. Section VII expands on this evidence.

The second operational anchor is the **Practice Flywheel**: the iterative cycle — **Identify → Activate → Prototype → Analyze → Transition** — that has now been demonstrated across multiple LE pilots run by the LENS partner team at JHU/APL. Each iteration produces working artifacts, sponsor-visible evidence, and curriculum material for the next cohort. The flywheel is what prevents the program from being a series of unconnected projects and makes it instead a compounding institutional asset — each cohort shortening the implementation gap for the next. The casebook itself is a flywheel artifact: the work of prior cohorts produces the reference dataset for the next.

The third operational anchor is **integrated student work** — the studio, practicum, and applied-project model that ensures every LENS graduate has, in addition to coursework, a portfolio of capability engineering deliverables produced inside real operational systems. The Summer 2025 PISTA incubator (Section VII) is the most recent example: a JHU/APL cohort of forty-four students collaborated across projects, with nine student fellows specifically shipping a working CBRNE (chemical, biological, radiological, nuclear, and explosives) decision-support prototype to DTRA leadership in ten weeks, with three full redesigns inside that window. The cases in this book are not the only operational record the program produces. Every graduate carries an addition to it. The credibility of the program is, in the end, the credibility of the artifacts its graduates have produced in the field.

VI · WHY THE SCHOOL OF EDUCATION

A reasonable reader asks: if learning engineering is engineering, why does this program live in a School of Education rather than a School of Engineering? The answer is that the binding constraint on the discipline is not the engineering, which is mature. It is the learning sciences, the educational measurement, the implementation science, and the equity-and-governance machinery — which live in Schools of Education and which the cases in this book establish as the load-bearing elements when capability interventions actually scale.

The learning sciences are the discipline that knows how skill, knowledge, and judgment develop, decay, and transfer across contexts. CRM was not an engineering achievement; the engineering of the cockpit was already mature in 1976. CRM was a **learning** achievement — a redesign of how aircrews learn to operate as a coordinated team, drawn from psychological research on group performance under stress. The same is true of TeamSTEPPS, the WHO Checklist's pause protocol, INPO's accreditation regime, and the Rickover oral examination tradition. The interventions that move the curves are interventions in how capability is learned, retained, and transferred — not in how the hardware is built.

Educational measurement is the discipline that knows how to construct the instruments that determine, validly and reliably, whether a person can do something. The cases in this book that turn on measurement failures — the VA wait-time scandal, Atlanta Public Schools, LIBOR, Wells Fargo, the F-35 readiness reporting — are measurement-instrument failures at heart. The capability to design measurement instruments that survive contact with the institutions measuring themselves is an educational-measurement competence. It lives in Schools of Education.

Implementation science — the field that studies the gap between evidence and practice — was developed largely in public-health schools and schools of education. Its frameworks (Fixsen et al., Aarons et al., Damschroder et al.) are the operational backbone of any serious capability-engineering effort, and Schools of Education have been their institutional home for two decades. A School of Engineering can teach the artifact. A School of Education teaches the artifact, the measurement instrument that decides if the artifact works, and the implementation pathway by which the artifact reaches scale. The casebook's central argument — that capability is a system parameter — requires all three.

Equity, governance, and ethics in education at scale are also, evidently, education problems. The inBloom, A-Level, COMPAS, Robodebt, and educational-predictive-analytics cases in this book are not failures of engineering competence. They are failures of the disciplines that know how to engineer measurement instruments, credentialing systems, and decision regimes that do not encode structural inequity. Those disciplines live in Schools of Education. The School of Engineering can build the model. The School of Education is where the construct gets validated, the consent framework gets designed, and the equity audit gets conducted.

The case for housing LENS in the Johns Hopkins School of Education is, finally, a case for treating capability engineering as the applied form of the educational sciences — engineering applied through learning, measurement, and equity, not engineering applied to learning as content. That is a real distinction. Schools of Engineering routinely build artifacts; the cases in this book show that the artifacts then need disciplines that those schools are not organized to provide. The placement is not a default. It is the shortest path between the discipline as it is and the discipline as it needs to be.

THE RECORD AT JOHNS HOPKINS

The argument above would be theoretical if Johns Hopkins did not already have an operating record. It does — and the deeper, longer half of that record is the School of Education's. The SoE has been studying the learner, designing the assessment, and engineering the intervention since long before the discipline of learning engineering had its current name. The disciplines this casebook argues are load-bearing — learning sciences, educational measurement, design-based research, implementation science, and equity-and-governance work — are disciplines the school's faculty have published in, taught, and applied at scale, in K–12, higher-education, clinical, military, and corporate settings, for decades.

Three durable lines of work make the point. The Center for Talented Youth, founded in 1979, has built one of the longest continuous evidence bases in U.S. education on the identification, assessment, and accelerated instruction of academically advanced learners; its measurement and program-evaluation work has been the empirical foundation of an entire sub-discipline. The Center for Research and Reform in Education has, since 2004, run large-scale evaluations of educational programs against rigorous evidence standards (Every Student Succeeds Act tiers, What Works Clearinghouse review), publishing the Best Evidence Encyclopedia and synthesizing what is known about which interventions actually work for which learners under which conditions. The Institute for Education Policy carries the same evidence-first orientation into the governance, curriculum-policy, and reform-design work that the case studies in Chapter 5 of this book name as the binding capability in modern education systems. These are not adjacent to the LDT program; they are the institutional environment that taught the program how to be rigorous.

The Learning Design and Technology program sits inside that environment and inherits it. For two decades the LDT M.Ed. and EdD tracks have graduated cohorts who entered as practitioners and left as designers and evaluators of learning systems — instructional designers and architects of learning experience, learning-analytics specialists, technology-mediated-learning researchers, professional developers, evaluators, and program leaders across the same K–12, higher-education, military, clinical, and corporate settings the cases in this book draw on. The program's intellectual core has always been the learning-engineering question — what does the system have to do in order to produce reliably capable people? — and its pedagogical core has always been design-based research: put the artifact in front of real learners, measure what happens, iterate, and publish what was learned about both the artifact and the learner. LENS is the formalization of that orientation as a specialization, not a departure from it.

Alongside that academic record runs a complementary thread of applied collaboration. Faculty and graduates of the LDT program have, over the past decade, partnered with operational organizations — including teams at the Johns Hopkins Applied Physics Laboratory — to study capability in real mission settings. Programs such as CIRCUIT (the workforce arm of the IARPA/MICrONS consortium, whose connectomics result appeared in Nature in 2025), the precision-medicine clinical-decision-trainer pilot reported in 2024, the DARPA AIE-supported TITAN work, and the DTRA-supported PISTA incubator in 2025 are useful proving grounds for the learning-engineering question because the deliverables are evaluated by sponsors who care whether the system worked. These collaborations have informed the program's case-based pedagogy and contribute several of the operational vignettes that follow. They are not the institutional home of the discipline; the School of Education is. They are how the school's disciplines meet operational settings, and how operational settings show up in the curriculum.

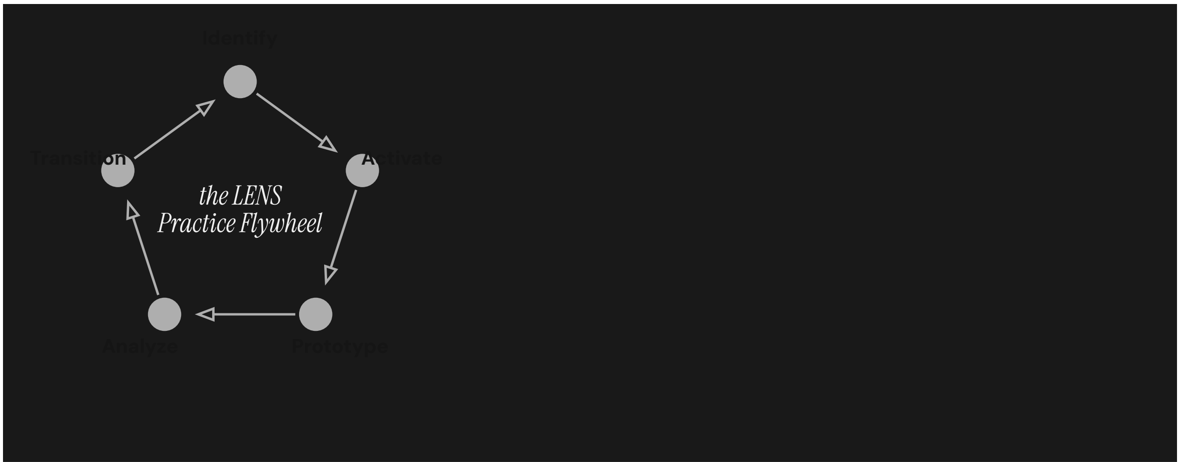
Other parts of the Johns Hopkins ecosystem do adjacent work the program draws on. The Armstrong Institute for Patient Safety and Quality at the School of Medicine — the institutional descendant of the Keystone ICU work foregrounded in Chapter 8 — is the most visible example: a continuing program of capability-engineering in clinical settings, with measurement and implementation methods that the casebook treats as canonical. The Bloomberg School of Public Health contributes the implementation-science tradition the program rests on, with departments and centers that have spent decades studying how interventions reach scale. The Whiting School of Engineering contributes human-systems engineering, AI methods, and cognitive-engineering research relevant to the human-AI teaming cases in Chapter 9. The Krieger School's Department of Cognitive Science and adjacent psychology programs contribute the learning- science methods the LDT program teaches. The Berman Institute of Bioethics contributes the governance-and-ethics framing that the cases in Chapter 5 require. None of these are formal partners in this volume; all of them are part of the institutional environment in which the discipline this casebook describes is recognizably at home.

That meeting is also embodied in the editors. One of us has spent a career in applied research environments thinking about how to make scientific and engineering teams reliably capable; the other has spent a career in the School of Education thinking about how to design, study, and evaluate the learning of learners. Both of us have, in practice, lived in the other half. The casebook is one product of that doubling. The discipline it describes is not new. The institutional home that lets it be taught at scale — the School of Education at Johns Hopkins, with its faculty, centers, doctoral programs, and the practitioner partnerships those carry — is where the work continues.

VII · PATHWAYS EVIDENCE · LE PILOTS

A program built on the claim that capability is engineerable should be able to point to the institutional evidence that it has done that work. LENS draws on a documented record of learning-engineering pilots — operated at the Johns Hopkins University Applied Physics Laboratory and across collaborating sites — that demonstrate the iterative LE process at operational scale before LENS itself enrolled its first cohort. The pilots span the same three domains the casebook treats as co-equal — science (instantiated here as connectomics and precision medicine), defense, and education — and they each carry independent evidence of the loop closing.

The Practice Flywheel



THE LENS PRACTICE FLYWHEEL — EACH ITERATION GENERATES EVIDENCE, ARTIFACTS, AND REFINED PRACTICE

The pilots share an operating cycle: **identify** a real challenge in an operational setting, **activate** a cohort of practitioners and the tools they need, **prototype** a working artifact under realistic constraints, **analyze** the evidence the prototype produces, and **transition** validated solutions to the sponsor or institution that asked the original question. Each iteration generates artifacts, evidence, and refined practice; each cohort inherits the prior cohort's tools and lessons. The flywheel makes the program compounding rather than serially episodic.

Pilot · CIRCUIT, MERIT, COMPASS — scientific and engineering impact at MICrONS / BRAIN CONNECTS scale

CIRCUIT — the **Cohort-based Integrated Research Community for Undergraduate Innovation and Trailblazing** — was built at JHU/APL as the best operational solution to a sponsor problem: the IARPA MICrONS connectomics program needed a proofreading workforce that could exercise expert biological judgment at scale, and CIRCUIT was the architecture that delivered it. The headline result is scientific. The trained cohorts contributed materially to the MICrONS reconstructions now published in **Nature**, with downstream contributions to FlyWire and H01 datasets, the BossDB and CAVE open-data platforms, and the NeuVue proofreading framework [23]. These are not auxiliary outputs of a workforce program. They are the foundational datasets and infrastructure on which contemporary connectomics depends.

The student training was a **byproduct** of the mission solution — but a deliberate, well-engineered byproduct that turned an operational requirement into a sustained opportunity to develop the next generation of computational neuroscientists. Doing well for the sponsor and doing good for the students compounded rather than competed. That structural alignment between scientific mission and human-capability development is the pattern LENS treats as the canonical positive case for how the Practice Flywheel operates at the boundary between science and workforce.

The model that produced those outputs was then **generalized into MERIT (Mentoring Exceptional Researchers to Innovate and Thrive)** — a replicable research-training framework built on holistic recruiting, the CIRCUIT eight pillars, the CCR competency framework, the hidden-curriculum literature, and the DoD KSAT alignment vocabulary [24]. MERIT operates across five developmental stages — Holistic Recruiting, Orientation and Research Foundations, Guided Research Practice, Independent Research with Mentorship, and Leadership / National Impact — and has been instantiated in computational neuroscience, AI, cybersecurity, threat assessment, and pandemic modeling. **COMPASS** is the layer of targeted mentor interventions that operates MERIT inside a given institutional setting — designed to help students uncover the **tacit and explicit knowledge gaps** that separate competent execution from independent research practice. COMPASS is not bound to a fixed workshop count. The COMPASS-Core curriculum is one well-documented instance, with an example set spanning orientation and SMART goal-setting, resilience and growth mindset, research navigation, networking and faculty engagement, scientific reading and writing, figures and oral presentations, metacognition and study skills, professional conduct and research ethics, STEM identity and belonging, and long-term career and funding strategy. The casebook treats the trio (CIRCUIT model, MERIT framework, COMPASS mentor interventions) as the load-bearing instance of the Practice Flywheel: requirements elicited from operational science, instrumented measurement of cohort progress, paired technical and cultural scaffolding, and an explicit iteration cycle that has now produced multiple cohorts and hundreds of student-authored contributions to the peer-reviewed scientific literature.

Pilot · TITAN (DARPA AIE)

TITAN — a DARPA Artificial Intelligence Exploration program — was implemented and explored by junior staff and student fellows when senior staff did not take it up, and produced both technical artifacts and a demonstrated proof of concept for the cohort-based incubator model the program now treats as a primary delivery mechanism. The pattern — talent identification, rapid prototyping under sponsor constraints, transition planning — is the same Practice Flywheel the casebook describes. TITAN is the earliest evidence that the model scales beyond the connectomics domain in which it was first piloted [25].

Pilot · PISTA (DTRA, Summer 2025)

PISTA — the **Perceptual Inference System for Tactical Agentic AI** — was a ten-week incubator delivering tactical AI decision support for the Defense Threat Reduction Agency's chemical, biological, radiological, nuclear, and explosives mission. Nine student fellows shipped PISTA, drawn from the broader Summer 2025 cohort of forty-four students at the JHU/APL AI Pathfinding Lab; all forty-four benefited from the shared cohort infrastructure, mentor pool, and cross-project collaboration even when working on other sponsor problems. The PISTA team delivered a working perception-reason-report agentic pipeline on a COTS stack (Gemini Flash, GPT-4o, Firebase) with deployed sensors on UGVs and small UAS. Three full redesigns inside ten weeks. Live demonstration to DTRA leadership. Follow-on funding under exploration. The pilot demonstrates the Practice Flywheel at compressed timescale in a defense context [26].

Pilot · ECHO (healthcare, culturally responsive provider training)

ECHO is an AI simulator for limited-English-proficiency patient encounters — a population of more than twenty-five million Americans whose access to safe care is structurally constrained by language barriers. The system pairs scenario-driven practice with rubric-based assessment and real-time coaching, designed for implementation inside existing medical-curriculum infrastructure. ECHO is the healthcare instance of the casebook's central argument: health equity is a **design commitment**, not an audit. The pilot shows the same flywheel operating in the clinical-training context.

Pilot · TeachMe EDDIE (methodology)

TeachMe EDDIE is the methodological instance of the pilot portfolio: an AI-augmented knowledge engineering system built around a 100-plus-node causal knowledge graph, a dual-axis learner model (declarative and procedural), and Mentor and Narrator agents bounded by a curated corpus. EDDIE operationalizes the difference between **retrieval** and **learning** — between an LLM that can produce a fluent answer and an instructional system that can move a learner from competence in one context to competence in another. Transfer is the north star. EDDIE is the LENS instance of the learning-engineering question that the rest of the dataset asks at scale.

Pilot · Precision-medicine community detection (clinical decision support)

Precision-medicine clinical decision tools represent the program's pilot in the application of evidence-based methodology to a high-consequence operational domain. Students and staff at APL contributed to community-detection methods for precision-medicine cohort identification — the kind of work that combines computational rigor with the clinical interpretive judgment LENS is organized to teach. The preprint is publicly available on medRxiv [27]. The same pattern recurs across the pilots: the technical artifact is necessary, the institutional scaffolding to use it under operational constraints is what makes it capability.

What the pilots demonstrate together

No single pilot would settle the LENS proposition. The cumulative record is what matters. Across pilots, the same set of LE practices recurs: explicit capability requirements; structured pedagogical scaffolding; instrumented measurement linked to operational outcomes; paired technical and cultural interventions; cohort-based delivery with deliberate inheritance of tools and lessons across cohorts; and an explicit Identify → Activate → Prototype → Analyze → Transition cycle. The same artifacts produced in those pilots — the rubrics, the prompts, the dashboards, the after-action reports, the published papers — become the curriculum material for the next cohort. The program is, in this sense, the institutional flywheel its case studies argue every high-consequence domain requires.

VIII · HOW TO READ THIS BOOK

Part I, "The Failure Modes," organizes one hundred cases under six recurring patterns: training gap, capability designed out, normalization of deviance, interface failure, governance failure, and knowledge loss. A seventh chapter — the Evidence Gap — anchors the measurement question. The taxonomy is not a theory. It is a finding: six categories that account for almost every well-documented case in the literature of preventable failure in complex sociotechnical systems, and that recur across healthcare, defense, education, energy, transportation, and government.

Part II, "What Works," collects the cases in which the capability parameter was engineered rather than left to drift. Every success case in the dataset shares a structural feature: a **paired intervention.** A technical artifact paired with a cultural authority. A measurement instrument paired with a governance body that listens to it. A curriculum paired with the institutional architecture to sustain it. Neither half alone moves the curve. The pair is irreducible. This is the strongest empirical pattern in the book and it directly informs the LENS pedagogical commitment to co-optimization across technical and human design.

Part III, "The Frontier," concerns the cases where the discipline is being asked to do work it has not done before — particularly the human-AI teaming cases that anticipate the next decade of capability engineering. The cases are deliberately less settled. The discipline is being asked to specify what good looks like before the catastrophe arrives. That is the work the program exists to teach.

Each case occupies a two-page spread. The left page tells what happened — narrative, evidence, attribution. The right page is the **Learning Engineering Lens:** a synthesis of what the case teaches, how the LENS curriculum addresses the pattern, and reflection questions designed for studio discussion. The casebook is built to be taught from, read straight through, or consulted as a reference. The curriculum crosswalk in the closing matter maps each case to the specific courses in the program that use it as a worked example.

IX · COURSE COVERAGE AND PEDAGOGICAL GAPS

The one hundred cases in this volume are mapped to the LENS courses they most directly inform. The distribution is itself diagnostic. The strongly covered topics are the ones the existing case literature most readily supports: bias and governance (LEN 7, 49 cases) and knowledge transfer (LEN 8, 45 cases); capability analysis (LEN 5, 36 cases) and evidence and measurement (LEN 4, 31 cases); human-AI teaming (LEN 2, 23 cases) and principles of learning engineering (LEN 1, 21 cases). These are not arbitrary numbers — they describe a dataset saturated with cases of **what goes wrong when ethics, governance, evidence, and institutional learning are not engineered.**

The thin coverage is also informative. The computational-methods topic (LEN 9, 10 cases) has the failure modes of algorithmic systems well-represented but the computational tools for capability instrumentation under-represented. The capstone (LEN 10, 19 cases) maps to fewer cases than the number of studio projects each LENS cohort undertakes. The systems-integration topic (LEN 3, 18 cases) and applied problem-solving topic (LEN 6, 13 cases) required focused retagging to reach defensible coverage; the work surfaced that operational systems-engineering content is **under-represented in the published case literature** relative to its curricular centrality, and that stakeholder-analysis and problem-framing have not historically been packaged as case material by other disciplines. These are pedagogical gaps the LENS studio sequence is organized to close — not by adding more failure cases, but by having students produce the missing artifacts in real operational settings and entering them into the dataset for the next cohort.

> *Capability is a designable, measurable, engineerable property of every complex system. The strongest argument for the discipline of learning engineering is the cumulative record of what its absence has cost. The strongest argument for its possibility is the cumulative record of what its presence has produced. Both records are in this book. The work of LDT and LENS at the Johns Hopkins School of Education is to add to the second record and shorten the first.*

HOW TO USE THIS BOOK

A field manual for capability engineering.

This book collects one hundred real incidents in which capability — what people could or could not do inside a complex system — determined whether the system worked. Some are failures: lives lost, systems wrecked, billions written off. Some are successes: deliberate interventions that engineered capability into the architecture of the work. Together they form an evidence base for the argument that capability is a designable, measurable property of every complex system.

Each case occupies a two-page spread. The left-hand page tells the story: what happened, what investigators found, and the documentary record. The right-hand page is the **Learning Engineering Lens** — what the case teaches about capability as a system parameter, how the LENS curriculum addresses the pattern, and a short set of reflection questions designed for studio discussion.

The Lens page also carries a **Who Builds This** note: the mix of expertise and tools a team would bring to engineer the fix, read off the case's failure modes. A Training-Gap case pulls in learning scientists and instructional designers; an Interface case, human- factors and interaction designers; a Governance case, policy, ethics, and measurement. Most cases need several at once, held together by domain experts and a learning engineer. No single discipline carries a case alone — reading the modes as a staffing list is itself a LENS skill, and what **you** bring is yours to decide.

READING THE TAXONOMY

Each case is tagged with one or more failure modes. Most cases involve several. The primary mode is listed first; contributing modes follow.

T Training Gap
Personnel were never taught what they needed to know.

D Designed Out
Human capability was deliberately removed from the system.

N Normalization of Deviance
Known risks were accepted as routine.

H Human-System Interface
Correct performance was made unreasonably difficult.

G Governance & Trust
Ethics, accountability, and stakeholder trust were afterthoughts.

K Knowledge & Institutional Memory
Organizational knowledge degraded or was never captured.

LENS COURSE CODES

Each case maps to one or more courses in the LENS specialization:

SHARED LDT FOUNDATIONS · ADDITIONAL CONTEXT

F1 — **Learning Sciences Studio.** How learning, motivation, and skill develop — applied to technology-mediated design.

F2 — **Critical Perspectives on Educational Technology.** How power, equity, and society shape — and are shaped by — learning tech.

LENS-DESIGNED COURSES

- **LEN 1** Principles of LE for Complex Systems
- **LEN 2** Human-AI Teaming and Adaptive Systems
- **LEN 3** Learning Engineering Systems
- **LEN 4** Evidence, Analytics, and Measurement
- **LEN 5** Human Capability Analysis and Requirements
- **LEN 6** Applied Problem Solving in LE
- **LEN 7** Bias, Risk, and Governance

LEN 8	Knowledge Transfer and Organizational Learning
LEN 9	Computational and AI Methods
LEN 10	Learning Engineering Project (capstone)

> *The strongest cases pair a catastrophic failure with a systematic capability intervention. Together they show both the cost of the gap and the tractability of the solution. The discipline that closes that gap is learning engineering.*

THE CASE MATRIX

One hundred cases at a glance

Gold numbers indicate failures and systemic conditions; teal numbers indicate paired-intervention successes and the open closing case.

1	USS Fitzgerald & McCain 2017	T·K·N	1·5·8
2	Boeing 737 MAX / MCAS 2018–19	D·T·H	1·2
3	Air France Flight 447 2009	T·H	1·5·2
4	Deepwater Horizon 2010	T·N·K	5
5	Three Mile Island 1979	T·H	1·5
6	Challenger & Columbia 1986/2003	N·K·G	1·8
7	Therac-25 1985–87	H·D·G	7·9
8	inBloom 2014	G	7
9	USMC INDOPACOM Training ongoing	T·K	5
10	Healthcare.gov 2013	K·T·G	7
11	V-22 Osprey 1991–	T·H·N	5
12	CRM + CAST 1981–	T·H·N	2
13	WHO Surgical Checklist 2008–	T·N	10
14	Keystone ICU (Pronovost) 2004–	T·N	4·10
15	Navy SWO Reform 2018–	T·K·N	10
16	INPO / Nuclear Training 1979–	T·K·G	8
17	Bhopal 1984	T·K·N·G	5·7

18	USS Vincennes 1988	H·T	2
19	Patriot Missile 1991	D·H·K	5·9
20	Grenfell Tower 2017	G·T·K·N	7
21	Summit Learning Rollout 2014–19	G·T·K	7·10
22	Tennessee Pre-K Study 2009–18	G·D	4·7·10
23	Korean Air Transformation 2000–	T·N	8
24	Toyota Andon Cord 1950s–	N·G	8
25	EHR/CPOE Implementation 2005–	H·D·G	7·9
26	F-35 Sustainment ongoing	T·K·D	5
27	TeamSTEPPS 2006–	T·N	8
28	Rickover Nuclear Navy 1954–	T·K·N	8
29	Uber ATG / Tempe 2018	T·N·G·H	2
30	Kegworth / BM 92 1989	T·H·K	5
31	Medical Errors (Makary) ongoing	T·H·N·K·G	4
32	VA Wait-Time Scandal 2014	G·K·N	4
33	Military Fratricide 1991–	T·H·K	5
34	ACGME Duty-Hour Reform 2003–17	T·K·N	5·4·10
35	UK A-Level Algorithm 2020	G·H·D	4·7
36	Australia Robodebt 2016–20	G·D·H	7·2
37	Algorithmic Bias in Ed ongoing	G·H·D	4·7·9
38	GIFT and the Adoption Gap 2012–	K·G·N	1·10·8
39	Georgia State Univ. 2012–	T·K	4

40	xAPI / TLA Gap ongoing	K·G	4·8
41	17-Year Implementation Gap ongoing	K·G·N	1·10·8
42	Cognitive Tutor 1990s–	T	1·4·9
43	Colgan Air 3407 2009	T	4·5·8
44	Asiana Airlines 214 2013	T·H	5·2
45	Mark 14 Torpedo 1941–43	T·K·G	10·7·8
46	Operation Eagle Claw 1980	T·K	5·8
47	Helios Airways 522 2005	T·H	5·2
48	AeroPerú 603 1996	H·T	5·2
49	Atlas Air 3591 2019	T·K	4·8
50	TransAsia 235 2015	T·H	5·2
51	Ford Pinto 1971–78	D·G	1·7
52	Takata Airbags 2008–23	D·G	4·7
53	GM Ignition Switch 2002–14	D·G	4·7·8
54	Mars Climate Orbiter 1999	D·K	5·8
55	Knight Capital 2012	D·K	5·9
56	Texas City BP Refinery 2005	N·T·K·G	4·7
57	Davis-Besse Reactor 2002	N·K·G	7·8
58	Mid Staffordshire NHS 2005–09	G·N·K	4·7
59	Sago Mine 2006	N·T·K	5·8
60	Upper Big Branch Mine 2010	N·G·K	4·7

61	Fukushima Daiichi 2011	N·G·K	7·8
62	Northeast Blackout 2003	H·K	2·8
63	Eastern 401 1972	H·T	5·2
64	Boeing 737 Rudder 1991, 1994	H·D	5·7
65	CrowdStrike Outage 2024	D·K·G	5·2
66	Petrov / 1983 False Alert 1983	H·T	2
67	TSB Bank IT Migration 2018	H·G	7·8
68	UK Post Office Horizon 1999–2015	G·H·K	7·2
69	Theranos 2003–18	G·D	4·7
70	Wells Fargo Accounts 2011–16	G·N	4·7
71	VW Dieselgate 2015	D·G	4·7
72	Cambridge Analytica 2014–18	G	1·7
73	Equifax Breach 2017	G·K	5·7
74	Hyatt Regency Walkway 1981	D·G	5·7
75	FIU Pedestrian Bridge 2018	D·G·N	5·8
76	Camp Fire / PG&E 2018	G·N·K	7·8
77	Texas Grid Freeze 2021	G·K·N	7·8
78	Saturn V Documentation 1972–	K	1·8
79	Boeing Starliner 2019–24	K·D	5·8
80	Ariane 5 Flight 501 1996	D·K·H	5·8
81	Tacoma Narrows Bridge 1940	D·K	1·8
82	Aliso Canyon 2015–16	G·N·K	7·8

Modes. T Training Gap · D Designed Out · N Normalization of Deviance · H Human-System Interface · G Governance & Trust · K Knowledge & Institutional Memory

LEN. The LDT / LENS courses each case is most relevant to. LEN 1 Foundations · LEN 4 Evidence & Measurement · LEN 10 Studio · LEN 5 Capability Analysis · LEN 7 Bias, Risk & Governance · LEN 2 Human-AI Teaming · LEN 8 Knowledge Transfer · LEN 9 Computational & AI Methods.

Parts. Part II — **What Works** — opens at Chapter 8 with the paired-intervention successes. Part III — **The Frontier** — closes the volume with the human-AI teaming cases and the open question at Case 100.

Chapter 1

Training Gap

They weren't taught what they needed to know.

Individuals and groups of individuals could no longer recognize that the processes designed to identify and communicate readiness were no longer working.

PARAPHRASING THE U.S. NAVY STRATEGIC READINESS REVIEW, 2017

CASE 1 DEFENSE 2017

USS Fitzgerald & USS John S. McCain

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT 17 sailors killed in two avoidable destroyer collisions in the Western Pacific within nine weeks

IN BRIEF

In the summer of 2017 two U.S. Navy destroyers of the forward-deployed Seventh Fleet collided with merchant ships nine weeks apart, killing seventeen sailors — seven on the Fitzgerald, ten on the John S. McCain. Their crews' seamanship, navigation, and console skills had eroded under a decade of relentless operational tempo and the self-study "training" that replaced the in-person school the Navy cut in 2003. Investigations by the Navy and the NTSB judged both collisions avoidable and traced them to insufficient training and weak oversight; on the McCain, a confusing touch-screen helm let a watch team believe it had lost steering it never actually lost. The case is this book's canonical training gap: stated readiness and real readiness diverged for years, and the system could no longer see the difference.

BACKGROUND

The destroyers belonged to the forward-deployed Seventh Fleet in Japan, kept at the fleet's highest tempo — where training was the first thing spent. Unlike home-ported ships, which rotate through a dedicated work-up cycle before deploying, the forward-deployed force was expected to be continuously available, so there was rarely a quiet stretch in which to recover a lapsed qualification. In 2003 the Navy replaced its in-person Surface Warfare Officers School course with self-study CD-ROMs, "SWOS in a Box."[0] By 2017 the GAO found 37% of the Japan-based ships' warfare certifications — including basic seamanship — expired, a fivefold rise since 2015, with the lapses routinely waived rather than fixed.[1]

WHAT HAPPENED

On 17 June 2017 the Fitzgerald, the give-way vessel in a busy shipping channel, took no early action and was struck by the container ship ACX Crystal off the coast of Japan; the sea poured into a berthing compartment where the crew slept, and seven sailors drowned before they could escape.[2] Nine weeks later the John S. McCain collided with the tanker Alnic MC near Singapore, killing ten: while shifting throttle control between consoles, a watchstander unknowingly handed off steering to another station, the ship turned across the strait's traffic, and no one on the bridge understood the touch-screen helm well enough to see what had happened.[3]

THE INVESTIGATION

The Navy's Comprehensive Review (2017) judged both collisions avoidable, citing failures in basic seamanship, navigation, and operating the ships' own equipment.[4] The NTSB found the McCain's probable cause to be a lack of Navy oversight that produced insufficient training and inadequate bridge procedures,[5] and faulted the design of the touch-screen helm, installed to modernize the bridge, whose blending of steering and throttle made an unintended transfer of control easy to trigger and hard to notice — a trap waiting for an under-trained crew.[6]

THE CAPABILITY GAP

The gap was invisible from inside. The Strategic Readiness Review (2017) described risks that "accumulated over time and did so insidiously," the system no longer able to see that the processes meant to surface shortfalls had themselves failed.[7] Each individual waiver was locally defensible — a deadline met, a deployment kept — but in aggregate they hollowed out the force while every readiness dashboard still reported green. Stated and actual readiness had diverged for a decade; the cost arrived as seventeen lives, not a failed inspection.

AFTERMATH & REFORM

The reforms were the deepest in a generation: the in-person officer pipeline was rebuilt and lengthened, a Ready-for-Sea Assessment and a forward-readiness command stood up, circadian watchbills adopted to fight fatigue, and the touch-screen helm slated for replacement by a conventional wheel and throttle.[8] Each change conceded that the training and the interface had been real requirements all along — and that trading them for tempo had only moved the cost onto two ships full of people.

SURFACE WARFARE OFFICER TRAINING, 2003 → 2017

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THE LEARNING ENGINEERING LENS

> *The risks that were taken in the Western Pacific accumulated over time and did so insidiously.*
>
> U.S. NAVY STRATEGIC READINESS REVIEW, 2017

LE INSIGHT

Fitzgerald/McCain is the canonical Training Gap case because the gap was invisible from inside the system. Operational tempo and self-study "training" combined to produce a fleet whose stated readiness and actual readiness diverged for more than a decade. Seven sailors died on Fitzgerald; ten on McCain. The cost of the divergence was paid in lives long after it could have been measured in dollars or inspections.

LENS APPROACH

LENS treats this as the worked example of LEN 5 (Human Capability Analysis and Requirements) interacting with LEN 4 (Evidence and Measurement). Students reconstruct the capability requirements for underway watchstanding from operational analysis and design the evidence system that would have surfaced the gap before the collisions. LEN 8 covers the institutional dynamics of deferred training.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify watchstanding proficiency as a measured deliverable — define the competency, instrument it, and gate qualification on demonstrated skill rather than sea time.
- Keep human-factors review in the procurement loop: validate that any interface change (e.g., the bridge console) is tested against operator performance before it is fielded.
- Design the readiness signal to report demonstrated capability, not self-attested course completion.

IN OPERATION / AFTER

- Audit the gap between reported and actual readiness with independent assessment (the Ready-for-Sea model) — with authority to pull a unit offline.
- Protect training time against operational tempo so the capability cannot erode silently when schedules tighten.
- Track leading indicators — qualification currency, near-miss reports — so divergence is visible before it is paid for in lives.

REFLECTION QUESTIONS

1. The Navy replaced classroom and simulator instruction with CD-ROM self-study in 2003. What measurement would have detected the capability shortfall before 2017?
2. The Strategic Readiness Review found the readiness-reporting system had itself stopped working. Design a capability-evidence pipeline that would not normalize its own gaps.
3. Identify a capability in your organization that is certified by completion (a checked box) rather than demonstrated performance. What measure would convert it to evidence — and who would hold the authority to act on a red signal?

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TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

LENS COURSES LEN 1 LEN 5 LEN 8 LEN 3

CASE 3 AVIATION 2009

Air France Flight 447

T Training Gap H Human-System Interface

IMPACT 228 killed; the deadliest accident in Air France's history

IN BRIEF

Air France Flight 447, an Airbus A330, fell into the South Atlantic on 1 June 2009, killing all 228 aboard — the deadliest accident in the airline's history. At cruise altitude the pitot probes iced over, the airspeed readings failed, and the autopilot handed the jet to a crew that had never trained to fly it by hand in that regime. The pilot flying held the nose up into a full aerodynamic stall and never recognized it; the aircraft descended for nearly four and a half minutes. The BEA traced the loss to the airspeed failure, the crew's inappropriate inputs, and a training system that taught stall prevention at low altitude but never stall recovery at altitude — a gap between the trained envelope and the operational one that reshaped global pilot-training rules.

BACKGROUND

AF447 left Rio for Paris on 31 May 2009 with 228 aboard an Airbus A330 — a highly automated jet with an excellent safety record. Its route crossed the equatorial storm band, and it carried a known vulnerability: pitot probes prone to brief icing at altitude. A replacement program was underway, but the accident aircraft had not yet been modified, so the vulnerability the program existed to close was still live on the very jet that crossed the storm band — the retrofit recognized as necessary but not yet fitted where it mattered.[0]

WHAT HAPPENED

At 35,000 feet the probes iced, the airspeed readings went invalid, and the autopilot and autothrust disconnected into a degraded control law. The pilot flying responded with sustained nose-up input; the jet climbed, stalled, and never recovered, falling some 38,000 feet into the ocean in about four and a half minutes. The stall warning sounded, then cut out at extreme angle of attack and resumed when the nose dropped — warning against the one input that would have begun a recovery, so that the cue meant to guide the crew instead punished the correct action and rewarded the fatal one, an inversion no amount of hand-flying instinct could resolve in the time available.[1]

THE INVESTIGATION

The BEA's 2012 final report named a chain: airspeed loss from pitot icing, inappropriate crew inputs, and the crew's failure to recognize and recover from the stall.[2] The pilots were not incompetent — they were outside anything their training had prepared them for, and the BEA said so: "the conditions under which pilots are trained and exposed to stalls... did not result in reasonably reliable expected behaviour patterns." The finding reframed the loss from a question of individual airmanship to one of training design: the crew had been drilled in a regime that never produced the responses the emergency demanded.[3]

THE CAPABILITY GAP

The gap was precise. Airlines trained stall recovery at low altitude; the simulators of the era could not reproduce a high-altitude stall, so crews never practiced the situation that arrived. Years of reliable automation had also let hand-flying skills atrophy, so when the autopilot handed back control the crew met an unfamiliar regime with rusty manual skills at the worst possible moment. The trained envelope and the operational envelope had quietly diverged, and AF447 fell into the gap between them — a gap no one had been positioned to see widen.[4]

AFTERMATH & REFORM

The BEA’s recommendations reached far beyond one airline: the pitot probes were replaced, and the report pressed for training in manual high-altitude flight, stall recovery, and unreliable-airspeed handling.[5] Regulators then made Upset Prevention and Recovery Training mandatory for airline pilots, closing at the regulatory level the gap that had been invisible at the airline level — moving the fix from a single carrier’s discretion to a binding standard every airline had to meet. The crew performed exactly as trained; the training was the wrong training, and only a system-wide mandate could keep that mismatch from recurring elsewhere.[6]

54 SECONDS: STALL WARNING SOUNDED, RESPONSE NEVER CAME

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THE LEARNING ENGINEERING LENS

> *The crew never understood that they were stalling and consequently never applied a recovery manoeuvre.*
>
> BEA, FINAL REPORT ON AIR FRANCE FLIGHT 447, JULY 2012

LE INSIGHT

AF447 is the canonical case of training that matched one envelope of operation perfectly and another not at all. Stall **prevention** training was excellent. Stall **recovery from cruise altitude** training did not exist because the simulators could not produce it. The pilots performed exactly as trained. The training was the wrong training.

LENS APPROACH

LENS uses AF447 in LEN 5 to teach the difference between the trained capability envelope and the operational capability envelope, and in LEN 2 to introduce human-AI handoff as a capability problem: the autopilot disconnection was a transition the crew had been trained to avoid rather than to handle. The case maps directly to canonical problem type three — Capability Cliff at Automation Boundary.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Define the operational capability envelope explicitly — including degraded-automation and high-altitude regimes — and train to its edges, not only to the routine center.
- Engineer warning logic so that cues never punish the correct recovery input; validate stall-warning behavior across the full angle-of-attack range before fielding.
- Treat manual-flying proficiency in degraded modes as a measured deliverable, not an assumed residual of automated operation.

IN OPERATION / AFTER

- Audit recurrent training against the actual operational record so the trained envelope cannot silently diverge from the regimes crews encounter.
- Monitor for skill atrophy where reliable automation reduces hands-on exposure, and refresh manual competence before it erodes.
- Sustain the reform at the regulatory level (mandatory upset recovery) so a fix proven in one carrier propagates to all rather than lapsing.

REFLECTION QUESTIONS

1. The simulators of 2009 could not produce high-altitude stall behavior. What is the equivalent gap in your domain — the operational regime your training environment cannot reproduce?

2. Design the recurrent-training curriculum that would have caught the AF447 gap. Be specific about cost, evidence, and what makes the curriculum falsifiable against the operational record.

3. The autopilot handed control to a crew at the one moment it was least prepared to take it. Identify an automation-to-human handoff in your domain that occurs precisely when the human is least ready, and design the trigger or warning that would change that.

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TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs

LENS COURSES LEN 1 LEN 5 LEN 2

CASE 5 ENERGY 1979

Three Mile Island

T Training Gap H Human-System Interface

IMPACT Partial meltdown of a Babcock & Wilcox PWR; minimal off-site dose; most serious accident in U.S. commercial nuclear history; catalyst for industry-wide reform

IN BRIEF

On 28 March 1979 a small cooling fault at Three Mile Island's Unit 2, a Babcock & Wilcox reactor near Harrisburg, escalated into a partial core meltdown — the most serious accident in U.S. commercial nuclear history. A relief valve stuck open while a control-room light reported it closed, and operators trained for dramatic design-basis ruptures misread the slow, ambiguous cascade and cut the cooling the core needed. A nearly identical near-miss at Davis-Besse eighteen months earlier had never been propagated to the fleet. The Kemeny Commission concluded the fundamental problems were people-related, not equipment. Off-site radiation was minimal, but the accident produced a system of reform — most enduringly INPO — making it the book's paired example of a failure that engineered lasting capability.

BACKGROUND

Three Mile Island's Unit 2, a Babcock & Wilcox reactor near Harrisburg, was run by operators trained for large, fast, design-basis ruptures — not the slow, ambiguous small-break sequence that actually came.[0] It should not have surprised anyone: eighteen months earlier a nearly identical stuck-open relief valve had occurred at Davis-Besse, but neither the utility, the vendor, nor the NRC grasped its significance or pushed the lesson out to the fleet — so the precise sequence that would later threaten the core had already been demonstrated and then quietly filed away rather than turned into a drill or a warning.[1]

WHAT HAPPENED

On 28 March 1979 a minor secondary-cooling upset tripped the reactor; a relief valve opened and then stuck open, draining coolant. The control-room light reported the **command** to close it, not its actual position, so the panel read "closed" while the valve stayed open. Misreading the rising pressurizer level, the operators throttled back the high-pressure injection the starving core depended on — taking the one action that turned a recoverable upset into a meltdown, precisely because the instrument they trusted was telling them a state that was not true. About half the core melted; off-site radiation, as it turned out, was minimal.[2]

THE INVESTIGATION

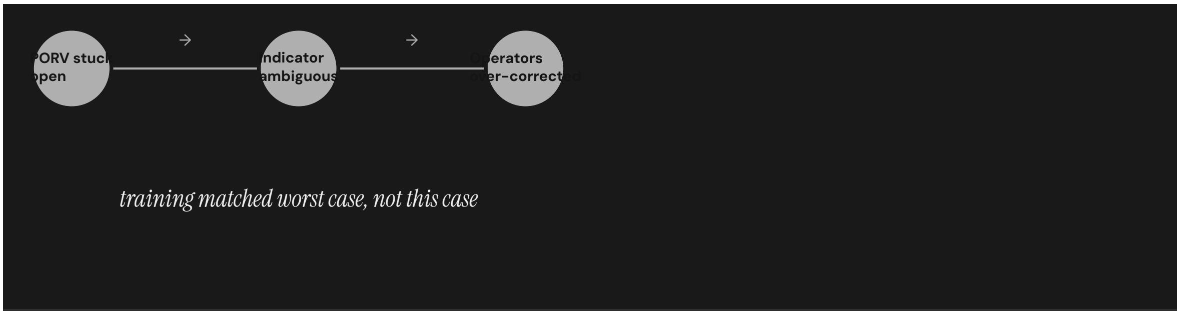
The Kemeny Commission (October 1979) inverted the industry's assumptions: "the fundamental problems are people-related problems and not equipment problems." The equipment was good enough that, but for the human failures, the accident would have been minor. The criticism reached past the operators to management, the utility, and the NRC — an institution-wide belief that serious accidents were effectively impossible, a complacency that had shaped training priorities, staffing, and oversight long before the relief valve ever stuck, so that the operators inherited a posture they had not chosen.[3]

THE CAPABILITY GAP

The gap was not intelligence but the right capability for the event that arrived. Operators drilled on design-basis ruptures had no model for an ambiguous cascade, and a control room reporting commands rather than states made correct diagnosis nearly impossible — the interface and the training were mismatched to the failure that actually arrived. Beneath that lay a second failure: the capacity to **learn** — to turn Davis-Besse into fleet knowledge — had itself broken down, so the industry kept making the same diagnosis blind, which is why the case resists any single-cause reading.[4]

AFTERMATH & REFORM

TMI produced not another accident but a system of reform. Within nine months the industry created INPO (December 1979) to set standards and force the sharing of operating experience the Davis-Besse failure had lacked.[5] The NRC overhauled licensing, required plant-referenced simulators and resident inspectors, and tied training to a national standard so that diagnosing ambiguous transients — not just textbook ruptures — became something crews practiced on equipment that behaved like their own plant. Each measure attacked one of the failures the accident had exposed: the training gap, the broken learning channel, and the gulf between what indicators displayed and what crews inferred. TMI is paired later with INPO (Case 16) as the book's strongest argument that failure can engineer durable reform.[6]



A MINOR COOLING FAULT, AN AMBIGUOUS INDICATOR, A MELTDOWN

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THE LEARNING ENGINEERING LENS

> *The fundamental problems are people-related problems and not equipment problems.*
>
> KEMENY COMMISSION REPORT ON THREE MILE ISLAND, 1979

LE INSIGHT

TMI is the textbook moment when an industry discovered that its capability assumptions did not survive contact with operational reality. Training to design-basis events left the operators blind to the genuinely ambiguous failures that complex systems actually produce. The human element is not a residual risk but the dominant one.

LENS APPROACH

LENS uses TMI in LEN 1 as the foundational problem-framing case: when trained scenarios and operational scenarios diverge, the divergence is the real risk. LEN 5 examines the control-room interface and the gap between what indicators displayed and what operators inferred; the pairing with INPO (Case 16) is the Failure-to-Reform arc this book treats as its strongest argument.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design control-room indicators to report actual system state, not the command issued, so operators diagnose from truth rather than intent.
- Build training around ambiguous, slow-onset cascades — not only dramatic design-basis ruptures — using simulators that mirror the operators' own plant.
- Engineer a learning channel that propagates precursor events across the fleet as drills, so a near-miss at one site becomes practiced knowledge everywhere.

IN OPERATION / AFTER

- Audit whether near-misses are actually reaching operators as changed procedure and training, and force-share operating experience through an independent standards body (the INPO model).
- Sustain plant-referenced simulators and resident oversight so diagnostic skill on ambiguous transients does not decay back to textbook drills.
- Monitor the institutional assumption that serious accidents are impossible, and treat its persistence as a measurable readiness risk.

REFLECTION QUESTIONS

1. TMI operators were trained for worst-case scenarios but failed in an ambiguous one. What is the equivalent training gap in your domain between the trained case and the messy case?
2. The Kemeny Commission called the human element the dominant risk. What evidence would you need to demonstrate the same conclusion in your own domain?
3. The Davis-Besse precursor occurred eighteen months earlier but never reached the operators who needed it. Design the channel in your domain that would turn a near-miss at one site into a drill at every other before the second event arrives.

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LENS COURSES LEN 1 LEN 5

CASE 9 DEFENSE ONGOING

Marine Corps Training in the INDOPACOM AOR

T Training Gap K Knowledge & Institutional Memory

IMPACT Structural readiness gap in DoD's stated top-priority theater

IN BRIEF

The 2022 National Defense Strategy names the Indo-Pacific the Pentagon's priority theater and China its "pacing challenge" — yet the theater so designated has among the least mature live-training infrastructure in the force. For nearly a decade the U.S. Marine Corps has been unable to meet its training requirements at Indo-Pacific ranges, papering over the gap with rotations back to U.S. ranges, virtual substitutes, and multinational exercises pressed into proxy duty. In May 2024 the GAO documented the decade-long shortfall and urged the Corps to analyze its unmet requirements and build a remediation plan. The case is this book's live, ongoing counterpart to its historical failures: a fully recognized, repeatedly documented capability gap that declared priority has not closed, because engineered priority — ranges, basing, certified units — is slow and expensive.

BACKGROUND

The 2022 National Defense Strategy names the Indo-Pacific the priority theater and China the "pacing challenge."[0] Set against that is an awkward fact: the theater so designated has among the least mature live-training infrastructure in the force — the place called most important is, where forces can actually rehearse, one of the least built out, so the strategy's top priority and the physical means to prepare for it point in opposite directions, a contradiction the briefings do not resolve.[1]

WHAT HAPPENED

The failure is a condition, not an event. For nearly a decade the Marine Corps has been unable to meet its training requirements at Indo-Pacific ranges, papering over the shortfall with rotations back to U.S. ranges, virtual substitutes, and multinational exercises pressed into proxy duty. The workarounds keep units partially trained; the structural gap does not close, because each substitute buys a single cycle of readiness without building the ranges, basing, or instrumented airspace that would let the theater train its own force — treating a permanent shortfall as a series of temporary ones.[2]

THE INVESTIGATION

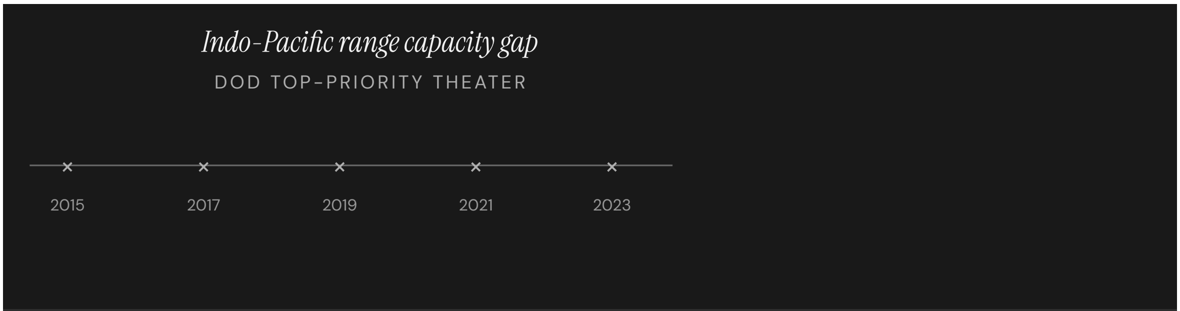
In May 2024 the Government Accountability Office documented the unmet requirements, the decade over which they had gone unmet, and the workarounds standing in for the real thing, and recommended the Corps formally analyze its unmet requirements and build a remediation plan for Indo-Pacific ranges; the Department concurred.[3] It was not the first warning — GAO has pressed the same readiness recommendations for years, and flagged the gap between overseas training and readiness reporting two decades ago, so the 2024 finding documents not a fresh discovery but the durability of a shortfall that survived repeated diagnosis, concurrence, and the passage of time without being engineered away.[4]

THE CAPABILITY GAP

What makes INDOPACOM diagnostic is the inverse correlation between stated and engineered priority. A theater can be named the pacing challenge in every briefing while its training infrastructure stays short for ten years, because capability follows where construction and dollars flow, not where strategy points. Declared priority is cheap; engineered priority — ranges, instrumented airspace, basing, certified units — is slow and expensive, and the gap between them is the real measure of intent. A strategy document can be rewritten in a season; a range complex takes years of construction and budget, so the decade-long persistence of the shortfall says more about resourcing than any reassertion of priority could.

AFTERMATH & REFORM

The reform remains mostly prospective: the GAO recommendations were open as of 2024, an analysis and a funded remediation plan still to come.[5] Whether the gap closes will be decided not by another strategy document but by whether the recommendation becomes programmed ranges, dollars, and schedule — converting a concurrence on paper into construction and certified units that a future review could actually measure. It sits at the front of this book as the live counterpart to its historical failures — a gap fully recognized, repeatedly documented, and still not engineered away.



DECADE OF UNMET TRAINING IN DOD'S PRIORITY THEATER

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THE LEARNING ENGINEERING LENS

> *Without meeting these requirements, the Marine Corps cannot ensure that its forces will be ready for combat.*
>
> PARAPHRASING GAO-24-107463, MILITARY READINESS, 2024

LE INSIGHT

INDOPACOM illustrates the difference between declared priority and engineered priority. A theater designated as DoD's pacing challenge while the training infrastructure to operate in it remains structurally short is not a resourcing oversight — it is a capability architecture failure. Capability follows where resources actually flow, not where briefings name as critical.

LENS APPROACH

LENS uses INDOPACOM in LEN 5 to teach the gap between stated requirements and engineered requirements, and in LEN 8 to examine the organizational dynamics that allow declared priorities to coexist with unfunded capability gaps for a decade. The case is also a live test for any student's claim about how to engineer cross-service capability at scale.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Tie any declared-priority designation to a funded engineering plan — ranges, basing, instrumented airspace, certified units — so strategy and resourcing cannot diverge unnoticed.
- Treat live-training infrastructure for a priority theater as a programmed deliverable with schedule and budget, not a workaround filled by rotations and exercises.
- Define the capability requirement for the theater explicitly, so the gap between what is needed and what is built is visible at the point of decision.

IN OPERATION / AFTER

- Audit declared versus engineered priority annually — measure where construction and dollars actually flowed against where strategy named as critical.
- Track recommendation status to closure with an authority that can escalate when a concurrence does not become programmed work.
- Monitor whether substitute training (rotations, virtual, multinational) is masking a structural shortfall rather than retiring it.

REFLECTION QUESTIONS

1. In your domain, what is the gap between **declared** priority and **engineered** priority? How would you measure it?
2. Construct the capability requirements artifact for a theater you do not currently operate in. What would it cost, and who would sign for it?
3. GAO documented this shortfall for years, the Department concurred, and the gap stayed open. Design the accountability mechanism that would convert a concurrence into programmed dollars and schedule, with a signal that fires when the plan slips.

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LENS COURSES LEN 5 LEN 8

CASE 26 DEFENSE ONGOING

F-35 Sustainment & Maintainer Shortage

T Training Gap K Knowledge & Institutional Memory D Designed Out

IMPACT **Fleet mission-capable rate about 55% (March 2023), far short of program goals; lifecycle cost exceeds $1.7T, with ~$1.3T in operating and support; maintainer, technical-data, and depot shortfalls are the binding readiness constraint (GAO-23-105341)**

IN BRIEF

The F-35 is the most expensive weapons program in history — roughly 2,500 jets planned and a lifecycle cost above $1.7 trillion, about $1.3 trillion of it in operating and support. The hard part was never the airplane but keeping a global fleet ready, and that half has lagged from the start. As of March 2023 the fleet's mission-capable rate was about 55%, far short of goal, with more than 10,000 components in the repair queue and depots behind schedule. GAO traced the shortfall to maintainer shortages, the military's lack of access to technical data, and contractor dependency, and urged a full reassessment of the sustainment strategy. It is the book's cleanest case of a platform fielded faster than the capability infrastructure to sustain it.

BACKGROUND

The F-35 is the most expensive weapons program in history: the Pentagon plans to buy nearly 2,500, at a lifecycle cost exceeding $1.7 trillion — roughly $1.3 trillion of it not the aircraft but the decades of operating and sustaining them.[0] The flyaway jet is the finite part; keeping a global fleet ready — maintainers, technical data, depots — is the open-ended part, and the part that lagged, because the cost that dominates the program is not buying the aircraft but the decades of sustaining them, the very work that received the least attention as the jets rolled off the line.

WHAT HAPPENED

The failure is a standing condition. As of March 2023 the fleet's mission-capable rate was about 55%, far short of goal; more than 10,000 components waited in the repair queue, and the depots averaged about 72 days per repair while still behind schedule in standing up the capacity to do the work at all — a backlog and a turnaround time that compound, since parts stuck in the queue keep jets grounded and the under-built depots cannot clear the queue fast enough to recover the mission-capable rate.[1]

THE INVESTIGATION

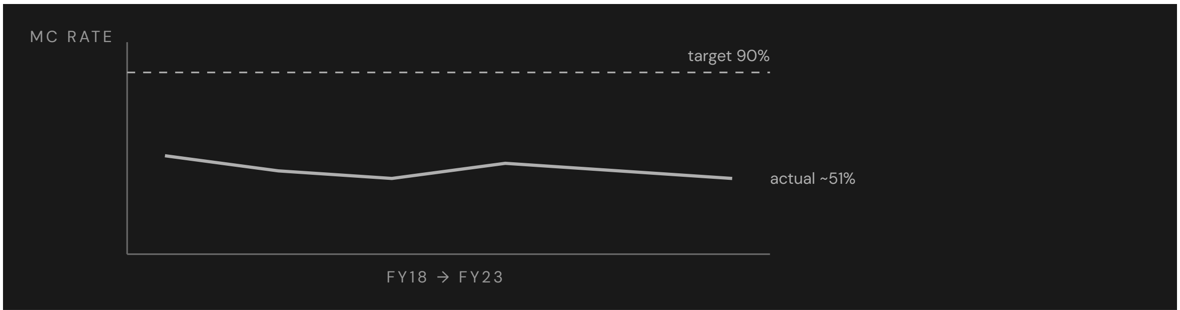
GAO's September 2023 review was bluntly titled: the Department and services **need to reassess the future sustainment strategy**. It traced the shortfall to maintainer shortages, the military's lack of access to the technical data needed to do its own repairs, and the resulting dependence on the prime contractor.[2] None of it was new — GAO has repeated the same diagnosis year after year, through a troubled logistics-software backbone and slow progress, even as procurement continued and readiness stayed flat, so the program kept buying more jets it could not fully sustain while the same three shortfalls were named in review after review without being closed.[3]

THE CAPABILITY GAP

The F-35 is the cleanest modern case of a platform fielded faster than the capability infrastructure to sustain it. Aircraft arrived on schedule; the maintainers, technical data, and depots were treated as follow-on costs rather than deliverables that had to field with the jets. The hardest part is the data: much of what is needed to repair the aircraft stayed controlled by the contractor, so the services cannot freely write procedures, qualify depots, or compete the work, which locks the fleet into a single source for the knowledge needed to keep it flying — merely expensive in peacetime, dangerous in war, when a contractor-dependent sustainment chain is exactly the kind of bottleneck an adversary would seek to exploit.[4]

AFTERMATH & REFORM

GAO urged a full reassessment of the sustainment strategy — government access to technical data, depot capacity, and a maintainer pipeline — rather than more patching.[5] Those recommendations remain a work in progress, with later reviews showing costs still rising and readiness still below goal. The F-35 sits in this book as the live argument for treating capability infrastructure — people, data, the means to sustain them — as a fielding gate, not an afterthought. The bill for skipping that gate does not disappear; it compounds, arriving as grounded jets, a swelling repair queue, and a sustainment chain the services cannot control rather than as a line item caught early enough to fix cheaply.



MISSION-CAPABLE RATE VS. FLEET TARGET

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THE LEARNING ENGINEERING LENS

> *Organizational-level maintenance has been affected by a number of issues, including a lack of technical data and training.*
>
> PARAPHRASING GAO-23-105341, F-35 AIRCRAFT, 2023

LE INSIGHT

F-35 is the live, current example of fielding a platform faster than its capability infrastructure can be built. The aircraft is the easy part; the maintainers are the hard part. A decade of program schedules treated maintainer training as a follow-on cost, not a fielding gate. The fleet is now operating at half of its design readiness, and the program of record is more than a trillion dollars over its 2018 estimate.

LENS APPROACH

LENS treats the F-35 in LEN 5 as the canonical case of **Capability- System Misalignment at Transition**, the program's first canonical problem type. Students design the capability infrastructure that should have accompanied each lot of aircraft delivered, including the training pipeline, technical-data deliverables, and depot establishment.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make sustainment infrastructure — maintainers, technical data, depot capacity — a contractual deliverable that fields with each lot of aircraft, not a follow-on cost.
- Secure government rights to the technical data needed to write procedures, qualify depots, and compete the work, so the fleet is not locked to a single source.
- Gate fielding on demonstrated sustainment capacity, not just aircraft delivery, so readiness is engineered alongside the platform.

IN OPERATION / AFTER

- Audit the mission-capable rate against repair-queue depth and depot turnaround, and act when the backlog signals a structural, not transient, shortfall.
- Track recurring GAO-style diagnoses to closure, refusing to let procurement outpace the sustainment fixes named year after year.
- Monitor contractor dependency as a wartime risk, and sustain a government maintainer pipeline and depot capacity that can hold under pressure.

REFLECTION QUESTIONS

1. Pick a current technology platform in your domain. Estimate the capability infrastructure that must field with it. What happens if half of that infrastructure is years behind the hardware?

2. The F-35 program treated maintainer training as follow-on cost. Design a fielding gate that would prevent that decision being available to a future program manager.

3. Much of the data needed to repair the F-35 stayed controlled by the contractor, foreclosing the government's ability to compete or qualify the work. For a platform in your domain, what data rights would you make a contractual deliverable up front, and how would you verify they were actually delivered?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · task analysis · design prototyping

LENS COURSES LEN 5 LEN 8 LEN 3

CASE 30 AVIATION 1989

Kegworth / British Midland 92

T Training Gap H Human-System Interface K Knowledge & Institutional Memory

IMPACT 47 killed and 74 seriously injured when the crew shut down the wrong engine

IN BRIEF

On 8 January 1989 British Midland Flight 92, a nearly new Boeing 737-400, crashed on the M1 motorway embankment near Kegworth, killing 47 and seriously injuring 74. After a fan blade fractured in the left engine, the crew shut down the right engine — the one still working. Reasoning from older 737s, on which the right engine fed the cabin air, they misattributed the smoke; new, harder-to-read electronic engine displays did not correct them, and the cabin crew who saw flames never told the flight deck. The pilots' mental model was right for the airplane they had flown the week before, but the brief conversion course never overwrote it. The AAIB issued 31 recommendations; Kegworth became the textbook case of capability degrading under system change.

BACKGROUND

On 8 January 1989 British Midland Flight 92, a nearly new Boeing 737-400 (G-OBME), left Heathrow for Belfast with 126 aboard. The -400 was a recent variant — bigger engines, a partly redesigned cockpit with new electronic engine instruments. The crew were experienced 737 pilots, but their conversion onto the variant had been brief and largely self-directed, the differences communicated as reading rather than drilled retraining — so the experience that made them confident on the type also seeded the prior model that would mislead them.[0]

WHAT HAPPENED

Climbing through 28,300 feet, a fan blade fractured in the left engine, filling the cabin with vibration and smoke. Reasoning from older 737s — on which the air-conditioning bleed drew from the right engine — the crew concluded the right engine was the source and throttled it back. The symptoms eased (partly because the autothrottle had also cut power on the failing left engine), seeming to confirm it, and they shut the right engine down. On final approach the damaged left engine failed completely; with the good engine off, the aircraft struck the M1 embankment short of the runway — 47 killed, 74 seriously injured. The brief calm after the shutdown was the trap: it appeared to confirm a diagnosis that was wrong, removing the doubt that might have prompted a reassessment.[1]

THE INVESTIGATION

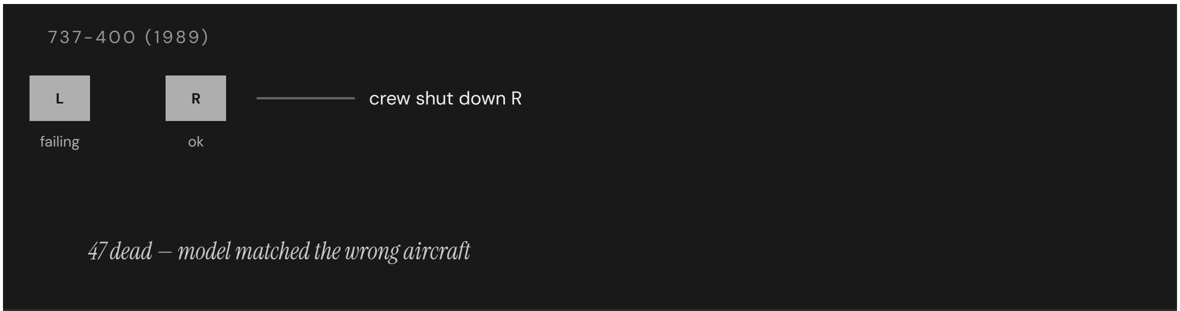
The Air Accidents Investigation Branch traced the disaster to the shutdown of the serviceable engine, and to a mental model carried from earlier 737s and applied to a -400 whose bleed-air configuration Boeing had changed — a difference the conversion training never disturbed, so the crew reasoned correctly from a configuration that no longer applied to the airframe they were flying.[2] The new electronic displays did not help: the vibration indicator that would have pointed at the failing engine was harder to read at a glance than the dials it replaced, so the one instrument that could have corrected the diagnosis was the least likely to be consulted under pressure.[3] And though cabin crew and passengers could see flames from the left engine, that never reached the flight deck — a crew-resource-management gap aviation would later train hard against, because those with the decisive evidence had no path to those making the decision.[4]

THE CAPABILITY GAP

The crew were skilled, and their mental model was correct — for the airplane they had flown the week before. The -400 was treated, for training, as an incremental change to a familiar type; under emergency pressure it behaved as a categorical one. Reading a manual note about revised bleed air is not the same as drilling a new reflex that fires before the old one, and the cockpit redesign had quietly removed perceptual cues a startled crew relies on. The change was filed as incremental precisely because nothing tested it until an emergency did. Capability had degraded under system change with no one noticing, because nothing failed until the day it did.[5]

AFTERMATH & REFORM

The AAIB issued thirty-one safety recommendations spanning conversion and difference training, engine-instrument and vibration-display design, cabin-to-flight-deck communication, and crashworthiness.[6] Kegworth became a standard human-factors teaching case and a reference for how a transition program should be built — so the differences between an old system and a new one are made hard to overlook, and the people closest to the evidence have a path to those making the decision. Each recommendation traced back to a failure the accident exposed — the unforced mental model, the unreadable instrument, the silent cabin — treated together rather than as isolated faults.



CREW MENTAL MODEL FROM OLDER VARIANT; CONFIGURATION CHANGED

REFERENCES

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2. AAIB Report 4/90 (1990) — accident sequence: left-engine fan-blade failure, shutdown of the serviceable right engine, and total power loss on approach; 47 killed, 74 seriously injured.
3. AAIB Report 4/90 (1990) — the older-737 bleed-air mental model misapplied to the 737-400's changed configuration.
4. AAIB Report 4/90 (1990) — the new electronic engine instrumentation and the readability of the vibration indicator.
5. AAIB Report 4/90 (1990) — the cabin-to-flight-deck communication gap (crew resource management).
6. AAIB Report 4/90 (1990) — the 31 safety recommendations on difference training, instrument design, CRM, and crashworthiness.
7. J. Reason, Human Error (Cambridge Univ. Press, 1990) — Kegworth as a case in the misapplication of a correct mental model to a changed system.

THE LEARNING ENGINEERING LENS

> *The crew's mental model of the older 737 was inappropriately applied to the 737-400 on which they were operating.*
>
> PARAPHRASING AAIB AIRCRAFT ACCIDENT REPORT 4/90 ON BRITISH MIDLAND 92, 1990

LE INSIGHT

Kegworth is the textbook example of Capability Degradation Under System Change. The pilots were excellent. Their mental model was excellent — for the airframe they had flown the previous week. The difference training had not been engineered to disturb the prior model with enough force to keep it from being misapplied. The new airframe was treated as an incremental change. The crew's response revealed it was a categorical one.

LENS APPROACH

LENS treats Kegworth in LEN 5 as the canonical **system-change** capability problem and in LEN 8 as the institutional version: how knowledge about what has changed travels from engineering to operations and what gets lost in transit. The case sits alongside Patriot (Case 19) in the canonical problem-type pair for system transition.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Classify any change that alters operator mental models — even a "minor" variant — as categorical, and gate it on drilled retraining rather than reading.
- Test new interface designs against startled-crew performance before fielding, so a redesign cannot quietly remove the perceptual cues operators rely on.
- Engineer a communication path from those closest to the evidence (cabin crew, passengers, sensors) to those making the decision, so decisive observations reach the flight deck.

IN OPERATION / AFTER

- Audit transition programs for capability that degraded silently — where nothing failed until an emergency tested the change — using simulator scenarios that force the new reflex.
- Monitor whether difference training actually overwrites the prior model, treating un-disturbed mental models as a measurable latent risk.
- Track that interface and communication recommendations are implemented together, not as isolated fixes, so the integrated failure mode does not recur.

REFLECTION QUESTIONS

1. The Kegworth crew's mental model was right for the previous variant. What change in your domain currently risks an analogous misapplication?
2. Difference training is a generic deliverable in transition programs. Design the artifact that would make differences hard to overlook rather than easy.
3. The cabin crew and passengers saw flames the flight deck never learned of. Identify a path in your domain by which the people closest to the evidence cannot reach the people making the decision, and design the channel that would carry it under emergency conditions.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs · knowledge bases · data pipelines

LENS COURSES LEN 5 LEN 8

CASE 33 DEFENSE 1991 – 2014

Military Fratricide — Desert Storm to Afghanistan

T Training Gap H Human-System Interface K Knowledge & Institutional Memory

IMPACT 24% of U.S. KIA in Desert Storm from friendly fire (35 of 146) — an order of magnitude above the historical baseline

IN BRIEF

Friendly fire killed an unusual share of coalition forces in the 1991 Gulf War: 35 of 146 U.S. combat deaths (24%) and 72 of 467 wounded (15%). (The often-cited "2% historical baseline" from Shrader's 1982 study is contested — later estimates run nearer 10–15%, and Shrader stepped back from it.) Post-war reviews blamed the chaos of combat, weak situational awareness and fire-control discipline, and combat-identification failures — and noted the military lacked a shared record to even study its own pattern. Fratricide is the failure of several systems to integrate; despite a generation of combat-ID investment, recurrences in Afghanistan and after show the rate never fell to a confidently low level, because integration is the hardest property to engineer by program.

BACKGROUND

Friendly fire is as old as war, but its true rate is hard to pin down. The most-cited estimate, from Charles Shrader's 1982 study Amicicide, put it under 2% of battle casualties — a figure later analysts challenged as far too low (nearer 10–15%), and one Shrader himself stepped back from. The disputed baseline mattered because it became the yardstick against which a modern war would be measured — and a yardstick set too low makes any later rate look like a catastrophe.[0]

WHAT HAPPENED

The 1991 Gulf War made the question grim and concrete: of 146 U.S. service members killed in action, 35 — about 24% — died by friendly fire, and 72 of 467 wounded (15%) were hit by their own side.[1] An A-10 strike on U.S. LAV-25s near Khafji killed seven Marines; a single A-10 attack on British Warrior vehicles killed nine — each an aircraft firing on friendly vehicles it had failed to identify, the recurring shape of the problem. Whatever the true baseline, a quarter of American combat deaths from one's own forces could not be waved off as the ordinary friction of battle.[2]

THE INVESTIGATION

Post-war reviews converged on familiar causes: the chaos of combat, inadequate situational awareness, weak adherence to fire-control measures, and combat-identification failures.[3] One finding cut deeper — the military lacked a comprehensive, shared record of fratricide incidents, so it could not study its own pattern, separate training failures from doctrine or equipment, or tell whether a fix was working. Without a common database, every incident was investigated in isolation and the aggregate signal that might have driven reform never formed. The capability to **learn** was itself missing, a second-order gap beneath the first.[4]

THE CAPABILITY GAP

Fratricide is not one problem but the failure of several systems to integrate — situational awareness, fire-control discipline, combat identification, joint coordination — each the subject of dedicated programs, yet the rate never fell to anything negligible. In Afghanistan a satellite-guided strike went wrong after a controller's device reset its coordinates, and in 2014 a B-1's targeting pod could not detect the infrared strobes marking U.S. troops, killing five. Each contributor was worked on; the integration **across** them — which is what actually keeps friendly forces from killing each other — is the hardest thing to engineer by program, because no single office owns it and no single procurement can deliver it — and it is where the capability kept failing.[5]

AFTERMATH & REFORM

The response was a generation of combat-identification investment: better IFF systems, blue-force-tracking networks, and changes to fire-control doctrine and training. The improvements were real, yet fratricide never dropped to a confidently low, stable rate, and even measuring it remained contested.[6] That is the lesson: where capability is an emergent property of many systems working together, no single program closes the gap — improving each contributor in isolation still leaves the integration between them unaddressed — and progress has to be measured against an honest baseline rather than a convenient one.

FRIENDLY-FIRE RATE, U.S. CASUALTIES, DESERT STORM VS. HISTORICAL

REFERENCES

1. C. R. Shrader, Amicicide: The Problem of Friendly Fire in Modern War (U.S. Army Combat Studies Institute, 1982) — the under-2% estimate from 269 incidents, a baseline later challenged as too low (with estimates nearer 10–15%).
2. "Friendly Fire: Facts, Myths and Misperceptions," U.S. Naval Institute Proceedings (June 1994) — Desert Storm: 35 of 146 KIA (24%) and 72 of 467 wounded (15%) by friendly fire; critique of the 2% "historical norm."
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6. Later-conflict recurrences (2001 coordinate-reset strike; 2014 B-1 infrared-strobe strike) and S. Snook, Friendly Fire (Princeton Univ. Press, 2000) on the 1994 Black Hawk shootdown — the systems-integration archetype.
7. "IFF Update: Stalled Again," U.S. Naval Institute Proceedings (June 1994) — the combat-identification and identification-friend-or-foe programs pursued after Desert Storm, and how slowly they matured.

THE LEARNING ENGINEERING LENS

> *The lack of a comprehensive and accessible automated database prevented thorough examination of the problem.*
>
> PARAPHRASING POST–DESERT STORM FRATRICIDE REVIEWS, C. 1993

LE INSIGHT

Fratricide is a multi-decade capability problem that resists single- intervention solutions. Each of the contributing causes — situational awareness, fire-control discipline, combat identification, joint coordination — has been the subject of dedicated programs. The rate persists because the integration across the contributors is itself the capability, and integration is the hardest property to engineer by program.

LENS APPROACH

LENS uses fratricide in LEN 5 to teach systems-of-systems capability analysis and in LEN 2 to introduce combat identification as a Human-AI Teaming problem (operators rely on automated IFF systems whose limitations are not in their training). LEN 8 examines why decades of awareness have not produced a sustained reduction.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat combat identification as a systems-of-systems requirement with a single owner for the integration, not as separate IFF, situational-awareness, and fire-control programs.
- Build a comprehensive, shared incident database from the outset so the pattern across events can be studied and fixes tested against it.
- Design fire-control discipline and identification into joint coordination, so the integration that prevents fratricide is engineered rather than left emergent.

IN OPERATION / AFTER

- Audit the friendly-fire rate against an honest baseline, distinguishing a temporary dip from a structural reduction before crediting any program.
- Monitor each contributor and the integration across them, since improving components in isolation can leave the joint failure mode untouched.
- Sustain the learning channel so later-conflict recurrences feed back into doctrine and training rather than being investigated in isolation.

REFLECTION QUESTIONS

1. Identify a capability problem in your domain that has been the subject of repeated programs without sustained improvement. What is the integration gap that the programs do not address?

2. Design a measurement system that would distinguish a temporary improvement in fratricide rate from a structural one.

3. The military lacked a shared incident database, so it could not study its own pattern. Identify a recurring harm in your domain that is investigated case-by-case but never aggregated, and design the shared record that would let the pattern become visible.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs · knowledge bases · data pipelines

LENS COURSES LEN 5 LEN 2 LEN 8

CASE 34 HEALTHCARE 2003–2017

ACGME 80-Hour Resident Duty-Hour Reform

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT **ACGME capped U.S. resident physician work hours at 80/week to reduce fatigue-related errors; subsequent RCTs (FIRST, iCOMPARE) showed mixed effects on patient outcomes and increased hand-off-related errors**

IN BRIEF

After the 1984 death of Libby Zion focused decades of concern on resident-physician fatigue, the ACGME capped resident work at 80 hours a week in 2003 and limited first-year shifts to 16 hours in 2011. The logic was intuitive: tired doctors err, so cut the hours. But hours were one input to a many-variable system; cutting them multiplied error-prone hand-offs and cost residents continuity and procedural repetitions. Two large randomized trials — FIRST (2016) and iCOMPARE (2019) — found flexible schedules non-inferior to the strict caps on patient outcomes, so the promised safety gain never appeared, and in 2017 the ACGME relaxed the intern cap. The case is the book's clearest example of a single-variable intervention into a capability system — and of evidence catching up with a well-meant policy.

BACKGROUND

After the 1984 death of Libby Zion — blamed on overworked, under-supervised residents — resident fatigue became a decades-long argument. New York's Bell Commission produced the first hours limits in 1989, and pressure for a national standard built from there — the reform's intuition, that exhausted physicians make errors, was strong enough that the simplest lever, capping the hours, became the obvious answer long before anyone tested what the long shift was producing.[0]

THE INTERVENTION

In 2003 the Accreditation Council for Graduate Medical Education capped resident work at 80 hours a week; in 2011 it limited first-year residents to 16-hour shifts. The logic was clean and intuitive — fatigue causes error, so reduce the hours. It was, in capability terms, a single-variable intervention: change one input and expect the outcome to move — a model that holds only if the rest of the system stays fixed, which in a teaching hospital it never does, since the hours removed had to be absorbed somewhere else.[1]

HOW IT WORKED

But hours were one input to a system with many. Cutting them redistributed the work and multiplied patient hand-offs — themselves a documented site of error — while reducing residents' continuity and the procedural repetitions that build skill; many reported feeling less prepared, not better rested. The long shift had quietly been doing work no one accounted for — sustaining a patient's care through one set of hands and accumulating the repetitions that turn a trainee into a clinician — and nothing was put in its place when it was cut.[2]

THE EVIDENCE

Two large randomized trials tested the policy. FIRST (Bilimoria et al., NEJM 2016), in surgery, found flexible duty hours non-inferior to the strict caps on patient outcomes and no worse for resident well-being — putting a randomized result against an intuition that had driven policy for years.[3] iCOMPARE (Silber et al., NEJM 2019), in internal medicine, reached a parallel result in a second specialty, making the finding harder to dismiss as an artifact of surgery.[4] Neither found the safety gain the cap had promised, and in 2017 the ACGME relaxed the 16-hour intern limit. The trials did not show fatigue is harmless — only that cutting one input, without rebuilding supervision and hand-offs, did not produce a safer system.[5]

WHAT TRANSFERRED

Duty-hour reform is the book's clearest case of a single-variable intervention into a multi-variable system. Read it against the successes here — the Keystone ICU project, crew resource management, the surgical safety checklist — which worked because they engineered supervision, hand-offs, and measurement **together with** the behavioral change, redesigning the surrounding architecture rather than pulling a single lever and hoping the rest would hold.[6] The lesson is not that fatigue does not matter; it is that capability is a property of the whole system, and a reform that moves one variable while leaving the others untouched will be judged, in the end, by what it actually produced rather than by the plausibility of its intuition.

ACGME DUTY-HOUR REFORM — CAPPING HOURS DID NOT, ON ITS OWN, ENGINEER THE CAPABILITY

REFERENCES

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6. ACGME, Common Program Requirements (2017 revision) — relaxation of the 16-hour first-year shift limit.
7. P. Pronovost et al., Keystone ICU intervention, NEJM 355: 2725–2732 (2006), doi:10.1056/NEJMoa061115 A. B. Haynes et al., surgical safety checklist, NEJM 360: 491–499 (2009), doi:10.1056/NEJMsa0810119 — integrated interventions that engineered the surrounding architecture.

THE LEARNING ENGINEERING LENS

> *Flexible, less-restrictive duty hour policies for first-year residents were associated with non-inferior patient outcomes and no significant difference in residents' satisfaction with overall well-being and education quality.*
>
> PARAPHRASING ICOMPARE TRIAL (SILBER ET AL., NEJM 2019)

LE INSIGHT

The clearest healthcare case in the dataset of a single-variable intervention into a multi-variable capability system. Pairs with Case 33 (fratricide) and Case 11 (V-22). The success cases — Keystone (14), CRM (12), Korean Air (23) — engineered supervisory, hand-off, and measurement architecture **together with** the behavioral change.

LENS APPROACH

LENS uses duty-hour reform in LEN 5 as the foundational capability-system case (what was the long-hour regime **producing** that was lost when the input was capped?), in LEN 4 to discuss measurement architecture (what FIRST and iCOMPARE measured, and what they did not), and in LEN 10 as a studio prompt for the integrated resident-training redesign that the reforms did not deliver.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Map the full set of variables the targeted input is coupled to — continuity, supervision, hand-offs, procedural reps — before changing one of them.
- Design the substitute for whatever the changed input was producing (e.g., structured hand-offs and supervision) into the same reform, not as a follow-on.
- Build the measurement that will test the reform's promised gain into the rollout, so the policy is falsifiable against the operational record.

IN OPERATION / AFTER

- Audit the intervention against patient and trainee outcomes with a controlled comparison, as FIRST and iCOMPARE did, rather than trusting the intuition.
- Monitor the variables that absorbed the change (hand-off frequency, procedural exposure) for the harms a single-lever fix can displace.
- Sustain a willingness to revise the policy when evidence catches up, as the 2017 relaxation did, rather than defending the original lever.

REFLECTION QUESTIONS

1. What capability is the long-hours / heavy-workload regime in your domain currently producing — supervisory exposure, continuity, procedural reps, tacit-knowledge transfer — that a simple cap would lose?

2. Design the integrated redesign — supervision, hand-off, measurement, exposure — that would substitute for the capability the input cap removes.

3. The reform was intuitive enough to set national policy years before FIRST and iCOMPARE tested it. Design the randomized or quasi-experimental check you would build into a future single-variable reform so its promised gain is measured before it is mandated, not after.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

LENS COURSES LEN 5 LEN 4 LEN 10 LEN 8

CASE 43 AVIATION 2009

Colgan Air Flight 3407

T Training Gap

IMPACT 50 killed near Buffalo; precipitated the FAA's 1,500-hour rule and the Pilot Records Database

IN BRIEF

On approach to Buffalo in February 2009, the captain of Colgan Air 3407 responded to a stall warning by pulling back on the controls instead of lowering the nose; the Bombardier Q400 stalled and crashed into a house in Clarence Center, killing all 49 aboard and one person on the ground. The NTSB found the captain had a documented history of training failures, and the first officer was ill, fatigued, and paid about $16,000 a year. The airline and the hiring pipeline knew the training history; the regulator did not. Victims' families mounted one of the most effective aviation-safety campaigns in a generation, producing the 2010 law that raised first-officer experience requirements to 1,500 hours and created the Pilot Records Database. The capability gap was visible to everyone except the system that licensed the pilots.

BACKGROUND

Colgan Air 3407, a Bombardier Q400 flying a regional route to Buffalo, was crewed by a captain with a documented history of training failures and a first officer who was ill, fatigued, and paid roughly $16,000 a year. The airline and the hiring pipeline knew the captain's record; the regulator did not — the information that should have shaped who sat in that seat lived inside the carrier's files and never crossed into the licensing system that was supposed to vouch for the crew.[0]

WHAT HAPPENED

On 12 February 2009, on approach to Buffalo, the aircraft slowed and its stall-warning stick shaker activated. The captain responded by pulling back on the control column — the opposite of stall recovery — and the Q400 stalled and crashed into a house in Clarence Center, killing all 49 aboard and one person on the ground. The reflex he reached for was exactly wrong for the situation, the kind of error a pattern of failed training events should have predicted long before he was handed the controls.[1]

THE INVESTIGATION

The NTSB found the probable cause to be the captain's inappropriate response to the stick shaker, against a backdrop of fatigue, weak airline training, and a hiring system blind to the captain's history. The data that should have flagged him existed — multiple failed training events — but did not flow to the decision that put him in the seat, so the hiring choice was made on a record that omitted the very facts that mattered most, an absence that looked like a clean slate.[2]

THE CAPABILITY GAP

Colgan is the canonical case for the gap between the information an institution holds about its operators and the information that reaches the decisions about those operators. The capability that was missing was not pilot skill in the abstract but a data flow: a way to make a pilot's documented training history actionable at the hiring and licensing decision, rather than buried in files no one consulted — a record can exist in full and still fail completely if it never reaches the moment and the person whose choice it was meant to inform.[3]

AFTERMATH & REFORM

The victims' families organized one of the most effective aviation-safety lobbying efforts in a generation, producing the Airline Safety and FAA Extension Act of 2010 — which raised the minimum experience for airline first officers to 1,500 hours and created the Pilot Records Database to make pilot history flow between carriers.[4] The 1,500-hour rule remains debated; the records database fundamentally restructured how the information that was missing in 2009 now moves, converting a buried file into a record a hiring carrier is required to retrieve — the difference between a fact that exists and one that arrives where the decision is made.

50

KILLED, CLARENCE CENTER NY

the data was in the system; the data flow was not

COLGAN 3407 — AND THE REFORM THAT FOLLOWED

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5. GAO-12-203, Pilot Records Database Implementation (2012); the Families of Continental Flight 3407 campaign.

THE LEARNING ENGINEERING LENS

> *The captain's inappropriate response to the activation of the stick shaker led to an aerodynamic stall from which the airplane did not recover.*
>
> NTSB AIRCRAFT ACCIDENT REPORT AAR-10/01, PROBABLE CAUSE, 2010

LE INSIGHT

Colgan is the canonical case for the gap between the information an institution holds about its operators and the information that reaches the decisions about whether those operators should be in that seat. The data was already in the system — multiple failed training events. The data flow that would have made that data actionable at the hiring decision did not exist.

LENS APPROACH

LENS uses Colgan in LEN 4 as a case for evidence-system design (the PRD as a designed information flow) and in LEN 8 as a case for advocacy-driven institutional change. Studio projects examine how families of victims became the load-bearing element of a reform the industry had resisted for years.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make a pilot's documented training history a required, structured input to the hiring and licensing decision, not a record held passively in carrier files.
- Design the data flow so that adverse signals (failed training events) are pushed to the decision point rather than waiting to be requested.
- Define record retention and transfer obligations across carriers up front, so a clean-looking application cannot conceal a known history.

IN OPERATION / AFTER

- Audit whether the records actually reach hiring decisions, treating a complete file that never arrives as a live failure, not a solved one.
- Monitor the pipeline for operators whose adverse history is not surfacing, and act before the gap is paid for in an accident.
- Sustain the records database against drift and gaps in coverage, since a data-flow remedy decays as fast as its weakest link.

REFLECTION QUESTIONS

1. What information about operators in your domain exists somewhere in the system but does not flow to the decisions that depend on it?
2. Design the data-flow architecture that would make a Colgan-equivalent visible **before** the accident rather than after.
3. The reform in 2009 was driven by victims' families, not the industry that had resisted it. What load-bearing constituency in your domain could force a stalled data-flow fix into existence, and what evidence would mobilize them?

WHO BUILDS THIS learning science & instructional design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models

LENS COURSES LEN 4 LEN 5 LEN 8

CASE 44 AVIATION 2013

Asiana Airlines Flight 214

T Training Gap H Human-System Interface

IMPACT 3 killed, 187 injured; Boeing 777 struck the seawall short of SFO runway 28L

IN BRIEF

On a clear afternoon at San Francisco in July 2013, the crew of Asiana 214 allowed their Boeing 777 to slow far below approach speed without recognizing that the autothrottle was not maintaining it. The aircraft struck the seawall short of runway 28L, killing three of the 307 aboard and injuring nearly two hundred. The NTSB found the crew had mismanaged the approach and inadequately monitored airspeed — but also that the complexity of Boeing's autoflight systems contributed: after the pilots disconnected the autopilot and pulled thrust to idle, the autothrottle reverted to a HOLD mode in which it would not wake to hold the selected speed. The captain believed it would. Asiana 214 is the most prominent recent case of automation surprise on a Western wide-body airliner.

BACKGROUND

Asiana 214 was a scheduled Boeing 777 flight from Seoul to San Francisco. The captain was experienced on other types but new to the 777 and flying it under the supervision of a training captain; the approach was a visual one, with the airport's instrument glideslope out of service for construction — so a pilot still learning the type's automation was hand-flying an approach stripped of the vertical guidance that would normally have anchored it, exactly the conditions under which an unfamiliar mode is most likely to bite.[0]

WHAT HAPPENED

On 6 July 2013, on final approach to runway 28L, the aircraft descended below the proper path and slowed. The crew had set an autopilot mode that left the autothrottle in HOLD, where it would not advance thrust to hold the selected speed. Airspeed decayed to about 103 knots against a target of 137; the 777 struck the seawall short of the runway, killing three and injuring nearly two hundred of the 307 aboard. The thirty-four-knot shortfall built up silently while the crew believed the automation was holding the speed, so the first unambiguous signal of trouble was the airframe itself running out of energy on short final.[1]

THE INVESTIGATION

The NTSB found the probable cause to be the crew's mismanagement of the approach and inadequate monitoring of airspeed, complicated by the complexity of the autothrottle and autopilot flight-director systems and by the crew's faulty mental model of the automation. The captain told investigators he had assumed the autothrottle would maintain speed; in that configuration, it does not — a gap between the system's documented behavior and the mental model the training had left him with, the kind of mismatch that surfaces only when the automation is asked to do what the operator wrongly believes it is already doing.[2]

THE CAPABILITY GAP

Asiana 214 is the textbook automation surprise on a Western wide-body: the mode the autothrottle was actually in and the mode the crew believed it was in diverged silently. The missing capability was not raw pilot skill but transparency — a system that made its own state, and the fact that the crew now owned the airspeed, salient at the moment it mattered, rather than leaving the handoff of responsibility to be inferred from a silent mode change the interface did nothing to announce.[3]

AFTERMATH & REFORM

The NTSB issued recommendations on autoflight training, energy-state monitoring, and the design of mode annunciation, and emphasized training that preserves manual-flying proficiency to counter automation dependence. The accident became a standard reference in airline automation-policy reviews and in the push toward "automation airmanship" curricula — shifting the lesson from one crew's error to a recurring design problem about how modes are annunciated and how monitoring is taught when the automation is doing most of the flying.[4]

EXPECTED	ACTUAL
137 kt	103 kt

the autothrottle was not maintaining the speed the crew believed it was

ASIANA 214 — APPROACH AIRSPEED AT THE SEAWALL

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THE LEARNING ENGINEERING LENS

> *Complexities of the autothrottle and autopilot flight director systems contributed to the accident.*
>
> NTSB AIRCRAFT ACCIDENT REPORT AAR-14/01, CONTRIBUTING FACTORS, 2014

LE INSIGHT

Asiana 214 is the aviation case for the LENS Human-AI Teaming proposition that automation transparency is a capability deliverable. The mode the autothrottle was actually in and the mode the crew believed it was in diverged silently — a textbook automation surprise. The training, the documentation, and the interface together did not surface the divergence.

LENS APPROACH

LENS treats Asiana 214 in LEN 2 as a worked case for automation transparency requirements. Students design the interface and training intervention that would have made the mode disconnect salient to the crew at the moment the aircraft slowed.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make automation transparency a design requirement: the system must announce its active mode and any silent reversion, not leave it to be inferred.
- Annunciate transfers of responsibility (e.g., when the crew now owns airspeed) explicitly at the moment they occur, so no handoff is implicit.
- Design and validate mode logic against operator mental models during development, so configurations that “do not do what pilots assume” are caught before fielding.

IN OPERATION / AFTER

- Train and audit energy-state monitoring so a slow, silent airspeed decay is caught by the crew before the airframe signals it.
- Preserve manual-flying proficiency through recurrent practice to counter the automation dependence that hid the decay.
- Monitor operations for mode-confusion events and feed them into automation-policy and annunciation-design reviews.

REFLECTION QUESTIONS

1. Identify a mode in an automated system you work with where the operator’s mental model and the system’s actual behavior can silently diverge. How would the operator know?
2. Design the interface change that would have alerted the Asiana crew to the mode mismatch before the airspeed decayed.
3. The autothrottle’s HOLD reversion handed airspeed back to a crew that did not know it now owned it. Design the annunciation that would make a transfer of responsibility between automation and operator unmissable at the instant it occurs.

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TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs

LENS COURSES LEN 5 LEN 2

CASE 45 DEFENSE 1941 – 1943

Mark 14 Torpedo Failures

T Training Gap K Knowledge & Institutional Memory G Governance & Trust

IMPACT Persistent torpedo failures across the first ~20 months of the Pacific War; resolved only after fleet-level testing forced acknowledgment of the defects

IN BRIEF

The U.S. Navy entered World War II with the Mark 14 torpedo, so expensive that the Bureau of Ordnance had effectively forbidden live testing in peacetime. Through 1942 submarine crews reported torpedoes running deep, failing to detonate, or exploding prematurely; the Bureau insisted the weapon was sound and blamed the operators. It took eighteen months and the personal intervention of Admiral Charles Lockwood — who ordered fleet-level live-fire tests — to confirm three separate defects: the torpedo ran about ten feet too deep, its magnetic exploder fired erratically, and its contact pin crushed on a square hit. The fixes were simple once the defects were acknowledged. The binding constraint was institutional: a bureau insulated from the operators who knew the weapon did not work.

BACKGROUND

The Mark 14 was the U.S. Navy's standard submarine torpedo at the start of the Pacific War. It had been so expensive to test that the Bureau of Ordnance had effectively forbidden live trials in the 1930s; the weapon went to war essentially unproven against realistic conditions, so the very decision meant to conserve a scarce and costly weapon guaranteed that its defects would first be discovered in combat, by the crews who could least afford them to surface there.[0]

WHAT HAPPENED

From early 1942, submarine crews reported a litany of failures: torpedoes that ran far below their set depth, magnetic exploders that detonated prematurely or not at all, and contact firing pins that crushed on a direct hit. The Bureau of Ordnance insisted the weapon worked and attributed the misses to crew error; captains who reported failures risked their careers — an arrangement that turned every field report into a self-accusation and so suppressed the very evidence that would have isolated the defects the Bureau refused to acknowledge.[1]

THE INVESTIGATION

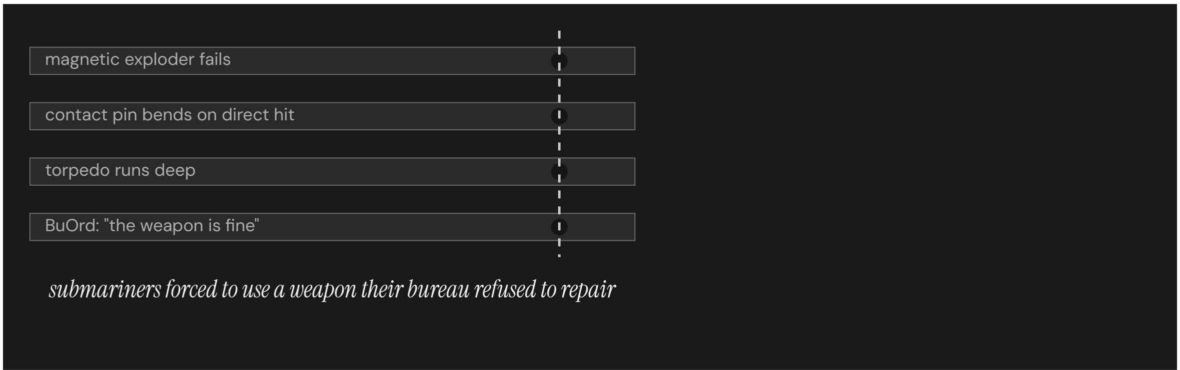
It took eighteen months and the intervention of Admiral Charles Lockwood, commander of the Pacific submarine force, who ordered fleet-level testing. A live-fire trial — and the USS Tinosa's July 1943 attack on the Tonan Maru, in which eleven torpedoes struck the stopped ship squarely and failed to detonate — forced the issue. The tests confirmed the torpedo ran about ten feet too deep, that the Mark 6 magnetic exploder failed routinely, and that the contact pin buckled on perpendicular impact — three independent defects that had been masking one another at sea, which is why a fleet commander's controlled trial, not another combat patrol, was finally able to separate and prove them.[2]

THE CAPABILITY GAP

The defects were real and separate, but the binding constraint was institutional. The capability that was missing was not at the front; it was a channel by which the bureau that owned the weapon could be made to hear what the boats already knew. The fixes — recalibrating depth, deactivating the magnetic exploder, redesigning the contact pin — were small. The refusal to believe the operators was the expensive part, measured not in engineering hours but in the months of patrols and the targets that escaped while a bureau defended a verdict the boats had already disproven.[3]

AFTERMATH & REFORM

By late 1943 the three defects were corrected and the Mark 14 became an effective weapon for the rest of the war. The episode entered U.S. Navy institutional history as the canonical case of a procurement bureau insulated from operator feedback, and is cited in modern organizational-learning literature on the cost of suppressing field reports — a reminder that the structure that decides whose evidence counts is itself a capability, and that its absence can cost more than any single technical defect it conceals.[4]



MARK 14 — MULTIPLE DEFECTS, SINGLE INSTITUTIONAL REFUSAL

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THE LEARNING ENGINEERING LENS

> *The Bureau certified the weapon; field reports of failure were treated as evidence of operator error.*
>
> PARAPHRASING BLAIR, SILENT VICTORY, 1975

LE INSIGHT

The Mark 14 is the canonical Navy case for the institutional refusal to believe operator feedback. The capability that was missing was not at the front. It was at the bureau that owned the weapon. The cost of the refusal was paid by the submariners forced to use it until the bureau yielded. The eventual fixes — re-setting depth calibration, replacing the magnetic exploder, redesigning the contact pin — were technical and small. The reform was institutional and slow. The capability that had to be built was the channel by which the bureau could be made to hear what the boats already knew.

LENS APPROACH

LENS uses the Mark 14 in LEN 7 as a governance failure case and in LEN 8 as an organizational-learning case in which the operators had the diagnosis and the institution lacked the structure to receive it. Studio projects (LEN 10) examine what the equivalent operator-to-institution feedback channel should look like.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Require realistic live testing before fielding, even for scarce and costly items, so defects surface in trials rather than in combat.
- Build an operator-feedback channel into the weapon's ownership from the start, with a path that does not put the reporter's career at risk.
- Separate the authority that certifies a system from the authority that investigates its field failures, so a bureau cannot judge its own product.

IN OPERATION / AFTER

- Audit field reports for suppressed or career-risking signals, and treat a pattern of "operator error" verdicts as a warning to test the system, not the crew.
- Empower an operational commander to order controlled trials when field reports conflict with the owner's certification.
- Sustain the feedback channel so multiple masking defects can be separated and proven before the cost compounds.

REFLECTION QUESTIONS

1. Identify a system in your domain whose owners are institutionally insulated from the operators using it. What feedback would they refuse to hear, and what would it cost?
2. Design the operator-feedback channel that the U.S. Navy Bureau of Ordnance should have had in 1941. Who signs, who receives, what triggers action?
3. The USS Tinosa fired a string of torpedoes that struck a stopped ship and failed to detonate before the bureau accepted the diagnosis. What is the operator-evidence threshold in your domain that would force the equivalent institutional acknowledgment — and how would you make sure it is reached before the cost is paid?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

LENS COURSES LEN 10 LEN 7 LEN 8

CASE 46 DEFENSE 1980

Operation Eagle Claw

T Training Gap K Knowledge & Institutional Memory

IMPACT 8 servicemembers killed in Iran; mission to rescue 52 American hostages aborted; catalyst for the founding of U.S. Special Operations Command

IN BRIEF

The April 1980 mission to rescue fifty-two American hostages held at the U.S. embassy in Tehran was aborted at a desert staging point when three of eight helicopters were lost to mechanical and weather problems. During the withdrawal a helicopter collided with a refueling C-130, killing eight servicemembers; the hostages remained captive for another nine months. The Holloway Commission found the operation had been assembled ad hoc: each service contributed its own units, equipment, and command relationships, the aircrews had not trained together, and there was no standing joint special-operations command to own the mission. The capability had to be improvised, and the improvisation failed. Eagle Claw catalyzed the creation of JSOC, the Goldwater-Nichols reforms, and ultimately U.S. Special Operations Command.

BACKGROUND

In November 1979, Iranian revolutionaries seized the U.S. embassy in Tehran and took fifty-two Americans hostage. After months of failed diplomacy, the Carter administration authorized a military rescue. No standing joint special-operations command existed to plan or execute it; the force had to be assembled from units drawn separately from the Army, Navy, Marines, and Air Force — each bringing its own equipment, procedures, and chain of command to a mission that demanded they act as one, with no institution whose job it was to make them cohere before the night of the raid.[0]

WHAT HAPPENED

On 24 April 1980, the mission staged at a remote site code-named Desert One. Three of the eight RH-53 helicopters were disabled by a dust storm and mechanical failure, dropping the force below the minimum needed; the commander aborted. During the withdrawal a helicopter collided with a C-130 tanker, and the resulting fire killed eight servicemembers. The mission failed before reaching Tehran — undone not by enemy action but by attrition and a chaotic withdrawal among aircraft and crews that had never operated together, the predictable cost of integration improvised under pressure rather than built and rehearsed in advance.[1]

THE INVESTIGATION

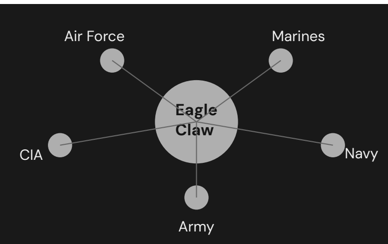
The Holloway Special Operations Review Group identified the operation's "ad hoc-ery" as central: each service contributed its own units, equipment, command relationships, and communications; the aircrews had not trained together as a unit; the RH-53D had been selected partly for a minesweeping cover story rather than for fitness for a desert rescue. There was no standing organization to own the mission end to end — no single authority responsible for the force's training, equipment fit, and command architecture as a whole, so each gap was someone's problem in part and no one's in full.[2]

THE CAPABILITY GAP

Each service was competent inside its own boundary. The capability that did not exist was the integration across them — a joint command, a common communications architecture, and a force that had trained together. That integration had to be improvised for a single high-stakes mission, and the improvisation could not hold — because cross-service cohesion is not summoned on demand but accrued through standing structure and repeated joint training, neither of which existed when the order came.[3]

AFTERMATH & REFORM

Eagle Claw produced the Holloway Commission, the creation of the Joint Special Operations Command in 1980, and — alongside the 1983 Beirut bombing and the Grenada invasion — the impetus for the Goldwater-Nichols Act of 1986 and the Nunn-Cohen Amendment establishing U.S. Special Operations Command in 1987. The reform built the institution the mission had needed and not had — converting a capability that had been improvised once and failed into a standing command with its own forces, training, and authority, so the next mission would inherit cohesion rather than assemble it from scratch.[4]



five organizations · no joint command · no shared training

THE ABSENCE OF AN INSTITUTION THAT COULD OWN THE MISSION

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THE LEARNING ENGINEERING LENS

> *The mission was ad hoc — assembled from units, equipment, and command relationships that had never operated together.*
>
> PARAPHRASING THE HOLLOWAY SPECIAL OPERATIONS REVIEW GROUP, 1980

LE INSIGHT

Eagle Claw is the canonical case in U.S. defense for the absence of institutionalized cross-service capability. Each service was competent inside its own boundary; the integration across them did not exist as a deliverable — until the reform that followed built the institution the mission had needed and not had.

LENS APPROACH

LENS uses Eagle Claw in LEN 5 as a worked case for cross- organizational capability requirements and in LEN 8 for the institutional reform that followed; it pairs with INPO (Case 16) as the defense analog of failure-driven institution building. The seven-year arc to Goldwater-Nichols (1986) and USSOCOM (1987) is itself a measurement: the institutional response time when the fix requires statutory action.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Stand up a single command that owns the cross-organizational mission end to end — its force, equipment fit, communications, and training — rather than assembling it per mission.
- Require the contributing units to train together as one force on a common communications architecture before they are committed.
- Select equipment for fitness to the actual mission, not for a convenient cover story or parent-service availability.

IN OPERATION / AFTER

- Audit whether a cross-organizational capability that exists on paper has ever actually operated as one force, and treat the absence of joint reps as an unfilled gap.
- Sustain the standing institution and its training pipeline so the next mission inherits cohesion rather than improvising it.
- Monitor for the recurrence of ad-hoc assembly, since the conditions that produced Desert One reappear whenever no single authority owns the joint mission.

REFLECTION QUESTIONS

1. Where in your domain does a cross-organizational capability exist on paper but not in practice? What would force its institutionalization?
2. Eagle Claw produced USSOCOM and Goldwater-Nichols six years later. Sketch the institutional design that an equivalent failure in your domain would force into existence.
3. The Holloway Commission named the mission's ad-hoc-ery as the diagnosis. What standing capability — institution, command, training pipeline — does your domain currently lack that an Eagle-Claw-class failure would force into existence?

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LENS COURSES LEN 5 LEN 8

CASE 47 AVIATION 2005

Helios Airways Flight 522

T Training Gap H Human-System Interface

IMPACT 121 killed; cabin failed to pressurize after a maintenance setting was left on "manual" and the crew misread the warning

IN BRIEF

Shortly after takeoff from Larnaca in August 2005, the crew of Helios 522 misread a warning horn. The same intermittent horn that signals an incorrect takeoff configuration on the ground sounds, in flight, when cabin altitude climbs past about 10,000 feet — and the pressurization selector had been left in “manual” after a leak check the night before. The cabin never pressurized. The crew, troubleshooting the wrong fault, lost consciousness to hypoxia; the Boeing 737 flew on autopilot until it ran out of fuel and crashed into a hillside near Athens, killing all 121 aboard. The investigation found a single cue carried two meanings with no differentiation, and the airline’s training had drilled only the ground meaning.

BACKGROUND

Helios 522 was a Boeing 737 departing Larnaca, Cyprus, for Athens. The night before, engineers had run a pressurization leak check and left the cabin-pressurization selector in the “manual” position. The pre-flight checks did not catch it, and the aircraft departed with its pressurization system not set to maintain cabin altitude automatically — a maintenance setting left in an abnormal position the following morning, the kind of latent configuration error that a pre-flight verification exists precisely to catch and here did not.[0]

WHAT HAPPENED

As the aircraft climbed on 14 August 2005, the cabin failed to pressurize and a warning horn sounded. On the 737 the same horn indicates an incorrect takeoff configuration on the ground and excessive cabin altitude in flight. The crew interpreted it as a configuration warning and spent minutes troubleshooting the wrong problem while hypoxia set in. By the time the aircraft neared Athens the flight crew was incapacitated; it circled on autopilot until fuel exhaustion and crashed, killing all 121 aboard. The crew spent their few useful minutes diagnosing a configuration fault that did not exist, the ambiguous horn having steered them away from the one problem — climbing cabin altitude — that was quietly disabling them.[1]

THE INVESTIGATION

The Hellenic Air Accident Investigation Board found the crew’s misidentification of the cabin-altitude warning as a takeoff-configuration warning to be a critical, irrecoverable error. The single horn carried two meanings with no differentiation between them, and the airline’s training had emphasized only the ground meaning; the in-flight case was an edge condition the crew had not been prepared to recognize — so an ambiguous cue and a one-sided training regime combined to make the wrong interpretation the natural one, with hypoxia closing the window before the error could be caught.[2]

THE CAPABILITY GAP

Helios 522 is the textbook example of a single cue carrying two meanings without differentiation — an interface that lied by ambiguity while the training filled in only one of the two meanings. The missing capability was a cue, or a training regime, that distinguished the deadly in-flight meaning from the routine ground one before the operators had to guess under hypoxia — a way to resolve the ambiguity at the interface or in training, rather than leaving it to be resolved by a crew already losing the cognition the decision required.[3]

AFTERMATH & REFORM

The investigation drove changes to Boeing's warning logic and to pressurization-related training and checklists across operators; later 737 variants distinguish the cabin-altitude warning from the takeoff-configuration warning. The accident is a standard case study in cue ambiguity, pre-flight verification, and the physiology of hypoxia in human-factors curricula — its remedies attacking the failure on both fronts at once, differentiating the warning at the interface and reinforcing the in-flight meaning in training so neither gap could be papered over by the other.[4]

ACTUAL	INTERPRETED
cabin alt.	T/O config.

one horn, two meanings · the training had drilled only one

HELIOS 522 — A CUE CARRYING TWO MEANINGS WITHOUT DIFFERENTIATION

REFERENCES

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THE LEARNING ENGINEERING LENS

> *The crew's interpretation of the cabin-altitude warning horn as a takeoff-configuration warning was a critical and irrecoverable error.*
>
> PARAPHRASING THE HELLENIC AAIB FINAL REPORT ON HELIOS 522, 2006

LE INSIGHT

Helios 522 is the textbook example of a single cue carrying two meanings without differentiation. The interface lied by ambiguity. The training filled in only one meaning. The combination produced a twenty-minute window in which the crew was solving the wrong problem.

LENS APPROACH

LENS uses Helios in LEN 5 as a case in cue ambiguity and in LEN 2 as a Human-AI Teaming case where the interface failed to communicate which of two possible system states it was warning about. Students redesign the cue.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Differentiate cues that carry distinct meanings in distinct contexts, so a single signal cannot be read as the wrong one under stress.
- Train both meanings of any dual-purpose cue, not just the routine one, so the rare-but-deadly interpretation is recognized first.
- Design pre-flight verification to catch abnormal maintenance configurations, so a setting left in "manual" cannot survive into operation.

IN OPERATION / AFTER

- Audit checklists and warning logic for cues whose meaning depends on phase or context, and resolve the ambiguity before an operator must guess.
- Monitor maintenance-to-operations handoffs for latent configuration errors that pre-flight checks have historically missed.
- Sustain training on the physiology that degrades the operator (e.g., hypoxia), so the window for self-correction is understood and guarded.

REFLECTION QUESTIONS

1. Identify a cue in your domain that carries two meanings in different operational contexts. How would the operator know which?
2. Redesign the Helios configuration warning to distinguish ground- and in-flight meaning without adding cognitive load.
3. A maintenance setting left in "manual" survived the pre-flight checks and reached the air. Design the pre-flight verification step in your domain that would make an abnormal maintenance configuration impossible to depart with unnoticed.

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LENS COURSES LEN 5 LEN 2

CASE 48 AVIATION 1996

AeroPerú Flight 603

H Human-System Interface T Training Gap

IMPACT 70 killed; Boeing 757 crashed into the Pacific after maintenance tape was left over static ports

IN BRIEF

While polishing the fuselage of a Boeing 757 in Lima in October 1996, maintenance staff taped over the static ports and failed to remove the tape before flight. With the static system blocked, the aircraft's altimeters, airspeed indicators, and altitude-alert and overspeed warnings all read inconsistently. The crew received a cascade of contradictory warnings — overspeed, stick shaker, ground proximity — and could not determine their true altitude or speed. Believing they were higher than they were, they flew into the Pacific, killing all 70 aboard. The investigation named the maintenance error as the primary cause but stressed that every cockpit instrument depended on the same blocked sensor: the apparent redundancy was an illusion, and the crew's training had no procedure for "everything you see is wrong."

BACKGROUND

AeroPerú 603 was a Boeing 757 night departure from Lima to Santiago. Before the flight, ground crew had covered the aircraft's static ports with adhesive tape during cleaning and polishing of the fuselage, and the tape was not removed. The static ports feed the instruments that tell the crew their altitude and airspeed — a single physical source upstream of nearly every primary cockpit display, so blocking it corrupted not one instrument but the whole air-data picture the crew would rely on in the dark.[0]

WHAT HAPPENED

On 2 October 1996, the aircraft took off into darkness over the ocean with its static system blocked. Altimeters, airspeed indicators, and the altitude-alert and overspeed systems all produced false and contradictory readings. The crew received simultaneous overspeed and stall warnings and ground-proximity alerts they could not reconcile. Believing they were higher than they were, they descended and struck the Pacific, killing all 70 aboard — the contradictory warnings offering no way to tell which, if any, instrument to trust, because every one of them drew on the same blocked source and so failed in concert rather than disagreeing usefully.[1]

THE INVESTIGATION

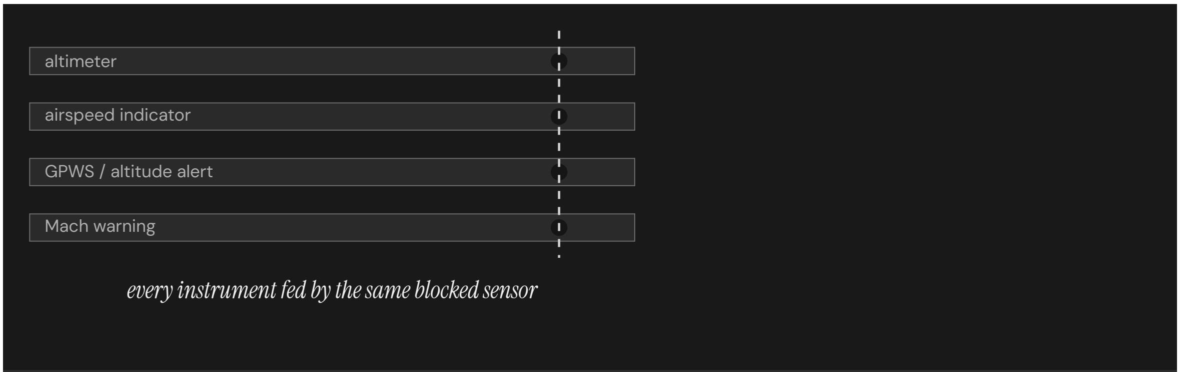
The Peruvian accident investigation identified the failure to remove the static-port tape as the primary cause, but emphasized that the cockpit had no independent reference by which to detect the inconsistency. Every instrument that should have flagged the failure drew on the same blocked source. The crew's training had no procedure for the case in which all primary instruments are simultaneously wrong — that failure had been assumed away rather than planned for, leaving the crew to improvise against a problem the system design had quietly made possible and the curriculum never named.[2]

THE CAPABILITY GAP

AeroPerú 603 is the case for redundancy that is not redundancy. The cockpit's apparent instrument redundancy was an illusion at the source — a common-cause failure that defeated every downstream display at once. The missing capability was both a maintenance control that made the blocked port impossible to miss and a procedure for flying when the instruments themselves cannot be trusted — the first preventing the common-cause failure at its source, the second giving the crew somewhere to stand when the apparent redundancy in front of them collapsed all at once.[3]

AFTERMATH & REFORM

The accident reinforced industry maintenance practices for conspicuous covering and mandatory removal-verification of static ports and pitot tubes, and entered the human-factors literature as a worked example of common-cause failure and the limits of crew training under total instrument corruption. It is paired with other pitot-static events in discussions of air-data integrity — its lesson being that redundancy counted at the display tells you nothing if the redundant paths share a single upstream point that can fail them all together.[4]



AEROPERÚ 603 — REDUNDANCY THAT WAS NOT REDUNDANCY

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THE LEARNING ENGINEERING LENS

> *The crew received contradictory indications they could neither reconcile nor override.*
>
> PARAPHRASING THE PERUVIAN DGAC / COMISIÓN INVESTIGADORA DE ACCIDENTES DE AVIACIÓN REPORT ON AEROPERÚ 603, 1996

LE INSIGHT

AeroPerú 603 is the case for redundancy that is not redundancy. Every cockpit indicator depended on the same physical sensor. The apparent redundancy in the cockpit was an illusion at the source. The training did not include the failure case because the failure case had been assumed not to occur.

LENS APPROACH

LENS uses AeroPerú in LEN 5 to teach the difference between apparent and actual redundancy and in LEN 2 for the human role when all interface data is unreliable. Studio projects examine the “trust nothing” procedure design.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Map redundant displays back to their upstream sources, and eliminate or instrument any common point whose failure would corrupt them all at once.
- Make critical sensor covers conspicuous and require verified removal, so a blocked static port cannot reach takeoff unnoticed.
- Provide an independent reference (e.g., a source not fed by the common component) so the crew can detect inconsistency at the source.

IN OPERATION / AFTER

- Audit maintenance controls for tasks that cover or block critical sensors, confirming each has a conspicuous marker and a removal-verification step.
- Train and rehearse a “trust nothing” procedure for total instrument corruption, the case the curriculum had assumed away.
- Monitor for common-cause failure modes whose probability was assumed negligible, and revisit that assumption when the consequence is catastrophic.

REFLECTION QUESTIONS

1. Identify a redundant interface in your domain whose redundancy depends on a common upstream component. What is the operator procedure when the upstream component fails?
2. Design the “trust nothing” procedure for the AeroPerú crew. What information remains reliable when all instruments are corrupted?
3. A taped-over static port survived to takeoff because nothing made it conspicuous or forced its removal to be verified. Design the maintenance control in your domain that would make a blocked or covered critical component impossible to overlook before operation.

WHO BUILDS THIS human factors & interaction design · learning science & instructional design — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · scenario simulation · competency models

LENS COURSES LEN 5 LEN 2

CASE 49 AVIATION 2019

Atlas Air Flight 3591

T Training Gap K Knowledge & Institutional Memory

IMPACT 3 killed; Boeing 767 cargo flight crashed in Texas after the first officer mistakenly activated go-around mode

IN BRIEF

On approach to Houston in February 2019, the first officer of Atlas Air 3591 — an Amazon Prime Air cargo flight — inadvertently triggered the go-around mode during a turbulent descent. Experiencing a somatogravic illusion that the aircraft was pitching up, he pushed the nose down sharply and dove the Boeing 767 freighter into Trinity Bay, killing all three aboard. The NTSB found the first officer had a long, undisclosed history of training failures across multiple carriers — and that Atlas could not see it, because the Pilot Records Database mandated by Congress after Colgan Air 3407 (Case 43) was not yet operational. Atlas relied on the older PRIA system, which surfaced only five years of history. The case is the live recurrence of Colgan: the information existed; the data-flow system was not yet complete.

BACKGROUND

Atlas Air 3591 was a Boeing 767 freighter operating for Amazon Prime Air from Miami to Houston. The first officer had failed training events at several previous employers and had not disclosed them on his Atlas application. At the time of hiring, the Pilot Records Database directed by Congress after Colgan Air 3407 was not yet operational; Atlas used the older PRIA system, which required only five years of records — a window too short to surface a pattern of training failures spread across several prior employers, so the record Atlas could legally obtain was structurally incapable of showing what it most needed to see.[0]

WHAT HAPPENED

On 23 February 2019, during a turbulent descent toward Houston, the first officer inadvertently activated the aircraft's go-around mode, which commanded a pitch-up. Experiencing a somatogravic illusion that the nose was rising into a stall, he pushed forward hard. The 767 entered a steep dive from which the crew did not recover and crashed into Trinity Bay, killing all three aboard. The startle and spatial disorientation that drove the fatal push were the same kind of breakdown his prior training failures had repeatedly recorded, had that history been visible to the carrier that hired him.[1]

THE INVESTIGATION

The NTSB found the probable cause to be the first officer's inappropriate response to the inadvertent go-around activation, driven by a startle and spatial-disorientation response, and identified his pattern of training deficiencies and Atlas's inability to access his full record as contributing factors. The records that would have informed the hiring decision existed but did not reach it — the same information-flow failure named after Colgan a decade earlier, recurring here because the remedy built to fix it was authorized but not yet carrying the full history.[2]

THE CAPABILITY GAP

Atlas Air 3591 is the iteration test of the Colgan reform. The reform partially worked — the Pilot Records Database had been authorized — but it was not yet operational and its coverage was incomplete, so the same information-flow gap that killed fifty people in 2009 was still partly open in 2019. Partial implementation of a remedy leaves a measurable harm inside the aperture — the years between authorization and full coverage are not a neutral transition but a window in which the original failure can recur on the cases the incomplete remedy does not yet reach.[3]

AFTERMATH & REFORM

The PRD final rule was published in 2021 and became effective for Part 121 carriers in 2022, with full historical coverage phasing in through 2024 — closing the gap the case exposed. Atlas 3591 is cited in implementation-fidelity discussions as evidence that a remedy is only as good as its completed coverage — that the date a rule is authorized and the date it actually reaches every carrier are different dates, and the distance between them is where the avoidable harm accumulates.[4]

THE COST OF PARTIAL IMPLEMENTATION BETWEEN COLGAN AND THE PRD'S FINAL COVERAGE

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5. GAO reports on Pilot Records Database implementation (2014–2024).

THE LEARNING ENGINEERING LENS

> *Atlas Air did not have access to portions of the first officer's training record that would have informed its hiring decision.*
>
> PARAPHRASING NTSB AIRCRAFT ACCIDENT REPORT AAR-20/02, 2020

LE INSIGHT

Atlas Air 3591 is the iteration test of the Colgan reform. The reform partially worked: the PRD exists. The reform was incomplete: not all carriers were covered. The case is a cautionary worked example of why partial implementation of a regulatory remedy leaves the original capability gap partially open.

LENS APPROACH

LENS uses Atlas Air 3591 in LEN 4 as a case for measurement-system coverage (the PRD had been built but its mandatory coverage was incomplete), and in LEN 8 for the iteration cycle of reform: an intervention that leaves an aperture creates a measurable harm inside the aperture.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the records system to require a full operator history, not a fixed-length window that can hide a multi-employer pattern.
- Plan the coverage phase-in explicitly, with interim controls protecting the cases the remedy does not yet reach.
- Specify the data the hiring decision needs and make its delivery mandatory before the decision, not optional or self-reported.

IN OPERATION / AFTER

- Audit coverage completeness, not just authorization, and treat the gap between the two as an active source of harm until closed.
- Monitor for adverse events occurring inside the aperture of a partially implemented remedy, and use them to accelerate full coverage.
- Sustain attention through the full phase-in, since a remedy authorized but not yet universal leaves the original gap measurably open.

REFLECTION QUESTIONS

1. Identify a regulatory remedy in your domain whose coverage is partial. What harm is occurring inside the aperture?

2. Design the deliverable that would close the PRD coverage gap in advance of a future Atlas Air 3591.

3. The PRD was authorized in 2010 but not fully effective until 2024, and Atlas 3591 fell inside that gap. For a remedy in your domain, design the interim control that would protect the cases the remedy does not yet reach while its coverage phases in.

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TOOLS scenario simulation · competency models · knowledge bases · data pipelines

LENS COURSES LEN 4 LEN 8

CASE 50 AVIATION 2015

TransAsia Airways Flight 235

T Training Gap H Human-System Interface

IMPACT 43 killed in Taiwan; crew shut down the only working engine after the other failed

IN BRIEF

Forty-three seconds after takeoff from Taipei Songshan in February 2015, the right engine of a TransAsia ATR 72 auto-feathered following a sensor fault. Working from memory under acute time pressure, the crew shut down the left engine — the one still producing thrust — leaving the aircraft with no power and too little altitude to recover. It clipped a viaduct, struck a taxi, and crashed into the Keelung River, killing 43 of the 58 aboard. The Aviation Safety Council found the captain had failed proficiency checks in the year before the accident and that the crew had not used the published engine-shutdown checklist, responding instead from a flawed memory of the procedure. The wrong-engine error mirrors Kegworth (Case 30) twenty-six years earlier.

BACKGROUND

TransAsia 235 was an ATR 72-600 twin-turboprop departing Taipei Songshan. The captain had failed simulator proficiency checks in the year before the accident and had been noted for weak handling of abnormal procedures. The published response to an engine failure was a checklist-driven procedure; under stress, crews tend to revert to memory — so a captain already flagged for weak abnormal-procedure handling was exactly the operator most likely to skip the checklist precisely when it mattered, a known weakness meeting a known failure mode.[0]

WHAT HAPPENED

On 4 February 2015, forty-three seconds after takeoff, the right engine's autofeather system activated following a faulty sensor signal — the engine itself was capable of producing power. The crew, identifying an engine problem, shut down the left engine, which was operating normally. With neither engine now producing useful thrust and the aircraft low and slow, it stalled, clipped a viaduct and a taxi, and crashed into the Keelung River, killing 43 of the 58 aboard. A recoverable single-engine event became unrecoverable the moment the working engine was shut down, leaving an aircraft with no thrust and no altitude in the seconds just after takeoff when both are least forgiving.[1]

THE INVESTIGATION

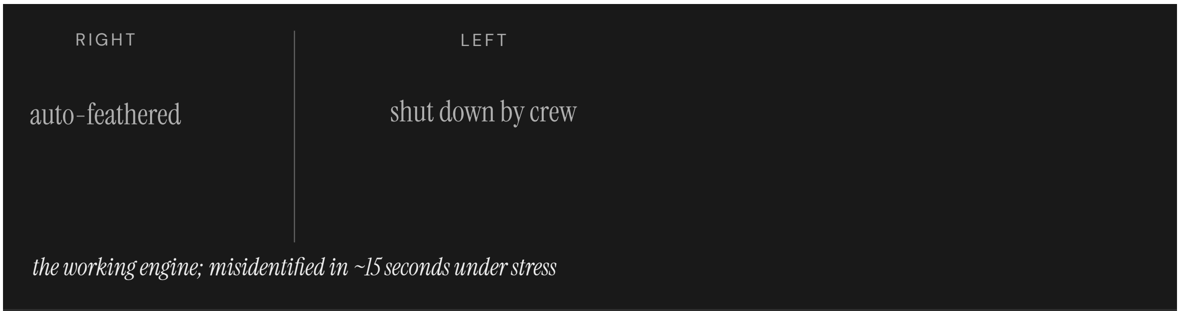
The Taiwan Aviation Safety Council found the crew failed to identify which engine had actually failed and shut down the wrong one. They did not follow the published engine-flameout and shutdown checklist, acting from memory under time pressure; the misidentification took roughly fifteen seconds. The captain's documented proficiency deficiencies and the airline's training and crew-resource-management shortfalls were contributing factors — the individual lapse sitting atop an organizational one, the carrier having left the very weaknesses its own proficiency checks had recorded uncorrected before the flight.[2]

THE CAPABILITY GAP

TransAsia 235 is the modern recurrence of the wrong-engine pattern that destroyed British Midland 92 at Kegworth in 1989 (Case 30). Crews under acute time pressure default to memory; if the memory is incomplete, the action follows the flawed memory rather than the checklist. The missing capability is a procedure design and training regime that keeps the verification step in the loop precisely when stress pushes operators to skip it — engineering the confirm-which-engine step so it fires under startle, rather than trusting it to a memory that time pressure is actively eroding.[3]

AFTERMATH & REFORM

The ASC issued recommendations on TransAsia's pilot training, proficiency-check standards, and crew-resource management; the airline ceased operations in 2016, after a second fatal accident the year before. The case is used to argue that the persistence of the wrong-engine pattern across a quarter-century reflects an un-engineered intervention, not merely individual error — the same failure recurring across crews, airframes, and decades points to a design and training gap that no amount of blaming the individual pilot has ever closed.[4]



TRANSASIA 235 — THE SAME WRONG-ENGINE PATTERN AS KEGWORTH (CASE 30)

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THE LEARNING ENGINEERING LENS

> *The flight crew was unable to identify which engine had failed and shut down the operating engine in error.*
>
> PARAPHRASING THE AVIATION SAFETY COUNCIL (TAIWAN) FINAL REPORT ON TRANSASIA 235, 2016

LE INSIGHT

TransAsia 235 is the modern recurrence of the Kegworth pattern under stress. Crews under acute time pressure default to memory; if the memory is incomplete or wrong, the action follows the memory. The capability gap was the same as in 1989. The intervention pattern needed is also the same.

LENS APPROACH

LENS uses TransAsia 235 in LEN 5 as a recurrence case for well-known capability problems (wrong-engine shutdown) and in LEN 2 to teach the difference between checklist-driven response and memory-driven response under stress.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the abnormal-procedure response so the confirm-which-unit verification step is forced under startle, not left to memory.
- Build proficiency checks whose red signals have the authority to ground an operator before the predicted failure, not merely to document a weakness.
- Train the checklist-driven response under realistic time pressure, so the trained reflex survives the stress that pushes crews to memory.

IN OPERATION / AFTER

- Audit whether documented proficiency deficiencies are actually acted on, treating an uncorrected red check as an organizational, not individual, failure.
- Monitor for recurrence of well-known patterns (wrong-engine shutdown) across crews and airframes as evidence the intervention is un-engineered, not merely human error.
- Sustain crew-resource-management and checklist discipline so the verification step does not erode back to memory under operational pressure.

REFLECTION QUESTIONS

1. Identify a procedure in your domain that is supposed to be checklist-driven but is actually memory-driven under stress. What is the gap?
2. Why has the wrong-engine shutdown pattern persisted across 25 years of aviation reform? What intervention has not yet been engineered?
3. The captain had failed proficiency checks in the year before the accident, yet kept flying. Design the proficiency-check regime in your domain whose red signal would actually remove an operator from duty before, not after, the failure it predicts.

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TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs

LENS COURSES LEN 5 LEN 2

Chapter 2

Designed Out

Capability was removed to save cost.

If we emphasize MCAS is a new function there may be greater certification and training impact.

BOEING INTERNAL MEETING MINUTES, 2013

CASE 2 AVIATION 2018 – 2019

Boeing 737 MAX / MCAS

D Designed Out T Training Gap H Human-System Interface

IMPACT 346 killed across two crashes — Lion Air 610 and Ethiopian Airlines 302

IN BRIEF

The Boeing 737 MAX was a re-engined 737 meant to fly without new pilot training — the commercial promise that sold it against the Airbus A320neo. Its bigger engines changed the jet's pitch behavior, so Boeing added software (MCAS) to mask the difference, then kept it out of the manuals and training and let it fire on a single angle-of-attack sensor. When that sensor failed on Lion Air 610 (2018) and Ethiopian 302 (2019), MCAS forced the nose down and crews who had never heard of it could not recover; 346 people died and the fleet was grounded. Investigations found the training omission was deliberate, to avoid certification cost. The MAX is the book's inverse case: human capability engineered out of a system to save the price of sustaining it.

BACKGROUND

The 737 MAX was Boeing's answer to the Airbus A320neo: a re-engined version of its best-seller that airlines could fly without retraining pilots on a new type. But the MAX's larger, forward-mounted engines changed how it pitched, so Boeing added software — MCAS — to make it handle like the older 737s, papering over an aerodynamic change with a control law so the airframe would feel identical from the cockpit. The whole commercial logic depended on that software staying invisible: the less of a "new function" MCAS seemed, the less the MAX would trigger costly new training, and the lower the airline's switching cost away from the A320neo. Boeing reportedly promised Southwest a rebate of about a million dollars per jet if simulator training proved necessary — a clause that turned the training requirement into a direct line on the program's ledger.[0]

WHAT HAPPENED

To keep MCAS in the background, it was left out of the manual and training, and allowed to fire on a single angle-of-attack sensor with no second source to cross-check it. When that sensor failed on Lion Air Flight 610 in October 2018, MCAS repeatedly forced the nose down; the crew, never told the system existed, fought it cycle after cycle until the jet dove into the Java Sea, killing all 189. Five months later Ethiopian Airlines Flight 302 repeated almost exactly the same sequence, killing 157 — 346 dead across the two crashes, and the entire MAX fleet grounded worldwide.[1]

THE INVESTIGATION

The investigations made the decisions legible. Internal 2013 minutes showed employees noting that calling MCAS a new function would bring "greater certification and training impact" — the cost the program was built to avoid, written down years before either crash;

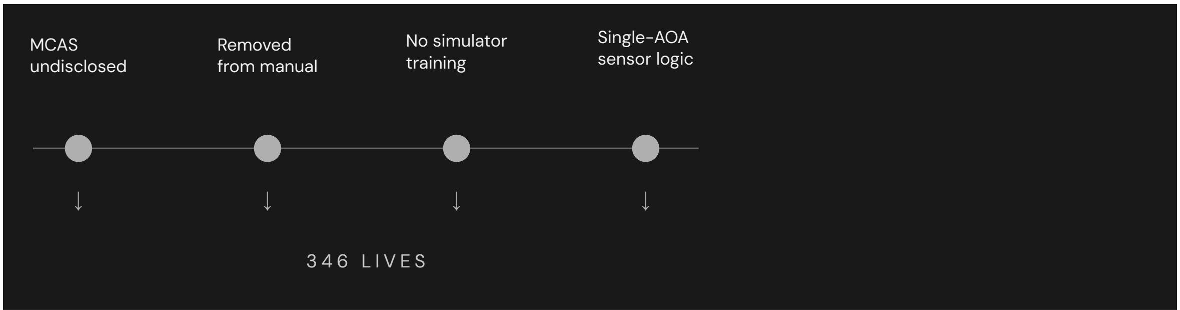
a 2016 survey found 39% of certification staff felt undue pressure. The House Transportation Committee's 2020 report concluded Boeing's assumption that simulator training was unnecessary "diminished safety, minimized the value of pilot training, and inhibited technical design improvements," naming the omission as a choice rather than an oversight.[2] The DOT Inspector General traced how the single-sensor design and the training omission passed through a certification process that delegated much of the safety judgment back to Boeing itself, so the company effectively reviewed its own most consequential trade-offs.[3]

THE CAPABILITY GAP

The MAX inverts this book's usual case. Human capability was not overlooked by accident; it was deliberately engineered out to avoid the cost of the training that would have created it. The pilots were not undertrained by oversight but by design — the absence of training was, in effect, a contract deliverable promised to customers and protected by a rebate. Seen that way, the crashes are not a training failure that befell a good airplane but exactly what the commercial and engineering decisions specified: a flight-control system that depended on pilots reacting correctly, in seconds, to a failure they had been guaranteed never to learn about.[4]

AFTERMATH & REFORM

The MAX was grounded for some twenty months. MCAS was redesigned to use both sensors, fire once, and with limited authority — restoring the margins the original had stripped away. The FAA's 2020 return-to-service required the simulator training the airplane had been built to avoid, and Congress tightened the FAA's oversight of the delegation system that had let Boeing self-certify so much of the design.[5] The reform conceded the point the program had spent years resisting: the training was a real requirement all along, and removing it from the paperwork only deferred the cost — and then moved it onto two airplanes full of people.



MCAS — CAPABILITY ENGINEERED OUT AT FOUR DECISION POINTS

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THE LEARNING ENGINEERING LENS

> *Boeing's assumption that simulator training was unnecessary diminished safety, minimized the value of pilot training, and inhibited technical design improvements.*
>
> HOUSE TRANSPORTATION COMMITTEE REPORT, 2020

LE INSIGHT

The 737 MAX is the documentary record of a design choice to remove human capability to avoid the regulatory and commercial cost of sustaining it. The pilots were not undertrained by accident — they were undertrained as a **requirement**. The omission of training was a contract deliverable. Once that is understood, the accident becomes legible as an engineering decision rather than a training failure.

LENS APPROACH

LENS treats capability requirements as system requirements: derived from operational analysis, traceable to design decisions, falsifiable against operational outcomes. A LENS-trained engineer in the MAX certification process would have produced a capability-requirements artifact mapping pilot tasks to system behaviors — and flagged the proposed manual omission as an unmitigated requirements gap.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat any control law that masks a handling change as a new function in its own right — document it, surface it to pilots, and size its training as part of the design, not after it.
- Forbid safety-critical authority from resting on a single sensor; require an independent source or a cross-check before software can command the flight surfaces.
- Quarantine commercial commitments — rebates, "no new training" promises — from the engineering judgment about what capability the airplane actually requires.

IN OPERATION / AFTER

- Audit delegated certification so the party making a trade-off is never the same party that signs off on its safety, restoring independent review of the riskiest decisions.
- Monitor in-service AOA-disagree and uncommanded-trim events as leading indicators, with a route that reaches engineering before a second airframe is lost.
- Re-validate the "no new training needed" claim against real fleet incidents, and reopen the training requirement the moment operational data contradicts the assumption.

REFLECTION QUESTIONS

1. If a customer contract required removing a training requirement you judged necessary, what artifact would you produce, who would you route it to, and what would you do if the routing failed?
2. Reconstruct the MAX accident as a **capability-requirements** failure: what was the elided requirement, at what life-cycle stage was it elided, and who had the authority to insert it?
3. MCAS was permitted to act on a single sensor with no cross-check. Identify a system you work on that takes a safety-critical action from one unverified input, and specify the redundancy or human confirmation it lacks.

WHO BUILDS THIS systems & human-factors engineering · learning science & instructional design · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS task analysis · design prototyping · scenario simulation · competency models · usability testing · cognitive walkthroughs

LENS COURSES LEN 1 LEN 5 LEN 2

CASE 7 HEALTHCARE TECHNOLOGY 1985 – 1987

Therac-25

H Human-System Interface D Designed Out G Governance & Trust

IMPACT **Six known massive radiation overdoses; at least three deaths; canonical case in software safety engineering**

IN BRIEF

The Therac-25, a radiation-therapy machine, massively overdosed at least six patients between 1985 and 1987, killing at least three. Its predecessors used hardware interlocks to stop the high-energy beam from firing without its target in place; to save cost, the Therac-25 removed them and trusted software — adapted from the older machines and never engineered for safety — to keep the beam modes separated. A race condition let a fast operator trigger the full beam with no target, while the console showed only a meaningless "Malfunction 54." Leveson and Turner's 1993 investigation, the founding case of software-safety engineering, found a systemic failure: a safeguard removed with nothing put in its place. The lesson — a safeguard you delete does not remove the hazard; it relocates it to whatever you failed to build.

BACKGROUND

The Therac-25 was a radiation-therapy linear accelerator. Its predecessors used hardware interlocks that physically blocked the high-energy beam unless the spreading target was confirmed in place — a mechanical backstop that could not be talked out of stopping the beam. To save cost and simplify the machine, the Therac-25 removed them and trusted software — adapted from the older machines and never engineered from the ground up for safety-critical use — to keep its two beam modes, a hundredfold apart in energy, safely separated. The safety case migrated silently from steel to code.[0]

WHAT HAPPENED

Between 1985 and 1987 the machine massively overdosed at least six patients. A race condition the engineers never knew about meant that if an operator entered a prescription, caught a mistake, and corrected it within about eight seconds — the speed an experienced operator naturally reaches — the full-power beam could fire with no target in place, delivering up to a hundred times the intended dose. The console showed only "MALFUNCTION 54," a code with no documented meaning, and offered to proceed; operators, assured the machine was safe and long accustomed to its cryptic faults, did. At least three patients died of the radiation burns.[1]

THE INVESTIGATION

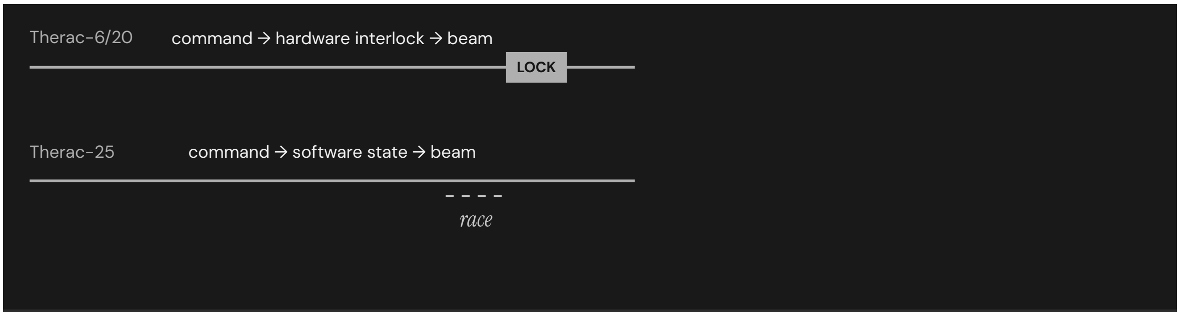
When patients reported searing pain, the manufacturer first insisted an overdose was impossible and treated each report as an isolated complaint rather than a signal. Nancy Leveson and Clark Turner's 1993 investigation — the founding case study of software-safety engineering — found the fault was systemic rather than a single bug: overconfidence in software, removal of the hardware safeguards without replacement, meaningless error messages, reused code never audited for safety, no independent review of the safety-critical logic, and an incident-reporting posture that dismissed the early warnings instead of compounding them into evidence.[2]

THE CAPABILITY GAP

The hardware interlock had not been redundant; it **was** the safety case, the one thing standing between a typing error and a lethal dose. Removing it put nothing in its place — no verified software check, no informed operator action, no independent monitor watching the beam. The operator stayed nominally in the loop but lost any means to detect what the machine was doing wrong, since the only feedback was a code that told them nothing, which made the human presence a formality rather than a safeguard. The question capability engineering would have forced — **what function now carries the interlock's load?** — was never asked of the redesign.[3]

AFTERMATH & REFORM

Therac-25 reshaped safety-critical software practice, making the case by counterexample for independent hazard analysis, safeguards that do not all rest on the same software, error messages that say what is wrong so the operator can act, and treating field reports as evidence to be aggregated rather than complaints to be closed. It remains the canonical teaching case across software, medical-device, and systems-safety curricula.[4] Its lesson is exact and portable: a safeguard you remove does not remove the hazard it guarded — it relocates the hazard to whatever you failed to put in its place, and waits there.



SOFTWARE RACE CONDITION REPLACED HARDWARE INTERLOCK

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THE LEARNING ENGINEERING LENS

> *Therac-25 illustrates the dangers of relying on software safety controls without rigorous engineering practices.*
>
> PARAPHRASING LEVESON & TURNER, IEEE COMPUTER, 1993

LE INSIGHT

Therac-25 is the moment when removing the human safety margin became visible as a design choice. The hardware interlock was not redundant — it was the safety case. When it was removed, no equivalent capability was put in its place. The operator was retained in the system but stripped of the ability to detect what it was doing wrong. Capability engineering would have asked, before removing the interlock, **what function takes its load?**

LENS APPROACH

LENS frames Therac-25 as a **Design-Out** failure made visible through Interface and Governance pathologies. Studio projects in LEN 5 ask students to produce capability-load diagrams tracing every safety function to a hardware backstop, a software check, or a trained operator action with the information needed to perform it. LEN 7 examines incident reporting as governance.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Before deleting any hardware interlock, write down the safety function it performs and explicitly reassign that function to a verified software check, an independent monitor, or an informed operator action.
- Do not let safety rest entirely on one software path; require an independent channel that does not share the same code, so a single defect cannot defeat the whole safety case.
- Specify error messages as a safety deliverable — each fault code must say what is wrong and what the operator should do, never offer to proceed past an undiagnosed condition.

IN OPERATION / AFTER

- Treat every field report of unexpected harm as evidence to be aggregated, not a complaint to be closed, with a standing path to halt the device when a pattern emerges.
- Audit any safety-critical code reused from a prior system against the new hazard set, since assumptions safe in the old context may be lethal in the new one.
- Instrument the machine so an independent monitor can detect a beam fired without its target and stop it, restoring the interlock's function even where the operator cannot see the fault.

REFLECTION QUESTIONS

1. Identify a system in your domain that migrated a safety function from hardware to software. Where did the human-capability load go, and who is accountable for sustaining it?
2. Therac-25 operators saw "MALFUNCTION 54" and continued treatment. Redesign that interface using LEN 2 principles so that the operator's correct action is the easiest action.
3. The Therac-25's safety-critical software was inherited from earlier machines and never re-audited. Identify reused code in a system you build that now carries a load it was never written for, and specify the review it should have received.

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LENS COURSES LEN 5 LEN 7 LEN 2

CASE 19 DEFENSE 1991

Patriot Missile / Dhahran

D Designed Out H Human-System Interface K Knowledge & Institutional Memory

IMPACT 28 U.S. soldiers killed in their barracks; roughly 100 wounded

IN BRIEF

On 25 February 1991 a Scud missile struck a U.S. barracks in Dhahran, Saudi Arabia, killing 28 soldiers and wounding about a hundred; the Patriot battery defending the area never engaged it. Designed for short engagements against Soviet aircraft in Europe, the Patriot tracked time in a register whose tiny rounding error grew with every hour of uptime. After about a hundred hours of continuous operation the Gulf War demanded, the radar's range gate was off by a third of a second — enough to look in the wrong place. Israeli operators had flagged the drift two weeks earlier, and a patch reached Dhahran the day after the strike. The capability to hold accuracy under sustained use was assumed away at design time, and no one carried that assumption forward when the mission changed.

BACKGROUND

The Patriot was built to defend Western Europe against Soviet aircraft in engagements of minutes — switched on, fired, switched off, then moved before it could be targeted. In the 1991 Gulf War it was doing something else: defending fixed sites in Saudi Arabia against ballistic missiles, in batteries left running continuously for more than a hundred hours because the threat came without warning and could not be scheduled. The mismatch between the original concept of operations and the new use was real, and invisible to the operators, because no one had told them the machine had been built around an assumption they were now violating.[0]

WHAT HAPPENED

The system tracked time in a 24-bit fixed-point register, and a tiny rounding error — invisible in any short engagement — accumulated with every hour of uptime. After about a hundred hours the timing was off by roughly a third of a second, which seems negligible until it is multiplied by a ballistic missile's speed: enough for the radar's range gate to look in the wrong place and reject the real track as noise. On 25 February 1991 an incoming Scud arrived where the Patriot was no longer searching, passed unengaged, and struck a barracks in Dhahran, killing twenty-eight soldiers of the 14th Quartermaster Detachment and wounding about a hundred.[1]

THE INVESTIGATION

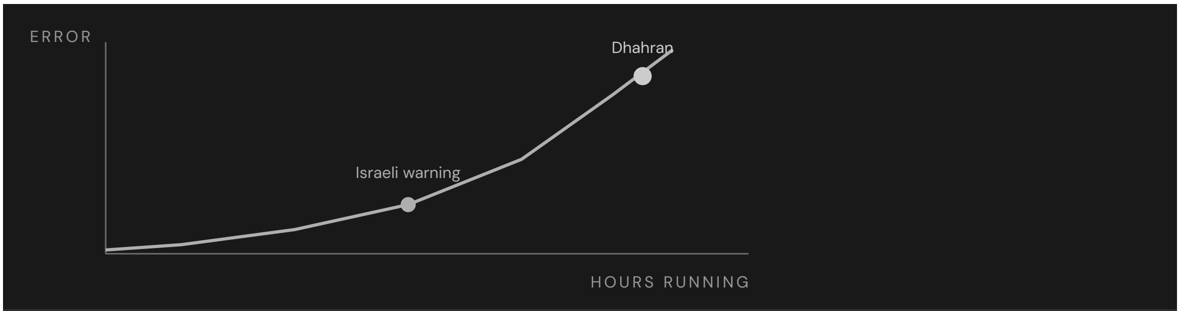
The drift was not unknown: Israeli operators had flagged it two weeks earlier from their own sustained use, and engineers had a patch already in hand — which reached Dhahran the day after the strike, too late to matter.[2] The only field mitigation was an advisory to reboot after "very long" run times, never defining "very long," so a crew could obey the instruction to the letter and still drift into the danger band. The General Accounting Office found the Army had simply presumed no one would run a battery continuously for so long, and so never treated the accumulating error as a hazard worth specifying or warning against.[3]

THE CAPABILITY GAP

The capability to hold accuracy under sustained operation was never built in, because the original concept of operations did not require it — a defensible omission for the system as first imagined, fighting in short bursts and moving on. The failure came when the concept changed and the assumption did not travel with it. Nothing carried the design's hidden premise forward to the soldiers in Dhahran; no artifact, briefing, or warning told them "this system was built assuming you would turn it off." The system did not malfunction — it did exactly what it was built to do; the transition between operating contexts did, with nothing in place to catch the broken premise.[4]

AFTERMATH & REFORM

The patch was distributed and concrete reboot intervals defined in place of the vague advisory, and the defect — a fixed-point truncation error growing without bound as uptime increased — became a staple of numerical-analysis and software-engineering teaching.[5] The harder lesson is the one this chapter turns on: a capability quietly assumed away at design time does not announce itself when the context shifts. It waits, intact and invisible, until the day the assumption is wrong — and by then the people relying on the system have no way to know the ground has moved beneath them, because the premise was never written where they could read it.



TRACKING ERROR GREW WITH EACH HOUR OF CONTINUOUS OPERATION

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3. GAO/IMTEC-92-26 (1992) — the prior Israeli warning, the software patch arriving in Dhahran on 26 February (the day after the strike), and the 28 killed.
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5. M. Barr / Barr Group, "An Update on the Patriot Missile Software Problem" — an engineering post-mortem of the accumulated-truncation defect.
6. GAO/IMTEC-92-26 (1992) and Skeel (1992) — the design assumption that failed to travel across the change in operational context, and the case's teaching legacy.

THE LEARNING ENGINEERING LENS

> *Army officials presumed that the users would not continuously run the batteries for such extended periods.*
>
> GAO/IMTEC-92-26, 1992

LE INSIGHT

Patriot is the canonical Designed-Out case from defense. The capability to maintain accuracy under sustained operation was omitted because the original concept of operations did not anticipate it. When the concept changed, the assumption did not travel. The system did not fail; the transition from one operational context to another did, and there was no capability infrastructure to catch it.

LENS APPROACH

LENS treats Patriot as the textbook example of **Capability Degradation Under System Change**. LEN 5 methods require operators of any transitioning system to have current capability-relevant documentation specifying what assumptions of the original design have changed. LEN 8 examines how organizational knowledge about design constraints travels from engineering to operations — and why, here, it arrived a day late.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make every load-bearing design assumption — here, that the system would be cycled off — an explicit, recorded requirement so a later change of use cannot silently violate it.
- Bound numerical error over the full intended operating envelope, including run times far beyond the original concept, rather than only the durations first imagined.
- Design the system to detect and bound its own accumulating drift, degrading or warning before the error reaches a magnitude that defeats the mission.

IN OPERATION / AFTER

- Issue concrete, testable operating limits — exact reboot intervals, not “very long” — derived from the design data and propagated to every crew.
- Establish a path for field reports of anomalous behavior, like the Israeli warning, to reach engineering and trigger fleet-wide action faster than a patch can lose a race with the threat.
- Require a capability-transition review whenever a system is redeployed to a new concept of operations, surfacing which original assumptions the new mission breaks.

REFLECTION QUESTIONS

1. The Patriot’s design assumed short engagements. What assumption in a current system you work with would become lethal if the operational context shifted, and how would operators learn of the shift?

2. Construct the capability-transition artifact that should have accompanied the redeployment of Patriot batteries from Europe to Saudi Arabia. What would it have said, and who would have signed it?

3. The field advisory said to reboot after “very long” run times without defining the term. Rewrite a vague operational instruction in your domain into a specific, testable limit, and name the design data the limit must be derived from.

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LENS COURSES LEN 5 LEN 8 LEN 9 LEN 6

CASE 51 INDUSTRIAL 1971 – 1978

Ford Pinto Fuel Tank

D Designed Out G Governance & Trust

IMPACT Fatalities attributed to rear-impact fuel-tank fires; foundational case in U.S. engineering ethics

IN BRIEF

The Ford Pinto, a rushed early-1970s subcompact, placed its fuel tank where rear-impact collisions could rupture it; fuel-fed fires killed and burned people, and in Grimshaw v. Ford (1981) a jury found Ford liable, having known of the hazard and not fixed it. A **common misconception**, popularized by a 1977 Mother Jones exposé, holds that Ford ran a cost-benefit calculation and chose to pay death settlements rather than install an $11 fix. The memo invoked for that story was actually a generic federal submission about rollover fires across the whole U.S. fleet — it never mentioned the Pinto. Strip away the myth and the real designed-out failure remains: rear-impact survivability lost to cost and packaging, and a known hazard went unfixed. It is the founding engineering-ethics case — and a lesson in accuracy.

BACKGROUND

The Pinto was Ford's rushed early-1970s answer to imported subcompacts, built to aggressive cost and weight targets on a compressed schedule that left little slack for late design changes. To save room and money, its fuel tank sat behind the rear axle with little crush space; in rear-impact tests it could be punctured and the filler neck torn away, releasing fuel near potential ignition sources. Engineers knew of the vulnerability from the company's own crash tests, and cheap fixes — a shield, a revised tank position — existed, but none was adopted before the car shipped to dealers.[0]

WHAT HAPPENED

In service, rear-impact collisions produced fuel-fed fires that killed and burned people, the precise failure mode the crash tests had shown. The reckoning came in court: in Grimshaw v. Ford — a 1972 crash that killed the driver and severely burned teenage passenger Richard Grimshaw — a jury found Ford liable, and the 1981 appellate court upheld liability, finding Ford had known of the hazard and not acted on it. Ford recalled some 1.5 million Pintos and Mercury Bobcats in 1978, and an Indiana prosecutor even charged it with reckless homicide (Ford was acquitted at trial). The car became the enduring emblem of corporate disregard for safety.[1]

THE INVESTIGATION

That emblem rests partly on a common misconception that has hardened into folklore. Mother Jones's 1977 exposé "Pinto Madness" popularized the claim that Ford had calculated it was cheaper to pay death settlements than install an eleven-dollar fix. The document invoked — the Grush-Saunby analysis — was actually a generic submission to regulators about fuel-fed fires in **rollovers** across the **entire** U.S. fleet; it valued societal costs using a government figure, never mentioned the Pinto by name, concerned a different crash mode, and postdated the car's design. Gary Schwartz later documented the gap in detail, including a death toll far below the hundreds the article implied.[2]

THE CAPABILITY GAP

Strip away the myth and the designed-out failure remains — which is why the case is worth keeping rather than discarding with the legend. Rear-impact survivability was a real capability that lost to cost, weight, and packaging, and a known vulnerability went unaddressed long enough for people to die and a jury to find Ford liable. The cartoon version — a villain coldly pricing lives against a cheap part — actually makes the lesson easy to dismiss as something that happens only to monsters. The truer one is ordinary and therefore more dangerous: safety capability erodes not through a single monstrous choice but through many small trades against cost and weight that no one owns as a requirement.[3]

AFTERMATH & REFORM

The Pinto drove enforcement of rear-impact fuel-system standards and helped shape the modern consumer-safety and punitive-liability regime, and for fifty years it has been the founding engineering-ethics case taught to new engineers. Its honest telling earns the place twice: as a lesson in how cost and weight quietly design out a safety capability, and as a caution in how a case itself gets mythologized into a shape more satisfying to repeat than accurate to learn from — the value of a statistical life is a real and legitimate regulatory tool, and the true objection was never that a number was used to weigh safety but that a known hazard went unfixed.[4]

$11

PER CAR · THE FIX OF LEGEND

the cost-benefit story is largely myth — the memo behind it was a fleet-wide rollover submission, not a Pinto ledger

PINTO — THE COST-BENEFIT STORY THAT BECAME MYTH

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THE LEARNING ENGINEERING LENS

> *Much of what people believe about the Pinto — a deliberate trade of lives against an eleven-dollar part — is myth; what is not myth is that a known fuel-system hazard went unfixed.*
>
> PARAPHRASING G. T. SCHWARTZ, "THE MYTH OF THE FORD PINTO CASE," 1991

LE INSIGHT

The Pinto is the founding Designed-Out ethics case — and a lesson in accuracy. The popular cost-benefit story is largely myth, but the designed-out capability (rear-impact survivability, traded against cost and packaging) was real, and Ford was held liable for it. LENS keeps the case for both halves: how cost quietly removes safety capability, and how a case can be mythologized into something easier to repeat than to learn from.

LENS APPROACH

LENS uses the Pinto in LEN 7 as the foundational engineering-ethics case and in LEN 1 as a problem-framing case for what happens when safety capability is treated as a cost line item.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make rear-impact survivability — or its equivalent safety capability — an explicit, owned requirement that cost and weight targets must be reconciled against, not silently traded away under schedule pressure.
- Treat the company's own crash-test evidence of a hazard as a design gate: a known vulnerability with a cheap fix cannot pass to production unaddressed.
- Bring an inexpensive mitigation (a shield, a revised tank position) into the design before ship rather than weighing it against expected litigation after.

IN OPERATION / AFTER

- Audit fielded products against the failure modes the company's testing already revealed, so a known hazard is fixed by recall before it accumulates a casualty record.
- Use the value-of-a-statistical-life calculation honestly — as a tool to justify fixing hazards, not to defend leaving them — and document the reasoning where it can be reviewed.
- Guard the case's lesson against its own myth: review failures for the ordinary accumulation of cost trades, not just for a single bad actor who is easy to disown.

REFLECTION QUESTIONS

1. Identify a current product in your domain where a safety capability is being priced against expected litigation. Whose signature would close that decision?
2. Design the regulatory deliverable that would have made the Pinto cost-benefit calculation unworkable in 1971.
3. The Pinto failure came not from one villainous choice but from many small trades against cost and weight that no one owned as a requirement. Map such an accumulation in a current program and name who should hold the safety capability as an explicit, non-tradable line.

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LENS COURSES LEN 1 LEN 7 LEN 6

CASE 52 INDUSTRIAL 2008 – 2023

Takata Airbag Inflators

D Designed Out G Governance & Trust

IMPACT More than 30 deaths and hundreds of injuries linked to inflator ruptures; largest automotive recall in history

IN BRIEF

Takata built its airbag inflators around ammonium nitrate — cheap and energetic but unstable, and used by almost no one else without a moisture-absorbing desiccant. Takata's inflators omitted the desiccant. Over years of heat and humidity the propellant degraded and could rupture the metal housing on deployment, spraying shrapnel at the driver; more than two dozen deaths and hundreds of injuries followed. Takata's own tests had shown the ruptures, but it reported them as isolated anomalies and, in places, falsified data; in 2017 it pleaded guilty to wire fraud, paid roughly $1 billion, and went bankrupt. The recall — over 100 million inflators across 19 automakers — is the largest in automotive history. Two capabilities were designed out: the desiccant, and the independent verification regulators never had.

BACKGROUND

An airbag inflates by burning a propellant inches from a person's face, so the propellant must be energetic **and** stable across the years and climates a car will see. To cut cost, Takata built its inflators around ammonium nitrate — cheap and energetic but notoriously unstable, shifting crystalline structure and absorbing moisture as temperature cycles day after day. The few competitors who used it at all added a desiccant to keep it dry over the life of the car; Takata's inflators, for years, shipped with none.[0]

WHAT HAPPENED

Over years of heat and humidity the propellant degraded, and a degraded charge can burn too fast, generating more pressure than the housing was built to contain. A firing inflator could then rupture its own metal housing and spray shrapnel into the cabin — turning the device meant to save a life into a fragmentation hazard aimed at the driver. More than two dozen deaths worldwide and hundreds of injuries followed. The recall grew to more than a hundred million inflators across some nineteen automakers — the largest in automotive history — and Takata went bankrupt; the toll keeps rising as unrepaired vehicles stay on the road and their inflators keep aging.[1]

THE INVESTIGATION

Takata's own testing had shown the ruptures, but internal documents revealed engineers raising the alarm internally while the company reported the failures to automakers and regulators as isolated anomalies rather than a systemic propellant problem — and, in places, manipulated the test data outright.[2] In 2017 Takata pleaded guilty to wire fraud and paid roughly a billion dollars in fine, restitution, and a victims' fund. The legal finding was pointed: not that a part had failed, but that the company had spent years misrepresenting what its own engineers knew.[3]

THE CAPABILITY GAP

Two capabilities were designed out, and the second matters as much as the first. The product capability was the desiccant — a stabilizing component competitors used, omitted to save cost, in plain view of anyone comparing designs side by side. The system capability was independent verification: the recall regime treated the manufacturer's representations about its own safety data as authoritative, with no independent pipeline to test inflator behavior across the fleet as it aged in the field — which let an obvious failure mode hide for years inside a process that kept receiving the evidence and reporting it back as noise.[4]

AFTERMATH & REFORM

The recall is still being worked off, vehicle by vehicle, long after the company that built the inflators ceased to exist — concrete proof that a designed-out capability can outlive the firm that removed it and become someone else's burden. The episode pushed regulators toward more aggressive, coordinated recall management that does not leave pace to each manufacturer. Its central lesson is the pairing: Takata is the modern Pinto in its product failure, and more in its system failure — the evidence pipeline a regulator relies on is itself a safety capability, and omitting it is as consequential as omitting the desiccant from the inflator.[5]

100M+

INFLATORS RECALLED · 19 AUTOMAKERS

the desiccant competitors used was the designed-out capability

TAKATA — THE LARGEST AUTOMOTIVE RECALL ON RECORD

REFERENCES

1. U.S. National Highway Traffic Safety Administration, Takata airbag inflator recall coordination materials — the ammonium-nitrate-without-desiccant design and the propellant-degradation rupture mechanism.
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3. U.S. Department of Justice, settlement and guilty plea with Takata Corporation (2017) — wire fraud, $1B in fine and restitution, and Takata's subsequent bankruptcy.
4. U.S. DOJ (2017) and Takata internal documents released in litigation — the sustained misrepresentation of inflator test data to automakers and regulators.
5. NHTSA Takata recall status reporting — the years-long completion of replacements and the continuing risk in unrepaired vehicles.
6. NHTSA Coordinated Remedy Program for the Takata recalls — the regulator's move to actively prioritize and manage replacement across nineteen automakers rather than leave pace to each manufacturer.

THE LEARNING ENGINEERING LENS

> *Takata engaged in a sustained pattern of misrepresenting inflator safety data to its automaker customers and to regulators.*
>
> PARAPHRASING THE U.S. DEPARTMENT OF JUSTICE SETTLEMENT WITH TAKATA, 2017

LE INSIGHT

Takata is the modern Pinto: an engineering choice to omit a stabilizing component, internal data showing the consequence, and a regulatory architecture that treated the manufacturer's representation as authoritative rather than as one input. The capability gap was at the regulator's evidence pipeline as much as at the manufacturer's bench.

LENS APPROACH

LENS uses Takata in LEN 4 as a measurement-system case (the manufacturer's data was reported but not interpreted as a class) and in LEN 7 as an industrial-governance failure spanning manufacturer, regulator, and customer auditors.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Qualify a safety-critical material against the full service life and climate range it will see, not just its as-delivered state, so slow degradation cannot turn a safe part lethal years later.
- When choosing a cheaper, less stable material, require the stabilizing measure competitors use — here a desiccant — rather than omitting it to save cost.
- Design inflator testing to detect the rupture failure mode as a class across temperature and humidity cycling, so it cannot be dismissed as an isolated anomaly.

IN OPERATION / AFTER

- Stand up an independent verification pipeline that tests fielded inflators across the aging fleet, so a regulator does not depend on the manufacturer's own representation of its safety data.
- Aggregate field rupture reports as a population signal rather than closing them one by one, with authority to trigger recall when a pattern appears.
- Manage the recall as a coordinated program that prioritizes replacement across automakers, since a designed-out hazard outlives the firm and keeps aging in unrepaired cars.

REFLECTION QUESTIONS

1. Where in your domain does a regulator receive manufacturer test data without an independent verification pipeline? What is the load-bearing trust assumption?
2. Design the verification regime that should have surrounded ammonium-nitrate inflator testing. Who funds it, who runs it, and what does it produce?
3. Takata's propellant degraded slowly with heat and humidity, so a part safe at delivery became lethal years later. Identify a component in your domain whose qualification testing does not cover its full service life, and specify the aging test that would close the gap.

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LENS COURSES LEN 4 LEN 7

CASE 53 INDUSTRIAL 2002 – 2014

GM Ignition Switch

D Designed Out G Governance & Trust

IMPACT 124 deaths attributed; ~2.6M vehicles recalled for the defective switch; $900M federal forfeiture (DOJ, 2015); the fix existed for eight years before the recall

IN BRIEF

A faulty ignition switch in several GM compact cars (the Chevrolet Cobalt, Saturn Ion) could slip from “run” to “accessory” while driving, cutting power steering and brakes and — fatally — disarming the airbags. GM engineers identified it in 2002. In 2006 an engineer approved a redesigned switch but did not change its part number, so the fix propagated as neither a revision nor a recall, and defective cars kept selling. The recall came only in 2014 — about 2.6 million vehicles, with 124 deaths attributed through GM’s compensation fund. The Valukas report found a culture that absorbed safety concerns, and GM paid a $900 million federal penalty. What was designed out was not a part but the pathway by which a known safety problem reaches the decision to recall.

BACKGROUND

The ignition switches in several GM compact cars — the Chevrolet Cobalt and Saturn Ion among them — had too little detent torque to hold their position, so a jostle, a bump in the road, or a heavy keychain could rotate the switch out of “run” while driving. That cut power steering and braking and, fatally, disarmed the airbags so they would not deploy in the very crash that loss of control often produced. GM engineers identified the problem in 2002, during development, before the cars ever reached customers.[0]

WHAT HAPPENED

In 2006 a GM engineer approved a redesigned switch but did not change its part number — and in an engineering organization the part number is the very mechanism by which the system knows something changed. The fix propagated as neither a revision nor a recall; defective cars kept selling, and crashes with non-deploying airbags were investigated piecemeal over years, no one connecting them to a switch the records said had “never changed.” The recall came only in 2014 — about 2.6 million vehicles, with 124 deaths attributed through the compensation fund. What finally broke the silence was a wrongful-death lawsuit, in which a family’s expert found the part had been quietly changed under the same number — the smoking gun GM’s own records had been built to miss.[1]

THE INVESTIGATION

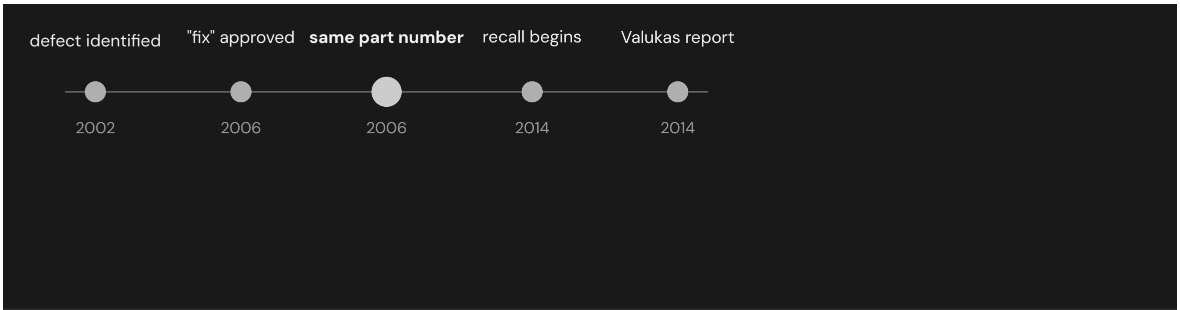
GM’s commissioned investigation by Anton Valukas (2014) found not a single villain but a culture that absorbed safety concerns until they dissipated — the “GM nod,” in which a room agrees a thing should happen and then no one acts, and the "GM salute," arms crossed and each person pointing elsewhere — and a fundamental failure to use the escalation processes the company already had on the books.[2] In 2015 GM paid a $900 million federal forfeiture for concealing the defect; total penalties and settlements ultimately exceeded $2.6 billion.[3]

THE CAPABILITY GAP

What was designed out of GM was not a part but a pathway — the institutional route by which a known safety problem reaches the decision to recall. The fix existed for eight years; the path from fix to recall did not, so the knowledge sat inert. The mechanism is mundane and all the more instructive for it: the part-number convention was the company's own way of seeing what had changed, and quietly breaking it blinded the organization to its own action. Suppressing a signal need not be a conspiracy; here it was one engineer taking the path of least resistance through a process no one was assigned to guard.[4]

AFTERMATH & REFORM

Mary Barra, who became CEO as the recall broke, used the Valukas report to restructure GM's safety decision-making — a global-safety leadership role, consolidated escalation channels, and a "Speak Up for Safety" program to give concerns a route upward that did not depend on one person's persistence to survive. The case became a standard teaching example in governance and psychological safety: an organization can hold the fix for the better part of a decade and still fail to act if the channel that carries bad news upward has been allowed to fail quietly, with no one accountable for keeping it open.[5]



GM IGNITION SWITCH — EIGHT YEARS BETWEEN THE FIX AND THE RECALL

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THE LEARNING ENGINEERING LENS

> *There was a fundamental failure to use the formal escalation processes that GM had established.*
>
> PARAPHRASING THE VALUKAS REPORT TO THE GM BOARD OF DIRECTORS, 2014

LE INSIGHT

The GM ignition switch case is the canonical example of a corporate organizational structure that suppressed the upward flow of safety information by procedural design. The fix existed for eight years before the recall. The institutional path between the fix and the recall did not.

LENS APPROACH

LENS uses GM in LEN 7 as a corporate-governance failure case, in LEN 8 to study Valukas-style retrospective institutional analysis, and in LEN 4 for the part-number-as-measurement-instrument story (changing the part without changing the number suppressed the signal).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the part number as a safety-critical signal: make changing a part without changing its number impossible by process, so the system cannot be blinded to its own revision.
- Specify the ignition switch's detent torque as a hold-position requirement and verify it, so a known under-torque defect cannot pass into production as a tolerable quirk.
- Build the route from "fix approved" to "recall decision" as an explicit deliverable, not a path that depends on one person choosing to escalate.

IN OPERATION / AFTER

- Correlate field crashes with non-deploying airbags as a population, so piecemeal incidents are connected to a common cause rather than investigated and closed one at a time.
- Run an escalation channel that carries bad news upward independent of any single person's persistence, with a named owner accountable for keeping it open.
- Audit whether approved fixes have actually propagated to a recall or revision, closing the gap in which a known defect keeps shipping under an unchanged record.

REFLECTION QUESTIONS

1. What information channel in your organization carries the same load that GM's part-number system did? Could it be silently bypassed?
2. Design the escalation deliverable that would have moved the GM ignition switch fix to a recall in 2006.
3. A single engineer changed the switch without changing its part number, and no one was assigned to guard that convention. Identify a process in your organization whose integrity depends on an unenforced convention, and name who should own enforcing it.

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LENS COURSES LEN 4 LEN 7 LEN 8 LEN 6

CASE 54 SPACE / AEROSPACE 1999

Mars Climate Orbiter — Unit Mismatch

D Designed Out K Knowledge & Institutional Memory

IMPACT ~$125M spacecraft lost on approach to Mars; ground software produced thrust output in pound-force while navigation expected newtons

IN BRIEF

The Mars Climate Orbiter, a NASA spacecraft built by Lockheed Martin and navigated by JPL, was lost on arrival at Mars in September 1999. Lockheed's ground software reported thruster impulse in pound-force seconds; JPL's navigation expected newton-seconds. The conversion — a factor of about 4.45 — was never applied at the boundary between the two teams, so every trajectory correction was mis-modeled and the error accumulated over the cruise. The orbiter arrived too low, into the atmosphere, and burned up; the orbiter and its $125 million were lost to a unit mismatch. The investigation found no individual blunder — both contractors did their own work correctly. What failed was the interface between them, which had no owner, specification, or verification step. It is the canonical case of interface-as-requirement.

BACKGROUND

The Mars Climate Orbiter, launched in December 1998 to study the Martian atmosphere, was one of NASA's "faster, better, cheaper" missions, built on a compressed budget that trimmed margins across the program. Lockheed Martin built the spacecraft and its ground software; JPL navigated it from Earth. Across the months of interplanetary cruise the two teams exchanged data so JPL could command the small trajectory corrections that keep a spacecraft on course — an exchange that crossed a software interface between the two organizations.[0]

WHAT HAPPENED

Lockheed's ground software reported thruster impulse in pound-force seconds; JPL's navigation expected newton-seconds. The conversion — a factor of about 4.45 — was never applied at the boundary where the two systems met. Each firing was mis-modeled by that factor, and the error accumulated steadily over the long cruise until the predicted and actual trajectories had quietly diverged. When the orbiter reached Mars on 23 September 1999 it arrived far too low, deep into the atmosphere it was built to study from orbit, and was destroyed. The orbiter, and its roughly $125 million, were lost to a unit conversion no one made.[1]

THE INVESTIGATION

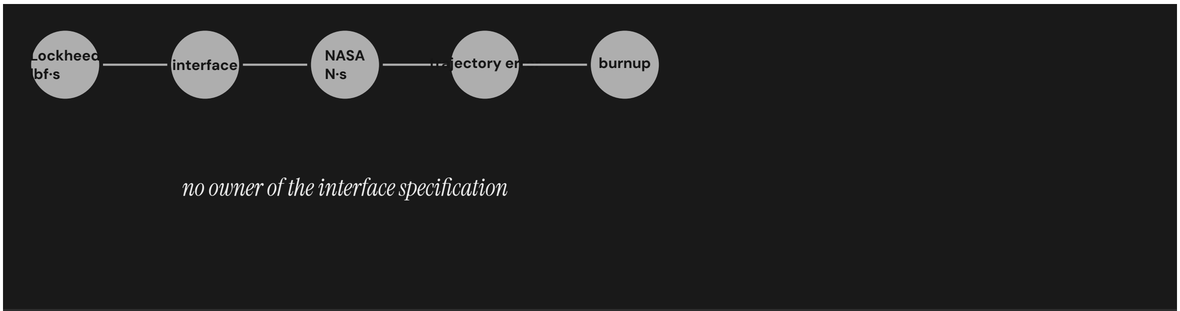
The Mishap Investigation Board put the proximate cause exactly there — the failed English-to-metric translation in ground software — but was careful to name the deeper one rather than stop at the bug. No individual blundered; both contractors did their own work correctly within their own assumptions. What failed was the boundary between them: there was no specified, verified interface fixing the units and checking that both sides agreed, and no end-to-end validation of the data flowing across the seam. Navigators had even noticed odd trajectory behavior in cruise, but the concern was never run fully to ground before arrival, and the chance to catch it passed.[2]

THE CAPABILITY GAP

The missing capability was ownership of the interface itself. Where a system is split across two organizations, the place they meet is not a documentation footnote but an engineering deliverable in its own right — with an owner, a specification, and a verification step. Here it had none of the three. Each team treated its own side as complete and the boundary as someone else's concern, so the one assumption that had to be shared and checked — what units are we speaking in? — was precisely the one no one verified. The spacecraft did not fail; it performed as built; the seam between the two halves of the organization did.[3]

AFTERMATH & REFORM

NASA tightened interface management and end-to-end verification and treated the loss as a cautionary tale about how far "faster, better, cheaper" could be pushed before the corners being cut turned out to be load-bearing. The orbiter became the canonical systems-engineering example of interface-as-requirement — the civilian, software parallel to the Patriot's untraveled assumption (Case 19): two competent halves of a system, a boundary nobody owned or verified, and a small unspecified thing at that boundary that quietly destroyed the whole.[4]



MARS CLIMATE ORBITER — THE UNOWNED INTERFACE

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THE LEARNING ENGINEERING LENS

> *The root cause was the failed translation of English units to metric units in a segment of ground-based, navigation-related mission software.*
>
> PARAPHRASING THE NASA MARS CLIMATE ORBITER MISHAP INVESTIGATION BOARD, 1999

LE INSIGHT

Mars Climate Orbiter is the textbook case for interface boundaries as engineering deliverables. The contractors did their work. The boundary between them did not have an owner. The capability that was missing was the interface-specification verification step.

LENS APPROACH

LENS uses Mars Climate Orbiter in LEN 5 to teach interface-as- requirement and in LEN 8 to discuss cross-contractor capability boundaries. The case is the foundational software-engineering parallel to Patriot (Case 19).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make the interface between two organizations an owned deliverable with a written specification of units and formats, not a footnote each side assumes the other handles.
- Fix and verify the unit convention at the boundary explicitly, requiring both sides to confirm agreement before data crosses the seam.
- Run end-to-end validation across the combined system, since each half can be correct within its own assumptions while their composition is wrong.

IN OPERATION / AFTER

- Treat in-cruise anomalies, like the noticed trajectory behavior, as signals to run to ground before a point of no return rather than concerns to revisit after arrival.
- Monitor predicted-versus-actual trajectory continuously so an accumulating boundary error surfaces while there is still time to correct it.
- Audit cross-contractor seams for unowned assumptions whenever a program trims margins under “faster, better, cheaper,” confirming the corners cut are not load-bearing.

REFLECTION QUESTIONS

1. Identify a contractor-to-contractor interface in your domain whose specification ownership is unclear. What would the equivalent unit-mismatch look like there?
2. Design the interface-verification deliverable that would have caught the pound-force / newton boundary before launch.
3. Navigators noticed odd trajectory behavior in cruise but the concern was never run to ground before arrival. Specify the threshold and process by which an in-flight anomaly in your domain must be resolved before a point of no return, rather than carried past it.

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LENS COURSES LEN 5 LEN 8

CASE 55 TECHNOLOGY 2012

Knight Capital Trading Loss

D Designed Out K Knowledge & Institutional Memory

IMPACT $440M loss in 45 minutes; firm forced to seek emergency capital and was effectively acquired within months

IN BRIEF

On 1 August 2012 Knight Capital, a major U.S. market maker, deployed new order-routing software to seven of its eight servers and missed the eighth. The new code reused a flag that, on the eighth server's old software, reactivated long-dead "Power Peg" test code never removed from the repository. At the opening bell it fired millions of unintended orders; in about 45 minutes Knight lost roughly $440 million — more than the firm was worth — and was effectively acquired within months. The SEC found Knight had no procedure to verify the deployment across all servers and no controls to halt the runaway orders. The capability designed out was deployment verification; the dead code was technical debt that exercised its option at the worst possible moment.

BACKGROUND

Knight Capital was one of the largest market makers in U.S. equities, routing an enormous share of retail order flow through automated systems wired directly into the market. On 1 August 2012 the New York Stock Exchange was launching a new Retail Liquidity Program, and Knight had updated its order-routing software to participate in it from the opening bell. The update went out to production the way countless updates had gone before — a routine deployment, on an ordinary morning, handled as unremarkable.[0]

WHAT HAPPENED

A technician deployed the new code to seven of the eight routing servers and missed the eighth, leaving one node running stale software. The new code reused a configuration flag that, on the old software still on that eighth server, had once activated long-dormant "Power Peg" test code — retired years earlier but never removed from the repository. At the open the dead code woke and began firing millions of unintended orders into the market. In about forty-five minutes Knight amassed a vast unwanted position; losses passed $170 million almost at once and reached about $440 million — more than the firm itself was worth. It survived only on emergency capital and was effectively acquired within months.[1]

THE INVESTIGATION

The Securities and Exchange Commission's 2013 order found Knight had no written procedure requiring a second technician to verify the deployment across all hosts, and no automated check that all eight servers were running the same code — nor controls able to recognize and halt the flood of erroneous orders once it began. Knight settled for a $12 million penalty.[2] The dead Power Peg code was the proximate trigger and the reused flag the match that lit it; but the underlying cause was the absence of the verification and risk controls that should surround any change to a system wired directly into the live market.[3]

THE CAPABILITY GAP

The capability designed out was deployment verification — confirming, every time, that what runs in production is exactly what was intended, on every node, before it touches live money. The dead code was technical debt in its most literal form: a retired function left in the repository is a standing option on a future failure, and reusing its flag exercised that option at the worst possible moment. As at the Mars Climate Orbiter's interface, the boundary that mattered — between "deployed" and "verified as deployed everywhere" — had no owner and no automated check, and a large institution kept walking across it routinely until the day the floor gave way.[4]

AFTERMATH & REFORM

Knight became the canonical case in modern software-operations practice for why deployment is itself an engineering deliverable, not a clerical step: automated verification that every host runs the intended build, disciplined removal of dead code from the repository, pre-trade risk limits, and kill switches that can stop a runaway process in seconds. Regulators sharpened their attention to automated market-access controls in response. The lesson rhymes with the orbiter's across a forty-year, civilian-to-financial gap: a small, unowned boundary inside a large automated system is precisely where the institution is most exposed, and least watching.[5]

$440M

LOSS IN 45 MINUTES · 7 OF 8 SERVERS UPDATED

dead code on the eighth server, repurposed flag, no deployment verification

KNIGHT CAPITAL — THE COST OF AN UNVERIFIED DEPLOYMENT

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THE LEARNING ENGINEERING LENS

> *Knight did not have written procedures requiring a second technician to verify the deployment.*
>
> PARAPHRASING THE SEC ORDER AGAINST KNIGHT CAPITAL, 2013

LE INSIGHT

Knight Capital is the financial-industry version of Mars Climate Orbiter (Case 54): a small, unspecified boundary inside a large system that took the institution down. The capability that was missing was deployment verification. The dead code was the proximate trigger; the absent procedure was the cause.

LENS APPROACH

LENS uses Knight Capital in LEN 5 to teach deployment-as-capability and in LEN 9 for the technical-debt argument: every line of dead code carries an option on a future failure. Students design the deployment deliverable that would have caught the eighth server.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Remove retired code and dormant flags from the repository as a matter of discipline, since every dead function left behind is a standing option on a future failure.
- Never reuse a configuration flag whose old meaning still exists somewhere in the fleet; treat flag semantics as a versioned, owned interface.
- Design pre-trade risk limits and a kill switch into any system wired to the live market, so a runaway process can be stopped in seconds rather than minutes.

IN OPERATION / AFTER

- Require automated verification that every host runs the intended build before a change touches live money, with a second technician confirming the deployment across all nodes.
- Monitor for the runaway condition — an abnormal order rate — and halt automatically, rather than relying on humans to notice and intervene mid-flood.
- Audit deployments for the orphaned node: confirm no host is left on stale code, closing the gap between “deployed” and “verified as deployed everywhere.”

REFLECTION QUESTIONS

1. Identify a deployment procedure in your domain whose verification step depends on convention rather than on a designed check. What is the eighth server?
2. Design the deployment deliverable that would prevent a Knight Capital–equivalent loss in your organization.
3. A retired “Power Peg” function left in the repository became the trigger years later when its flag was reused. Identify dead code or a dormant feature flag in a system you run that is still an option on a future failure, and specify the policy that should have removed it.

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TOOLS task analysis · design prototyping · knowledge bases · data pipelines

LENS COURSES LEN 5 LEN 9

PART I — THE FAILURE MODES

Chapter 3

The Slow Erosion

Known risks became routine.

NASA culture allowed flying with flaws when problems were defined as normal and routine.

COLUMBIA ACCIDENT INVESTIGATION BOARD, 2003

CASE 4 ENERGY 2010

Deepwater Horizon

T Training Gap N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT 11 killed; largest marine oil spill in U.S. history; $65B+ in damages

IN BRIEF

On 20 April 2010 the Macondo well blew out beneath the Deepwater Horizon rig in the Gulf of Mexico, killing 11 workers and releasing roughly 4.9 million barrels of oil — the largest marine spill in U.S. history, and more than $65 billion in eventual costs to BP. The crew had misread the negative-pressure test, the primary check of well integrity, and kept working a well that was no longer sealed. The National Commission found human error, not equipment, the root cause, and judged the failures so systematic they cast doubt on the safety culture of the whole industry. Every defense — procedure, training, supervision, contractor coordination, equipment — had drifted independently until none caught the others. It is the book's canonical multi-layer normalization failure.

BACKGROUND

The Deepwater Horizon was drilling BP's Macondo exploratory well in the Gulf of Mexico when, on 20 April 2010, the well blew out. Eleven workers were killed, the rig burned and sank, and roughly 4.9 million barrels of oil flowed into the Gulf over 87 days — the largest marine oil spill in U.S. history, eventually costing BP more than $65 billion.[0] The well sat at the end of a long chain of contractors and schedules, each operating to its own tolerance, so that the 87 days of uncontrolled flow stand as a measure not of one mistake but of how far the accumulated drift had to be unwound before the Gulf was sealed again.

WHAT HAPPENED

The proximate trigger was a misread safety check. The crew ran a negative-pressure test — the primary test of whether the well was sealed — got anomalous readings, accepted an incorrect explanation for them, and proceeded to displace the heavy drilling mud. The well was not sealed; hydrocarbons surged up the riser to the rig and ignited.[1] The anomalous pressure was the well speaking plainly that it remained live, but the explanation the crew accepted let the displacement proceed on schedule, and once the heavy mud was gone nothing stood between the reservoir and the rig but a barrier that had already failed.

THE INVESTIGATION

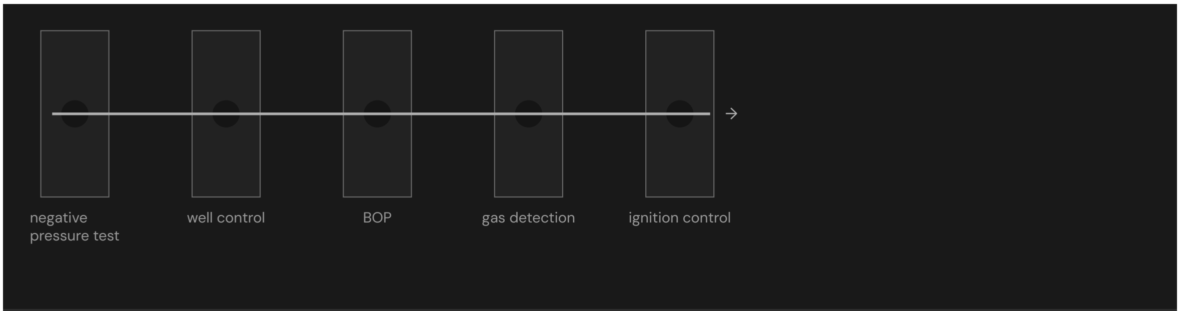
The National Commission concluded that human error, not mechanical failure, was the root cause, and that the mistakes revealed "such systematic failures in risk management that they place in doubt the safety culture of the entire industry."[2] Government and academic reviews found training that had not covered the well-control situation the crew faced, an unclear chain of command, and a cascade of failed defenses — "a complex and interlinked series of mechanical failures, human judgments, engineering design, operational implementation and team interfaces.”[3] That the Commission reached past the rig to the whole industry’s safety culture marked the failure as structural rather than local: the same drift, it judged, was latent wherever the same pressures and the same defenses operated unexamined.

THE CAPABILITY GAP

Deepwater Horizon is the canonical multi-layer failure: procedure, training, supervision, contractor coordination, and equipment had each drifted independently until none caught the others. The negative-pressure test was the trigger, but the system had been carrying accumulated procedural debt for years, and the capability to recognize an unsafe well-state was simply not present at the moment it was needed.[4] Because each defense had degraded inside its own tolerance, no single review of any one layer would have flagged a crisis; the danger lived in the correlation between layers, which is precisely the property no individual procedure was designed to watch.

AFTERMATH & REFORM

The blowout drove a restructuring of offshore regulation — the Minerals Management Service was broken up and replaced, and drilling-safety and well-integrity rules were tightened — while BP paid tens of billions in penalties and settlements. The deeper lesson is the normalization one: no single layer failed catastrophically on its own; each had quietly drifted within tolerance until the day the tolerances lined up.[5] Splitting the regulator conceded that an agency both leasing acreage and policing safety carried a built-in conflict, and tightening the well-integrity rules conceded that the negative-pressure test had been a real barrier all along — one the system had been quietly permitted to misread.



CASCADING DEFENSES — EACH FAILED IN TURN

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THE LEARNING ENGINEERING LENS

> *The immediate causes of the Macondo well blowout can be traced to a series of identifiable mistakes ... that reveal such systematic failures in risk management that they place in doubt the safety culture of the entire industry.*
>
> NATIONAL COMMISSION, DEEP WATER (REPORT TO THE PRESIDENT), 2011

LE INSIGHT

Deepwater Horizon is the canonical multi-layer failure: training, procedure, supervision, contractor-coordination, and equipment all drifted independently until none caught the others. The negative-pressure test was the proximate trigger, but the system as a whole had been operating with accumulated procedural debt for years. The capability to recognize an unsafe well-state simply was not present at the moment it was needed.

LENS APPROACH

LENS uses Deepwater Horizon in LEN 5 as the canonical case for multi-layer capability analysis. Students reconstruct each defense layer, identify the deviation accumulated in each, and design the measurement system that would have surfaced the drift before April 20, 2010. LEN 8 examines contractor-coordination as a capability boundary problem.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make the negative-pressure test a hard, instrumented gate: define the pass criteria so an anomalous reading stops work rather than inviting a rationalizing explanation.
- Specify a single, unambiguous well-control chain of command across operator and contractors before drilling begins, so no decision falls between organizations.
- Build well-control training around the ambiguous small-anomaly case the crew actually faced, not only the textbook blowout.

IN OPERATION / AFTER

- Audit each defense layer for drift independently — and audit the correlation between them, since the danger lived where tolerances aligned.
- Track accumulated procedural debt as a standing metric so silent degradation surfaces before tolerances coincide.
- Structurally separate the regulator's leasing and safety roles so revenue pressure cannot erode well-integrity enforcement.

REFLECTION QUESTIONS

1. Identify a multi-defense process in your domain. For each layer, what is the procedural debt that has accumulated since it was designed?
2. The negative-pressure test was the proximate trigger. Design a capability check that would have caught the misinterpretation in real time.
3. The Commission judged the drift industry-wide, not rig-specific. What measure in your domain would reveal whether a single failure is a local lapse or a sample from a systemic distribution?

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LENS COURSES LEN 5 LEN 8 LEN 3

CASE 6 SPACE / AEROSPACE 1986 / 2003

Challenger & Columbia

N Normalization of Deviance K Knowledge & Institutional Memory G Governance & Trust

IMPACT 14 astronauts killed (7 per accident); 17 years between identical organizational pathologies

IN BRIEF

NASA lost two Space Shuttle crews to the same organizational pathology seventeen years apart: Challenger in 1986, when O-ring seals failed in cold weather, and Columbia in 2003, when foam debris breached the wing's thermal protection — fourteen astronauts in all. Both flaws had been seen repeatedly and accepted as routine. Sociologist Diane Vaughan named the mechanism "normalization of deviance" from the Challenger investigation; the Columbia Accident Investigation Board found the same culture intact and concluded NASA's organizational culture had as much to do with the accident as the foam. The pair is the strongest single-institution evidence that culture is an engineerable property of a system — and that a pathology diagnosed but left unrepaired will recur, at the same cost.

BACKGROUND

The Space Shuttle flew with two flaws its engineers knew about. The solid-rocket booster joints sealed with O-rings that stiffened in cold; the external tank shed foam insulation that struck the orbiter on ascent. Both had appeared on flight after flight without catastrophe, and both were progressively reclassified from hazards to accepted, "routine" features of flying.[0] Each survived flight became evidence that the flaw was tolerable, so the very absence of disaster fed the reclassification — the safety margin redefined downward not by decision but by the accumulating habit of getting away with it.

WHAT HAPPENED

On 28 January 1986 Challenger launched on an unusually cold morning; an O-ring failed to seal, and the vehicle broke apart 73 seconds after liftoff, killing all seven aboard. Seventeen years later, on 1 February 2003, foam that had struck Columbia's wing on ascent had opened a breach in its thermal protection; the orbiter disintegrated on reentry, killing its seven.[1] Both flaws were the ones engineers had already flagged and the institution had already filed as routine, so each accident was less a surprise than the arrival of a bill the system had decided, repeatedly, not to pay.

THE INVESTIGATION

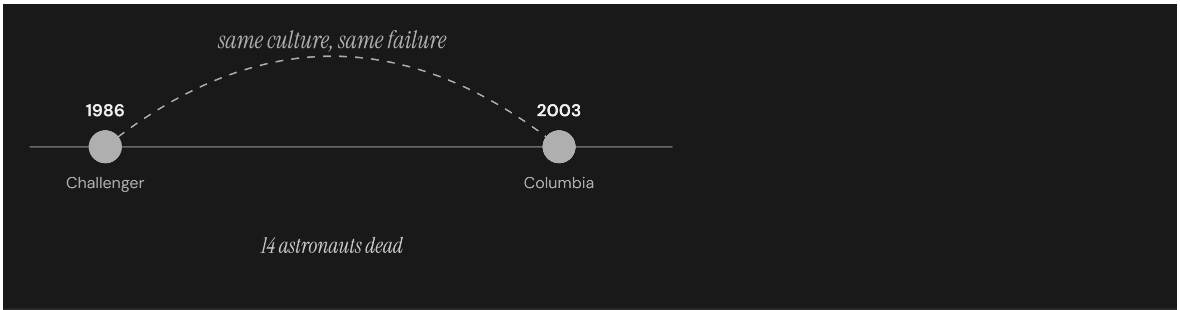
The Rogers Commission traced Challenger to the O-ring and to a launch decision that overrode engineers' cold-weather warnings.[2] Seventeen years later the Columbia Accident Investigation Board found the same patterns intact — flying with flaws defined as routine, and a structure that suppressed the upward flow of safety concerns — and concluded that “the NASA organizational culture had as much to do with this accident as the foam.”[3] That the same board found the same structure seventeen years on is the case’s sharpest point: the first reform had fixed the hardware and the schedule pressure but not the mechanism by which a flaw becomes redefined as acceptable.

THE CAPABILITY GAP

Diane Vaughan’s “normalization of deviance,” developed from Challenger, named the mechanism: deviations from the safety baseline become acceptable through production pressure, weak communication, and habit, one small step at a time. Columbia validated the concept against its author’s intent. The pathology had been diagnosed in 1986 and never engineered away — which is the point: culture is a system property, and a diagnosis without a remediation decays.[4] Naming the mechanism did not arrest it, because a name lives in a report while the production pressure and the suppressed warnings live in the daily flow of work, where they kept doing what they had always done.

AFTERMATH & REFORM

Each accident produced reform — the Rogers Commission’s redesign of the booster joint, the CAIB’s call to treat culture as a safety variable and rebuild independent technical authority — and the Shuttle was retired in 2011. The pair stands as the book’s strongest evidence that organizational culture is engineerable, and that leaving it unengineered is a choice with a recurring, lethal cost.[5] The CAIB’s insistence on independent technical authority conceded the deeper lesson the booster-joint redesign alone had missed in 1986: the upward path for a dissenting engineer is itself a piece of safety hardware, and one that has to be rebuilt and defended rather than assumed.



17 YEARS, SAME PATTERN — CHALLENGER TO COLUMBIA

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THE LEARNING ENGINEERING LENS

> *These repeating patterns mean that flawed practices embedded in NASA's organizational system continued for 20 years and made substantial contributions to both accidents.*
>
> COLUMBIA ACCIDENT INVESTIGATION BOARD, 2003

LE INSIGHT

Challenger/Columbia is the strongest documentary evidence that organizational culture is an engineerable property of a system, and that diagnoses without engineered remediations decay. The same pathology, twice, seventeen years apart, in the same institution, with the same human cost — and with the diagnosis already on the record. Capability engineering treats culture as a deliverable.

LENS APPROACH

LENS uses the pair in LEN 1 to introduce normalization of deviance as a systems concept, in LEN 8 to examine why organizational learning failed across a 17-year interval, and in LEN 7 to address the governance failure that allowed a known pathology to persist.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Hold known flaws as open hazards with explicit owners, so a clean flight cannot silently reclassify a deviation as routine.
- Build independent technical authority into the launch decision, giving dissenting engineers a path that production pressure cannot override.
- Define the safety baseline quantitatively at design time, so any drift below it is a flagged change rather than an unremarked habit.

IN OPERATION / AFTER

- Re-audit every diagnosed-but-unrepaired pathology on a fixed cadence, verifying remediation in practice rather than on paper.
- Track the upward flow of safety concerns as a measured signal — count and resolve dissents — so suppression becomes visible.
- Treat the cultural mechanism, not just the hardware fix, as the deliverable that must persist between major incidents.

REFLECTION QUESTIONS

1. What is the equivalent "diagnosed but not repaired" pathology in your domain? What evidence would close the loop?
2. The CAIB called culture engineerable. Sketch the engineering deliverable for the cultural intervention you would propose in your domain — including its measurement signal.
3. Seventeen years separated two accidents with the same root pathology. What mechanism in your organization would verify that a past diagnosis has actually been remediated rather than merely documented?

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LENS COURSES LEN 1 LEN 7 LEN 8 LEN 3

CASE 11 DEFENSE 1991 – PRESENT

V-22 Osprey

T Training Gap H Human-System Interface N Normalization of Deviance

IMPACT ~65 killed across ~17 hull-loss accidents since 1991; serious-mishap rate above comparable fleets (GAO-26-108905, 2025); some fixes stretch to the 2030s

IN BRIEF

The V-22 Osprey — the tiltrotor flown by the Marines, Air Force, and Navy — has had about 17 hull-loss accidents and roughly 65 fatalities since 1991, including 19 Marines in a single 2000 test crash and 8 airmen off Yakushima, Japan, in 2023. The Yakushima crash traced to cracks in a transmission gear (a flaw in the X-53 steel alloy) and a pilot's decision to keep flying through warnings. In December 2025, GAO and a NAVAIR review found the V-22's joint program office had failed for years to assess and address mounting safety risks, that serious-mishap rates exceeded comparable fleets, and that some fixes stretch toward
2034. The Osprey is the steady-state form of normalization of deviance:

a documented, reviewed shortfall accepted as the cost of flying the airframe.

BACKGROUND

The V-22 Osprey tilts its rotors to take off like a helicopter and cruise like a turboprop — an ambitious capability shared awkwardly across three services and a joint program office. Since development began in the 1980s it has suffered about 17 hull-loss accidents and roughly 65 fatalities, including 19 Marines in a single crash during 2000 testing.[0] The same configuration that makes the tiltrotor uniquely useful makes it uniquely demanding to sustain, and the joint arrangement diffused ownership of that burden across three services that each carried the airframe but none of which fully owned its safety trajectory.

WHAT HAPPENED

On 29 November 2023 a CV-22B broke up off Yakushima, Japan, killing all eight airmen aboard. The Air Force traced it to cracks in a transmission gear — rooted in metallic inclusions in the X-53 steel alloy used for the gears — and to the pilot's decision to keep flying despite warnings to land. The crash grounded Osprey fleets worldwide for months.[1] The materiel flaw and the human decision to press on through warnings were the same paired factors earlier reviews had named, so Yakushima read less as a novel failure than as one more draw from a pattern the program had already characterized but not closed.

THE INVESTIGATION

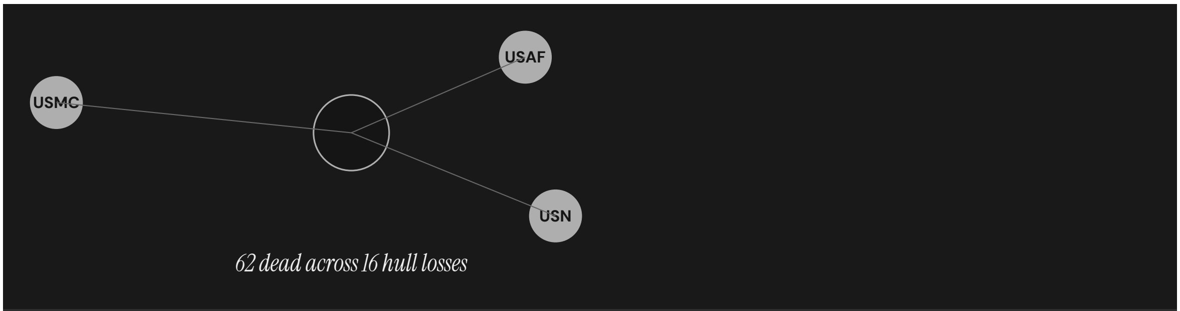
In December 2025 the Government Accountability Office and a NAVAIR review reported together that the joint program office had failed for years to adequately assess and address mounting safety risks, even as service members died.[2] Serious-mishap rates generally exceeded those of comparable Navy and Air Force fixed- and rotary-wing fleets from FY2015 to FY2024 and spiked in 2023–2024; a gearbox flaw dating to 2006 was not evaluated until 2024, and full fixes for some issues are not expected until the 2030s.[3] An eighteen-year gap between a gearbox flaw arising and its being evaluated is the timescale of normalization made literal: the deviation persisted long enough to become the airframe's accepted background condition rather than an open defect demanding action.

THE CAPABILITY GAP

The V-22 is the steady-state form of normalization of deviance. The shortfall is not unknown — it has been documented across successive reviews — but the system built to remediate it has accepted its own incompleteness as the cost of operating the airframe. Each crash carries a familiar set of factors (materiel failure, human error, weak coordination across the three services), and each is followed by adjustments that do not converge.[4] Convergence is the missing property: three services adjusting in parallel, none holding final authority over the whole, produce motion without resolution, so the mishap rate stays elevated while every individual response looks reasonable on its own terms.

AFTERMATH & REFORM

Groundings, gear-inspection regimes, and a redesign program have followed, with full fixes for some gearbox issues stretching toward 2034. The case is instructive precisely because the problem is recognized: a known, reviewed capability gap can persist for decades when each incident is treated as an isolated event rather than a sample from a distribution the program keeps drawing from.[5] Fixes that stretch toward 2034 mean the airframe will keep flying for years inside a margin already judged worse than comparable fleets — the program in effect electing to operate the distribution it has been told it is drawing fatalities from.



PERSISTENT CAPABILITY GAPS ACROSS THREE SERVICES

REFERENCES

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2. U.S. Air Force Accident Investigation Board findings, via USNI News (Aug. 2024) — the 29 Nov. 2023 Yakushima CV-22B crash: transmission-gear cracks (X-53 inclusions) and continued flight despite warnings.
3. U.S. GAO, Osprey Aircraft: Additional Oversight and Information Sharing Would Improve Safety Efforts, GAO-26-108905 (Dec. 2025) — the joint program office's failure to assess and address risks.
4. GAO-26-108905 (2025) — serious-mishap rates exceeding comparable Navy/Air Force fleets (FY2015–FY2024); the 2006 gearbox flaw evaluated only in 2024; fixes into the 2030s.
5. NAVAIR independent review of the V-22 (Dec. 2025) — materiel and cross-service-coordination factors and unresolved catastrophic parts issues.
6. USNI News, V-22 program coverage (2024–2025).

THE LEARNING ENGINEERING LENS

> *Materiel failure and human-error factors were the most frequently cited causal factors in serious Osprey accidents.*
>
> PARAPHRASING THE NAVAIR INDEPENDENT REVIEW OF THE V-22, 2025

LE INSIGHT

V-22 demonstrates the steady-state version of normalization of deviance: a platform whose shortfall has been documented, reviewed, and acted on across multiple administrations without producing sustained improvement. The capability gap has itself been normalized. Each incident is treated as an event rather than as a sample from a distribution the program continues to draw from.

LENS APPROACH

LENS treats V-22 in LEN 5 as a multi-service capability-coordination problem and in LEN 8 as a study in long-cycle organizational learning failure. Students design the evidence system that would distinguish a true reduction in mishap rate from the natural variation of a chronically marginal platform.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Assign one accountable owner for the airframe's safety trajectory rather than diffusing it across three services with no final authority.
- Treat each known materiel flaw as an open defect on a clock, so a problem like the gearbox cannot quietly become accepted background condition.
- Define an absolute mishap-rate threshold against comparable fleets that triggers mandatory action, not merely review.

IN OPERATION / AFTER

- Aggregate incidents as samples from one distribution, not isolated events, so the elevated rate itself becomes the thing being managed.
- Audit whether parallel service-level adjustments are actually converging on a lower rate, and escalate when they are not.
- Gate continued operation on remediation progress, so fixes stretching toward 2034 do not silently license years of degraded margin.

REFLECTION QUESTIONS

1. Identify a platform or process in your domain that has been operating in a documented shortfall for years. What measurement would have to change for the shortfall to become unacceptable?
2. The V-22's three services do not converge on remediation. Design the governance structure that would force convergence.
3. A gearbox flaw went eighteen years between arising and being evaluated. What mechanism in your domain converts a long-tolerated defect back into an open item that demands action?

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TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs · incident analysis · safety dashboards

LENS COURSES LEN 5 LEN 8 LEN 3

CASE 56 ENERGY INDUSTRIAL 2005

Texas City BP Refinery Explosion

N Normalization of Deviance T Training Gap K Knowledge & Institutional Memory G Governance & Trust

IMPACT 15 killed, 180 injured at BP's Texas City refinery; CSB found systemic safety-culture decay

IN BRIEF

On 23 March 2005 a startup at BP's Texas City refinery overfilled a distillation tower — operators were working from instruments that had malfunctioned for years — and venting hydrocarbon vapor found an ignition source and exploded, killing 15 workers in trailers parked beside the unit and injuring 180. The Chemical Safety Board found accumulated safety-culture decay, deferred maintenance, and a celebrated cost-cutting program. Its central finding reshaped U.S. industrial safety: BP's **personal**-safety metrics were among the best in the industry, while its **process**-safety state — the integrity of the hazardous process itself — had been drifting, unmeasured. Texas City is the canonical evidence that the wrong measurement, reported as a win, is worse than no measurement at all.

BACKGROUND

BP's Texas City refinery, one of the largest in the U.S., had absorbed years of deferred maintenance and corporate cost-cutting, with instruments and alarms tolerated in a degraded state. On the surface it looked safe: its personal-injury rates were among the best in the industry.[0] The strong injury numbers actively reassured the organization, because the metric the company trusted most was the one that carried no information about the degrading instruments and alarms on which a safe startup actually depended.

WHAT HAPPENED

On 23 March 2005, during a unit startup, operators overfilled a distillation tower far past its safe level, working from a level indicator that had malfunctioned for years and alarms that did not sound. Hydrocarbon vapor vented, drifted across the site, found an ignition source — an idling truck — and exploded. Fifteen workers in temporary trailers parked beside the unit were killed and about 180 injured.[1] Every contributor had been tolerated as routine for years — the broken indicator, the silent alarms, the trailers parked beside a hazardous unit — so the startup was run blind to a danger the site had long since stopped seeing.

THE INVESTIGATION

The U.S. Chemical Safety Board's investigation became the most influential in the agency's history.[2] It found accumulated safety-culture decay, deferred maintenance, broken instruments tolerated as routine, trailers sited dangerously close to a hazardous unit, and a cost-cutting program celebrated internally — and it drew the distinction that would reshape the field: "indicators of personal safety are not indicators of process safety."[3] The Board's force came from naming the deeper error as one of measurement: the company had not failed to measure but had measured the wrong dimension and then trusted the reassuring number it produced.

THE CAPABILITY GAP

Texas City is the foundational evidence that an organization can measure the wrong thing and report excellent performance while its real capability gap widens. Personal-safety metrics — slips, trips, recordable injuries — carried no information about the integrity of the hazardous process, so the signal regime was blind in the very dimension that killed people. The wrong measurement, trusted, is worse than none.[4] A blank dashboard at least invites suspicion; a confident green reading on the wrong axis manufactures false assurance, which is why the excellent injury record made the process-safety drift harder to see rather than easier.

AFTERMATH & REFORM

The Baker Panel review followed, BP invested heavily in process safety, and the process-safety/personal-safety distinction — developed in the CCPS literature and codified in OSHA's 1992 process-safety-management standard — moved into mainstream U.S. industrial regulation after 2005. The case's lasting contribution is a measurement lesson: count the thing that can kill you, not the thing that is easy to count.[5] That the distinction had been available in the CCPS literature and the OSHA standard before the explosion underscores the point: the knowledge existed, but the refinery's reporting had not been built to carry it upward where the hazard actually lived.

PERSONAL SAFETY	PROCESS SAFETY
best in class	drifting

the measurement system was blind in the dimension that killed people

TEXAS CITY — THE WRONG MEASUREMENT, REPORTED AS A WIN

REFERENCES

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THE LEARNING ENGINEERING LENS

> *Indicators of personal safety are not indicators of process safety.*
>
> U.S. CHEMICAL SAFETY BOARD, BP TEXAS CITY FINAL INVESTIGATION REPORT, 2007

LE INSIGHT

Texas City is the foundational evidence that organizations can be measuring the wrong thing and reporting excellent performance while their actual capability gap widens. Personal-safety metrics had no information about process-safety state. The signal regime was blind in the dimension that killed people.

LENS APPROACH

LENS uses Texas City in LEN 4 as the canonical wrong-measurement case and in LEN 7 for the governance failure that allowed years of cost-cutting to be reported as wins. Studio projects design the process-safety measurement deliverable.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Instrument process-safety state directly — barrier integrity, alarm health, instrument validity — rather than inferring safety from personal-injury rates.
- Treat degraded instruments and silent alarms as startup-blocking conditions, not routine items to defer past the next run.
- Set facility-siting rules that keep occupied trailers away from hazardous units as a design constraint, not a tolerated exception.

IN OPERATION / AFTER

- Audit whether the headline metric actually carries information about the hazard that can kill, and retire reassuring numbers that do not.
- Track deferred maintenance and tolerated-defect counts as process-safety leading indicators reported to executives.
- Verify that the process-vs-personal-safety distinction is wired into the reporting chain so it reaches the layer that funds maintenance.

REFLECTION QUESTIONS

1. Identify a “personal safety vs. process safety” equivalent in your domain. What capability gap is invisible to the metric your institution currently reports?
2. Design the process-safety dashboard that BP Texas City’s executives should have been receiving in 2003.
3. A confident green reading on the wrong axis manufactured false assurance. What headline metric in your organization might be reassuring precisely because it measures the wrong dimension?

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TOOLS incident analysis · safety dashboards · scenario simulation · competency models · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

LENS COURSES LEN 4 LEN 7 LEN 3

CASE 57 ENERGY 2002

Davis-Besse Reactor Head Corrosion

N Normalization of Deviance K Knowledge & Institutional Memory G Governance & Trust

IMPACT Football-sized cavity discovered in the reactor pressure-vessel head; near-miss; ~$600M recovery and extended outage

IN BRIEF

During a 2002 refueling outage at the Davis-Besse nuclear plant in Ohio, workers found a football-sized cavity eaten through six inches of the carbon-steel reactor-vessel head by leaking boric acid, leaving only a thin layer of stainless cladding between the cavity and reactor coolant at 2,200 psi — a serious near-miss for a loss-of-coolant accident. The leakage had been visible for years and inspections deferred; FirstEnergy had won an NRC deferral of a mandatory inspection to fit its refueling schedule, and the NRC's Inspector General later found the decision had weighted economics over safety. Davis-Besse is the canonical post-Three-Mile-Island near-miss — evidence that even an industry that built INPO to engineer safety can let regulator-operator dynamics erode it.

BACKGROUND

After Three Mile Island, U.S. commercial nuclear power had built — through INPO and a strengthened NRC — a reputation for engineering safety as a system. Davis-Besse, a pressurized-water reactor in Ohio, operated inside that regime, with a known industry-wide hazard: boric acid leaking from cracked nozzles could corrode the carbon-steel reactor-vessel head.[0] The hazard was generic and understood across the fleet, which is what makes the case sharp: the danger was not a surprise mechanism but a recognized one the post-TMI regime was precisely supposed to keep inspected and contained.

WHAT HAPPENED

During a refueling outage in early 2002, workers found a cavity about the size of a football eaten clean through six inches of the carbon-steel head, leaving only a thin layer of stainless-steel cladding between the corrosion and reactor coolant at 2,200 psi. Had the cladding given way, the result would have been a medium-break loss-of-coolant accident with a badly degraded safety margin.[1] The thin remaining cladding was the entire distance between routine operation and the kind of accident the entire regulatory regime existed to prevent — a margin measured in a fraction of an inch of unintended material rather than in any engineered barrier.

THE INVESTIGATION

The boric-acid leakage had been observable for years, and inspections had been deferred. FirstEnergy had requested — and the NRC had granted — a deferral of a mandatory inspection (NRC Bulletin 2001-01) so it would align with the plant's February 2002 outage.[2] The

NRC's Office of Inspector General later found the agency had inappropriately weighted the utility's economic arguments over safety — its oversight, the OIG concluded, had not been adequate to ensure that safety would not be compromised.[3] The deferral was not a covert lapse but a documented decision both parties signed off on, which is the unsettling part: the erosion happened through the regulator's normal process, in the open, agreed to be safe to wait.

THE CAPABILITY GAP

Davis-Besse is a near-miss in the dimension above operations: the regulator-operator relationship. The same industry that had shown, through INPO, that it knew how to engineer safety let a known corrosion mechanism go uninspected because inspecting it was inconvenient and expensive. Regulatory capture — the regulator adopting the operator's economic frame — is a capability failure at the institutional layer above the plant.[4] The plant's own engineering competence was never the missing piece; what failed was the independence of the layer meant to overrule a utility's schedule when safety required it, and that layer had quietly adopted the schedule as its own.

AFTERMATH & REFORM

INPO and the NRC restructured significant parts of their inspection regimes, and reactor-head inspection requirements were tightened across the fleet. The lesson pairs with the book's TMI / INPO arc: institutional capability is not built once. It erodes if it is not re-engineered — and the erosion is quietest where the regulator and the regulated agree it is safe to wait.[5] Tightening the head-inspection requirement across the fleet conceded that a mandatory inspection had been a real barrier all along — one the deferral process had been allowed to treat as negotiable against a refueling schedule.

1/4″

OF STAINLESS CLADDING REMAINED

between a 6-inch corrosion cavity and reactor coolant at 2,200 psi

DAVIS-BESSE — THE POST-TMI NEAR-MISS

REFERENCES

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2. NRC event records and OIG (2002) — the 6-inch corrosion cavity, the remaining cladding, and the 2,200-psi coolant margin.
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THE LEARNING ENGINEERING LENS

> *The NRC's actions were not adequate to ensure that safety would not be compromised.*
>
> PARAPHRASING THE NRC OFFICE OF INSPECTOR GENERAL DAVIS-BESSE REPORT, 2002

LE INSIGHT

Davis-Besse is the case for how regulator-operator dynamics can erode the capability of an industry that had previously demonstrated, through INPO, that it knew how to engineer safety. Regulatory capture is a capability failure at the institutional layer above operations.

LENS APPROACH

LENS uses Davis-Besse in LEN 7 to study regulatory-capture dynamics and in LEN 8 to compare with INPO's earlier success. The case is a reminder that institutional capability requires sustained re-engineering.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make safety-critical inspections of known generic hazards non-deferrable, so a mandatory check cannot be traded against a refueling schedule.
- Require any deferral decision to be argued on safety margin alone, with the utility's economic case explicitly excluded from the record.
- Preserve the independence of the oversight layer so it can overrule an operator's timeline rather than adopt it.

IN OPERATION / AFTER

- Audit granted deferrals for whether the regulator reasoned from safety or from the operator's economics, and flag drift toward the latter.
- Track observable degradation (like boric-acid leakage) against inspection currency, so a known mechanism cannot run uninspected for years.
- Re-engineer institutional safety capability on a cadence, treating post-incident regimes like INPO as maintained, not permanent.

REFLECTION QUESTIONS

1. Identify a regulator-operator relationship in your domain in which the regulator may be at risk of accepting the operator's economic argument over its own safety judgment. What signal would surface it?
2. Design the institutional control that would prevent a Davis-Besse-style deferral from being granted.
3. The erosion here happened openly, through the regulator's normal process. What audit would distinguish a defensible deferral from one in which the regulator has quietly adopted the operator's schedule as its own?

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LENS COURSES LEN 7 LEN 8

CASE 58 HEALTHCARE 2005 – 2009

Mid Staffordshire NHS Foundation Trust

G Governance & Trust N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT Excess deaths at Stafford Hospital documented across years; the Francis Inquiry produced 290 recommendations and restructured UK patient-safety governance

IN BRIEF

Between roughly 2005 and 2009, Stafford Hospital, run by the Mid Staffordshire NHS Foundation Trust, subjected patients to appalling neglect — left without food, water, or basic care — while mortality ran above expected. The trust had been chasing financial targets that depended on staffing cuts. Robert Francis QC's public inquiry produced a 2,000-page report and 290 recommendations, identifying the structural problem as the gap between the trust's reported performance and patients' actual experience: every governance layer above the ward received reports that targets were being met, and none checked those reports against what was happening to patients. Mid Staffs is the dataset's strongest case for the harm done when measurement and reality diverge over years.

BACKGROUND

The Mid Staffordshire NHS Foundation Trust ran Stafford Hospital in England. Pursuing Foundation Trust status and the financial targets that came with it, the board cut staffing — and the cuts fell on the wards. The institution's reported performance, the numbers that travelled upward, stayed on target.[0] The targets the board chased were financial and procedural, so cutting ward staff improved the very figures the trust was measured on even as it removed the people on whom patient care directly depended.

WHAT HAPPENED

On the wards the reality was appalling. Patients were left in their own excrement, denied food and water, given the wrong medication or none, for years; mortality ran substantially above expected. The harm was not a single incident but a sustained condition — visible to anyone on the ward and invisible in the reports that left it.[1] Because the suffering was a continuous state rather than a nameable event, it never generated the kind of discrete incident a reporting system is built to catch, and so it accumulated for years beneath numbers that stayed reassuringly on target.

THE INVESTIGATION

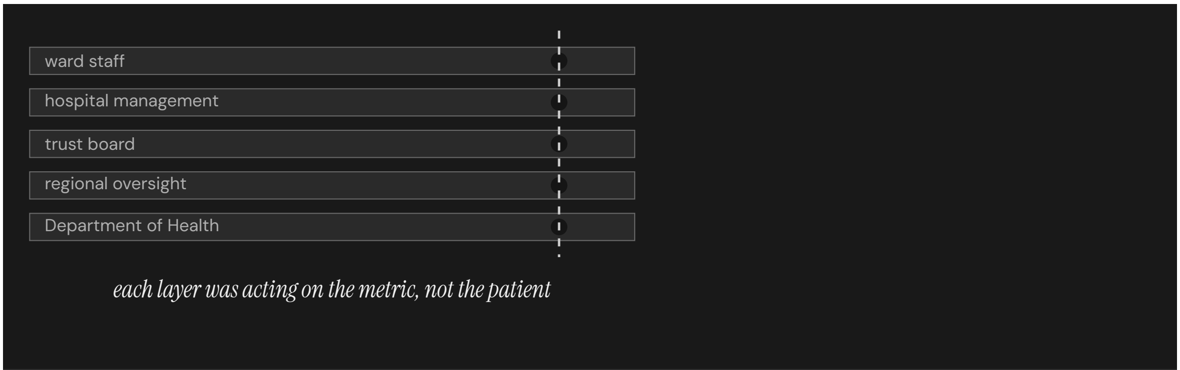
The Healthcare Commission investigated; the layers above did not. Robert Francis QC's public inquiry ran to some 2,000 pages and 290 recommendations.[2] Its structural finding was the divergence between reported performance and patient experience: every governance layer above the ward had received reports that the hospital was meeting its targets, and none had checked them against what was happening to patients. "The system as a whole failed in its most essential duty — to protect patients from unacceptable risks of harm."[3] The phrase "the system as a whole" located the failure deliberately above any single ward or manager: no one layer was solely at fault, because each had trusted the layer below to be reporting reality rather than targets.

THE CAPABILITY GAP

Mid Staffs is the British analog of the VA wait-time scandal (Case 32): measurement and reality diverged over years while every layer of governance acted on the measurement. The capability that was missing was not clinical skill on the ward but the institutional habit of testing whether the numbers corresponded to the patients — a check no layer above the ward performed.[4] Each layer reasonably assumed the check belonged to someone else, so the verification that the report matched the patient fell into the gap between layers — exactly the place a reporting chain built only to pass numbers upward is structurally unequipped to look.

AFTERMATH & REFORM

The Francis Inquiry restructured how the NHS treats patient safety, and the Berwick review that followed pressed for a culture of learning over targets. The lesson is the measurement one in its starkest form: a reporting chain can run clean from ward to Department of Health while, underneath it, the thing being reported on quietly fails.[5] Berwick's framing — learning over targets — named the deeper correction: as long as the target is the thing the institution rewards, the report will describe the target, and only a culture that prizes finding the gap will keep checking the report against the patient.



MID STAFFS — MEASUREMENT AND REALITY DIVERGED ACROSS YEARS

REFERENCES

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THE LEARNING ENGINEERING LENS

> *The system as a whole failed in its most essential duty — to protect patients from unacceptable risks of harm.*
>
> ROBERT FRANCIS QC, REPORT OF THE MID STAFFORDSHIRE NHS FOUNDATION TRUST PUBLIC INQUIRY, 2013

LE INSIGHT

Mid Staffordshire is the British analog of the VA wait-time scandal (Case 32). Measurement and reality diverged over years; every layer of governance above the operating environment was acting on the measurement; patients paid the cost of the divergence. The Francis Inquiry recommendations changed how the NHS thinks about patient safety as an institutional capability.

LENS APPROACH

LENS uses Mid Staffs in LEN 4 for the divergence-of-measurement problem and in LEN 7 for the governance failure across multiple layers. Studio projects examine the Francis recommendations as a template for institutional reform deliverables.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design targets so that gaming them (e.g., cutting ward staff) cannot improve the metric while degrading the outcome it stands for.
- Build a direct patient-experience signal — independent of the financial and procedural targets — into the reporting chain from the start.
- Assign explicit ownership of the report-versus-reality check at each layer, so verification cannot fall into the gap between them.

IN OPERATION / AFTER

- Audit sustained conditions, not just discrete incidents, since continuous harm never trips an event-based reporting system.
- Cross-check upward reports against ground truth at the ward periodically, treating a clean report as a hypothesis to be tested.
- Reward a culture of learning over target attainment, so the institution keeps looking for the gap rather than describing the target.

REFLECTION QUESTIONS

1. Identify a multi-layer reporting chain in your domain. What would it take for the top layer to know whether the reports correspond to reality?
2. The Francis Inquiry produced 290 recommendations. Pick five that you think were most load-bearing and explain why.
3. The verification fell into the gap between layers because each assumed it belonged to someone else. In your domain, who explicitly owns the check that a report matches reality — and how would you know if no one does?

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LENS COURSES LEN 4 LEN 7 LEN 3

CASE 59 ENERGY INDUSTRIAL 2006

Sago Mine Disaster

N Normalization of Deviance T Training Gap K Knowledge & Institutional Memory

IMPACT 12 killed in a West Virginia coal-mine explosion; emergency-response failures compounded the initial event; MINER Act of 2006

IN BRIEF

On 2 January 2006 lightning ignited methane in a sealed area of the Sago Mine in West Virginia; the seals failed, the explosion propagated, and thirteen miners were trapped behind it. Twelve died of carbon-monoxide poisoning over the hours that followed; only one, Randal McCloy Jr., survived. A communications failure briefly told the nation — and the families — that twelve had been found alive, when the opposite was true. The MSHA investigation found seals built to an inadequate design, an inadequate emergency plan, and lapsed self-rescue training — each marginal for years, all inadequate together in the one window that mattered. Sago drove the MINER Act of 2006, strengthening mine-rescue requirements and mandating underground refuge chambers.

BACKGROUND

The Sago Mine in West Virginia had sealed off a mined-out area behind barriers built to a design that had not kept pace with current standards. Its emergency plan and the miners' self-rescue training were, like the seals, marginally adequate — good enough on an ordinary day, untested against the worst one.[0] On any ordinary day each of these margins was invisible precisely because it was never tested at its edge, so the mine could run for years with every defense quietly thin and nothing to signal that the thinness was accumulating.

WHAT HAPPENED

On 2 January 2006 lightning ignited methane in the sealed area. The seals failed and the explosion propagated, trapping thirteen miners behind it. Over the hours that followed twelve died of carbon-monoxide poisoning; only one, Randal McCloy Jr., survived. Compounding the tragedy, garbled communications briefly told the nation and the families that twelve had been found alive — the opposite of the truth.[1] The hours of carbon-monoxide exposure were exactly the window the emergency plan and self-rescue provisions existed to bridge, so the marginal defenses failed in the one stretch of time their adequacy was supposed to guarantee.

THE INVESTIGATION

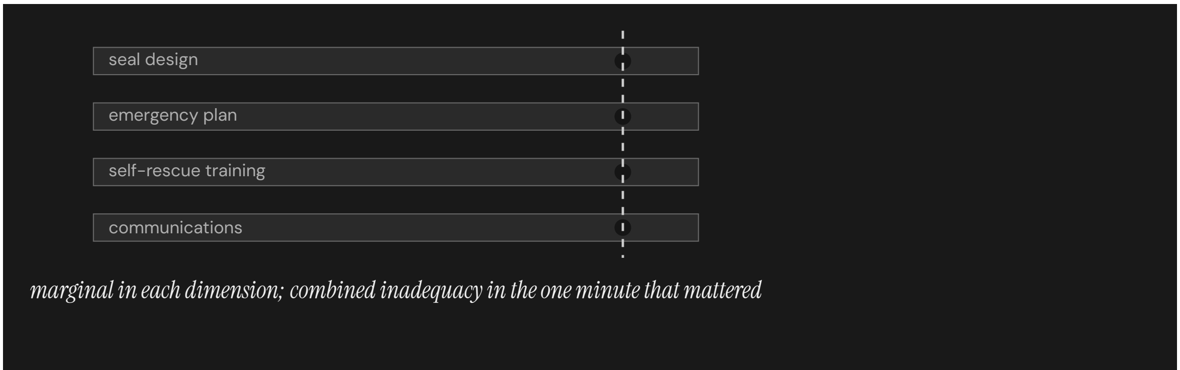
The Mine Safety and Health Administration found the seals had been built to a design that did not meet then-current standards, the mine's emergency plan was inadequate, and self-rescue training had not been kept current.[2] The miners "faced multiple equipment, training, and emergency-response shortcomings that compounded their initial trapping" — no single failure decisive, the combination lethal.[3] That each shortcoming was real but none was solely decisive is the finding's whole weight: an investigation looking for one nameable cause would have found several survivable ones and missed the lethal way they combined.

THE CAPABILITY GAP

Sago is the cumulative-inadequacy pattern. Each defense — seal design, emergency plan, self-rescue training, communications — was marginally adequate for years and recoverable on its own; none was the dramatic, nameable cause. They failed together in the only minute that mattered, which is exactly how normalization works: a system drifts within tolerance on several fronts until the tolerances align.[4] The hazard lived not in any one margin but in their simultaneity, which no inspection of a single defense could surface, because each looked acceptable on its own and the danger was a property only of the set.

AFTERMATH & REFORM

Sago drove the federal MINER Act of 2006, which strengthened mine-rescue requirements, tightened seal standards, improved communications and tracking, and mandated breathable-air refuge chambers underground.[5] The reform addressed the combination rather than a single cause — the right response to a failure whose lesson is that marginal-everywhere is itself a system-level hazard, even when no single margin looks alarming. The refuge-chamber mandate in particular conceded that survivable air over those carbon-monoxide hours had to be engineered in advance, not left to the chain of marginal defenses that failed together at Sago.



SAGO — CUMULATIVE INADEQUACY ACROSS MULTIPLE DEFENSES

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THE LEARNING ENGINEERING LENS

> *The miners faced multiple equipment, training, and emergency-response shortcomings that compounded their initial trapping.*
>
> MINE SAFETY AND HEALTH ADMINISTRATION, SAGO INVESTIGATION REPORT, 2007

LE INSIGHT

Sago is the case for the cumulative inadequacy pattern in industrial accidents. No single failure caused the disaster. Each failure on its own would have been recoverable. The combination was not, and the combination had been the operating condition of the mine for years.

LENS APPROACH

LENS uses Sago in LEN 5 for cumulative-inadequacy analysis and in LEN 8 for the legislative-reform arc that followed. Studio projects compare Sago and Upper Big Branch (Case 60) as paired cases.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Assess defenses jointly against the worst-case window, since each margin looks acceptable alone and the hazard lives in their simultaneity.
- Engineer a guaranteed survivable resource — like breathable air over the rescue window — rather than relying on a chain of marginal provisions.
- Keep seal design, emergency plans, and self-rescue training current to standards as a coupled set, not as independently deferred items.

IN OPERATION / AFTER

- Track how many defenses sit at the margin simultaneously, treating marginal-everywhere as a measurable system-level hazard.
- Stress-test the emergency response against the exact window it exists to bridge, so untested margins are exposed before an event.
- Investigate near-misses for combined inadequacy, not a single nameable cause, so survivable contributors are not dismissed individually.

REFLECTION QUESTIONS

1. Identify a process in your domain that is marginally adequate across multiple parameters. What is the cumulative failure mode?
2. Sago produced the MINER Act. What legislative change would your domain require if a Sago-equivalent occurred?
3. The danger was a property of the set of defenses, not any single one. What assessment in your domain would evaluate defenses jointly rather than one at a time?

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LENS COURSES LEN 5 LEN 8

CASE 60 ENERGY INDUSTRIAL 2010

Upper Big Branch Mine Explosion

N Normalization of Deviance G Governance & Trust K Knowledge & Institutional Memory

IMPACT 29 killed in West Virginia coal mine; MSHA found systematic falsification of safety records; first U.S. mining-industry CEO convicted of a federal mine-safety charge (misdemeanor)

IN BRIEF

On 5 April 2010 coal dust and methane ignited at Massey Energy's Upper Big Branch mine in West Virginia, killing 29 miners — the worst U.S. mine disaster in forty years. Investigators found Massey had kept two sets of records: an internal log of actual conditions and a separate, clean log for federal inspectors. Foremen were directed to suppress methane readings, disable monitors, and falsify pre-shift examinations — not as an aberration but as a stable routine across years and management layers. CEO Don Blankenship was convicted of a misdemeanor conspiracy to violate mine-safety standards — the first U.S. mining CEO criminally convicted on such a charge. Upper Big Branch is the dataset's clearest case of a measurement system engineered deliberately to defeat the regulator.

BACKGROUND

Massey Energy's Upper Big Branch mine in West Virginia operated under federal safety rules enforced through inspections and the records the mine kept. Massey kept two: an internal log of actual conditions, and a separate, clean log maintained for the inspectors.[0] The enforcement regime depended entirely on the records reflecting reality, so a second, sanitized set of books did not merely break a rule — it disabled the very mechanism by which the regulator was supposed to see the mine at all.

WHAT HAPPENED

On 5 April 2010 accumulated coal dust and methane ignited and tore through the mine, killing twenty-nine miners — the worst U.S. coal disaster in forty years. The conditions that fed the blast — high methane, inadequate ventilation, dust not rendered inert — were the very ones the clean, inspector-facing records had been built to hide.[1] The records had done their intended work right up to the blast: they kept the conditions that killed twenty-nine men out of the regulator's view precisely while those conditions were building toward ignition.

THE INVESTIGATION

MSHA's investigation and a parallel U.S. Department of Justice probe found foremen instructed to suppress methane readings, disable monitors, and falsify pre-shift examinations.[2] Massey CEO Don Blankenship was eventually convicted of a misdemeanor count of conspiring to willfully violate mine-safety standards — the first U.S. mining-industry CEO criminally convicted on a mine-safety charge — though acquitted on the felony counts.[3]

That the instructions ran down to foremen and the conviction reached up to the CEO marks the practice as vertical, not local: the deception was operated at the working level and conceived above it, spanning the management layers in between.

THE CAPABILITY GAP

The dual-records architecture was not the work of a few bad foremen; it was a stable institutional practice sustained across years and multiple management layers. The capability gap was not in the miners but in the executive ranks that designed and operated a measurement system specifically to defeat the regulator — which makes it the book's clearest case of measurement engineered as deception.[4] Where other cases show measurement drifting or pointed at the wrong dimension, here it was deliberately constructed to mislead, so no improvement in the regulator's reading of the official records could ever have helped — the records were the lie.

AFTERMATH & REFORM

The case anchors the argument that decision-grade evidence needs structural protection from the institution that produces it, when that institution has a stake in what the evidence says. Blankenship's conviction set a marker for corporate-officer accountability — and left open the regulator-side question: what architecture would have detected two sets of books before twenty-nine people died?[5] Holding a CEO criminally accountable raised the cost of designing such a system, but accountability after the fact is not detection; the unanswered question is what independent signal could have exposed the divergence between the two logs while the mine was still running.

INSPECTOR LOG	INTERNAL LOG
compliant	actual

two sets of records, sustained for years as institutional practice

UPPER BIG BRANCH — MEASUREMENT ENGINEERED AGAINST THE REGULATOR

REFERENCES

1. U.S. MSHA, Internal Review of MSHA's Actions at Upper Big Branch and the accident investigation (2011–2012) — the dual records and the 29 deaths.
2. Governor's Independent Investigation Panel (J. McAtee, 2011) — mine conditions: methane, ventilation, and coal-dust inerting.
3. MSHA and U.S. DOJ findings — suppressed methane readings, disabled monitors, and falsified pre-shift examinations.
4. United States v. Blankenship (S.D.W. Va., 2015–2016) — the misdemeanor conviction and felony acquittals.
5. H. Berkes / NPR investigative reporting on Massey Energy.
6. A. Hopkins, Disastrous Decisions: The Human and Organisational Causes of the Gulf of Mexico Blowout (2012) — corporate decision-making and safety (comparative).

THE LEARNING ENGINEERING LENS

> *Massey kept two sets of books — one for inspectors, one for itself.*
>
> PARAPHRASING FEDERAL INVESTIGATORS, MSHA UPPER BIG BRANCH REVIEWS (2011 – 2012)

LE INSIGHT

Upper Big Branch is the case for deliberately engineered measurement deception. The dual-records practice was not error. It was capability design — by management, against the regulator. The case anchors the LENS argument that decision-grade evidence requires structural protection against the institution that produces it having a stake in what it reports.

LENS APPROACH

LENS uses Upper Big Branch in LEN 4 as the most explicit case for measurement-system protection and in LEN 7 as a corporate-criminal accountability case. Studio projects examine what regulator-side architecture would have detected the dual-records system earlier.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Source safety-critical data from monitors the operator cannot disable or edit, so a clean official log cannot be constructed by hand.
- Design measurement so the producing institution has no unilateral control over the record the regulator relies on.
- Build cross-checks that compare independent streams, making a single sanitized set of books detectably inconsistent.

IN OPERATION / AFTER

- Establish whistleblower and anomaly-detection channels that can surface a divergence between actual and reported conditions while operations continue.
- Audit for the structural conditions that make dual records possible — operator-controlled monitors, no independent verification — not just for violations.
- Pair corporate-officer accountability with detection architecture, since raising the cost of deception does not by itself reveal it in time.

REFLECTION QUESTIONS

1. Where in your domain could two sets of records plausibly be kept? What architectural change would make that impossible?
2. What does it mean for a measurement system to be "structurally protected" from the institution that produces it?
3. Accountability after the fact is not detection. What independent signal could have exposed the divergence between the two logs while the mine was still running?

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LENS COURSES LEN 4 LEN 7

CASE 61 ENERGY 2011

Fukushima Daiichi

N Normalization of Deviance G Governance & Trust K Knowledge & Institutional Memory

IMPACT Three reactor meltdowns following the Tōhoku earthquake and tsunami; ~160,000 people displaced; cleanup projected at $200B+

IN BRIEF

On 11 March 2011 the Tōhoku earthquake and the tsunami it spawned overwhelmed TEPCO's Fukushima Daiichi plant: the wave topped the seawall, flooded the emergency diesel generators, and cut cooling to the reactors. Three of the six cores melted down and hydrogen explosions spread radioactive material, displacing some 160,000 people; cleanup is projected above $200 billion. The independent Diet commission (NAIIC), chaired by Kiyoshi Kurokawa, called it a disaster "made in Japan" — the product of regulatory capture and a culture reluctant to challenge utility assumptions; evidence of large historical tsunamis had been discussed internally but never forced a change to the seawall. Other reviews stressed under-estimated external hazards. Fukushima is the post-TMI evidence that safety institutions like INPO must be deliberately built, not assumed.

BACKGROUND

Fukushima Daiichi sat on Japan's northeast coast, its reactors and their backup diesel generators protected by a seawall sized to a design-basis tsunami. Evidence of much larger historical waves — back to the ninth-century Jōgan event — had been discussed in TEPCO's own internal assessments, but no institutional path turned that evidence into a higher seawall.[0] The gap was not in the data but in the conversion: the larger-wave evidence lived inside the utility's own assessments, where it could be discussed indefinitely without ever becoming a binding requirement to raise the wall it implicated.

WHAT HAPPENED

On 11 March 2011 the Tōhoku earthquake struck and the tsunami that followed topped the seawall. The plant lost off-site power; the diesel generators meant to keep cooling running were inundated. Cooling failed, three of the six reactor cores melted down, and hydrogen explosions spread radioactive material across the region. Some 160,000 people were displaced, and cleanup is projected above $200 billion.[1] Siting the backup generators where a wave that overtopped the seawall would reach them tied the entire cooling chain to that single design assumption, so once the wall was topped the loss of cooling followed almost mechanically from the layout itself.

THE INVESTIGATION

The independent investigation chaired by Kiyoshi Kurokawa for the National Diet (NAIIC) called the accident "made in Japan" — the product of regulatory capture, deference to authority, and an institutional reluctance to challenge utility assumptions.[2] Other major reviews — the Hatamura government commission (2012) and the IAEA Director General's report (2015) — emphasized the under-estimation of external hazards and defense-in-depth assumptions over the cultural critique; the Kurokawa framing is the most-cited but not the only consensus reading.[3] That serious independent reviews diverged on emphasis — cultural capture versus under-estimated external hazards — is itself part of the record, and the book treats Kurokawa's as the most-cited reading rather than the settled one.

THE CAPABILITY GAP

Fukushima is the post-TMI case showing that the INPO pattern (Case 16) is not self-executing. The U.S. industry built INPO to force operating discipline and shared learning; the Japanese industry did not build an equivalent with the independence to override a utility's optimistic assumptions. The internal tsunami evidence existed; the institutional capability to act on it did not.[4] Evidence without an independent body empowered to act on it is inert: the larger-wave assessments could be acknowledged and shelved indefinitely because no institution stood outside the utility with the standing to convert them into a mandated change.

AFTERMATH & REFORM

The accident drove a restructuring of Japanese nuclear regulation — a new, more independent Nuclear Regulation Authority — and a global re-examination of external-hazard margins and station-blackout protection. Paired with INPO and Davis-Besse, it triangulates the book's claim: sustained nuclear-safety capability is an institution that must be deliberately built and rebuilt, not a property a competent industry simply has.[5] Creating a more independent regulator after the fact conceded the structural diagnosis directly: the missing piece had been an authority sitting outside the utility, and the reform was precisely to build the independence that the pre-2011 arrangement had lacked.

3 of 6

REACTORS MELTED DOWN

tsunami exceeded design basis the institutional evidence had already questioned

FUKUSHIMA DAIICHI — "MADE IN JAPAN," PER THE DIET INQUIRY

REFERENCES

1. National Diet of Japan Fukushima Nuclear Accident Independent Investigation Commission (NAIIC; K. Kurokawa, chair), Report (2012) — the internal tsunami evidence and the regulatory-capture finding.
2. NAIIC (2012) — the accident sequence: seawall overtopping, generator inundation, and three core meltdowns.
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6. Y. Funabashi & K. Kitazawa, Fukushima in Review (2012); cf. INPO (Case 16) and Davis-Besse (Case 57).

THE LEARNING ENGINEERING LENS

> *What must be admitted — very painfully — is that this was a disaster "Made in Japan."*
>
> NATIONAL DIET OF JAPAN FUKUSHIMA NUCLEAR ACCIDENT INDEPENDENT INVESTIGATION COMMISSION, 2012

LE INSIGHT

Fukushima is the post-TMI case that establishes that the INPO pattern (Case 16) is not self-executing. The U.S. industry built INPO; the Japanese industry did not. The cost of the difference, paid in 2011, is the strongest available evidence that capability institutions must be deliberately built, not assumed.

LENS APPROACH

LENS uses Fukushima in LEN 8 as the cross-cultural comparison to INPO and in LEN 7 for the regulator-utility dynamics study. The case is paired with INPO and Davis-Besse to triangulate what sustained nuclear-safety capability requires.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Build an institutional path that converts an acknowledged hazard (like the larger-wave evidence) into a binding design requirement, not a discussable assessment.
- Site critical backups so a single overtopped barrier cannot disable the whole cooling chain at once.
- Establish an oversight body independent of the utility, with standing to override optimistic in-house assumptions about external hazards.

IN OPERATION / AFTER

- Audit whether internally held hazard evidence has actually driven design changes, treating shelved assessments as an open finding.
- Re-examine external-hazard margins on a cadence rather than freezing them at a design-basis figure set once.
- Verify that the regulator retains the independence to act, since institutional safety capability erodes if it is assumed rather than rebuilt.

REFLECTION QUESTIONS

1. Identify a regulatory regime in your domain whose effectiveness depends on a cultural willingness to challenge authority. What if the culture changes?
2. INPO is U.S.-specific. Design the structural features that would have to be in place for a comparable institution to function in a different national context.
3. The tsunami evidence was discussed but never converted into a binding requirement. What institutional path in your domain turns an acknowledged hazard into a mandated change rather than a shelved assessment?

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LENS COURSES LEN 7 LEN 8 LEN 3

CASE 62 ENERGY 2003

Northeast Blackout

H Human-System Interface K Knowledge & Institutional Memory

IMPACT 50 million people without power across eight U.S. states and Ontario; $6B+ economic loss; FERC Order 693 followed

IN BRIEF

On 14 August 2003 a high-voltage transmission line in Ohio sagged into a tree and tripped — an event the grid should have absorbed. But FirstEnergy's control-room alarm system had been silently failing for over an hour, so operators did not know the line was gone. Further lines tripped, and a cascade swept across the Eastern Interconnection; within minutes 50 million people across eight U.S. states and Ontario lost power, at a cost above $6 billion. The U.S.-Canada task force found FirstEnergy lacked situational awareness, its alarm system had failed without notice, vegetation management was poor, and the regional coordinator could not intervene. The reforms made reliability standards mandatory and enforceable (FERC Order 693). The gap sat at the automation-operator boundary: a silent failure left operators blind.

BACKGROUND

The Eastern Interconnection is built to ride through the loss of a single transmission line, and control rooms watch the grid through software and alarms. FirstEnergy, the Ohio utility at the center of the story, ran a control room whose alarm system had — unknown to anyone — been silently failing for over an hour.[0] The interconnection's single-line resilience assumes the operators can see which line is gone; a silent alarm broke that assumption at its root, leaving a grid designed to tolerate one fault blind to the fault it was tolerating.

WHAT HAPPENED

On 14 August 2003 a 345 kV line in Ohio sagged into a tree and tripped. With the alarms silent, operators did not know the line was gone and so did not take the corrective steps that would have contained it. Further lines tripped and a cascade swept across the interconnection; within minutes 50 million people across eight states and Ontario lost power, at a cost above $6 billion.[1] The first trip was the routine single-line loss the grid was built to absorb; what turned it into a cascade was not the tree but the hour in which the operators acted on a picture of the grid that no longer matched the grid.

THE INVESTIGATION

The U.S.-Canada Power System Outage Task Force found FirstEnergy operators "did not have adequate situational awareness," that the alarm system had failed without notice, that vegetation management was inadequate, and that the regional reliability coordinator lacked the authority and information to intervene.[2] The reforms produced FERC Order 693, which for the first time made compliance with reliability standards mandatory and enforceable rather than voluntary.[3] Making the standards mandatory addressed the deeper finding that a voluntary regime had let vegetation management and operator awareness drift: when compliance is optional, the practices that prevent a cascade are exactly the ones a cost-pressured utility lets slip.

THE CAPABILITY GAP

The gap sat at the boundary between automation and operator. When the alarm system stopped doing its job, it did so silently — and the operators had no independent signal that they were losing control of their grid segment. The missing capability was the meta-monitor: the watch on the watchman, the check that tells you when the thing that is supposed to tell you has itself stopped.[4] Silence is the most dangerous failure mode because it is indistinguishable from a quiet, healthy grid; the operators were not ignoring a warning but trusting an absence of warning that the broken system could no longer guarantee.

AFTERMATH & REFORM

FERC Order 693 and the new mandatory reliability regime — enforced by an Electric Reliability Organization with audits and penalties — treated grid reliability as a deliverable rather than a best practice. The 2003 blackout endures as the canonical case of silent automation failure: a system that quits without announcing it, leaving the humans nominally in charge with nothing to act on.[5] Backing the standards with audits and penalties conceded that reliability could not be left to good intentions: the practices that would have kept the operators sighted had to be enforceable obligations, not best practices a utility could quietly defer.

NORTHEAST BLACKOUT — SILENT ALARM FAILURE ABSORBED THE OPERATOR'S AWARENESS

REFERENCES

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2. Task Force (2004) — 50 million people affected across eight states and Ontario; the minute-by-minute sequence.
3. Task Force (2004) — FirstEnergy "did not have adequate situational awareness," plus the vegetation-management and reliability-coordinator findings (quoted).
4. FERC Order No. 693, Mandatory Reliability Standards for the Bulk-Power System (2007) — enforceable standards.
5. North American Electric Reliability Council reports (2004) and the creation of the Electric Reliability Organization.
6. M. R. Endsley (1995), situation-awareness theory — the human-factors frame for silent-automation failure.

THE LEARNING ENGINEERING LENS

> *FirstEnergy ... did not have adequate situational awareness of conditions on its system.*
>
> U.S.-CANADA POWER SYSTEM OUTAGE TASK FORCE, FINAL REPORT ON THE AUGUST 14 2003 BLACKOUT, APRIL 2004

LE INSIGHT

The 2003 blackout is the canonical case for silent automation failure: a system that stops doing its job without telling its operators. The capability that was missing was the meta-monitor — the system that watches the monitor. The reform that followed treated grid-reliability compliance as a deliverable rather than as a best practice.

LENS APPROACH

LENS uses the 2003 blackout in LEN 2 as a Human-AI Teaming case for silent-automation-failure handling and in LEN 8 for the legislative-reform arc that produced enforceable reliability standards.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design monitoring with a positive heartbeat so a silent system is distinguishable from a healthy one, never assuming no alarm means no problem.
- Build a meta-monitor that watches the alarm system itself and announces when the watchman has stopped.
- Give the regional reliability coordinator the authority and information to intervene across utility boundaries before a cascade forms.

IN OPERATION / AFTER

- Audit vegetation management and operator situational awareness against enforceable standards, since voluntary practice drifts under cost pressure.
- Test alarm and monitoring systems for detectable failure, verifying that an outage of the monitor itself raises an alert.
- Back reliability obligations with audits and penalties so the practices that keep operators sighted cannot be quietly deferred.

REFLECTION QUESTIONS

1. Identify an automated monitoring system in your domain whose silent failure would not be detected by its operators. How would they know?
2. Design the meta-monitor that should have been watching FirstEnergy's alarm system in 2003.
3. Silence was indistinguishable from a healthy grid. What positive heartbeat signal in your domain would let operators tell a quiet system from a dead one?

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LENS COURSES LEN 2 LEN 8

Chapter 4

The Interface Problem

Correct performance was made impossible.

The shootdown of Flight 655 showcases the chaos of combat environments, but it also reveals lessons for technology adoption and its use in stressful situations.

PARAPHRASING U.S. NAVAL PROCEEDINGS RETROSPECTIVE ON THE VINCENNES INCIDENT, 2018

CASE 18 DEFENSE 1988

USS Vincennes & Iran Air Flight 655

H Human-System Interface T Training Gap

IMPACT 290 civilians killed — the deadliest shootdown of a commercial airliner by a military force

IN BRIEF

On 3 July 1988, during a surface skirmish with Iranian gunboats in the Persian Gulf, the USS Vincennes shot down Iran Air Flight 655 — a civilian Airbus A300 climbing on a scheduled route — killing all 290 aboard, the deadliest airliner shootdown by a military force. The Aegis combat system worked: its radar correctly showed the aircraft ascending. Yet operators in the Combat Information Center, primed to expect a hostile attack, read the contact as a descending F-14 and fired. The Navy's inquiry attributed the tragedy to human error under extreme stress — confirmation bias and the unconscious distortion of data — not equipment failure. Vincennes is the book's foundational human-AI-teaming case: the most advanced system afloat failed because its interface did not support correct performance under combat stress.

BACKGROUND

In the closing weeks of the Iran-Iraq War, U.S. warships patrolled a tense Persian Gulf where civilian airliners and hostile military aircraft shared the same crowded sky. The USS Vincennes, a cruiser with the Navy's most advanced Aegis combat system, was in a surface fight with Iranian gunboats when an aircraft took off from a nearby airfield used by both civil and military traffic.[0] The dual-use field meant a single track could be either threat or scheduled flight, so the crew's reading of the contact carried the whole burden of identification — and they were already maneuvering hard against the gunboats, the kind of divided attention under which a wrong call becomes far easier to make than a right one.

WHAT HAPPENED

The aircraft was Iran Air Flight 655, a civilian Airbus A300 climbing on its scheduled route. The Vincennes' crew identified it as a descending, hostile F-14 and fired two surface-to-air missiles; all 290 people aboard were killed — the deadliest shootdown of a commercial airliner by a military force.[1] An ascending airliner and a diving attack jet are opposite behaviors, yet the crew converged on the second while the radar reported the first; the engagement consumed only minutes, collapsing identification, decision, and launch into a window too narrow for anyone to slow down and reconcile the contradiction.

THE INVESTIGATION

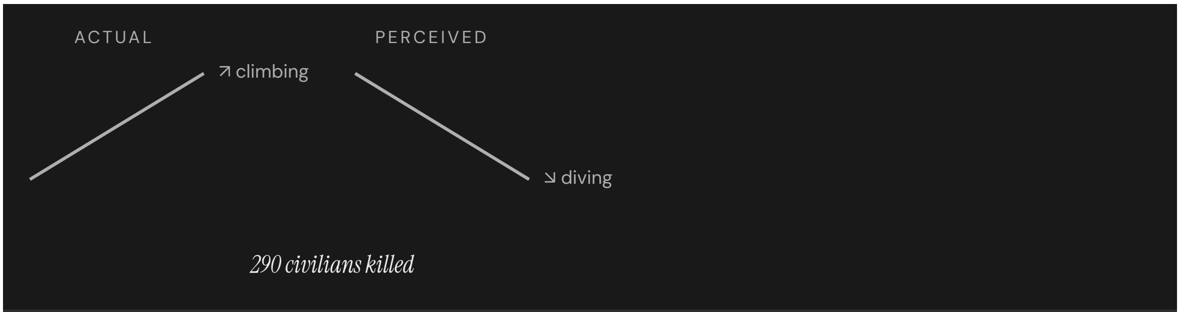
The Aegis system had not malfunctioned: its radar correctly showed the aircraft ascending. Yet several crew members in the Combat Information Center independently believed it was descending,[2] a shared error that is more damning than a single misread, because it shows the interface offered no cross-check strong enough to break a wrong reading once the team had settled into it. The Navy's Fogarty inquiry attributed the tragedy to human error under extreme stress — to confirmation bias and the "stress and unconscious distortion of data" — as operators read every indication through the lens of a presumed hostile attack, because the framing arrived before the data did, so each new return was fitted to the expected threat rather than weighed against it.[3]

THE CAPABILITY GAP

The system did not lie, and the operators did not act in bad faith. The interface and the operational framing combined to make a particular misreading not merely possible but likely. Correct performance was possible in principle and unsustainable in practice — and the gap between those two is the engineering problem. An interface that presents data without guarding against the crew's pre-formed expectation has not been designed for the stress it will actually meet; the burden of overriding a presumed hostile attack was left entirely to the operator's discipline, exactly when combat had stripped that discipline of the time and calm it needed to work.[4]

AFTERMATH & REFORM

Decades later the case became a central reference in the human-AI-teaming conversation: a 2018 Naval Institute retrospective placed it at the heart of how operators should be teamed with automated decision aids under stress.[5] Its lesson is that interface design is a capability deliverable, not an aesthetic one — and that the most advanced system afloat is only as good as the human reading it under fire. The retrospective reframed the loss not as a one-off error but as a predictable outcome of teaming a person with a decision aid that displayed truth without defending it, a pattern that recurs wherever automation is fast and the human is the last check.



AEGIS RADAR DATA CORRECTLY DISPLAYED; FIVE OPERATORS READ IT WRONG

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THE LEARNING ENGINEERING LENS

> *The shootdown of Flight 655 ... reveals lessons for technology adoption and its use in stressful situations.*
>
> PARAPHRASING U.S. NAVAL PROCEEDINGS RETROSPECTIVE ON THE VINCENNES INCIDENT, 2018

LE INSIGHT

Vincennes is the canonical case for human-AI teaming under stress. The system did not lie. The operators did not act in bad faith. The interface and the operational framing combined to make a particular misreading not just possible but likely. Correct performance was possible in principle and unsustainable in practice — and that gap is the engineering problem.

LENS APPROACH

LENS uses Vincennes in LEN 2 as the foundational case for human-AI teaming under operational stress, and in LEN 5 to teach the analysis of capability requirements in conditions the original interface designers never expected. The case sits at the center of the program's argument that interface design is a capability deliverable, not an aesthetic one.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the contact display to make ascent-versus-descent unmistakable at a glance, so the cue that contradicts a presumed threat cannot be passed over under time pressure.
- Test the interface against the worst case it will actually meet — confirmation bias during a simultaneous surface engagement — not against the calm conditions of acceptance testing.
- Build a mandatory disconfirmation step into the engagement sequence so identifying a hostile track requires actively ruling out the civilian one.

IN OPERATION / AFTER

- Audit live engagements and exercises for cases where the crew's reading diverged from sensor data, treating each as a near-miss that exposes an interface gap.
- Train and drill operators specifically on the stress regime — divided attention, presumed-hostile framing — rather than only on nominal track identification.
- Track whether decision aids are trusted past their evidence, and feed that signal back into interface and procedure revisions.

REFLECTION QUESTIONS

1. Identify a high-stress interface in your domain. What framing arrives before the data, and how does it shape what operators see?
2. Vincennes' operators acted under tunnel vision. Design the procedural intervention that would have forced one of the five to call out the contradiction.
3. The Aegis radar reported the truth and the crew still read its opposite. What would an interface have to make impossible — not merely visible — to keep a pre-formed expectation from overriding correct data under stress?

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LENS COURSES LEN 5 LEN 2

CASE 25 HEALTHCARE 2005 – PRESENT

EHR / CPOE Implementation

H Human-System Interface D Designed Out G Governance & Trust

IMPACT ~$30B federal investment under HITECH; documented increase in pediatric ICU mortality post-CPOE at one institution; ongoing usability harm at scale

IN BRIEF

The 2009 HITECH Act poured roughly $30 billion into accelerating electronic health records, and adoption surged — but the systems had been built to billing and administrative specifications, not clinical workflow. New categories of harm followed. A disputed but canonical 2005 study found that deploying a commercial order-entry (CPOE) system at one pediatric ICU was associated with a near-doubling of mortality; later deployments showed mixed or improved results, yet the warning stuck. A 2023 survey across 200-plus hospitals found EHR usability is clinicians' top complaint, and that usability scores track patient-safety outcomes. Specific harms recur — a missed abnormal result hidden behind a default, a fatal overdose an unactivated alert would have caught. There is still no regulatory framework monitoring EHR safety. It is the book's live interface-mismatch case.

BACKGROUND

The 2009 HITECH Act authorized roughly $30 billion in incentives to accelerate electronic health record adoption, and adoption surged. But the systems clinicians now had to use had been built to administrative and billing specifications, not to clinical workflow or human-factors specifications — a mismatch baked in before the first order was entered.[0] Because the incentives rewarded adoption rather than usability, hospitals raced to install whatever cleared the bar, and the specification gap propagated to the bedside at national scale before anyone had to demonstrate that the tools supported the work of care rather than merely the work of billing.

WHAT HAPPENED

New categories of harm appeared alongside the new tools. A 2005 study reported that deploying a commercial computerized order-entry (CPOE) system at one pediatric ICU was associated with a near-doubling of mortality — a single-institution result that provoked debate and that later deployments elsewhere did not reproduce, but that became canonical as a warning that powerful tools without workflow integration can disrupt care at the moment of greatest acuity, when the seconds the interface adds to an order are the seconds a critical patient does not have.[1] Specific harms recur: a cancer treatment delayed for years because a default surfaced an old normal result instead of a recent abnormal one; a baby killed by an overdose an alert would have caught had it been switched on — each a case where the interface's defaults and toggles, not the clinician's competence, determined whether the right information reached the decision.[2]

THE INVESTIGATION

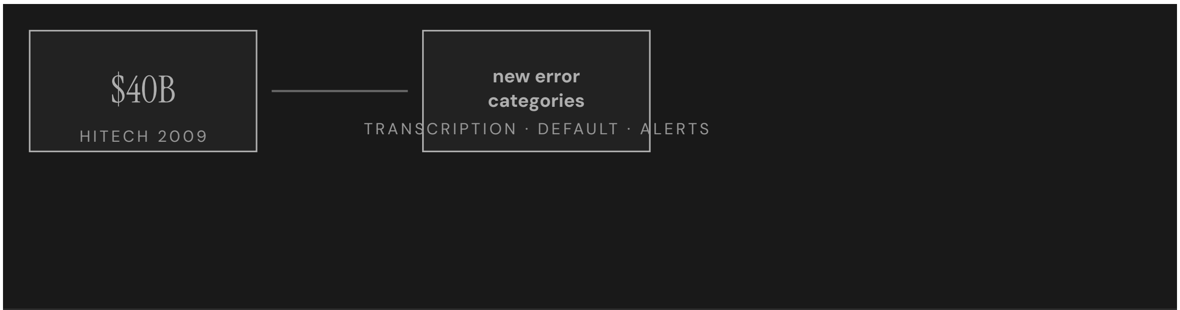
A 2023 KLAS Arch Collaborative survey across more than two hundred hospitals found EHR usability is consistently the top complaint of physicians, nurses, and pharmacists, and that end-user experience scores correlate with patient-safety outcomes — turning a stream of anecdotes into longitudinal evidence that usability is now itself a patient-safety variable. Spanning some two hundred hospitals, the survey converts what could be dismissed as one site's grievance into a reproducible signal, the kind of measurement that makes an interface problem legible as a safety problem rather than a satisfaction one.[3]

THE CAPABILITY GAP

EHR/CPOE is the canonical ongoing case of an interface designed for one set of requirements and deployed against another. The systems work for billing and administration because that was the design constraint; they fail for clinical workflow because clinical workflow was not. Decades and tens of billions of dollars later, usability remains among the largest contributors to in-system harm in U.S. healthcare — a cost that persists precisely because it was never a design requirement and so was never engineered out, only worked around by clinicians absorbing the friction shift after shift.[4]

AFTERMATH & REFORM

Usability and safety have risen on the agenda — AMA/Pew/MedStar usability work, KLAS benchmarking, safety-focused design guidance — but there is still no regulatory framework that monitors deployed EHR safety the way drugs or devices are monitored. The lesson is that buying the tool is not the same as engineering the capability: an interface deployed against the wrong specification keeps extracting a cost no one is formally counting, and without a monitoring regime the harm stays diffuse — spread across millions of encounters, attributable to no single failure, and therefore easy for the system to keep paying indefinitely.[5]



$40B INVESTED; NEW ERROR PATHWAYS EMERGED FROM INTERFACE DESIGN

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THE LEARNING ENGINEERING LENS

> *Reports of new types of errors directly related to EHR implementation — errors that can compromise quality of care and patient safety — have emerged.*
>
> PARAPHRASING SITTIG & SINGH, EHR-RELATED SAFETY RISKS, 2013

LE INSIGHT

EHR/CPOE is the canonical ongoing case of an interface designed for one set of requirements and deployed against another. The systems work for billing and administration. They fail for clinical workflow because clinical workflow was not the design constraint. Forty billion dollars later, usability remains the single largest contributor to in-system harm in U.S. healthcare.

LENS APPROACH

LENS treats EHR/CPOE in LEN 7 as the live, ongoing example of Design-Out and Interface failure under governance opacity, in LEN 2 as a human-AI teaming problem (alert fatigue, default surfacing), and in LEN 9 as a computational systems problem (the alerting architecture itself is a learnable model). The Han 2005 pediatric ICU finding is the durable warning; the KLAS 2023 surveys across two hundred hospitals are the contemporary, longitudinal evidence that usability is now itself a patient-safety variable.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Write clinical workflow and human-factors performance into the procurement specification, so a system that serves billing but not bedside care cannot clear the buy.
- Validate defaults, alerts, and result-surfacing against real clinical scenarios before deployment, since it was the default and the unactivated alert — not clinician skill — that determined the harm.
- Tie any adoption incentive to demonstrated usability, so the money rewards a tool that supports care rather than merely one that is installed.

IN OPERATION / AFTER

- Stand up a post-deployment safety-monitoring regime for fielded EHRs comparable to how drugs and devices are surveilled, with thresholds and authority to force changes.
- Track end-user experience scores as a patient-safety variable, using the cross-institution evidence that usability correlates with outcomes.
- Audit recurring interface-driven harms — surfaced defaults, disabled alerts — and feed them back into vendor design requirements rather than absorbing them as clinician burden.

REFLECTION QUESTIONS

1. What is the equivalent system in your domain that was designed for one specification and deployed against another? How would you measure the harm?

2. Design the regulatory architecture that would surface EHR safety harms at scale. Be specific about signal, threshold, and authority.

3. The Han 2005 mortality result was disputed; the 2023 KLAS surveys are consistent across two hundred hospitals. What ongoing measurement architecture would have to exist for the equivalent emerging clinical-AI deployment in your domain to be evaluated honestly while in use?

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LENS COURSES LEN 7 LEN 2 LEN 9

CASE 29 AUTONOMOUS SYSTEMS 2018

Uber ATG / Tempe Fatality

T Training Gap N Normalization of Deviance G Governance & Trust H Human-System Interface

IMPACT One pedestrian killed — the first fatality involving a self-driving vehicle striking a pedestrian

IN BRIEF

On the night of 18 March 2018 a modified Volvo running Uber's self-driving system struck and killed Elaine Herzberg as she crossed a road in Tempe, Arizona — the first pedestrian killed by an autonomous vehicle. The NTSB found the safety operator had been watching a video on her phone, but placed heavy blame on Uber: it had not recognized the risk of automation complacency, trained for it, or enforced its own no-phone policy. The system itself was programmed not to brake when a crash was deemed unavoidable, and could not classify a pedestrian who was not near a crosswalk. The human was kept in the loop as a passive monitor — a role the NTSB noted is chronically unperformable — with no infrastructure to make it work. Uber ATG is the book's defining human-AI-teaming case.

BACKGROUND

Uber's Advanced Technologies Group tested self-driving cars on public roads with a safety operator behind the wheel, present to take over if the automation failed. The role was passive surveillance: watch a system that drove itself well almost all of the time, and intervene in the rare moment it did not.[0] That structure asks a person to stay vigilant for an event that almost never comes, the precise condition under which human attention is known to lapse — so the role was set up to demand exactly the kind of sustained monitoring that people are least able to deliver.

WHAT HAPPENED

On 18 March 2018 a modified Volvo SUV in autonomous mode struck and killed Elaine Herzberg as she crossed a road at night in Tempe, Arizona — the first pedestrian killed by a self-driving vehicle. The safety operator was looking down at a video on her phone in the seconds before impact.[1] The phone was not an aberration but the predictable filling of an attention vacuum: a role with nothing to do for hours invites exactly that drift, and nothing in the car's design or the monitoring around the seat pulled the operator's eyes back to the road when it finally mattered.

THE INVESTIGATION

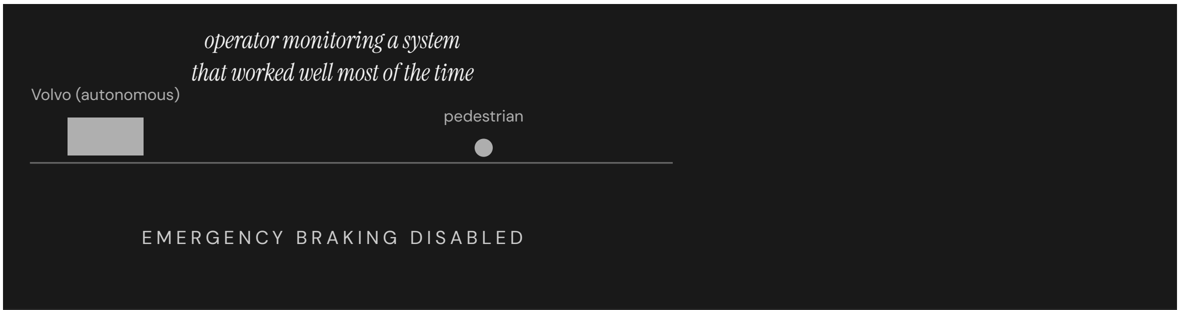
The NTSB's probable cause centered on the operator's failure to monitor the road, but it placed heavy blame on Uber, which "did not adequately recognize the risk of automation complacency and develop effective countermeasures": training was inadequate and the no-phone policy unenforced.[2] Naming the company alongside the operator was the board's way of locating the fault in the design of the role rather than the lapse of the person filling it — the policy existed on paper but had no mechanism behind it to make compliance the default. The system compounded the human gap. It was programmed not to apply emergency braking when a crash was judged unavoidable — removing the automated backstop — and it could not classify an object as a pedestrian unless it was near a crosswalk, so the very situation on the road that night fell into a blind spot the software was not built to see.[3]

THE CAPABILITY GAP

A human was retained not because the designers believed a person could meaningfully catch the failure, but because the regulatory and public posture required one present. The role of "monitor" was assigned without the interface, training, or authority to make it performable — a placeholder for safety rather than an instrument of it. As the NTSB chairman put it, "humans tend to tune out when tasked with monitoring automated systems that work well most of the time." The design was safe only on the assumption that the failure case would not arrive — until it did, and the assumption that had quietly held the whole arrangement together was paid for with a life.[4]

AFTERMATH & REFORM

Uber suspended testing, later exited self-driving, and the case reshaped how the industry and regulators treat safety drivers — toward two-operator teams, driver-monitoring systems, and honest accounting of what a monitor can and cannot do.[5] Each of those reforms is a concession that the single passive observer had been an unsupported role all along: a second operator shares the vigilance burden, and driver-monitoring closes the attention vacuum the original design left open. Its lasting contribution is the reframing of passive monitoring as a role that must be engineered to be performable — or not assigned at all.



DESIGNED AS PASSIVE MONITOR; EQUIPPED ONLY TO FAIL

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THE LEARNING ENGINEERING LENS

> *Repeatedly, humans tend to tune out when tasked with monitoring automated systems that work well most of the time.*
>
> NTSB CHAIRMAN ROBERT SUMWALT, UBER TEMPE HEARING REMARKS, 2019

LE INSIGHT

Uber ATG is the defining case for the LENS Human-AI Teaming competency. A human was retained in the system not because the designers believed a human could meaningfully act, but because the regulatory architecture required a human be present. The role of "monitor" was assigned without the capability infrastructure to support it. The system that resulted was performable only on the assumption that the failure case would not arrive — until it did.

LENS APPROACH

LENS treats this case in LEN 2 as the live exemplar of monitoring as an unsupportable role. Students reconstruct the capability requirements for the safety operator and design the interface, training, and authority structure that would have made the role performable — or made the case for not retaining the role at all.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Engineer the monitoring role to be performable — give the operator an active task, a usable interface, and the authority to act — or do not assign a human backstop you do not expect to work.
- Keep the automated emergency-braking backstop active rather than suppressing it, so the system does not silently remove its own last line of defense.
- Validate object classification against the real operating environment, including pedestrians away from crosswalks, before fielding the system on public roads.

IN OPERATION / AFTER

- Enforce the no-phone and attention policies with driver-monitoring that detects and corrects drift in real time, not a written rule alone.
- Audit safety-operator attention data continuously, treating sustained lapses as a design failure of the role rather than a fault of the individual.
- Deploy two-operator teams or equivalent redundancy so the vigilance burden does not rest on a single person doing an unsupportable job.

REFLECTION QUESTIONS

1. Identify a passive-monitor role in your domain. What evidence would tell you the role is or is not performable as designed?
2. The Tempe vehicle was programmed not to brake when a crash was unavoidable. Reconstruct the design rationale and propose the deliverable that should have prevented that decision.
3. Uber had a no-phone policy with nothing to enforce it. What is a rule in your domain that exists on paper but lacks the mechanism to make compliance the default — and how would you engineer that mechanism?

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LENS COURSES LEN 2

CASE 63 AVIATION 1972

Eastern Air Lines Flight 401

H Human-System Interface T Training Gap

IMPACT 101 killed in the Everglades; the entire flight crew fixated on a landing-gear indicator bulb while the autopilot silently disengaged

IN BRIEF

On 29 December 1972 an Eastern Air Lines L-1011 descended into the Florida Everglades on a night approach to Miami, killing 101 of the 176 aboard. The cause was attention, not mechanics: all three flight-deck crew became absorbed in a landing-gear indicator bulb that had failed to light, and while they worked it, the autopilot's altitude hold was inadvertently disengaged. The aircraft sank slowly; an altitude alert chimed but went unheeded in the cockpit clutter, and no one was monitoring the flight path. The NTSB findings launched decades of research into attention, monitoring, and cockpit-alert design, and the accident is a formative event behind Crew Resource Management. Eastern 401 is the canonical case of a low-priority task crowding out a life-critical one.

BACKGROUND

Eastern Air Lines Flight 401, a wide-body Lockheed L-1011, was on a night approach to Miami on 29 December 1972 with a full flight deck — captain, first officer, and flight engineer.[0] Three crew was the era's safeguard against any one person being overloaded, the assumption being that more eyes meant more coverage; the night would show that without a designed division of those eyes, three people could converge on a single trivial problem as easily as one.

WHAT HAPPENED

The crew noticed that the landing-gear indicator light had not illuminated, and all three focused on the bulb. While they worked the problem, one of them inadvertently nudged the controls and disengaged the autopilot's altitude hold. The TriStar began a slow descent into the Everglades; an altitude-warning chime sounded but, in the auditory clutter of the cockpit, no one registered it. The aircraft hit the swamp, killing 101 of the 176 aboard.[1] The descent was gentle enough to go unfelt, and the single chime carried no more urgency than the routine sounds around it — so the one cue that could have broken the fixation was indistinguishable, by design, from the background it sounded against.

THE INVESTIGATION

The NTSB found the crew had become so engrossed in the landing-gear difficulty that they failed to monitor the flight path.[2] A burned-out indicator bulb had absorbed the attention of an entire qualified crew while a wide-body airliner descended unwatched — the dispro-portion between the trigger and the outcome is exactly what made the case so durable a teaching example. The findings inaugurated decades of research into attention, monitoring, and the design of cockpit alerts — the accident is cited in nearly every introductory cognitive-engineering course as the example of how a low-priority task can crowd out a life-critical one when attention is undivided among the available channels.[3]

THE CAPABILITY GAP

The missing capability was a designed division of attention across the crew, and an alert that carried the priority its message deserved. Three competent people attended to a burned-out bulb while no one watched the altitude; the chime existed but did not cut through. The flight deck had the people and the information — it lacked the design that would have kept one channel of attention on the thing that could kill them. Attention had been treated as something the crew would naturally allocate well, rather than as a resource the cockpit had to be built to protect, and that unexamined assumption was the gap.[4]

AFTERMATH & REFORM

Eastern 401 was one of the formative events behind Crew Resource Management (Case 12) — the explicit allocation of monitoring and cross-checking roles — and behind modern standards for prioritized cockpit alerting.[5] Both reforms encode the same correction: CRM assigns someone to keep flying the aircraft while others troubleshoot, and prioritized alerting ensures the most dangerous condition is the loudest one — turning the night's two failures into permanent structural defenses. Its lesson is that attention is a designable parameter, not a personal virtue: a system that lets all eyes converge on one task must guarantee the critical channel is still watched.

EASTERN 401 — THREE CREW, ONE TASK, NO MONITOR

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6. Eastern 401 as a formative case behind CRM (Case 12) and prioritized cockpit alerting standards.

THE LEARNING ENGINEERING LENS

> *The crew became so engrossed in the landing-gear difficulty that they failed to maintain altitude.*
>
> PARAPHRASING NTSB AIRCRAFT ACCIDENT REPORT NTSB-AAR-73-14 ON EASTERN 401, 1973

LE INSIGHT

Eastern 401 is the canonical attention-failure case. The capability that was missing was a designed division of attention across the crew. CRM exists because Eastern 401 happened, and the discipline of cockpit alert design exists because the altitude warning chime did not carry the priority weight it needed to.

LENS APPROACH

LENS uses Eastern 401 in LEN 5 to teach attention as a designable parameter and in LEN 2 to introduce alert prioritization as a capability deliverable. The case anchors the CRM origin story.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Build an explicit division of attention into the operating procedure, so at least one role is always assigned to the life-critical channel while others troubleshoot.
- Design alerts with a priority hierarchy, ensuring the most dangerous condition is also the most salient rather than one indistinguishable chime among many.
- Treat attention as a resource the system must protect by design, not a virtue the operators are assumed to supply on their own.

IN OPERATION / AFTER

- Drill crews on scenarios where a trivial problem competes with a critical one, and verify the monitoring role actually holds under that pressure.
- Audit whether alerts in service are heard and acted on, retiring or redesigning any that get lost in routine clutter.
- Track distraction-related near-misses so attention-displacement failures are visible before they cause a loss.

REFLECTION QUESTIONS

1. Identify a low-priority task in your domain that could plausibly absorb all of an operator's attention. What is the life-critical task that would be displaced?

2. Redesign the altitude warning chime of an L-1011 so that it cuts through a focused troubleshooting conversation.

3. Eastern 401 had three qualified people and still left the flight path unwatched. How would you assign and verify a "someone is always watching the critical channel" role so it cannot collapse when the whole team is drawn to one problem?

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LENS COURSES LEN 5 LEN 2

CASE 64 AVIATION 1991, 1994

Boeing 737 Rudder Hardovers

H Human-System Interface D Designed Out

IMPACT 157 killed across United 585 (Colorado Springs, 1991) and USAir 427 (Pittsburgh, 1994); rudder Power Control Unit malfunctions traced to a thermal-shock condition

IN BRIEF

Two Boeing 737s fell out of controlled flight when their rudders reversed — moving opposite to the pilots' input: United 585 near Colorado Springs in 1991 (25 killed) and USAir 427 near Pittsburgh in 1994 (132 killed), 157 in all. The NTSB investigation took years because the failure was rare and unrecoverable. It traced the cause to a thermal-shock condition in the rudder's power-control-unit servo valve — cold-soaked hydraulic fluid hitting a hot valve — that could jam the valve and reverse the rudder. The pilots had no procedure to recognize or recover the failure; the manuals and training never anticipated it, and in that state no available control input could save the aircraft. The 737 rudder cases are the book's case for an unrecoverable failure mode hidden inside a working aircraft.

BACKGROUND

The Boeing 737, the most-produced jetliner in history, steers in yaw with a rudder driven by a hydraulic power control unit (PCU). The PCU's servo valve was a dual-concentric design intended to be fault-tolerant — a reassuring assumption that had passed certification.[0] Because the design was believed to be fault-tolerant, a single jam was treated as something the redundant geometry would contain; that confidence is precisely what let the failure mode hide in plain sight across a fleet flying millions of hours without incident.

WHAT HAPPENED

Twice, 737s rolled and dove out of level flight unrecoverably: United Airlines Flight 585 near Colorado Springs in 1991 (25 killed) and USAir Flight 427 near Pittsburgh in 1994 (132 killed) — 157 deaths between them. In each, the rudder had swung hard over, opposite to what the crew commanded.[1] A control that moves opposite to its input is the cruelest failure a pilot can face: every corrective action deepens the upset, and the three years separating the two losses meant the first crash yielded no usable answer in time to protect the second.

THE INVESTIGATION

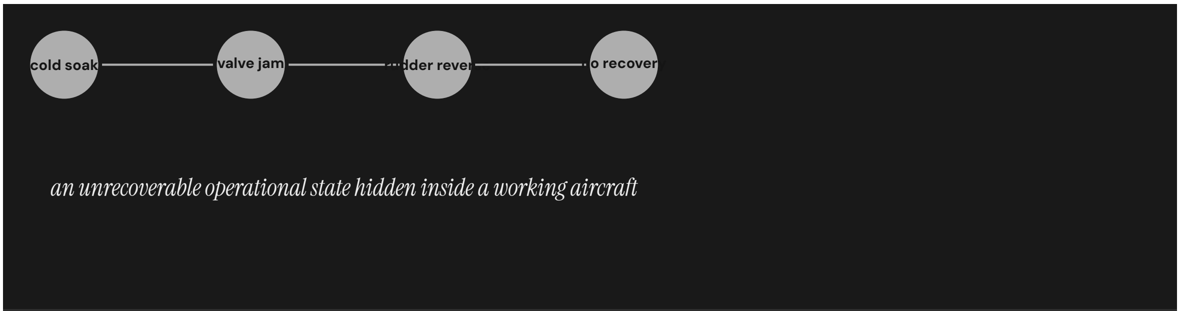
The NTSB investigation took years, because the failure was both rare and unrecoverable, leaving little to reconstruct.[2] An unrecoverable event tends to destroy its own evidence, and a rare one offers no pattern to work from, so the board had to reason toward a mechanism the wreckage could only hint at. It eventually identified a thermal-shock condition in the rudder servo valve — cold-soaked hydraulic fluid striking a hot valve under specific conditions — that could jam the secondary valve and let the rudder move opposite to commanded input, a convergence of cold, heat, and timing so narrow it had eluded every test the design had been put through.[3]

THE CAPABILITY GAP

The gap was not in the pilots: in the failure mode no control input available to them could recover the aircraft, and they had no procedure even to recognize a rudder reversal, because the manuals and training never anticipated it. The missing capability sat upstream — in a certification process that had not surfaced the failure mode, and maintenance procedures that did not test for it. When no human action is recoverable, capability engineering must move to the design and certification, not the cockpit — no amount of pilot skill or training can close a gap that lives in a part the crew cannot reach and a state no procedure names, which is why the only real fix was upstream of the flight deck entirely.[4]

AFTERMATH & REFORM

Boeing redesigned the rudder PCU and the fleet was retrofitted, and the cases reshaped how rare, unrecoverable failure modes are hunted in certification and flight test.[5] Retrofitting the entire fleet conceded that the original fault-tolerant assumption had been wrong, and the changed approach to certification testing accepted that a part passing its tests is not the same as a part proven safe across every condition it will meet. The 737 rudder sits at the intersection of mechanical design, certification testing, and crew capability — all three would have had to fail together for the accidents to occur, and all three did.



737 RUDDER — THE FAILURE MODE CERTIFICATION DID NOT ANTICIPATE

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THE LEARNING ENGINEERING LENS

> *The aircraft was operated for years with a design feature that, under specific conditions, was unrecoverable.*
>
> PARAPHRASING NTSB AIRCRAFT ACCIDENT REPORT NTSB-AAR-99-01 ON THE 737 RUDDER PCU, 1999

LE INSIGHT

The 737 rudder cases are the case for an unrecoverable operational state hidden inside an apparently working system. The capability gap was not in the pilots — there was no capability that would have recovered the aircraft. The gap was in the certification process that had not surfaced the failure mode and in the maintenance procedures that did not test for it.

LENS APPROACH

LENS uses the 737 rudder cases in LEN 5 as a capability-limits case (when no human action is recoverable, capability engineering must move upstream) and in LEN 7 for certification governance.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Test components against the full envelope of conditions they will meet in service — including rare combinations like thermal shock — rather than only the nominal cases certification asks for.
- Treat any “fault-tolerant” claim as a hypothesis to be falsified in flight test, not an assumption that lets a single jam go unexamined.
- Provide crews a procedure to recognize and respond to control reversals, while accepting that for unrecoverable modes the real fix must sit in the design.

IN OPERATION / AFTER

- Reopen and reconcile investigations of rare, unexplained events instead of leaving them undetermined, since a buried first case left the second crew unwarned.
- Build maintenance procedures that actively test for the surfaced failure mode across the fielded fleet, not just the redesigned units.
- Hunt for unrecoverable failure modes proactively in service data, treating their absence in the record as unproven rather than as evidence of safety.

REFLECTION QUESTIONS

1. Where in your domain might an unrecoverable failure mode exist that has not yet manifested? How would you find it?
2. Design the certification-test specification that should have caught the 737 rudder thermal-shock failure in development.
3. The 737 rudder passed certification yet was not safe across every condition it met. What is a component in your domain certified as fault-tolerant on an assumption no one re-tests — and how would you challenge that assumption before it fails?

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LENS COURSES LEN 5 LEN 7 LEN 3

CASE 65 TECHNOLOGY 2024

CrowdStrike Falcon Outage

D Designed Out K Knowledge & Institutional Memory G Governance & Trust

IMPACT 8.5 million Windows machines crashed; airlines, hospitals, broadcasters, and banks affected simultaneously; largest IT outage on record

IN BRIEF

On 19 July 2024 CrowdStrike pushed a content update to its Falcon endpoint sensor that contained a logic error in a small configuration file. On every Windows machine running the affected sensor, the file drove the kernel driver to crash, looping the blue screen of death and requiring hands-on recovery of each device. Roughly 8.5 million machines went down at once — across hospitals, airlines, banks, broadcasters, and governments worldwide — the largest IT outage on record. CrowdStrike's post-incident review found the content update had not been put through the same automated testing or staged rollout as code: the pipeline treated "content" as a lesser category than "code," though the operating system did not. CrowdStrike is the cybersecurity-vendor analog of Knight Capital, six orders of magnitude larger.

BACKGROUND

CrowdStrike's Falcon sensor runs deep inside Windows — in the kernel — to detect threats, and it updates constantly: not only its code, but "content," the rapid-response detection configuration pushed to customers continuously. The deployment pipeline treated that content as a lighter category than code, with less testing and no staged rollout.[0] The distinction had a logic: content shipped fast and often, precisely so the sensor could keep pace with new threats, and slowing it down with full code-grade testing seemed to defeat its purpose. That speed was the very reason the safety gate was lowered on the artifact that could still crash a kernel.

WHAT HAPPENED

On 19 July 2024 a content update contained a logic error in a small configuration file. On every Windows machine running the affected sensor, the file drove the kernel driver to read out of bounds and crash, looping the blue screen of death and requiring manual recovery of each device. Roughly 8.5 million machines failed at once — hospitals, airlines, banks, broadcasters, and governments worldwide — the largest IT outage on record.[1] Because the file went to every affected sensor simultaneously with no staged rollout, there was no first wave to catch the fault and no blast radius short of the whole install base; the requirement for hands-on recovery of each device turned a single bad push into weeks of physical labor across the world.

THE INVESTIGATION

CrowdStrike's own post-incident review found the content update had not been put through the same depth of testing and staged rollout as its code releases. The fault was not exotic: a category boundary in the deployment pipeline — content treated as safer than code — that did not match the operational reality, in which a bad content file could crash the kernel exactly as bad code could. The boundary was an organizational convenience rather than a technical truth, and the kernel, which executes whatever reaches it, recognized no such distinction at all.[2]

THE CAPABILITY GAP

The missing capability was the recognition that, for deployment safety, content **is** code: anything that can crash the kernel must clear the same testing and staged-rollout gates. CrowdStrike's customers had trusted the vendor's deployment safety, and that trust turned out to be load-bearing for the operation of a large slice of the global economy on a single morning. Each customer had implicitly outsourced a safety gate to the vendor's pipeline, so the one missing gate inside that pipeline was multiplied across every institution that ran the sensor, with no independent check standing between a bad push and their kernels.[3]

AFTERMATH & REFORM

CrowdStrike moved content updates onto staged rollouts and stronger validation, Microsoft revisited kernel-level access for security vendors, and the episode prompted scrutiny of concentration risk in endpoint security.[4] Each response targets a different layer of the same failure: staged rollout limits the blast radius of any one push, reconsidering kernel access limits how much a vendor fault can break, and the concentration-risk scrutiny acknowledges that a single vendor had become a shared point of failure for much of the economy. It is the cybersecurity-vendor analog of Knight Capital (Case 55) — an unverified deployment to a system wired into critical operations — at a scale six orders of magnitude larger.

8.5M

MACHINES · SINGLE CONFIGURATION FILE

content treated differently from code in the deployment pipeline

CROWDSTRIKE — THE COST OF A CATEGORY BOUNDARY IN A DEPLOYMENT

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THE LEARNING ENGINEERING LENS

> *Our content configuration update process did not include the same depth of testing and staged rollout as our code releases.*
>
> PARAPHRASING CROWDSTRIKE PRELIMINARY POST-INCIDENT REVIEW, JULY 2024

LE INSIGHT

The CrowdStrike outage is the live case for what happens when a deployment safety architecture treats categories of artifact differently than the operational system actually does. Content looked operationally identical to code; it was treated as different in the deployment pipeline. The mismatch became the largest IT outage in history.

LENS APPROACH

LENS uses CrowdStrike in LEN 5 as a categories-and-boundaries capability case and in LEN 2 for the vendor-customer trust architecture: customers trusted CrowdStrike's deployment safety; that trust was load-bearing for the operation of the global economy on a single day.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Define deployment-safety gates by operational impact, not artifact category: anything that can crash the kernel clears the same testing and staged rollout as code.
- Make staged rollout mandatory for every update, so a faulty push is caught by a first wave instead of reaching the entire install base at once.
- Design for recovery, not just prevention — assume a bad update will ship and ensure it does not require hands-on intervention at every device.

IN OPERATION / AFTER

- Audit vendor deployment-safety practices your institution depends on, rather than treating the outsourced safety gate as trustworthy by default.
- Map and reduce concentration risk so a single vendor fault cannot take down a large slice of critical operations simultaneously.
- Run post-incident reviews that interrogate category boundaries in the pipeline and feed the findings back into validation gates.

REFLECTION QUESTIONS

1. Identify a vendor relationship in your domain whose deployment-safety practice your institution does not audit. What would the audit reveal?
2. Design the cross-vendor staged-rollout protocol that should be standard for endpoint security software.
3. CrowdStrike's pipeline treated content as safer than code, but the kernel did not. What category boundary in one of your systems is an organizational convenience that the operational reality ignores — and what would it cost if it broke?

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TOOLS task analysis · design prototyping · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

LENS COURSES LEN 5 LEN 2

CASE 66 DEFENSE 1983

Stanislav Petrov / 1983 False Alert

H Human-System Interface T Training Gap

IMPACT Soviet early-warning system reported five incoming U.S. ICBMs; Lt. Col. Petrov correctly assessed the signal as false; retaliation averted

IN BRIEF

On the night of 26 September 1983 the Soviet Oko early-warning system reported five U.S. intercontinental missiles inbound. The duty officer, Lt. Col. Stanislav Petrov, judged it a false alarm — a real first strike would involve hundreds of missiles, not five — and reported it as such rather than up the launch-decision chain. He was right: sunlight glinting off high-altitude clouds had fooled the satellite's infrared sensors. Petrov's judgment is widely credited with averting nuclear war. The case is the book's strongest positive evidence for human-in-the-loop capability: the automation failed in a mode its designers never anticipated, and a person with contextual judgment and the latitude to override it was the recoverability the system had. Keeping humans in some loops is not nostalgia — it is capability engineering.

BACKGROUND

Soviet nuclear command rested partly on Oko, a satellite early-warning system meant to detect U.S. missile launches and feed a launch-on-warning posture. The duty officer at the Serpukhov-15 bunker was responsible for verifying any alert and passing it up the chain.[0] A launch-on-warning posture compresses the time between detection and decision to almost nothing, so the duty officer's verification was not a formality but the single human checkpoint standing between a satellite reading and the machinery of retaliation.

WHAT HAPPENED

On the night of 26 September 1983 Oko reported a U.S. intercontinental ballistic missile launch — then, minutes later, four more, for five in all. Lt. Col. Stanislav Petrov, the duty officer, assessed the signal as a false alarm — a genuine first strike, he reasoned, would involve hundreds of missiles, not five — and reported it as such. He was correct.[1] The reasoning that saved the situation came from outside the system entirely: Oko could report what it saw, but it could not weigh five launches against the doctrine of a real first strike, and that mismatch between the alert and the strategic picture was exactly the judgment the machine had no way to make.

THE INVESTIGATION

The cause was an unanticipated automation failure: sunlight reflecting off high-altitude clouds at a particular geometry had fooled the Oko satellite's infrared sensors into reading launches that were not there.[2] It was a failure of the rarest kind — a benign natural phenomenon the sensors had never been designed to discount — which is precisely why no automated check existed to catch it. Later investigation identified the satellite-geometry failure mode and modified the algorithm; the scenario is now permanently archived in early-warning training as the canonical false positive, a permanent reminder that the system's worst error was one it could not have flagged for itself.[3]

THE CAPABILITY GAP

The automation had failed in a mode its designers had not imagined, and no automated check could catch it. What caught it was a human with the contextual judgment to weigh the alert against what a real attack would look like, and the institutional latitude to override the system rather than simply relay it. Petrov's "funny feeling" was a career's worth of judgment doing the work the automation could not — and crucially, the chain of command had left him room to act on it rather than forcing him to pass the alert upward unfiltered, so the recoverability lived as much in the authority structure as in the man.[4]

AFTERMATH & REFORM

Petrov's decision is widely credited with averting nuclear war, and the case is permanently studied in command-and-control training.[5] Preserving the episode in training is itself a design choice: it keeps the failure mode and the human override alive as institutional memory, so the lesson does not erode the way the original sensor's blind spot had. It is the book's strongest argument that retaining a human in some loops is capability engineering, not nostalgia: the human role is the recoverability of an automation failure the designers did not anticipate — which is exactly why fully automating a strategic decision removes the one element that can catch the unimagined error.

SYSTEM	OPERATOR
5 ICBMs	false alarm

the human in the loop was the recoverability of an automation failure

PETROV, 1983 — HUMAN-AI TEAMING AT THE HIGHEST POSSIBLE STAKES

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4. The subsequent Oko algorithm modification and the incident's use in early-warning training.
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THE LEARNING ENGINEERING LENS

> *I had a funny feeling in my gut.*
>
> STANISLAV PETROV, QUOTED IN THE WASHINGTON POST, 1999

LE INSIGHT

Petrov is the canonical positive case for human-in-the-loop nuclear command-and-control. The automation failed in a mode its designers had not anticipated. The human's contextual judgment was the backstop that allowed the failure to be recovered before it became catastrophic. The case is the strongest argument in this book that the design choice to retain humans in some loops is not nostalgia but capability engineering.

LENS APPROACH

LENS uses Petrov in LEN 2 as the positive Human-AI Teaming case at the highest possible stakes: the human role is the recoverability of an automation failure that the designers did not anticipate. The case anchors arguments against full-automation of strategic decisions.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Where automation feeds an irreversible decision, design the human role to be a genuine override — with the context, judgment cues, and authority to refuse a faulty alert, not merely relay it.
- Give operators a strategic picture against which to weigh an alert, so an anomalous signal can be tested against what a real event would look like.
- Treat unimagined failure modes as a design assumption: keep a human in the loop precisely where no automated check can cover the unanticipated.

IN OPERATION / AFTER

- When a failure mode surfaces, modify the algorithm and archive the scenario in training so the lesson persists as institutional memory.
- Audit whether human-in-the-loop roles still carry real override authority over time, or have decayed into ceremonial relays.
- Preserve and refresh the contextual judgment override depends on through deliberate training, so the capability does not erode between rare events.

REFLECTION QUESTIONS

1. Identify an automated system in your domain where retaining a human-in-the-loop is genuinely capability-engineering rather than ceremonial. How would you tell the difference?

2. Petrov's "funny feeling" was contextual judgment built across a career. Design the training that produces it deliberately.

3. Petrov's recoverability lived as much in his latitude to override as in his judgment. Map an automated decision in your domain where the operator has the knowledge to catch a failure but lacks the authority to act on it — and redesign the authority structure.

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LENS COURSES LEN 2

CASE 67 TECHNOLOGY 2018

TSB Bank IT Migration

H Human-System Interface G Governance & Trust

IMPACT 1.9 million UK customers locked out of accounts; £330M+ in compensation and remediation; CEO resigned

IN BRIEF

In April 2018 TSB Bank tried to migrate some five million customer accounts off its former parent Lloyds' systems onto a new platform from its current owner, Sabadell, over a single weekend. When services came back online, nearly every component failed: 1.9 million customers were locked out, some saw strangers' accounts, mortgages vanished, payments bounced. Recovery took months and cost over £330 million; the CEO resigned and the regulator fined the bank. The independent review found the platform had been tested under conditions that did not approximate real load, certified ready by a process that did not challenge the certification, and pushed live against technical recommendations that it was not ready. TSB is the financial-sector analog of Healthcare.gov: a migration shipped without adequate testing because schedule pressure overrode the technical signal.

BACKGROUND

TSB Bank, spun out of Lloyds and acquired by Spain's Sabadell, needed to move some five million customer accounts off Lloyds' systems onto a new Sabadell-built platform. The cutover was scheduled for a single weekend.[0] Compressing a five-million-account migration into one weekend left no room for partial failure: the schedule itself became a forcing function, framing readiness as a date to be hit rather than a condition to be proven, and that framing would later prove decisive when the technical signal said the platform was not ready.

WHAT HAPPENED

When customer-facing services came back online that Sunday evening in April 2018, nearly every component of the new platform had problems. About 1.9 million customers were locked out; some saw other people's accounts, mortgages disappeared, and card payments failed. The recovery took months, cost more than £330 million in compensation and remediation, and the chief executive resigned.[1] That nearly every component failed at once points away from a single defect and toward a platform that had never been exercised under real load — the kind of systemic breakdown that follows when a system is proven only in conditions it will never actually meet.

THE INVESTIGATION

The Slaughter and May independent review found the migration had been tested under conditions that did not approximate real customer load, and that the platform had been certified ready by a process that did not adequately challenge the certification — a certification that confirmed readiness rather than interrogating it, which is how a system that would fail under real conditions could be signed off as fit.[2] Decisively, the executive decision to proceed had been taken against technical recommendations that the platform was not ready; the Financial Conduct Authority later fined TSB, treating the override of a known technical objection as a failure of governance and not merely of engineering.[3]

THE CAPABILITY GAP

The technical signal existed — the platform was not ready, and people knew it. But the decision authority sat at the executive layer, where the signal arrived weakened by passage through intermediate layers, and the institutional architecture gave the technical layer no way to halt the migration. The missing capability was not testing knowledge but a governance structure in which a "not ready" could stop a scheduled go-live. Knowing a system is unready is worthless if the knowledge cannot reach the decision with its force intact and the authority to act on it; here the truth was present but powerless.[4]

AFTERMATH & REFORM

TSB rebuilt its testing and migration governance, paid out and was penalized, and the case entered the literature as a study in technical-decision authority.[5] Rebuilding governance rather than merely the platform was the right diagnosis: the failure had been one of who could say "stop" and be heeded, so the durable fix had to live in the decision structure rather than the code. It is the financial-sector analog of Healthcare.gov (Case 10): a large migration shipped without the testing the institution knew it needed, because schedule pressure overrode a technical signal that had no authority to win.

TSB — READINESS SIGNAL ARRIVED ATTENUATED AT THE DECISION

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THE LEARNING ENGINEERING LENS

> *The migration proceeded notwithstanding clear signals that the platform was not ready.*
>
> PARAPHRASING THE SLAUGHTER AND MAY INDEPENDENT REVIEW OF THE TSB MIGRATION, 2019

LE INSIGHT

TSB is the canonical case for schedule pressure overriding technical signal in a regulated industry. The technical signal existed. The decision authority was at the executive layer where the signal arrived weakened by passage through multiple intermediate layers. The institutional architecture did not allow the technical layer to halt the migration.

LENS APPROACH

LENS uses TSB in LEN 7 as a corporate-governance case and in LEN 8 for the institutional structure of technical-decision authority. Studio projects compare TSB and Healthcare.gov.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Test the platform under conditions that approximate real production load, since a system proven only in unrepresentative conditions will fail when it meets the real ones.
- Make certification adversarial — a process that tries to disprove readiness — rather than a sign-off that confirms it.
- Avoid single-weekend big-bang cutovers where feasible; stage the migration so partial failure is survivable rather than catastrophic.

IN OPERATION / AFTER

- Build a governance structure in which a technical “not ready” can halt a scheduled go-live, so the signal reaches the decision with its force and authority intact.
- Ensure the decision authority hears the technical signal directly rather than through intermediate layers that attenuate it.
- After any failure, fix the decision structure that allowed a known objection to be overridden, not just the system that broke.

REFLECTION QUESTIONS

1. Where in your organization does a technical signal arrive at the executive layer attenuated by intermediate layers? What is the cost of the attenuation?
2. Design the institutional structure that would allow a technical lead to halt a migration like TSB’s without resigning.
3. TSB’s certification confirmed readiness rather than interrogating it. Examine a sign-off process in your domain that rubber-stamps rather than challenges — what would it take to make the certification adversarial enough to catch an unready system?

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LENS COURSES LEN 7 LEN 8

Chapter 5

Governance Failures

Trust and accountability were afterthoughts.

The fire at Grenfell Tower was the culmination of decades of failure by central government and other bodies in positions of responsibility.

GRENFELL TOWER INQUIRY, PHASE 2 REPORT, 2024

CASE 8 EDUCATION AT SCALE 2014

inBloom

G Governance & Trust

IMPACT $100M initiative collapsed in ~14 months; 9 states withdrew; data infrastructure for education set back years

IN BRIEF

inBloom was a $100-million, Gates-funded shared data infrastructure for U.S. student records — and the technology worked: no bug, no breach, no performance failure. What killed it in about fourteen months was everything around the technology. It launched without consent frameworks, community engagement, transparency about data use, or a way for parents to participate in decisions about their children's data. Parent-privacy groups organized opposition state by state, and nine states withdrew. Analysts read inBloom as the failure of technocratic reform — the assumption that technically sound infrastructure generates its own legitimacy. It is the purest governance failure in the dataset, and the book's clearest argument that in education at scale, stakeholder trust and governance are not optional features but load-bearing structure.

BACKGROUND

inBloom was an ambitious shared data infrastructure for U.S. K-12 student records, backed by $100 million from the Gates Foundation, hosted on commercial cloud, and built by enterprise engineers. The premise was that a common store would spare districts from rebuilding the same plumbing and let applications interoperate across systems that had never spoken to one another. Technically it was sound — no bug, no breach, no performance failure ever undid it, and that very soundness is what makes the case instructive.[0]

WHAT HAPPENED

What undid it was everything around the technology. inBloom launched without adequate consent frameworks, without meaningful community engagement on data governance, without transparency about what was collected and why, and without any way for parents to participate in decisions about their children's data. Each omission read, to a worried parent, as a decision made about their child without them in the room. Parent groups organized opposition state by state, and nine states — among them New York, Louisiana, and Illinois — withdrew as the political cost of staying overtook any promised efficiency. Within about fourteen months the $100-million initiative collapsed.[1]

THE INVESTIGATION

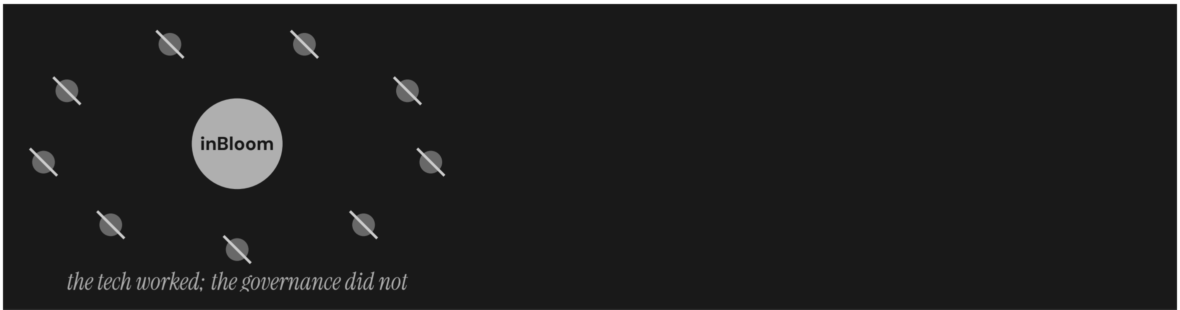
Analysts at Data & Society read inBloom as the failure of technocratic education reform: the assumption that technically sound infrastructure generates its own legitimacy proved catastrophically wrong. Legitimacy, on this reading, is earned from the stakeholders a system acts upon, not conferred by the quality of its engineering. The technology was never the problem; the governance was — the consent, transparency, and trust that had been treated as add-ons rather than as the foundation the whole effort needed before a single record moved.[2]

THE CAPABILITY GAP

inBloom is the purest governance failure in this dataset: nothing technical was wrong, and everything sociotechnical was. The missing capability was the design of stakeholder trust — consent, accountability, and a voice for the families whose data was at stake — treated as a precondition for deployment rather than a feature to add later. Once opposition formed, no patch could retrofit the trust that should have been built in from the start, because trust withheld at launch cannot be engineered back in under fire. In education at scale, those are load-bearing elements, not optional ones.[3]

AFTERMATH & REFORM

inBloom's collapse set back shared education-data infrastructure for years and became a standard cautionary tale; it also helped drive a wave of state student-data-privacy laws that codified, after the fact, the consent and transparency the project had skipped.[4] The lesson the book takes from it is that ethics-as-design-constraint is not ideology but engineering — and inBloom is the $100-million empirical test of what happens when you skip it, paid not in downtime but in a dead initiative and a chilled field.



$100M OF WORKING TECHNOLOGY; NINE STATES WITHDREW

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THE LEARNING ENGINEERING LENS

> *The technology was not the problem. The governance was the problem.*
>
> PARAPHRASING THE ANALYSIS IN BULGER, MCCORMICK & PITCAN, DATA & SOCIETY, 2017

LE INSIGHT

inBloom is the purest governance failure in this dataset. Nothing technical was wrong. Everything sociotechnical was. The case is the strongest argument in the book for treating ethics-as-design- constraint not as ideology but as engineering: a $100M empirical test of what happens when you do not.

LENS APPROACH

LENS uses inBloom in LEN 7 as the canonical governance failure and in LEN 1 as a stakeholder-analysis case. The case anchors the LENS threading commitment that equity and accountability are design commitments, not modules. Studio projects (LEN 10) require students to produce a stakeholder-trust deliverable as a precondition for deployment.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat consent, transparency, and a parent-facing voice as load-bearing requirements gathered before the data store is built, not features bolted on after launch.
- Engage the families and districts whose data is at stake as design stakeholders from the outset, so the governance questions surface in requirements rather than in opposition.
- Make legitimacy an explicit deliverable: document who must agree, on what terms, before any student record moves into shared infrastructure.

IN OPERATION / AFTER

- Audit live deployments for the gap between technical soundness and stakeholder trust, since a clean system can still be losing the political ground it stands on.
- Monitor state-by-state consent and withdrawal signals as a leading indicator, treating organized parent opposition as data about a governance defect, not noise.
- Sustain a standing transparency channel so families can see what is collected and why throughout operation, not only at adoption.

REFLECTION QUESTIONS

1. What is the equivalent unbuilt governance infrastructure in your domain? What would the $100M empirical test of its absence look like?
2. Design the stakeholder-trust deliverable that a future inBloom-equivalent should have to produce before deployment.
3. inBloom's engineers were excellent and its technology never failed, yet the project collapsed. Where in your own domain is technical soundness being mistaken for legitimacy — and who would have to consent before a system you build could claim it?

WHO BUILDS THIS governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits

LENS COURSES LEN 1 LEN 10 LEN 7 LEN 6

CASE 10 GOVERNMENT 2013

Healthcare.gov Launch

K Knowledge & Institutional Memory T Training Gap G Governance & Trust

IMPACT ~27,000 federal-marketplace enrollments in the first month vs. a 7M first-year target; hundreds of millions in remediation

IN BRIEF

When Healthcare.gov launched on 1 October 2013 it collapsed under load it had never been tested for — about 27,000 enrollments through the federal marketplace in its first month against a seven-million first-year target. The people assembled to build it knew insurance markets and government programs but not technology product launches; key technical roles went unfilled, no office clearly owned the integration (CMS thought the contractor CGI was the lead integrator; CGI did not), and no end-to-end test was run before launch. The fix-it team that rescued the site became the U.S. Digital Service. At root it was a capability mismatch at scale: the organization lacked the human capabilities the system required, and the governance chain surfaced no signal of the gap before launch. It is the rare governance failure that produced lasting institutional reform.

BACKGROUND

Healthcare.gov was the federal insurance marketplace at the center of the Affordable Care Act — a high-visibility online system that millions would hit on day one, with a political deadline that could not slip. The people assembled to build it understood insurance markets and large government programs, but not the launch of a consumer technology product at that scale, and key technical positions went unfilled — so the very expertise the launch most needed was the expertise the team most lacked.[0]

WHAT HAPPENED

There was no clear division of responsibility among the many government offices involved; CMS believed the contractor CGI was the lead system integrator, and CGI did not share that understanding — so the single most important role on the program was one no party believed it held. No end-to-end test was run before launch, meaning the assembled pieces were never exercised together as a user would exercise them. The site went live on 1 October 2013 and immediately collapsed under load it had never been validated for: about 27,000 enrollments through the federal marketplace in its first month against a seven-million first-year target.[1]

THE INVESTIGATION

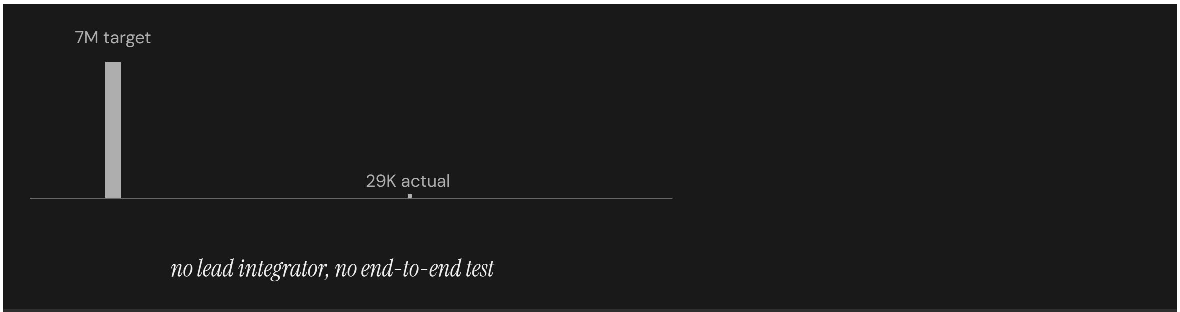
Reviews by the GAO and the HHS Inspector General found that no single person had a clear understanding of the project's status, and that the governance chain had no signal that would surface the readiness gap before launch. With ownership diffused across offices and no integration test to fail, the chain could report progress at every level while the whole remained untested — a structure in which bad news had nowhere to enter and no one positioned to act on it.[2]

THE CAPABILITY GAP

Healthcare.gov is a capability failure wearing a technology costume. The site was salvageable in weeks once the right people arrived, which is the clearest proof that the code was never the binding constraint; the original failure was that the wrong people had been assembled, and that the governance chain meant to catch the mismatch had no mechanism to see it. The missing capability was the matching of human capability to system requirement — and the institutional signal that would have flagged its absence before a deadline locked the launch in place.[3]

AFTERMATH & REFORM

The rescue effort pulled together the team that became the U.S. Digital Service — a permanent institution born from a failure visible on the news every night, its mandate built directly from what the launch had lacked.[4] It is the rare case in this book of a governance failure that produced durable organizational reform, and a reminder that the technical narrative ("the website crashed") often hides the real one: the wrong capability was assembled, unnoticed, by a chain that had no way to notice.



29,000 ENROLLMENTS VS. 7,000,000 TARGET — FIRST TWO MONTHS

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THE LEARNING ENGINEERING LENS

> *No single person had a clear understanding of the project's status.*
>
> PARAPHRASING THE HHS OFFICE OF INSPECTOR GENERAL REVIEW OF HEALTHCARE.GOV, 2016

LE INSIGHT

Healthcare.gov is a capability case wearing a technology costume. The technology was salvageable in weeks once the right people arrived. The original failure was that the wrong people had been assembled in the first place, and the governance chain that should have noticed the mismatch had no signal to surface it. The aftermath — USDS — is the rare case in this book of a governance failure that produced permanent institutional reform.

LENS APPROACH

LENS uses Healthcare.gov in LEN 1 as a problem-framing case (the technical narrative obscures the capability narrative), in LEN 5 to teach capability-requirements analysis for a large government program, and in LEN 8 to discuss the founding of USDS as organizational learning that survived.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Match assembled human capability to the system's real requirements at staffing time — fill the technical roles a consumer-scale launch demands before committing to the deadline.
- Name a single accountable integration owner and confirm every party shares that understanding, so no critical role is one each office assumes someone else holds.
- Gate launch on a full end-to-end test that exercises the assembled system the way users will, not on per-component sign-offs.

IN OPERATION / AFTER

- Build a governance signal that can surface a readiness gap upward, with someone holding the authority to delay a politically fixed launch date.
- Independently audit project status against demonstrated capability, since a diffuse chain can report green at every level while the whole remains untested.
- Institutionalize the rescue capability — as USDS did — so the talent that fixes a launch outlives the crisis that summoned it.

REFLECTION QUESTIONS

1. Healthcare.gov shipped without end-to-end testing. What is the equivalent missing deliverable in a current high-stakes deployment in your domain?
2. USDS was born from the Healthcare.gov failure. What is the institutional capability your domain still lacks, and what failure would have to occur to produce it?
3. No single office believed it owned integration, so the gap stayed invisible until launch day. In a project you know, who actually owns the seam between components — and what signal would let them, rather than the news, learn the system is not ready?

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LENS COURSES LEN 1 LEN 5 LEN 7 LEN 6

CASE 17 INDUSTRIAL 1984

Bhopal

[T] Training Gap [K] Knowledge & Institutional Memory [N] Normalization of Deviance [G] Governance & Trust

IMPACT ≈ 15,000–20,000 killed; ≈ 500,000 injured; worst industrial disaster in history

IN BRIEF

On the night of 2–3 December 1984, about forty tons of methyl isocyanate gas escaped from a Union Carbide pesticide plant in Bhopal, India — the worst industrial disaster in history. Thousands died within hours; estimates of total deaths run to 15,000–20,000, with roughly half a million exposed or injured. Safety systems had been off-line for months, the plant was understaffed, and workers were inadequately trained to recognize or handle the emergency. Investigators traced the catastrophe to operating errors, design flaws, maintenance failures, training deficiencies, and cost-cutting that endangered safety. Bhopal catalyzed the creation of the U.S. Chemical Safety Board and reshaped industrial-safety regulation for decades. It is the largest-magnitude capability-and-governance failure on record.

BACKGROUND

Union Carbide's pesticide plant in Bhopal, India, stored methyl isocyanate (MIC) — an extraordinarily toxic intermediate — in bulk, holding a lethal hazard in tanks beside a populated city. By 1984 the plant was running under heavy cost pressure: understaffed, with several key safety systems out of service for months, and workers inadequately trained to handle an MIC emergency or read its warning signs. Each economy was individually defensible on a ledger; together they thinned every layer of defense the process depended on.[0]

WHAT HAPPENED

On the night of 2–3 December 1984, water entered an MIC storage tank and triggered a runaway reaction; the safety systems that should have contained it were non-operational, so the one event the plant existed to prevent met no working barrier on its way out. About forty tons of gas vented over the sleeping city. Thousands died within hours; estimates of total deaths run to 15,000–20,000, and roughly half a million people were exposed or injured — the worst industrial disaster in history, its toll set by who happened to be downwind.[1]

THE INVESTIGATION

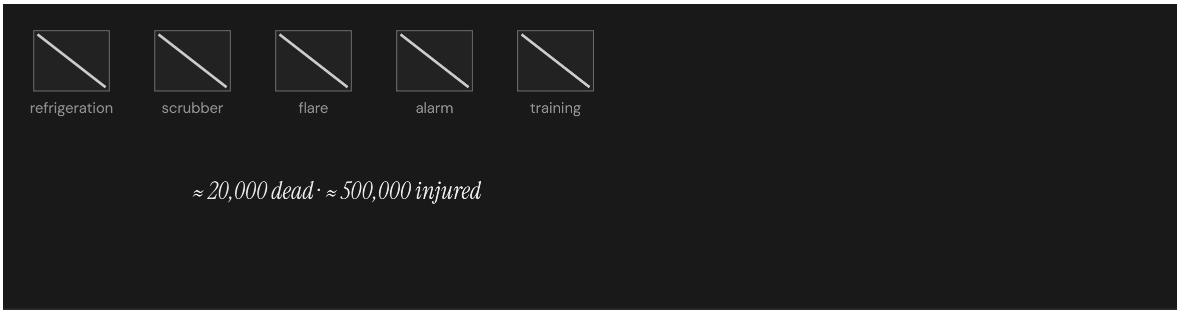
Investigations found the catastrophe "resulted from operating errors, design flaws, maintenance failures, training deficiencies and economy measures that endangered safety" — a list with no single villain, which is precisely what made it hard to govern.[2] Human-factors analysis placed Bhopal alongside Three Mile Island in its neglect of the human element, and the U.S. Chemical Safety Board would later find ineffective employee training an underlying cause in nine of its first twenty-three chemical-incident investigations — a pattern that traces to Bhopal and shows the same gap recurring long after the lesson was available.[3]

THE CAPABILITY GAP

Bhopal is the largest-magnitude capability-and-governance failure on record, and a multi-layer one: training, maintenance, design, staffing, and oversight had all degraded together, and no layer above the plant was accountable for the whole. Because the erosion was spread across layers, no single inspection or metric saw it, and each degraded layer made the next one matter more. The capability to operate an extraordinarily hazardous process safely had been hollowed out by cost-cutting, and the governance that should have caught the hollowing did not exist to ask whether the whole was still safe.[4]

AFTERMATH & REFORM

The disaster reshaped industrial safety worldwide and, in the United States, catalyzed the creation of the Chemical Safety Board — an INPO-equivalent for industrial chemistry — and the process-safety regime that followed, an oversight layer built to own exactly the whole that no one had owned at Bhopal.[5] The book's recurring arc runs through Bhopal in its starkest form: a catastrophe forces into being the institution the industry should have built before it, at a price no institution-building exercise should ever have to cost.



SAFETY SYSTEMS OFF-LINE FOR FOUR MONTHS BEFORE THE RELEASE

REFERENCES

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THE LEARNING ENGINEERING LENS

> *Operating errors, design flaws, maintenance failures, training deficiencies and economy measures that endangered safety.*
>
> NEW YORK TIMES INVESTIGATION, 1985

LE INSIGHT

Bhopal is the largest-magnitude capability-and-governance failure on record. It is also the catalyst for the creation of the U.S. Chemical Safety Board, an INPO-equivalent for industrial chemistry. The pattern that follows nearly every case in this book — diagnose, then engineer remediation at scale — runs through Bhopal in canonical form.

LENS APPROACH

LENS treats Bhopal in LEN 5 as a multi-failure capability analysis case and in LEN 7 as the largest-magnitude governance failure in the dataset. The Failure → Reform arc to the CSB demonstrates the institution-building pattern at industry scale.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat each safety system, staffing level, and training requirement as a non-negotiable layer of defense, not a cost line available for trimming under pressure.
- Assign a single accountable owner for the integrated hazard, so degradation spread across layers cannot fall between everyone's responsibilities.
- Require that operators be trained to recognize and respond to the specific emergency the process can produce before the process is run with that hazard in bulk.

IN OPERATION / AFTER

- Audit the whole defense-in-depth posture together rather than layer by layer, since each degraded barrier silently raises the stakes on the next.
- Track how long any safety system stays off-line and gate continued operation on its restoration, treating extended downtime as an unacceptable condition.
- Sustain an oversight layer — an INPO- or CSB-equivalent — with authority and reach above the individual plant to own the integrated risk.

REFLECTION QUESTIONS

1. Bhopal produced the CSB. What is the institution your domain has not yet built that a comparable disaster would force into existence?

2. Identify a plant or facility in your domain that has had safety systems off-line for an extended period. What is the measurement gap that allowed it?

3. At Bhopal training, maintenance, design, staffing, and oversight all degraded together while each looked acceptable alone. Whose job in your organization is to judge whether the whole is still safe — and what would they have to see to declare that it is not?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards · governance frameworks · bias & evidence audits

LENS COURSES LEN 5 LEN 7 LEN 3

CASE 20 INDUSTRIAL 2017

Grenfell Tower

G Governance & Trust T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT 72 killed in a residential tower fire in London; decades of regulatory failure

IN BRIEF

On 14 June 2017 a fire spread up the exterior of Grenfell Tower, a London public-housing block, killing 72 people. It raced because the tower had been wrapped in combustible aluminium-composite cladding during a refurbishment. The public inquiry found the disaster the culmination of decades of failure: cladding firms engaged in “systematic dishonesty,” marketing combustible materials as safe; regulators and inspectors missed effectively banned products across sixteen site visits; and the London Fire Brigade, whose “stay put” advice proved fatal, was unprepared for a cladding fire whose risks earlier incidents had already shown. The failure spanned manufacturers, regulators, inspectors, and responders — each contributing a piece, none owning the whole. Grenfell is the book’s case for capability failure distributed across many hands.

BACKGROUND

Grenfell Tower was a 1970s public-housing block in West London, refurbished in 2015–16 with new exterior cladding intended to improve the building’s appearance and efficiency. The cladding chosen used a combustible aluminium-composite material — installed despite safety experts’ cautions that it was unsuitable for a high-rise, a warning that sat between the people who issued it and the people who specified the panels without ever stopping the decision.[0]

WHAT HAPPENED

On 14 June 2017 a kitchen fire broke out and, instead of staying contained as a tower’s compartment design assumes, climbed the building’s exterior on the combustible cladding, wrapping the tower in flame within minutes and defeating the very principle the building was meant to rely on. Residents, following long-standing “stay put” advice premised on that containment, remained in their flats as the route to safety closed around them; 72 people died.[1]

THE INVESTIGATION

The Grenfell Tower Inquiry found the fire the culmination of decades of failure by central government and every body responsible. Cladding companies had engaged in “systematic dishonesty,” marketing combustible products as safe and corrupting the very test data buyers relied on; inspectors visited the site sixteen times and none noticed that effectively banned materials were in use, so sixteen chances to catch the hazard each passed it by.[2] The London Fire Brigade was unprepared: the risks of rapidly developing cladding fires were known from prior incidents — Knowsley Heights, Garnock Court, Shepherd’s Court — but "this knowledge had not informed firefighting policies, practices or training," so each near-miss taught no one whose job was to act on it.[3]

THE CAPABILITY GAP

Grenfell is the book's strongest evidence that capability failure can be distributed across many actors, each contributing a small piece and none accountable for the whole. Manufacturer fraud, regulatory capture, inspection incompetence, training gaps, and lost institutional memory all converged on one building, and because each actor saw only its fragment, each could regard its own part as tolerable. The inquiry called it a "grey elephant" — a danger known but ignored — and the missing capability was anyone owning the integrated risk that everyone could see in part but no one held in full.[4]

AFTERMATH & REFORM

The inquiry's Phase 2 report (2024) and the government response (2025) drove an overhaul of building-safety regulation, cladding remediation, and fire-service doctrine — reforms that, between them, tried to assign the ownership the original system had left vacant.[5] Grenfell's lesson is the governance one this chapter turns on: when responsibility for a known risk is split across dozens of actors, the risk has, in effect, no owner — and a system with no owner for its gravest hazard will eventually pay for it, in the currency of the people living inside it.

LEVELS OF FAILURE: MANUFACTURE, REGULATION, INSPECTION, RESPONSE

REFERENCES

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THE LEARNING ENGINEERING LENS

> *This knowledge had not informed firefighting policies, practices or training.*
>
> GRENFELL TOWER INQUIRY, PHASE 1, 2019

LE INSIGHT

Grenfell is the strongest evidence in the dataset that capability failure can be distributed across many actors, each of whom contributes a small piece, none of whom is accountable for the whole. The inquiry's "grey elephant" framing — known but ignored — describes a pattern that LENS treats as a primary governance problem in any high-consequence domain.

LENS APPROACH

LENS treats Grenfell in LEN 7 as the canonical multi-actor governance failure case, in LEN 8 to discuss institutional memory of warnings across the LFB and other agencies, and in LEN 10 as a studio prompt for designing the integrated capability-and-governance deliverable that would have caught the cladding decision.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Assign one accountable actor to own the building's integrated fire risk end to end, so no hazard can fall into the gaps between manufacturers, inspectors, and responders.
- Verify combustible-material claims against independent evidence rather than trusting vendor marketing, treating "systematic dishonesty" as a threat the process must defeat.
- Preserve the containment principle the building relies on: gate any cladding choice on whether it keeps a compartment fire from climbing the exterior.

IN OPERATION / AFTER

- Route every site inspection and near-miss into a shared record so sixteen visits cannot each miss the same banned material in isolation.
- Feed prior-incident knowledge into firefighting policy, training, and "stay put" doctrine, so lessons from earlier cladding fires actually change practice.
- Sustain a single line of accountability for the integrated hazard through the building's life, not only at refurbishment.

REFLECTION QUESTIONS

1. What is the "grey elephant" — the well-known risk that nobody owns — in your domain?
2. Design the deliverable that forces a single actor to own the integration risk that Grenfell distributed across dozens.
3. Sixteen inspections, prior fires, and expert cautions each touched a fragment of the Grenfell hazard, yet none assembled it. What mechanism in your domain could gather scattered partial warnings into one picture in front of someone empowered to stop the work?

WHO BUILDS THIS governance, ethics & measurement · learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

LENS COURSES LEN 10 LEN 7 LEN 8 LEN 3

CASE 21 EDUCATION AT SCALE 2014–2019

Summit Learning / Personalized Learning Rollout

G Governance & Trust T Training Gap K Knowledge & Institutional Memory

IMPACT Personalized-learning platform deployed across ~380 U.S. schools; parent revolts in Brooklyn, Cheshire, McPherson, Kennebunk; multiple districts withdrew within two years

IN BRIEF

Summit Learning, a personalized-learning platform from Summit Public Schools backed by the Chan Zuckerberg Initiative, was offered free to U.S. districts from 2015 and reached roughly 380 schools and 80,000 students by 2018. By 2019 prominent adopters were withdrawing under parent and student pressure — opt-out campaigns in Brooklyn, cancellations in Cheshire, Kennebunk, and McPherson — amid walkouts and complaints about screen time, disengagement, and data privacy. The pedagogy itself (competency-based progression, projects, mentoring) was defensible and often effective; what failed was deployment governance. There was no evaluation framework districts could read before adopting, no parent-facing data agreement, and no exit path independent of the vendor. Summit is a clean test of the book's claim that the governance architecture must be engineered alongside the tool.

BACKGROUND

Summit Learning was a personalized-learning platform developed by Summit Public Schools with technical and financial support from the Chan Zuckerberg Initiative, offered free to U.S. districts from 2015 — a price that lowered the bar to adoption while raising no governance questions at the door. Its pedagogy — competency-based progression, self-directed projects, mentor check-ins — was defensible and in many places effective, which is why the eventual revolt could not be blamed on the instructional design.[0]

WHAT HAPPENED

By 2018 the platform reached roughly 380 schools and an estimated 80,000 students, scaling fast on the strength of a free offer and a well-funded sponsor. By 2019 its most visible adopters were withdrawing under parent and student pressure: Brooklyn's MS 442 ran an organized opt-out; districts in Cheshire, Kennebunk, and McPherson cancelled or scaled back after parent meetings where the unanswered governance questions surfaced all at once. Walkouts and complaints about screen time, eye strain, disengagement, and data privacy converged into a revolt that was not about the instructional design at all.[1]

THE INVESTIGATION

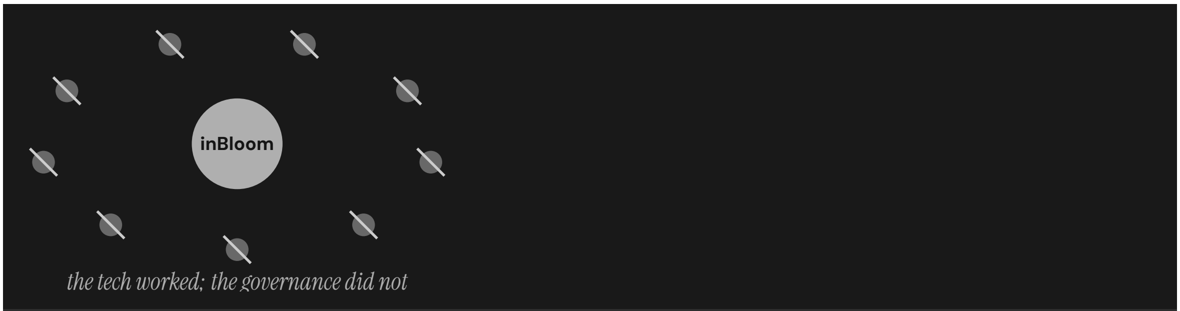
Press coverage and later analyses located the failure in deployment governance, not pedagogy: there was no evaluation framework a district could read before adopting, no parent-facing data-handling agreement, and no exit pathway that did not depend on the vendor's goodwill — three absences that each became a grievance the moment families looked for them. The implementation never surfaced the governance questions parents would ask, so when those questions arrived they arrived as opposition rather than as design input, and the argument was lost before it started.[2]

THE CAPABILITY GAP

Summit is a clean test of the book's central claim: a technology that worked at the pedagogical level still failed because the governance architecture — consent, evidence, measurement, exit — had not been engineered alongside it. A working tool with no accountability contract is a liability waiting for the first organized objection. The pattern recurs across the ed-tech dataset (inBloom, Case 8): a well-intentioned tool, a well-funded rollout, and no institutional contract with the families and teachers operating inside it — the same omission producing the same collapse in a second case.[3]

AFTERMATH & REFORM

Several districts withdrew or rebranded their use, CZI and Summit revised their outreach, and the episode became a standard caution in ed-tech adoption — a reputational cost paid for governance work that would have been cheaper to do first.[4] Its lesson for the field is concrete: an adoption decision should have to produce a public evidence summary at parent reading level, a data-handling agreement at the same resolution, and a documented exit path — governance artifacts that make a tool's deployment legitimate, not just its design sound, and that turn the questions parents will ask into inputs gathered before launch rather than weapons raised after it.



$100M OF WORKING TECHNOLOGY; NINE STATES WITHDREW

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THE LEARNING ENGINEERING LENS

> *The tools were free. The accountability architecture had not been built.*
>
> EDITORS' SYNTHESIS OF SUMMIT LEARNING ROLLOUT COVERAGE (NEW YORK TIMES, WIRED, EDUCATION WEEK, 2018–2019)

LE INSIGHT

Summit Learning is a clean test of the book's central claim: technology that worked at the pedagogical level still failed because the **governance** architecture (consent, evidence, measurement, exit) had not been engineered alongside it. The pattern — well-intentioned tool, well-funded rollout, no institutional contract with the families and teachers operating inside it — recurs across the educational-technology dataset (Cases 8, 38, 42) and is the educator's-side analog of the governance failures in Cases 35 and 36.

LENS APPROACH

LENS uses Summit Learning in LEN 7 as the foundational consent-and-evidence case for educational technology, and in LEN 10 as a studio prompt for the governance artifacts that any educational-technology adoption decision should produce: a public evidence summary at parent reading level, a data-handling agreement at the same resolution, and a documented exit pathway that does not depend on the vendor's goodwill.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Engineer the governance architecture — consent, evidence, measurement, exit — in lockstep with the pedagogy, so a sound tool ships with a sound accountability contract.
- Produce, before adoption, a public evidence summary and data-handling agreement at parent reading level that answer the questions families will raise.
- Build a documented exit pathway that does not depend on the vendor's goodwill, so a district can leave without being captured.

IN OPERATION / AFTER

- Treat organized parent and student objections as governance signal about a missing contract, not as resistance to the instructional design.
- Monitor adopting districts for the early grievances — screen time, privacy, disengagement — that precede a withdrawal, and route them to a decision-maker.
- Maintain a re-adoption pathway so a withdrawn district can return only after completing the governance work that the first rollout skipped.

REFLECTION QUESTIONS

1. What is the equivalent of the "free tool, free of governance" pattern in your domain — the offer that bypasses the accountability architecture because it does not yet exist?
2. Design the parent-reading-level governance artifact that a district should require before adopting an educational-technology platform.
3. Summit's withdrawals in Brooklyn, Cheshire, Kennebunk, and McPherson were led by parents, not regulators. What is the equivalent local constituency in your domain that institutional accountability has not yet accommodated, and how would they be heard before deployment rather than after?

WHO BUILDS THIS governance, ethics & measurement · learning science & instructional design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · scenario simulation · competency models · knowledge bases · data pipelines

LENS COURSES LEN 7 LEN 10 LEN 8

CASE 22 EDUCATION AT SCALE 2009–2018

Tennessee Voluntary Pre-K Study

G Governance & Trust D Designed Out

IMPACT Vanderbilt longitudinal RCT of a state-funded universal pre-K program: early gains faded by third grade; sixth-grade outcomes were worse than the control group on several measures

IN BRIEF

Tennessee's Voluntary Pre-K program enrolled some 18,000 four-year-olds a year, and Vanderbilt researchers studied it with a rare large-scale randomized controlled trial — children admitted by lottery against those who applied but didn't enroll. Through kindergarten the pre-K children showed the expected gains in literacy and vocabulary. By third grade the gains had faded, and by sixth grade the pre-K group did somewhat worse than controls on several academic and behavior measures. The unwelcome result was contested, its methods attacked, and the field largely declined to absorb it. The case is in the book not because pre-K is bad — other programs show durable effects — but because the measurement was unusually rigorous and the discipline's capacity to act on an inconvenient finding was tested and largely failed.

BACKGROUND

The Tennessee Voluntary Pre-Kindergarten Program served roughly 18,000 four-year-olds a year. Demand exceeded supply, and that scarcity created an ethical lottery — a fair way to ration scarce seats that doubled as a clean randomizer. It let Vanderbilt's Peabody Research Institute run something rare in education: a randomized controlled trial, following children admitted by lottery against children who applied but did not enroll, the kind of design rarely available in a live policy setting.[0]

WHAT HAPPENED

Through kindergarten the pre-K children showed the expected gains — stronger letter knowledge, vocabulary, early literacy — exactly the early result the program had been funded to produce. By third grade the gains had faded. By sixth grade, the researchers reported, the pre-K children were doing somewhat **worse** than the control group on several state academic measures and on teacher-reported behavior — a reversal that turned an expected success story into an uncomfortable one the longitudinal design had been built to detect.[1]

THE INVESTIGATION

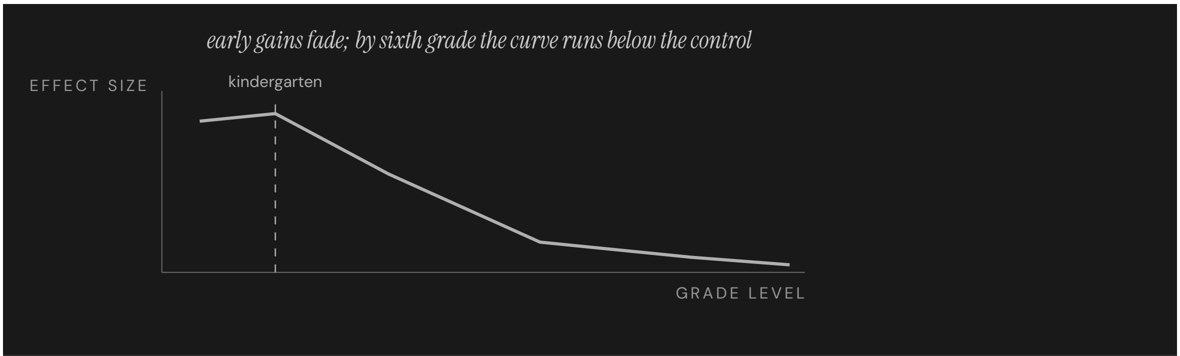
The result contradicted policy consensus and provoked an unusual response: the study was attacked, its methods contested, and the field largely declined to internalize the findings, defending the policy rather than interrogating it. Other rigorous programs — Perry Preschool, Abecedarian — remain durable, so the lesson is not that pre-K fails; it is that a discipline met an unwelcome, well-measured answer and mostly looked away, treating the inconvenient evidence as an attack to repel rather than a finding to absorb.[2]

THE CAPABILITY GAP

Tennessee Pre-K is the cleanest case in the dataset for what happens when a field has not engineered its own capacity to update on contrary evidence. The measurement instrument worked — a real RCT in a real policy setting, the rare study whose design could not easily be waved off. The institutional architecture for acting on what it found did not, so a strong instrument fed a discipline with no pathway to receive its answer. A measurement that returns an inconvenient result is only as valuable as the discipline's willingness to absorb it, and here the willingness was the part that was missing.[3]

AFTERMATH & REFORM

The debate continued for years through follow-up studies and counter-analyses, and the episode became a touchstone in the methodology of early-childhood research — argued over more than acted upon.[4] Its place in this book is as a governance-of-evidence case: the capability that needed engineering was not a better study but an implementation-science pathway that could route an unwelcome finding into program redesign rather than rejection, so the cost of the study bought a course correction instead of a controversy.



TENNESSEE PRE-K — MEASUREMENT ARCHITECTURE WORKED; THE FIELD'S RESPONSE DID NOT

REFERENCES

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THE LEARNING ENGINEERING LENS

> *By sixth grade, children who had attended TN-VPK were doing somewhat worse on academic achievement and discipline measures than children in the control group.*
>
> DURKIN, LIPSEY, FARRAN & WIESEN (2022), VANDERBILT PEABODY RESEARCH INSTITUTE

LE INSIGHT

Tennessee Pre-K is the cleanest case in the dataset for what happens when a discipline has not engineered its own capacity to update on contrary evidence. The measurement instrument worked. The institutional architecture for acting on what it found did not. Compare to Case 7 (Makary methodology debate) and Case 32 (VA Wait-Time): in each, a measurement that returned an unwelcome answer was contested rather than absorbed.

LENS APPROACH

LENS uses Tennessee Pre-K in LEN 4 as a measurement-architecture case (longitudinal RCT in a real policy context), in LEN 7 to discuss the institutional politics of unwelcome findings, and in LEN 10 as a studio prompt for designing the implementation-science pathway that would absorb such findings into program redesign rather than rejection.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Commit in advance to a decision rule for how an unwelcome but well-measured result will change the program, so the response is designed before the finding arrives.
- Build the longitudinal measurement to detect fade-out and reversal, not just the early gains a program is funded to show.
- Establish an implementation-science pathway that can route a contrary finding into redesign, giving the evidence somewhere to go.

IN OPERATION / AFTER

- Audit how the discipline actually receives inconvenient evidence, treating reflexive method-attacks as a symptom of a missing absorption pathway.
- Sustain follow-up measurement through later grades so a faded or reversed effect cannot be obscured by an early success.
- Protect the independence of the measurement function so a strong instrument is not dismantled for returning an answer the field did not want.

REFLECTION QUESTIONS

1. What measurement instrument in your domain has returned an unwelcome answer, and how did the discipline respond?

2. Design the implementation-science pathway that would absorb a Tennessee-Pre-K-style finding into program redesign rather than rejection.

3. The Tennessee RCT was strong enough that its methods were attacked rather than its conclusion accepted. What decides, in your field, whether a rigorous but inconvenient result is absorbed or contested — and who holds the authority to act on it?

WHO BUILDS THIS governance, ethics & measurement · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · task analysis · design prototyping

LENS COURSES LEN 4 LEN 7 LEN 10

CASE 35 EDUCATION AT SCALE 2020

UK A-Level Algorithm / Ofqual

G Governance & Trust H Human-System Interface D Designed Out

IMPACT ≈ 280,000 A-level entries downgraded; disadvantaged students disproportionately harmed; government U-turn within days

IN BRIEF

When COVID-19 cancelled the UK's 2020 exams, the regulator Ofqual used an algorithm to "standardize" teacher-predicted A-level grades against each school's historical performance. About 39% of grades were adjusted downward and roughly 280,000 entries were downgraded; high-achieving students at historically low-performing state schools were systematically capped by their school's past results, while small-cohort private-school students kept their predicted grades. The algorithm encoded existing structural inequality and amplified it nationwide in a single day. After public outcry — Hannah Fry called it the first time an entire nation felt an algorithm's injustice at once — the government withdrew the algorithm within days and reverted to teacher assessment. It is the book's clearest case of an algorithm deployed without the governance to catch its discriminatory effect.

BACKGROUND

When COVID-19 cancelled the UK's 2020 exams, the qualifications regulator Ofqual needed grades without exams and a defensible way to produce them at national scale. Rather than accept teacher-predicted grades — which tend to run optimistic — it built an algorithm to "standardize" them against each school's historical performance, on the logic that a school's past results were the best available anchor for its present cohort.[0]

WHAT HAPPENED

Roughly 39% of A-level grades were adjusted downward from teacher assessment — about 280,000 entries — with some dropped two or more grades, each one a university place or career path put at risk. High-achieving students at historically low-performing state schools were systematically capped by their school's past results; small-cohort private-school students, by contrast, kept their predicted grades because small classes escaped the standardization. The pattern fell hardest on those whose individual achievement was hidden inside a school average, punishing exactly the students the system claimed to serve.[1]

THE INVESTIGATION

Analysts identified the trap quickly: the algorithm assumed schools that had historically underperformed — often for reasons of funding, not student capability — would continue to, and used that assumption to suppress individual grades, so a student's result was decided partly by who had attended their school in prior years. It encoded structural inequality and amplified it at national scale in a single release.[2] The mathematician Hannah Fry observed it was "the first time that an entire nation has felt the injustice of an algorithm simultaneously," a harm whose very visibility forced the reckoning.[3]

THE CAPABILITY GAP

The missing capability was governance: an equity and impact assessment that would have surfaced, before deployment, that the model traded individual fairness for aggregate calibration and would hit disadvantaged cohorts hardest — a tradeoff that was a property of the design, not a surprise it sprang. The mathematics was no mystery; what was absent was any process with the authority to ask whether the result was just before it was applied to hundreds of thousands of students, and any appeal route for the students it would harm.[4]

AFTERMATH & REFORM

Within days the government withdrew the algorithm and accepted teacher predictions, the speed of the reversal matching the speed and visibility of the harm, and the episode became a landmark in algorithmic- accountability debate and policy.[5] Its lesson is the chapter's: an algorithm that distributes life-shaping outcomes is a governance instrument, and deploying one without an equity check and an appeal path is a governance failure no amount of statistical sophistication redeems — the model can be flawless at the task it was set and still unjust in the task that mattered.

AN ALGORITHM ENCODING HISTORICAL INEQUALITY AT NATIONAL SCALE

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THE LEARNING ENGINEERING LENS

> *I think it's the first time that an entire nation has felt the injustice of an algorithm simultaneously.*
>
> HANNAH FRY, 2020

LE INSIGHT

A-Level is the defining national-scale algorithmic-bias case. The algorithm worked exactly as specified; the specification encoded inequality. The harm was visible, immediate, and felt simultaneously by an entire cohort. The case stands as the strongest argument that **construct definition** — what the algorithm is trying to predict — is itself a capability-engineering decision.

LENS APPROACH

LENS uses A-Level in LEN 7 as the foundational case for bias, risk, and governance, in LEN 4 as a measurement-instrument case (an instrument that encodes the bias of its training data), and in LEN 10 as a technical case for bias detection and mitigation. The case is paired in this book with Robodebt (Case 36) and educational algorithmic bias (Case 37).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Run an equity and impact assessment before deployment that tests how the model distributes harm across cohorts, not just whether it calibrates in aggregate.
- Treat the tradeoff between individual fairness and aggregate calibration as an explicit design decision requiring sign-off, not a buried property of the method.
- Build an appeal path for affected individuals into the system from the start, so a person can contest an outcome the model imposed.

IN OPERATION / AFTER

- Audit live algorithmic decisions for disparate impact on disadvantaged groups, since a model can be statistically sound and socially unjust at once.
- Monitor outcomes against the populations the instrument claims to serve, watching for the pattern where high achievers are capped by a group average.
- Keep a fast withdrawal-and-revert capability ready, so a system distributing life-shaping outcomes can be pulled when its injustice surfaces.

REFLECTION QUESTIONS

1. Identify a measurement instrument in your domain that encodes structural inequality. What would it take to surface that bias before deployment?
2. The A-Level algorithm worked as specified. Redesign the **specification** — not the algorithm — to remove the bias.
3. Ofqual's model traded individual fairness for aggregate calibration, a choice baked into its design. Find a decision system in your domain that makes the same tradeoff implicitly, and name who should hold the authority to ask whether its output is just before it is applied.

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LENS COURSES LEN 4 LEN 7 LEN 9

CASE 36 GOVERNMENT 2016 – 2020

Australia Robodebt

G Governance & Trust D Designed Out H Human-System Interface

IMPACT Roughly 1 million debt notices issued; ~470,000 found wholly or partially unlawful for ~433,000 people; A$1.8B Prygodicz class-action settlement (debts zeroed/refunded plus interest); Royal Commission found "venality, incompetence and cowardice"

IN BRIEF

From 2016 the Australian government automated welfare-debt recovery with an income-averaging algorithm that compared welfare records against annual tax data spread evenly across the year. That assumption — steady, year-round employment — fit only about 7% of recipients; the rest, with irregular work, were misclassified as overpaid. Roughly a million debt notices went out, about 470,000 of them later found wholly or partly unlawful, with the burden of proof reversed onto recipients and no human review. The scheme was linked to deaths, including suicides. A 2023 Royal Commission found it sustained by “venality, incompetence and cowardice,” and a class action settled for A$1.8 billion. Robodebt is the canonical case of full automation deployed without a human in the loop and without lawful basis.

BACKGROUND

From 2016 the Australian government sought to automate welfare-debt recovery, drawn by the promise of recovering money at a fraction of the administrative cost. Its Robodebt scheme compared recipients’ welfare records against annual tax data, averaging income evenly across the year to estimate overpayments — and issuing debts on that basis at scale, replacing the individual case review the work had previously required.[0]

WHAT HAPPENED

The averaging assumed stable, year-round employment, which fit only about 7% of recipients; the other 93%, with irregular work, were misclassified as overpaid — so the assumption at the core of the method was wrong for almost everyone it touched. Roughly a million debt notices were issued, default judgments raised without human review, and the burden of proof reversed onto recipients to disprove debts the government had invented, demanding records many of the poorest could not produce. The scheme was linked to deaths, including by suicide.[1]

THE INVESTIGATION

A 2023 Royal Commission found the scheme unlawful and sustained by “venality, incompetence and cowardice,” with ministers failing to ensure it was lawful before unleashing it on a million people.[2] Analysts noted the algorithm was “comparatively simple” and its harms "entirely predictable from the outset" — meaning no technical advance was needed to foresee them, only the will to look; about 470,000 debts were found wholly or partly unlawful, and a class action settled for A$1.8 billion, with debts zeroed or refunded plus interest, the public paying back what the automation had wrongly taken.[3]

THE CAPABILITY GAP

Robodebt is the canonical case of full automation without a human in the loop and without lawful basis. The algorithm was not sophisticated and the harm was not technical: the scheme survived four years because the governance architecture treated automation as an efficiency mechanism rather than a decision regime requiring accountability, legality, and a path to contest. Calling it "automation" disguised that every notice was still a government decision to take money from a citizen. Reversing the burden of proof onto the citizen was a governance choice, not a coding one, made by people who could have chosen otherwise.[4]

AFTERMATH & REFORM

The debts were zeroed and refunded, the settlement paid, and the Royal Commission's findings referred for further action; "Robodebt" entered Australian political memory as a byword for automated administrative harm, a label now reached for whenever a new scheme proposes to automate a consequential decision.[5] Its lesson is exact: a government decision that takes money from people is a decision, however it is computed — and automating it does not relieve the state of the duty to make it lawfully, reviewably, and with the burden of proof where the law puts it, because the duty attaches to the decision, not to the method.

INCOME-AVERAGING ASSUMPTION MATCHED 7% OF RECIPIENTS

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THE LEARNING ENGINEERING LENS

> *A costly failure of public administration, in both human and economic terms.*
>
> ROYAL COMMISSION INTO THE ROBODEBT SCHEME, 2023

LE INSIGHT

Robodebt is the canonical case of full automation without human-in-the-loop and without lawful basis. The algorithm was not sophisticated and the harm was not technical: the system survived four years because the governance architecture treated automation as an efficiency mechanism rather than as a decision regime that required accountability.

LENS APPROACH

LENS treats Robodebt in LEN 7 as the canonical case for automated decision-making without human oversight and in LEN 2 as a Human-AI Teaming failure at the institutional scale: the human role was designed out of the decision loop entirely. The case anchors the program's argument that automated decision systems require deliverable-grade accountability artifacts before deployment.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Establish lawful basis and confirm it before deployment, treating an automated decision regime as bound by the same legal duties as a human one.
- Validate the model's core assumption against the population it will act on, since an averaging rule that fits 7% of recipients is a designed-in error.
- Keep a human in the loop for consequential decisions and leave the burden of proof where the law puts it, not on the citizen.

IN OPERATION / AFTER

- Audit issued decisions against ground truth and watch for the predictable harms — wrongful debts, contested notices — that signal a flawed assumption at scale.
- Provide an accessible path to contest and review every automated decision, and monitor whether affected people can actually use it.
- Track downstream harm, including welfare and safety signals, so a scheme linked to severe consequences is halted rather than left to run for years.

REFLECTION QUESTIONS

1. Robodebt automated a decision that legally required individual review. Identify a current process in your domain at risk of equivalent automation-without-due-process.
2. Design the accountability artifact that would have to be signed before a Robodebt-equivalent could be deployed.
3. Robodebt's income-averaging assumption fit only about 7% of recipients, yet ran for four years. What assumption in an automated process you rely on would be wrong for most of the people it touches — and who is responsible for testing it before deployment?

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LENS COURSES LEN 7 LEN 2

CASE 37 EDUCATION AT SCALE ONGOING

Algorithmic Bias in Educational Predictive Analytics

G Governance & Trust H Human-System Interface D Designed Out

IMPACT Predictive "at-risk" models show racial calibration bias that can misdirect support away from Black and Latinx students; a large majority of U.S. public colleges now use predictive analytics

IN BRIEF

Most U.S. public colleges now use predictive analytics to flag “at-risk” students for early support. Research finds these models carry racial calibration bias: they miscalibrate by race in ways that can misclassify Black and Latinx students — and so misdirect the very support the prediction is meant to trigger. The magnitude depends heavily on how “at-risk” is defined, making this partly a construct-definition problem: the choice of what to predict is itself a capability decision with equity consequences. The models inherit historical patterns of discrimination from their training data, and a “flag” can confirm a biased instructor’s low expectations rather than prompt help. It is the educational analog of the UK A-level case — an algorithm built to do good that can allocate help away from the students who need it most.

BACKGROUND

Colleges increasingly use predictive analytics to identify students at risk of failing or dropping out, so advisors can intervene early — a genuinely well-meant aim, to reach struggling students before they vanish. A large majority of U.S. public colleges now use some form of these models, which makes any bias they carry a quiet, system-wide feature of how support is allocated rather than an isolated experiment.[0]

WHAT HAPPENED

The intervention is well-intentioned, but research finds the models carry racial calibration bias: they miscalibrate by race in ways that can misclassify Black and Latinx students relative to their actual outcomes — and so misdirect the support the flag is meant to trigger, turning a tool built to help into one that can steer help away. Crucially, the magnitude of the bias depends on how “at-risk” is defined, which makes it partly a construct-definition problem rather than a coding bug — a flaw in what the model was asked to predict, not in how it computes the prediction.[1]

THE INVESTIGATION

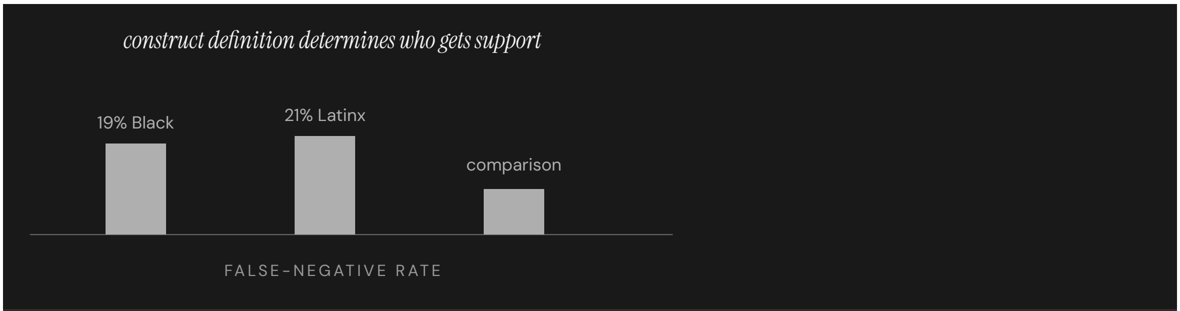
Researchers studying equity in completion-prediction models have documented these calibration gaps and traced them to training data that encodes historical patterns of discrimination — so the model learns the past's inequities and projects them forward as predictions.[2] As Baker and Hawn put it, algorithmic bias in education "poses significant threats to educational equity, potentially amplifying existing social and economic disparities" — and the harm compounds when an instructor with deficit assumptions reads an "at-risk" flag as confirmation rather than a cue to help, letting the prediction become a self-fulfilling label.[3]

THE CAPABILITY GAP

The bias is not in the math; it is in the definition of "at-risk." Define it one way and support flows to one population; define it another way and it flows to another, so the construct silently sets who the system decides to help. The choice of what the model predicts — the construct — is a capability-engineering decision with measurable equity consequences, and it is the part most often made implicitly, by whoever assembles the training labels, rather than governed deliberately by anyone accountable for where help is sent.[4]

AFTERMATH & REFORM

Unlike the discrete failures elsewhere in this chapter, this one is ongoing and quiet — embedded in advising dashboards at hundreds of institutions — which is what makes it dangerous: there is no single collapse to force a reckoning, only a steady misallocation no headline announces.[5] Its lesson, pushed upstream, is the chapter's: governing an algorithm's fairness begins not at deployment but at construct definition and label choice, with an equity audit of what the model is asked to predict and for whom the prediction allocates help — catching the bias where it is introduced rather than where it surfaces.



PREDICTIONS OVER-FLAG THE VERY STUDENTS WHO SUCCEED

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THE LEARNING ENGINEERING LENS

> *Algorithmic bias in educational systems poses significant threats to educational equity, potentially amplifying existing social and economic disparities.*
>
> BAKER & HAWN, 2021

LE INSIGHT

Educational predictive analytics is the ongoing live case for algorithmic bias at the construct level. The bias is not in the sigmoid; it is in the definition of "at-risk." Defining at-risk in one way allocates support to one population; defining it in another way allocates support to another. The choice of definition is a capability-engineering decision with measurable equity consequences.

LENS APPROACH

LENS treats this case as the positive counterpart to Georgia State (Case 39). LEN 4 examines construct definition as the load-bearing measurement choice. LEN 7 examines the governance architecture that determines whose construct gets adopted. LEN 9 covers the technical bias-mitigation methods.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Govern the construct definition deliberately: have an accountable owner decide and document what "at-risk" means and whom the prediction will route help toward.
- Audit training data for encoded historical discrimination before fitting, so the model does not learn past inequities as future predictions.
- Test calibration across racial groups during development and treat disparate misclassification as a design defect, not a tolerable residual.

IN OPERATION / AFTER

- Monitor deployed dashboards for the quiet, distributed misallocation that has no single failure event to announce it.
- Train advisors and instructors to read an "at-risk" flag as a cue to help rather than confirmation of a deficit assumption.
- Sustain a recurring equity audit of what each model predicts and for whom, since a bias embedded across hundreds of institutions persists until someone is tasked to find it.

REFLECTION QUESTIONS

1. Pick a predictive analytic in your institution. Reconstruct the construct definition behind it. What is the equity consequence of that definition?

2. Design the governance review that a new predictive model should pass before it allocates resources to or away from a population.

3. This failure is ongoing and quiet, with no single collapse to force a reckoning. What would it take to make a slow, distributed misallocation visible enough that someone with authority had to act on it?

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LENS COURSES LEN 4 LEN 7 LEN 9

CASE 68 TECHNOLOGY GOVERNMENT 1999 – 2015

UK Post Office Horizon Scandal

G Governance & Trust H Human-System Interface K Knowledge & Institutional Memory

IMPACT ~900 sub-postmasters wrongfully prosecuted; many imprisoned; documented suicides; described as the most widespread miscarriage of justice in UK history

IN BRIEF

The UK Post Office's Horizon accounting system, built by Fujitsu and rolled out in 1999, generated phantom shortfalls in sub-postmasters' branch ledgers. Rather than accept the software was at fault, the Post Office prosecuted them — around 900 over two decades — for theft and false accounting; people were imprisoned, bankrupted, and driven to suicide, in what is now called the most widespread miscarriage of justice in UK history. Internal documents showed engineers had known about Horizon bugs throughout. Convictions began to be quashed in 2021, and a public inquiry continues. The failure ran through the prosecutor and the courts: each accepted "the computer said so" as authoritative because no actor had the standing or expertise to challenge it. Horizon is the book's case for institutional deference to a flawed algorithm.

BACKGROUND

The UK Post Office ran thousands of branches through sub-postmasters — local operators personally liable for any shortfall in their accounts, a liability that put each operator's livelihood behind the numbers the system reported. In 1999 it deployed Horizon, an accounting system built by Fujitsu, to track every branch's ledger, making the software the single authority on whether a branch's books balanced.[0]

WHAT HAPPENED

Horizon produced systematic accounting errors — phantom shortfalls that appeared in branch ledgers where no money was actually missing. The Post Office treated the shortfalls as real and the sub-postmasters as thieves: over two decades it prosecuted around 900 for theft and false accounting, refusing to accept the system itself was at fault even as the same pattern recurred branch after branch. People were imprisoned, lost homes, went bankrupt, and some died by suicide — the human cost of trusting the ledger over the person.[1]

THE INVESTIGATION

Documents later released through litigation showed Fujitsu and Post Office engineers had known about Horizon bugs throughout the period — the knowledge of fallibility existed inside the institution even as it prosecuted people for the system's errors.[2] The courts began quashing convictions in 2021, and the public inquiry under Sir Wyn Williams found that senior employees "knew, or at the very least should have known, that Legacy Horizon was capable of error" — establishing it as the most widespread miscarriage of justice in UK history, sustained precisely because that internal knowledge never reached the people on trial.[3]

THE CAPABILITY GAP

The gap was at the regulator, the prosecutor, and the courts: each accepted Fujitsu's representation that Horizon was reliable, despite documentation to the contrary, because no institutional actor had the standing or expertise to interrogate it, so the claim of reliability passed unchallenged through every layer that could have tested it. "The computer said so" became, for two decades, a sufficient basis for criminal conviction — the governance hazard of treating automated output as authoritative rather than as evidence to be challenged, with a person's account on the other side of the scale.[4]

AFTERMATH & REFORM

Convictions have been overturned — some by an exceptional act of Parliament, a measure of how far the ordinary appeal routes had failed — compensation schemes established, and Fujitsu and the Post Office called to account before the continuing inquiry.[5] Horizon's lesson is the chapter's in its bluntest form: an automated system's output is not testimony, and any institution that lets "the computer said so" stand unchallenged against a human's account has built a machine for manufacturing injustice, one that runs for as long as no one is empowered to switch it off.

900

WRONGFUL PROSECUTIONS ACROSS 20+ YEARS

"the computer said so" was an institutionally sufficient basis for conviction

HORIZON — INSTITUTIONAL DEFERENCE TO AN ALGORITHM KNOWN TO BE FLAWED

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THE LEARNING ENGINEERING LENS

> *A number of senior, and not so senior, employees of the Post Office knew, or at the very least should have known, that Legacy Horizon was capable of error.*
>
> SIR WYN WILLIAMS, POST OFFICE HORIZON IT INQUIRY, VOLUME 1, JULY 2025

LE INSIGHT

Horizon is the canonical case for institutional deference to automated systems whose internal evidence was already known to be flawed. The capability gap was at every layer that took the software's output as authoritative — including the courts. "The computer said so" became, for two decades, an institutionally sufficient basis for criminal prosecution.

LENS APPROACH

LENS uses Horizon in LEN 7 as the canonical example of institutional deference to algorithmic output and in LEN 2 for the most extensive multi-decade automation-bias case in the dataset. Studio projects examine what evidentiary architecture would require **interrogating** automated output before acting on it.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design automated output to be treated as challengeable evidence, not authoritative testimony, especially where a person's liability rides on it.
- Build a route by which engineers' knowledge of system bugs reaches anyone acting on the output, so internal fallibility cannot stay hidden.
- Give some institutional actor the standing and expertise to interrogate the system's reliability before its output is used against a person.

IN OPERATION / AFTER

- Audit the recurring-error pattern across branches or cases, treating the same fault appearing repeatedly as evidence of the system, not the operators.
- Maintain an appeal path that can challenge automated output without requiring an act of Parliament to overturn a wrong decision.
- Sustain independent review of the system's accuracy throughout its operating life, so a claim of reliability cannot pass unexamined for decades.

REFLECTION QUESTIONS

1. Identify a decision in your domain currently made on the strength of "the computer said so." What evidentiary architecture should sit beside the output?
2. Design the institutional check that would have made Horizon's reliability subject to genuine challenge in 2005.
3. Engineers inside the Post Office and Fujitsu knew Horizon was capable of error, yet that knowledge never reached the courtroom. What pathway in your domain carries — or fails to carry — known system fallibility to the people relying on the output?

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LENS COURSES LEN 7 LEN 2

CASE 69 HEALTHCARE TECHNOLOGY 2003 – 2018

Theranos

G Governance & Trust D Designed Out

IMPACT $9B valuation collapsed; thousands of patients given unreliable results; founder convicted on multiple counts of wire fraud

IN BRIEF

Theranos claimed a blood-testing device that could run hundreds of tests from a finger-stick drop. It did not work. Internal data showed accuracy far below what the company told investors, partners, and patients; to keep up appearances, Theranos ran most patient samples on conventional commercial analyzers and reported them as its own device's results. It reached a $9-billion valuation and put unreliable tests in front of real patients through a Walgreens partnership. The fraud exploited a regulatory seam — the FDA had not validated the device, while the CLIA regime governing the lab did not match the company's product claims — and neither the board nor investors had the depth to challenge it; a journalist did. Elizabeth Holmes was convicted of wire fraud in 2022. Theranos is the book's case for fraud exploiting a governance gap between regulatory regimes.

BACKGROUND

Theranos claimed to have built a blood-testing platform — the "Edison" — that could perform hundreds of laboratory tests from a single finger-stick drop of blood, a promise that would have upended a diagnostics industry built on venous draws and large analyzers. It rode that claim to a $9-billion valuation and a partnership putting its tests in Walgreens stores, carrying the unproven device straight to retail patients.[0]

WHAT HAPPENED

The device did not work. Internal data showed accuracy far below what the company represented to investors, partners, and patients — the gap between the claim and the instrument was known inside the company. To preserve the appearance of a working product, Theranos ran most patient samples on conventional commercial analyzers and reported the results as though they had come from its proprietary device, a substitution that kept the fiction alive while putting unreliable results in front of real patients making real medical decisions.[1]

THE INVESTIGATION

The fraud exploited a regulatory seam: the FDA had not validated the device, while the CLIA regime that governs laboratory operation did not match the architecture of the company's product claims, and neither layer validated those claims independently — so the device fell into a gap each regulator assumed the other covered. The board, the investors, and Walgreens lacked the technical depth to challenge them, and a celebrity board offered prestige in place of scrutiny; the journalist John Carreyrou, and CMS inspectors who revoked

Theranos's CLIA certificate in 2016, were what finally surfaced the truth from outside the failed oversight chain.[2]

THE CAPABILITY GAP

The capability gap sat in the regulatory architecture — at the boundary between two regimes, where a deliberate fraud could live because neither the FDA nor CLIA owned end-to-end validation of a novel diagnostic making clinical claims, and an unowned boundary is exactly the shelter a fraud needs. The governance lesson is that the seam between regulators is itself a place that must be engineered, because it is exactly where a bad actor will operate — choosing the gap precisely because no one is watching it.[3]

AFTERMATH & REFORM

Theranos collapsed, its CLIA certificate was revoked, and Elizabeth Holmes was convicted of multiple counts of wire fraud in 2022, the conviction closing a chapter the regulators had been slow to open.[4] The case is canonical in business education for fraud-as-product-strategy and in health regulation for the gap that let unvalidated clinical tests reach patients — a reminder that "disruptive" claims in a regulated domain demand more validation, not less, precisely because the novelty is what tempts the oversight regimes to defer to one another.

CLAIM	REALITY
Edison	Siemens

the device did not work; the regulatory seam between FDA and CLIA hid that fact

THERANOS — FRAUD EXPLOITING A REGULATORY SEAM BETWEEN FDA AND CLIA

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THE LEARNING ENGINEERING LENS

> *The company misrepresented to investors, regulators, and ultimately patients the accuracy of its blood-testing technology.*
>
> U.S. V. HOLMES, INDICTMENT, 2018

LE INSIGHT

Theranos is the canonical case for fraud exploiting the seam between two regulatory regimes. The FDA had not approved the device; CLIA accepted the laboratory operation. Neither layer had the capability to validate the claims independently. The capability gap was at the regulatory architecture.

LENS APPROACH

LENS uses Theranos in LEN 7 for the regulatory-seam failure and in LEN 4 for the measurement-validation gap. The case demonstrates that capability engineering at the boundary between regulatory regimes is itself a governance deliverable.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Assign explicit end-to-end validation ownership across the FDA–CLIA-style seam, so a novel diagnostic cannot fall into a gap each regulator assumes the other covers.
- Require independent verification of accuracy claims against the actual device before clinical use, not against a substituted commercial analyzer.
- Staff the oversight chain — board, partners, regulators — with the technical depth to interrogate the claims, treating prestige as no substitute for expertise.

IN OPERATION / AFTER

- Audit whether reported results actually came from the validated instrument, watching for the substitution pattern that hides a non-working device.
- Monitor the regulatory seam as a standing risk, since a determined fraud will choose precisely the boundary no single regime owns.
- Protect external scrutiny — journalists, inspectors, whistleblowers — as the backstop that surfaces what a captured oversight chain misses.

REFLECTION QUESTIONS

1. Identify a regulatory seam in your domain where two regimes meet without an explicit handoff. What could exploit it?
2. Design the validation regime that would have caught Theranos in 2013.
3. Theranos's board and investors had prestige but not the technical depth to challenge the device's claims. Where in your domain does reputational authority stand in for the expertise needed to validate what is being approved?

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LENS COURSES LEN 4 LEN 7

CASE 70 TECHNOLOGY GOVERNMENT 2011 – 2016

Wells Fargo Fake Accounts

G Governance & Trust N Normalization of Deviance

IMPACT ~3.5 million unauthorized accounts opened; ~$3B in penalties; CEO resigned; the Federal Reserve capped the bank's assets

IN BRIEF

To meet aggressive sales quotas, Wells Fargo employees opened roughly 3.5 million unauthorized customer accounts over years. The behavior was widespread and visible to internal risk and compliance functions, but the bank's response was to fire individual "bad apples" while leaving the incentive structure intact — and that structure was the actual cause: sales targets most employees could not meet by ethical means. The 2016 CFPB and OCC actions brought $3 billion in penalties, the CEO resigned, and the Federal Reserve took the unprecedented step of capping the bank's assets. Wells Fargo is the canonical case of an incentive architecture that manufactured the misconduct the institution then prosecuted at the front line, while insisting the misconduct was individual.

BACKGROUND

Wells Fargo built its retail strategy on "cross-selling" — opening many products per customer — and drove it with aggressive sales quotas pushed down to branch employees, whose pay and jobs depended on hitting them. The metric the bank chose to manage by thus became the thing every front-line employee was structurally compelled to maximize, whatever it took.[0]

WHAT HAPPENED

Unable to meet the quotas honestly, employees opened roughly 3.5 million unauthorized accounts in customers' names — the rational response to a target most could not reach by legitimate means. The behavior was widespread and longstanding, and visible to internal risk and compliance functions for years; the institutional response was to discipline individual employees as bad apples while leaving the incentive structure intact, fixing the symptom and preserving the cause.[1]

THE INVESTIGATION

The 2016 CFPB and OCC actions brought roughly $3 billion in penalties, and the bank's own independent-directors investigation (2017) documented how the sales-target architecture drove the misconduct — locating the cause in the design, not the people executing it.[2] Investigators tied the misconduct directly to the bank's incentive-compensation structure, which had made such sales practices a foreseeable result rather than an aberration. The CEO resigned, and the Federal Reserve imposed an unprecedented cap on the bank's assets, reaching past individual penalties to constrain the institution itself.[3]

THE CAPABILITY GAP

Wells Fargo is the strongest evidence in the dataset that incentive architecture is a capability-engineering deliverable. The measurement system used to manage employees produced exactly the behavior the institution then prosecuted — the same structure that demanded the result punished the people who delivered it. The gap was not at the front line but at the governance layer that designed the incentives and then, for years, treated the predictable result as individual moral failure, mistaking a designed outcome for a character flaw.[4]

AFTERMATH & REFORM

The asset cap constrained the bank's growth for years — a rare, structural consequence that hit the institution where it would feel it — and the case became the standard teaching example of measurement-gaming and incentive design.[5] Its lesson is that any quota becomes a target to be gamed, and that an institution is accountable for the behavior its measurement system makes rational, not just the behavior endorsed in its values statement — because employees respond to the incentive they are paid on, not the value they are told to honor.

3.5M

UNAUTHORIZED ACCOUNTS

the incentive architecture made misconduct rational for the front line

WELLS FARGO — THE MEASUREMENT SYSTEM PRODUCED THE BEHAVIOR THE INSTITUTION THEN PROSECUTED

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THE LEARNING ENGINEERING LENS

> *Wells Fargo's sales practices were a foreseeable consequence of its incentive compensation structure.*
>
> PARAPHRASING THE 2016 REGULATORY AND INDEPENDENT-DIRECTORS FINDINGS ON WELLS FARGO

LE INSIGHT

Wells Fargo is the strongest available evidence that incentive architecture is a capability-engineering deliverable. The measurement system used to manage employees produced the behavior the institution then prosecuted. The gap was not at the front line. It was at the governance layer that had designed the incentives.

LENS APPROACH

LENS uses Wells Fargo in LEN 4 as a measurement-gaming case and in LEN 7 for the corporate-governance dynamics that allow such incentives to persist for years. Studio projects redesign the incentive architecture.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design incentive and quota structures against the behavior they will rationally produce, treating the measurement system as a capability deliverable in its own right.
- Set targets that are achievable by legitimate means, since a quota most employees cannot reach honestly is an instruction to cheat.
- Pair every performance metric with a countervailing measure that detects gaming before it becomes systemic.

IN OPERATION / AFTER

- Audit for the gap between the behavior the measurement system rewards and the conduct the institution claims to value, and treat divergence as a design fault.
- Empower risk and compliance to escalate a pattern of gaming as a structural cause, not to absorb it as isolated misconduct.
- Hold the governance layer that designed the incentives accountable for predictable outcomes, rather than disciplining only the front line.

REFLECTION QUESTIONS

1. Identify a measurement system in your organization that is currently producing the behavior the organization claims to deplore. What is the gap between the measurement and the goal?

2. Design the incentive architecture for a bank's branch sales force that does not produce a Wells-Fargo-equivalent failure.

3. Wells Fargo's misconduct was visible to risk and compliance for years before it was addressed at the structural level. What lets an organization see a pattern of gaming and still treat it as individual bad apples — and who would have to be empowered to name the incentive as the cause?

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LENS COURSES LEN 4 LEN 7

CASE 71 INDUSTRIAL GOVERNMENT 2015

Volkswagen Dieselgate

D Designed Out G Governance & Trust

IMPACT ~11 million vehicles equipped with defeat-device software; $33B+ in penalties and remediation; multiple criminal convictions

IN BRIEF

Volkswagen engineered a “defeat device” into its diesel emissions software: code that detected when a car was on a regulatory test bench and switched on full emissions controls only then. On the road the controls were largely disabled, producing nitrogen-oxide emissions up to forty times the legal limit, across roughly 11 million vehicles for years. The deception was uncovered not by regulators but by a small West Virginia university team comparing real-world to lab measurements. Internal documents showed the defeat device was an institutional decision — a deliberate response to a standard VW’s engineers did not believe they could meet within cost — not a rogue act. VW paid more than $33 billion in penalties and remediation, with criminal convictions. Dieselgate is the book’s case for organized evasion of a measurement system.

BACKGROUND

Volkswagen had staked its U.S. growth on “clean diesel,” promising cars that met strict nitrogen-oxide limits without sacrificing performance or cost — a marketing position that left no room to admit the engineering could not deliver all three at once. Its engineers did not believe they could actually meet the standard within the platform’s cost constraints, putting the company’s public promise on a collision course with its own technical reality.[0]

WHAT HAPPENED

So they cheated by design. VW’s emissions software detected when a vehicle was on a regulatory test bench — by its steering, speed, and duration patterns, the telltale signature of a lab rather than a road — and switched on full emissions controls only during the test. On the road the controls were largely disabled, producing emissions up to forty times the legal limit, across roughly 11 million vehicles for years, so the pollution the standard existed to prevent flowed freely everywhere except where it was measured.[1]

THE INVESTIGATION

The deception was uncovered not by the regulator but by a small university research team in West Virginia comparing real-world emissions to lab results — the gap between road and bench being exactly what the regulator’s own test could never reveal.[2] Internal documents then showed the defeat device had been authorized inside VW’s engineering hierarchy as a deliberate institutional response to a standard the team could not meet — not the work of a rogue engineer, but a decision made up the chain of command. VW pleaded guilty, paid more than $33 billion in penalties and remediation, and saw multiple executives convicted.[3]

THE CAPABILITY GAP

The exploitable gap was at the regulator's instrument: the emissions test ran in a regime the vehicle could detect, so a manufacturer determined to evade could tune its behavior to the test rather than to the road, optimizing for the measurement instead of the goal it stood for. Dieselgate is the canonical case of institutionally engineered evasion of a measurement system — and a reminder that any test the measured party can recognize is a test it can defeat, because a detectable test invites the very gaming it is meant to catch.[4]

AFTERMATH & REFORM

The reform closed the gap at the instrument: the EU introduced real-world driving-emissions testing, moving the measurement off the predictable bench and onto the road the standard actually cared about, making it far harder to game.[5] The pattern parallels Takata (Case 52) — a manufacturer's fraud meeting a regulator whose evidence pipeline trusted the manufacturer's test conditions, and a fix that had to upgrade the measurement itself, not just punish the cheat, because punishing the cheat leaves the exploitable instrument in place for the next one.

TEST BENCH	REAL WORLD
compliant	up to 40×

the protocol operated in a regime the vehicle could detect

DIESELGATE — ENGINEERED EVASION OF A MEASUREMENT SYSTEM

REFERENCES

1. U.S. EPA, Notice of Violation to Volkswagen (2015) — the defeat device and emissions exceedances.
2. West Virginia University / ICCT real-world diesel-emissions study (2014) — the discovery comparing road to lab.
3. U.S. Department of Justice Plea Agreement with Volkswagen AG (2017) — the institutional decision, $33B+ in penalties, and convictions (quoted).
4. Volkswagen internal documents released through litigation — authorization of the defeat device within the engineering hierarchy.
5. EU Real Driving Emissions (RDE) testing regulation — the post-Dieselgate measurement reform.
6. J. Ewing, Faster, Higher, Farther: The Volkswagen Scandal (2017); cf. Takata (Case 52).

THE LEARNING ENGINEERING LENS

> *The defeat device was the product of a long-standing institutional decision to evade emissions standards.*
>
> PARAPHRASING THE U.S. DEPARTMENT OF JUSTICE PLEA AGREEMENT WITH VOLKSWAGEN AG, 2017

LE INSIGHT

Dieselgate is the canonical case for institutionally engineered evasion of a measurement system. The capability gap was at the regulator's test protocol: it operated in a regime the vehicle could detect. The reform — real-world emissions testing — was a capability deliverable upgrade at the regulator's instrument boundary.

LENS APPROACH

LENS uses Dieselgate in LEN 4 as the canonical case for measurement-system evasion and in LEN 7 for the corporate- governance dynamics of engineered fraud. The reform pattern parallels Takata (Case 52).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design measurement protocols the measured party cannot detect and tune to, so behavior is optimized for the goal rather than the test.
- Surface the gap between a public commitment and the engineering reality early, so an unmeetable standard is renegotiated rather than secretly evaded.
- Build a governance check between an impossible target and the team asked to meet it, so the response is escalation, not a defeat device.

IN OPERATION / AFTER

- Audit real-world behavior against test-condition behavior, treating a divergence between road and bench as the signature of engineered evasion.
- Upgrade the measurement instrument itself after a discovered cheat, not just penalize the cheat, since the exploitable test invites the next one.
- Sustain independent, outside-the-loop verification — as the university team provided — to catch fraud the regulator's own protocol cannot see.

REFLECTION QUESTIONS

1. Identify a measurement protocol in your domain that operates in a regime detectable by the entity being measured. What is the evasion potential?
2. Design the upgrade to a regulatory test protocol that makes Dieselgate-style evasion structurally infeasible.
3. VW's defeat device was authorized up the engineering hierarchy, not the act of a rogue engineer. What allows an institution to sanction fraud as a deliberate response to an unmeetable standard — and what governance check should sit between an impossible target and the team asked to meet it?

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LENS COURSES LEN 4 LEN 7

CASE 72 TECHNOLOGY 2014 – 2018

Cambridge Analytica / Facebook

G Governance & Trust

IMPACT ~87 million Facebook profiles harvested without informed consent; FTC $5B penalty; foundational data-governance reform

IN BRIEF

A Cambridge University researcher's personality-quiz app, taken by about 270,000 people, exploited Facebook's then-permissive Graph API to collect not only the quiz-takers' data but their friends' too — roughly 87 million profiles. The dataset was passed to Cambridge Analytica for political micro-targeting. Facebook's permission architecture had been designed and tested for delivering social experiences, not red-teamed against systematic harvesting; its API contract assumed benevolent developer intent. The architecture worked exactly as designed — the design assumption was wrong. The scandal accelerated the EU's GDPR, helped spur the California Consumer Privacy Act, and produced a $5-billion FTC penalty and consent decree. Cambridge Analytica is the book's case for platform-governance failure: a load-bearing assumption about how an interface would be used, never stress-tested against abuse.

BACKGROUND

Facebook's Graph API let third-party apps request user data to build social experiences — and, at the time, an app could collect not only its own users' data but their friends' data too, so a single consenting user opened a window onto people who had never agreed to anything. The permission architecture had been designed and tested against ordinary use cases, the friendly developers it imagined rather than the hostile ones it would eventually attract.[0]

WHAT HAPPENED

A Cambridge University researcher built a personality-quiz app taken by about 270,000 people; through the friends-permission it harvested roughly 87 million profiles — a return of more than three hundred profiles for every person who actually used the app. The dataset was passed to Cambridge Analytica, which used it for political-campaign micro-targeting across multiple elections — none of the 87 million having meaningfully consented, most never even aware the app existed.[1]

THE INVESTIGATION

Investigations by the UK Information Commissioner and the U.S. FTC, prompted by Guardian reporting, established the scope of the harvesting and Facebook's responsibility for it — the platform, not just the researcher, was found answerable for what its design permitted.[2] The platform's API had never been red-teamed against systematic data extraction; its contract simply assumed developers would behave, a trust extended at the scale of a social network and never tested against someone willing to abuse it.[3]

THE CAPABILITY GAP

The gap was not technical — the architecture worked exactly as designed, which is what makes it a governance case rather than a bug. It was a governance gap: a load-bearing design assumption ("developers are benevolent") that no one had stress-tested against a determined abuser, an assumption holding up the whole permission model without ever being named as one. On a platform at societal scale, an unexamined assumption about how an interface will be used is itself a capability deliverable — and an unexamined one is a latent failure waiting for the first actor willing to exploit it.[4]

AFTERMATH & REFORM

The scandal accelerated the EU's GDPR, helped spur the California Consumer Privacy Act, and produced a $5-billion FTC penalty and a consent decree under which Facebook still operates — the abuse of one design assumption reshaping data law across two jurisdictions.[5] Its lesson for platform governance is that permission architectures must be designed against the worst plausible developer, not the typical one — because at scale, the worst plausible developer will arrive, and the friends-permission that looked harmless against ordinary use becomes a harvesting tool in the wrong hands.

87M

PROFILES · 270K QUIZ-TAKERS

the Graph API contract assumed benevolent developer intent

CAMBRIDGE ANALYTICA — DESIGN ASSUMPTION AS LOAD-BEARING CAPABILITY

REFERENCES

1. U.S. FTC, In the Matter of Facebook, Inc., Consent Order (2019) — the $5B penalty; Facebook gave developers "far more user data than was necessary" (quoted).
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3. C. Cadwalladr & E. Graham-Harrison, The Guardian investigation (2018) — the disclosure.
4. The 87 million profiles collected via the friends-permission Graph API from 270,000 quiz-takers.
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THE LEARNING ENGINEERING LENS

> *Facebook gave developers access to far more user data than was necessary for the apps they built.*
>
> PARAPHRASING THE U.S. FTC ORDER, IN THE MATTER OF FACEBOOK INC., 2019

LE INSIGHT

Cambridge Analytica is the canonical case for platform-governance failure: an API contract that assumed benevolent intent and was not engineered against systematic abuse. The capability gap was not technical. The architecture worked exactly as designed; the design assumption was wrong.

LENS APPROACH

LENS uses Cambridge Analytica in LEN 7 as the foundational case for platform governance and data ethics, and in LEN 1 to teach that "design assumption" is a load-bearing capability deliverable.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Make every load-bearing design assumption explicit and red-team it against a determined abuser before launch, treating "developers are benevolent" as a hypothesis to test.
- Scope data access to what an app genuinely needs, so one user's consent cannot silently reach hundreds of non-consenting others.
- Design permission architectures against the worst plausible developer, since at societal scale that developer will eventually arrive.

IN OPERATION / AFTER

- Monitor API usage for systematic extraction patterns that depart from the ordinary social use the interface was built for.
- Audit third-party developers' actual data flows against their stated purpose, rather than trusting a contract that assumes good behavior.
- Sustain the platform's own accountability for what its design permits, treating downstream abuse as a governance defect to remediate, not a third party's sole fault.

REFLECTION QUESTIONS

1. Identify an API or interface in your domain whose contract assumes benevolent intent. What is the systematic-abuse case it was not red-teamed against?

2. Design the platform-governance deliverable that should accompany the launch of a new third-party developer API.

3. The friends-permission let one consenting user expose hundreds who had not agreed. Where in your domain does one person's consent silently extend to others, and what governance would make that reach visible and accountable?

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LENS COURSES LEN 1 LEN 7 LEN 6

CASE 73 TECHNOLOGY GOVERNMENT 2017

Equifax Data Breach

G Governance & Trust K Knowledge & Institutional Memory

IMPACT 147 million Americans' personal data exposed; CEO resigned; ~$700M settlement; foundational U.S. data-breach case

IN BRIEF

In 2017 attackers exploited a known, two-month-old vulnerability in Apache Struts on an Equifax web portal — a patch had been available, and Equifax's own security team had told IT to apply it; no one did. Over the next 2.5 months the attackers exfiltrated the personal data of 147 million Americans, and Equifax did not disclose the breach until September. A Senate investigation found systematically inadequate patching, an incomplete asset inventory (so the company did not know which systems needed the fix), and an incident-response function treated for years as a cost center. The CEO resigned and Equifax settled for about $700 million. No single failure caused the breach; cumulative inadequacy across routine cybersecurity work did. Equifax is the data-breach analog of the Sago mine disaster (Case 59).

BACKGROUND

Equifax, one of the three U.S. credit bureaus, held the most sensitive financial data on virtually every American adult — a concentration of identity information that made any breach catastrophic by definition. One of its public web portals ran a version of Apache Struts with a known critical vulnerability for which a patch had been available for two months — and which Equifax's own security team had flagged for IT to apply, so the warning and the fix both existed inside the company and simply went unacted-upon.[0]

WHAT HAPPENED

The patch was not applied. Attackers exploited the vulnerability beginning in May 2017 and quietly exfiltrated personally identifying information on 147 million Americans over the next two and a half months, a window long enough that the theft was less a break-in than a sustained occupation. Equifax did not publicly disclose the breach until September, so the people whose identities were taken learned of it only well after the fact.[1]

THE INVESTIGATION

The U.S. Senate Permanent Subcommittee on Investigations found Equifax's patching practices systematically inadequate and the company lacking "a comprehensive IT asset inventory" — so it could not reliably know which systems needed the patch, leaving the security team's warning with no map to act on.[2] Monitoring was weak, response delayed, and the incident-response architecture had been treated for years as a cost center rather than a capability worth funding; the CEO resigned and Equifax settled for roughly $700 million.[3]

THE CAPABILITY GAP

Equifax is the institutional-cybersecurity case for cumulative inadequacy in routine work: patching, asset inventory, monitoring, response. Each function was below standard; none alone produced the breach; the combination did, the marginal weaknesses compounding into a single open door. The capability gap was the management of unglamorous, universally-agreed-necessary maintenance — exactly the work easy to defer because deferring it usually costs nothing, until the one time it costs everything, and a function starved as a cost center has no slack left on that day.[4]

AFTERMATH & REFORM

The settlement funded consumer compensation and credit monitoring, and the breach pushed patching discipline, asset inventory, and breach-disclosure timelines up the corporate agenda — elevating, after the loss, the unglamorous work that had been deferred before it.[5] It is the data-breach analog of the Sago mine disaster (Case 59): no dramatic single cause, just several routine defenses each left marginally inadequate, failing together on the day a determined attacker arrived to test all of them at once.

EQUIFAX — CUMULATIVE INADEQUACY IN ROUTINE CYBERSECURITY WORK

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2. The breach record — 147 million affected, exploitation from May 2017, public disclosure in September 2017.
3. Senate PSI (2019) — "Equifax lacked a comprehensive IT asset inventory" (quoted).
4. U.S. FTC / CFPB / state settlement ($700 million, 2019) and the CEO's resignation.
5. U.S. GAO, Actions Taken by Equifax and Federal Agencies in Response to the 2017 Breach, GAO-18-559 (2018).
6. Apache Struts CVE-2017-5638 advisory; cf. the Sago mine disaster (Case 59).

THE LEARNING ENGINEERING LENS

> *Equifax lacked a comprehensive IT asset inventory.*
>
> U.S. SENATE PERMANENT SUBCOMMITTEE ON INVESTIGATIONS, HOW EQUIFAX NEGLECTED CYBERSECURITY, MARCH 2019

LE INSIGHT

Equifax is the canonical institutional-cybersecurity case for cumulative inadequacy in routine work: patching, inventory, monitoring, response. Each function was below industry standard. None alone produced the breach. The combination was the breach.

LENS APPROACH

LENS uses Equifax in LEN 5 for cumulative-inadequacy analysis in cybersecurity and in LEN 7 for the institutional dynamics that allow routine work to be chronically underfunded.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Fund routine security maintenance — patching, monitoring, response — as a capability, not a cost center, so the unglamorous defenses have slack on the bad day.
- Build and maintain a comprehensive asset inventory so a known vulnerability can be mapped to every system that runs it.
- Wire the security team's warnings to an owner with the authority and resources to apply a flagged patch promptly.

IN OPERATION / AFTER

- Audit the routine defenses together — patching, inventory, monitoring, response — since each marginally inadequate layer raises the odds the combination fails.
- Monitor for sustained, low-noise exfiltration, treating a months-long quiet window as the failure mode to detect, not just a single break-in.
- Hold breach-disclosure timelines short, so affected people learn of a theft of their identity without months of delay.

REFLECTION QUESTIONS

1. Identify a piece of routine work in your domain that is chronically deferred. What is the cumulative-inadequacy threshold?
2. Equifax did not know which assets ran Struts. Design the asset-inventory deliverable that an organization the size of Equifax should be able to produce on demand.
3. Equifax's security team had flagged the patch, but the warning had no asset map to act on and no funded function to carry it out. What turns a known, agreed-upon fix into deferred work in your organization — and what would force it to be done before the one time it costs everything?

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LENS COURSES LEN 5 LEN 7

CASE 74 INDUSTRIAL 1981

Hyatt Regency Walkway Collapse

D Designed Out G Governance & Trust

IMPACT 114 killed and 216 injured in Kansas City when suspended walkways collapsed; foundational U.S. engineering-ethics case

IN BRIEF

During a crowded tea dance in the atrium of the Kansas City Hyatt Regency on 17 July 1981, two suspended walkways collapsed, killing 114 and injuring 216 — then the deadliest structural collapse in U.S. history. The cause was a connection detail changed during construction: the original design hung the walkways from single long rods; the as-built version used a two-rod arrangement that doubled the load on the upper connection. The structural engineer's office approved the change without recalculating the load, and the connection had in fact been overstressed from the start. Missouri revoked the responsible engineers' licenses, and the case became the foundational engineering-ethics example. The capability gap was institutional: nothing required a change to a load-bearing connection to be re-analyzed by the engineer of record.

BACKGROUND

The Kansas City Hyatt Regency's atrium featured walkways suspended from the ceiling, carrying crowds above an open public space where any failure would drop directly onto people below. The original engineering design hung them from single continuous steel rods, a configuration in which each rod carried one walkway's load to the structure above.[0]

WHAT HAPPENED

During construction the connection was changed — for ease of assembly, to avoid threading a single long rod through both walkways — to a two-rod arrangement that effectively doubled the load on the upper walkway's connection, and the structural engineer's office approved the change without recalculating it, treating an assembly convenience as if it carried no structural consequence. On 17 July 1981, during a crowded tea dance, the over-stressed connection let go; two walkways fell onto the atrium floor, killing 114 people and injuring 216 — at the time the deadliest structural collapse in U.S. history.[1]

THE INVESTIGATION

The National Bureau of Standards found the as-built connection carried roughly twice its intended load and had been inadequate even under the building code's requirements — so the original single-rod design had itself been marginal, and the field change pushed an already-thin connection past failure.[2] The Missouri licensing board revoked the licenses of the responsible engineers, fixing accountability on the engineer of record, and the case became the foundational engineering-ethics example taught to undergraduates: how a small construction change, accepted casually, can turn a design that works into one that fails.[3]

THE CAPABILITY GAP

The capability gap was at the institutional review of construction changes. Nothing required a change to a load-bearing connection detail to be re-analyzed by the engineer of record before it was built — so a modification that doubled a critical load passed through the system without anyone computing what it did, the absence of a required check letting a fatal change look like a routine one. The missing capability was change control with the engineer's calculation in the loop, a gate that re-derives the consequence before the field accepts the change.[4]

AFTERMATH & REFORM

The collapse reshaped how the profession treats shop-drawing review and the engineer of record's responsibility for connection design, hardening into rule the accountability the failure had exposed, and it anchored modern engineering-ethics teaching.[5] Its lesson generalizes well beyond steel: a change that looks trivial at the point of assembly can be catastrophic at the point of load, and the only defense is a review that re-derives the consequence rather than trusting that "it's a small change" — because triviality at assembly is no guarantee of triviality under load.

AS DESIGNED	AS BUILT
1× load	2× load

a small construction change, accepted without re-analysis

HYATT REGENCY — CONNECTION DETAIL CHANGED IN THE FIELD

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THE LEARNING ENGINEERING LENS

> *The change in the rod configuration doubled the load on the connection that failed.*
>
> PARAPHRASING THE NATIONAL BUREAU OF STANDARDS INVESTIGATION, NBSIR 82-2465, 1982

LE INSIGHT

The Hyatt Regency case is the canonical engineering-ethics case for institutional review of construction changes. The change looked small; the load implication was catastrophic. The capability gap was at the review process that should have caught the doubled load.

LENS APPROACH

LENS uses Hyatt Regency in LEN 5 for change-control deliverables and in LEN 7 for the engineering-licensure architecture that holds the engineer of record accountable. Studio projects examine the change-control deliverable.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Require that any change to a load-bearing detail be re-analyzed by the engineer of record before it is built, with the calculation in the loop.
- Treat an assembly-convenience change as a structural decision, not a routine one, so ease of construction never substitutes for analysis.
- Set the original design margin so a connection is not already marginal under code before any field change touches it.

IN OPERATION / AFTER

- Audit as-built connections against the as-designed intent, catching field substitutions that altered a critical load path.
- Route shop-drawing and field changes through a review that re-derives the consequence rather than trusting “it’s a small change.”
- Hold the engineer of record accountable for connection design through construction, so responsibility for the integrated load does not dissolve in the field.

REFLECTION QUESTIONS

1. Identify a “small change during construction” pattern in your domain. What is the institutional review that should accompany it?
2. Design the change-control deliverable that would have surfaced the doubled-load issue at the Hyatt Regency before construction.
3. The two-rod change was approved for ease of assembly by an office that never recalculated its load. What in your domain lets a convenience at the point of assembly pass without anyone re-deriving its consequence at the point of load?

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LENS COURSES LEN 5 LEN 7

CASE 75 INDUSTRIAL 2018

FIU Pedestrian Bridge Collapse

D Designed Out G Governance & Trust N Normalization of Deviance

IMPACT 6 killed in Miami when a pedestrian bridge collapsed during construction; NTSB found systemic design and review failures

IN BRIEF

On 15 March 2018 a pedestrian bridge under construction at Florida International University collapsed onto the road below, killing six. The NTSB found the design firm had made calculation errors — underestimating demand and overestimating capacity at a critical truss connection — that the independent peer review failed to catch. In the days before, large cracks appeared at that connection; they were inspected, discussed in a meeting the morning of the collapse, and judged not a safety concern, even as crews worked on the span and traffic ran beneath it. The errors were the timeless ones — a connection under-designed, a review that missed it, warning signs rationalized away — in a structure built with state-of-the-art tools. FIU is the contemporary proof that better design software does not retire the old failure modes.

BACKGROUND

The FIU-Sweetwater pedestrian bridge over Southwest Eighth Street in Miami was a modern, computationally designed concrete truss, built to be assembled beside the road and lifted into place — engineered with tools more capable than any prior era's, the kind of software that lets a designer model a structure in detail no slide rule ever could. That sophistication framed the bridge as a showcase of modern method.[0]

WHAT HAPPENED

It collapsed during construction on 15 March 2018, dropping onto the road and the cars stopped beneath it, killing six — the failure landing on the very traffic the bridge would one day span. The NTSB found the design firm had made calculation errors at a critical truss connection — underestimating the load demand and overestimating the capacity, an error in both directions at once — and that the independent peer review, the safeguard meant to catch exactly such mistakes, had not caught them.[1]

THE INVESTIGATION

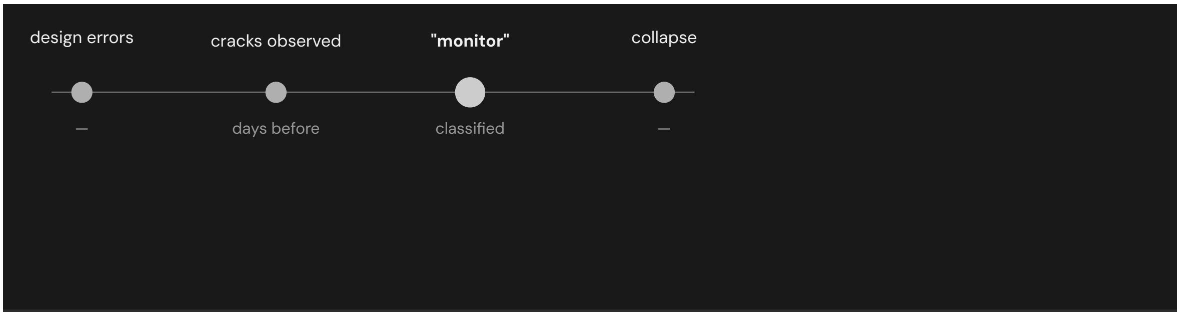
The warnings were there. Large cracks had appeared at that connection in the days before, were inspected, and were judged not to threaten safety — a judgment reaffirmed in a meeting the morning of the collapse, even as workers operated on the span and traffic ran underneath, so the structure was failing visibly while being declared safe.[2] The errors that produced the collapse were the timeless ones: a load case inadequately considered, a review that did not catch the gap, and visible distress rationalized as not safety-critical — each a failure mode older than the software that drew the bridge.[3]

THE CAPABILITY GAP

FIU is the contemporary reminder that more capable design tools do not retire the old failure modes. The bridge was designed with sophisticated software, yet failed for the same reasons structures have failed for a century — because the human and institutional capabilities around the tools (independent review that re-derives the answer, a rule that visible cracking on a critical member stops work) had not kept pace with the software's sophistication, leaving the newest tools guarded by the oldest, weakest procedures.[4]

AFTERMATH & REFORM

The NTSB faulted the design firm, the peer reviewer, and the parties who kept the road open beneath a cracking structure — spreading the responsibility across every layer that could have intervened — and the case sharpened scrutiny of accelerated-bridge-construction methods and independent design review.[5] Its lesson pairs with the Hyatt Regency (Case 74) across forty years: the computer drew a better bridge, but the governance of the calculation — who checks it, and what a crack is allowed to mean — is still where safety lives, unchanged by four decades of better software.



FIU — OBSERVED CRACKS CLASSIFIED AS NOT SAFETY-CRITICAL

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4. NTSB HAR-19/02 (2019) — the failure of the independent design peer review.
5. OSHA report on the FIU bridge collapse (2019); Engineering News-Record coverage.
6. Cf. the Hyatt Regency collapse (Case 74); accelerated-bridge-construction review practice.

THE LEARNING ENGINEERING LENS

> *The probable cause of the collapse was the load and capacity calculation errors made by FIGG Bridge Engineers in its design of the main span truss.*
>
> NTSB HIGHWAY ACCIDENT REPORT HAR-19/02 ON THE FIU PEDESTRIAN BRIDGE, 2019

LE INSIGHT

The FIU bridge is the canonical modern engineering case for cumulative review failure. Each layer of review failed in its own way. The collapse was not predicted but was predictable. The cracks that had been observed were classified as not safety- critical because the institutional protocol for "potentially significant crack on a new structure" defaulted to "monitor."

LENS APPROACH

LENS uses FIU in LEN 5 to teach review-process deliverables and in LEN 8 to compare with the Hyatt Regency (Case 74) case from forty years earlier — same failure mode, different decade.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Require an independent peer review that re-derives the load and capacity at critical connections, rather than re-reading the original firm's calculation.
- Keep the human and institutional checks — review, escalation, stop-work authority — abreast of the design tools, so sophisticated software is not guarded by weak procedures.
- Set a rule that visible cracking on a critical member halts work, so distress cannot be rationalized away while crews and traffic remain exposed.

IN OPERATION / AFTER

- Audit observed field distress against design intent, treating a crack on a new structure as potentially significant until re-analysis proves otherwise.
- Route field observations to a decision-maker with authority to stop work, rather than resolving them in a meeting that defaults to "monitor."
- Sustain scrutiny of accelerated-construction methods and independent review, so the same century-old failure mode does not recur a decade later.

REFLECTION QUESTIONS

1. The FIU and Hyatt Regency cases are 40 years apart and structurally similar. What has not been engineered in the discipline that would change that?
2. Design the field-observation escalation protocol that would have brought the FIU cracks to a decision-maker with the authority to halt work.
3. At FIU the most capable design software in history was guarded by a peer review that missed the error and a crack-response rule that defaulted to "monitor." Where in your domain has tooling outrun the human and institutional checks meant to catch its mistakes?

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LENS COURSES LEN 5 LEN 8

CASE 76 ENERGY 2018

Camp Fire / PG&E

G Governance & Trust N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT 85 killed in Paradise, California; deadliest U.S. wildfire in a century; PG&E pleaded guilty to 84 counts of involuntary manslaughter

IN BRIEF

On 8 November 2018 a worn hook on a nearly century-old PG&E transmission line failed in high winds and drought, igniting the Camp Fire, which swept into the town of Paradise faster than people could evacuate. Eighty-five died — the deadliest U.S. wildfire in a century — and PG&E later pleaded guilty to 84 counts of involuntary manslaughter. Investigators found PG&E had known for years that its transmission infrastructure across high-fire-risk areas was deteriorating, and had deferred maintenance to fund other priorities, under a regulatory regime that let the deferrals continue. Infrastructure built for one climate was operating in another. Camp Fire is the book's foundational climate-era case for utility capability under changed conditions, and it restructured how California regulates utility wildfire risk.

BACKGROUND

PG&E operated aging transmission lines across the wildfire-prone foothills of Northern California — some hardware approaching a century old — in a climate growing hotter and drier than the one the grid had been built for, so the operating environment had drifted away from the assumptions the infrastructure was designed against.[0]

WHAT HAPPENED

On 8 November 2018 a worn C-hook on a PG&E transmission tower failed in high winds, dropping a live line and igniting a fire under severe drought conditions — a single piece of aged hardware setting off a catastrophe the dry, windy conditions stood ready to amplify. The fire moved into the town of Paradise faster than its evacuation routes could clear it; 85 people died — the deadliest U.S. wildfire in a century. PG&E later pleaded guilty to 84 counts of involuntary manslaughter, accepting criminal responsibility at a scale rare for a utility.[1]

THE INVESTIGATION

CalFire's investigation and the Butte County District Attorney's report found that PG&E had known for years about the deteriorating condition of its transmission infrastructure in high-fire-risk areas, and had deferred the maintenance to fund other corporate priorities — so the hazard was not unknown but a recognized risk repeatedly postponed.[2] The gap was simultaneously at the utility's asset-maintenance decisions and at the regulatory architecture that had allowed the deferrals to continue, neither side holding a line that would have forced the work.[3]

THE CAPABILITY GAP

Camp Fire is the climate-era case for utility-capability failure under changing risk. The infrastructure had been designed and maintained for one set of conditions and was operating in another, more dangerous one, so the safety margins the original design assumed had quietly eroded as the climate shifted beneath them. The capability to update operations — inspection cadence, vegetation management, de-energization, replacement — to match the actual risk did not exist as an institutional deliverable, on either the utility's side or the regulator's, leaving no one tasked with closing the widening gap.[4]

AFTERMATH & REFORM

PG&E entered bankruptcy under tens of billions in wildfire liability, and California restructured how it regulates utility wildfire-risk planning — mandatory mitigation plans, inspections, and public-safety power shutoffs, turning the deferred work into requirements with teeth behind them.[5] Paired with the Northeast Blackout (Case 62), Camp Fire shows utility capability failing in a second way: not a silent control-room failure but a slow, known erosion of physical infrastructure against a rising hazard the institution declined to fund against — a failure measured in years of deferral rather than seconds of cascade.

85

KILLED IN PARADISE · SINGLE TRANSMISSION HOOK

infrastructure designed for one climate, operating in another

CAMP FIRE — CAPABILITY MISMATCH UNDER CHANGED CONDITIONS

REFERENCES

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2. The Camp Fire record — 85 killed; PG&E's guilty plea to 84 counts of involuntary manslaughter (2020).
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5. PG&E bankruptcy and California wildfire-mitigation reforms (2019–).
6. Cf. the Northeast Blackout (Case 62); climate-and-infrastructure literature.

THE LEARNING ENGINEERING LENS

> *PG&E knew its equipment was failing in high-fire-risk areas and continued operating without remediation.*
>
> PARAPHRASING THE BUTTE COUNTY DISTRICT ATTORNEY'S CAMP FIRE REPORT, 2020

LE INSIGHT

The Camp Fire is the canonical climate-era case for utility- capability failure under changing risk conditions. The infrastructure was designed for one set of conditions and operated in another. The capability to update operations to match actual conditions did not exist as an institutional deliverable.

LENS APPROACH

LENS uses the Camp Fire in LEN 7 for utility regulatory governance and in LEN 8 for institutional response to changing operating conditions. The case is paired with the Northeast Blackout (Case 62) as utility-capability failures of different kinds.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Re-derive the infrastructure's safety margins against current conditions, not the historical climate it was designed for, and treat the gap as a hazard to close.
- Make updating operations to match rising risk — inspection cadence, vegetation management, replacement — an explicit institutional deliverable with an accountable owner.
- Build de-energization and equipment-replacement triggers tied to known high-risk hardware in high-fire-risk areas.

IN OPERATION / AFTER

- Audit deferred maintenance against the hazard it guards against, so a recognized, deteriorating risk cannot be postponed year after year.
- Hold the regulator's line with mandatory mitigation plans, inspections, and penalties, so the cost of resilience is forced before the disaster rather than after.
- Monitor aging hardware in the highest-risk corridors as a standing priority, treating a worn critical component as an active threat, not a backlog item.

REFLECTION QUESTIONS

1. Identify infrastructure in your domain that was designed for one set of conditions and is now operating in another. What is the capability deliverable to bridge the gap?
2. Design the regulatory architecture that would prevent a utility from deferring critical wildfire-risk maintenance.
3. PG&E knew its hardware was deteriorating in high-fire-risk areas yet deferred the work for years, and the regulator let it. What in your domain lets a recognized, rising hazard be postponed indefinitely — and who would have to hold the line that forces the spending before the catastrophe?

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LENS COURSES LEN 7 LEN 8

CASE 77 ENERGY 2021

Texas Grid Freeze

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT ~246 official deaths (Texas DSHS); academic excess-mortality estimates range 700–1,000; ~$130B in damages; foundational U.S. grid-resilience case

> IN BRIEF
>
> In February 2021 Winter Storm Uri drove temperatures across Texas far below what the ERCOT grid's generators and gas supply were built to withstand. Gas wells froze, gas-fired generation (the bulk of the lost capacity) tripped, and coal, nuclear, and wind also fell; over roughly four days ERCOT shed load and millions lost power in blackouts that, in many cases, never rolled. Texas officially recorded 246 deaths; peer-reviewed excess-mortality estimates run 700–1,000, with about $130 billion in damage. The proximate failure was the absence of winterization requirements for Texas generators — and ERCOT had been warned, after similar storms in 1989, 2011, and 2014, that winterization was inadequate. The warnings never produced a mandate, because the cost would pass to consumers and the rare event was judged acceptable.

BACKGROUND

Texas runs its own electric grid under ERCOT, largely outside federal reliability jurisdiction — a structure that prized low cost and light regulation, and that kept federal reliability mandates at arm's length. Its generators and the gas supply feeding them were built for Texas heat, not sustained deep cold, and winterization was not required, so cold-weather hardening was left to each operator's discretion and largely skipped.[0]

WHAT HAPPENED

In February 2021 Winter Storm Uri drove temperatures far below those design assumptions, exposing every unhardened system at once. Gas wells and wellheads froze, gas-fired generation — the bulk of the lost capacity — tripped offline, and coal, nuclear, and wind also lost output, the cold defeating each fuel source in its own way. Over roughly four days ERCOT directed load shed, and millions of Texans lost power in rolling blackouts that, in many places, did not roll but simply stayed off. Officially 246 died; peer-reviewed excess-mortality estimates run 700–1,000, with around $130 billion in damage.[1]

THE INVESTIGATION

The FERC–NERC joint inquiry identified the absence of winterization requirements as the proximate capability failure — the single missing requirement behind a cascade of frozen equipment.[2] And it was no surprise: ERCOT and Texas generators had been warned after similar cold-weather events in 1989, 2011, and 2014 that winterization was inadequate — warnings that never produced a mandate, because the cost would pass to consumers and the rare event was judged acceptable, so three decades of foresight changed nothing about the build.[3]

THE CAPABILITY GAP

The Texas freeze is a designed-out capability meeting a changing climate. Winterization was treated as optional because the rare event was deemed acceptable — but the acceptable rate of the rare event changed while the designed-out capability did not, so a bet that once looked reasonable kept being renewed long after the odds had moved. Four warnings across three decades did not move the institution, because nothing in its structure forced the cost of resilience to be paid before the catastrophe rather than after, and a warning with no mechanism behind it is only a forecast of the bill.[4]

AFTERMATH & REFORM

After 2021 Texas finally mandated weatherization of generation and key gas facilities, with inspections and penalties behind it — converting, after the deaths, the discretionary hardening that three decades of warnings had failed to compel.[5] The case sits beside the chapter's other slow-erosion failures: a known, repeatedly flagged vulnerability accepted as routine until the tolerances aligned — here, with a population-scale death toll the warnings had, in effect, predicted, the cost of resilience finally paid in full and after the fact.

TEXAS GRID — FOUR WARNINGS ACROSS THREE DECADES, NO WINTERIZATION MANDATED

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3. FERC–NERC (2021) — the prior cold-weather warnings (1989, 2011, 2014).
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5. University of Texas at Austin Energy Institute analyses (Busby et al., 2021).
6. Texas weatherization legislation (SB 3, 2021) and the PUCT/ERCOT reforms.

THE LEARNING ENGINEERING LENS

> *Texas generators had been warned across three decades that winterization was inadequate. The warnings did not produce winterization.*
>
> PARAPHRASING THE FERC-NERC JOINT INQUIRY REPORT ON THE FEBRUARY 2021 TEXAS EVENT

LE INSIGHT

The Texas Grid Freeze is the case for designed-out capability under changing climate conditions. Winterization was treated as optional because the rare event was acceptable. The acceptable rate of the rare event changed; the designed-out capability did not.

LENS APPROACH

LENS uses the Texas Grid Freeze in LEN 7 as a regulatory-governance case and in LEN 8 for institutional learning across repeated near- misses (1989, 2011, 2014) that did not produce remediation.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Re-test the “acceptable rare event” assumption against current climate odds, since a bet that looked reasonable decades ago may no longer hold.
- Build winterization or equivalent resilience into the design where the hazard reaches, rather than leaving hardening to each operator’s discretion.
- Create a structural mechanism that forces the cost of resilience to be paid up front, so a warning is backed by a mandate rather than left as a forecast.

IN OPERATION / AFTER

- Audit repeated cold-weather or near-miss warnings to confirm they actually produced hardening, treating an unactioned warning as an open liability.
- Mandate weatherization with inspections and penalties behind it, so compliance does not depend on each operator absorbing a cost it can pass to consumers.
- Track the changing rate of the rare event so a designed-out capability is reconsidered when the odds move, not only after a catastrophe.

REFLECTION QUESTIONS

1. Identify a capability in your domain that has been designed out because the relevant rare event was judged acceptable. Is the rate still acceptable?
2. The Texas grid was warned four times in three decades. What institutional architecture allows that pattern, and what would change it?
3. Winterization stayed optional because its cost would pass to consumers while the rare event was judged acceptable. Where in your domain is a resilience cost being deferred onto a future catastrophe, and what would force it to be paid before rather than after?

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LENS COURSES LEN 7 LEN 8

Chapter 6

Knowledge Lost

Institutional memory degraded.

Schedulers are among the top ten highest-turnover positions in the VA.

DEBRA DRAPER, GAO DIRECTOR OF HEALTH CARE, TESTIMONY TO THE HOUSE COMMITTEE ON VETERANS' AFFAIRS, 2019

CASE 32 GOVERNMENT 2014

VA Wait-Time Scandal

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Veterans died waiting for care; 300,000+ on waiting lists or waiting 6+ months; staff falsified records

IN BRIEF

In 2014 the VA Inspector General found that staff at the Phoenix VA — and then nationwide — had created secret waiting lists and falsified appointment data to hide that veterans were waiting weeks or months for care; some died waiting while the system reported success. A 14-day access target, unrealistic given staffing, was met by hiding reality rather than surfacing it. The warning signs ran back fifteen years: GAO had flagged data-reliability problems since 2000, the IG since 2005, with no systemic fix — and schedulers, the staff operating the measurement, are among the VA's highest-turnover roles, so the institution kept losing the knowledge even to see it was lying to itself. The VA case is the book's canonical example of normalization of deviance applied to measurement itself.

BACKGROUND

The Veterans Health Administration measured access to care against a 14-day appointment target — a target that, given staffing, was often unrealistic to meet honestly. Schedulers, the staff who operate that measurement, are among the VA's highest-turnover positions.[0] The target functioned as the headline number leadership watched, so the pressure to show 14-day compliance bore down hardest on the very front-line role least equipped, through constant churn, to record the access data accurately or to question what the number was leaving out.

WHAT HAPPENED

In 2014 the VA Office of Inspector General found that staff at the Phoenix VA — and then across the system — had created secret waiting lists and falsified appointment data to hide that veterans were waiting weeks or months for care. Some veterans died waiting, inside a system that, by its own metrics, was succeeding.[1] The secret lists were the mechanism by which an impossible target was reconciled with reality: official records showed appointments inside 14 days while the true wait accumulated off the books, so the more the metric was gamed, the more confidently the reporting line above it declared the access problem solved.

THE INVESTIGATION

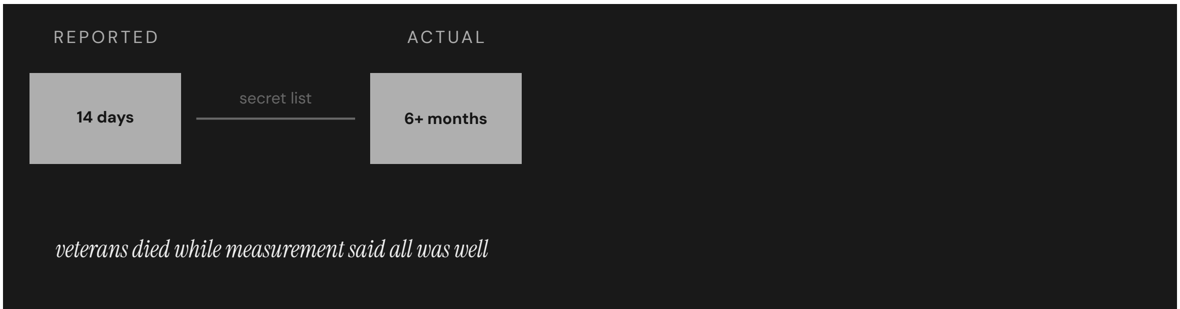
The warning signs went back fifteen years: GAO had flagged data-reliability concerns since 2000, and the VA IG had identified problems in 2005, 2007, and 2008, with the incoming administration warned in 2008 — none of which produced systemic change.[2] GAO named scheduler training as a root cause: with schedulers among the top-ten highest-turnover roles, the institution perpetually lost the knowledge required even to run the measurement honestly, and five years on still reported data-reliability concerns.[3] Each warning had named a real defect in how access was recorded, yet because the schedulers who held the practical knowledge of the system kept turning over, every wave of findings landed on a workforce that had to relearn the instrument from scratch, so the same defect resurfaced report after report.

THE CAPABILITY GAP

This is normalization of deviance applied to measurement itself. When the measurement system cannot capture reality — and the people operating it churn before the gaming can be unlearned — the gap between reported and actual performance becomes invisible. Veterans died inside a system whose numbers said it was fine, which is the lethal form of the evidence problem this book treats as a design failure, not a reporting one.[4] An institution that cannot retain the staff who run its measurement loses not only accuracy but the memory that the number was ever wrong, so the gaming stops registering as deviance at all and hardens into the ordinary way the work is done.

AFTERMATH & REFORM

The scandal forced resignations, the Veterans Access, Choice, and Accountability Act (2014), and a long, uneven effort to rebuild scheduling and data integrity.[5] Its lesson is that decision-grade evidence is a design requirement: an institution must be able to surface its own failures without relying on the very people incentivized — and too transient — to hide them. The Choice Act bought access by routing care outside the VA, but the underlying capability — a measurement the institution could trust even as its schedulers churned — was the harder thing to rebuild, and the data-reliability concerns that persisted for years afterward show why.



SCHEDULERS FALSIFIED RECORDS; THE MEASUREMENT SYSTEM HID THE GAP

REFERENCES

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THE LEARNING ENGINEERING LENS

> *Schedulers are among the top ten highest-turnover positions in the VA.*
>
> DEBRA DRAPER (GAO DIRECTOR OF HEALTH CARE), HOUSE VA COMMITTEE TESTIMONY, GAO-19-687T, JULY 2019

LE INSIGHT

The VA case is the canonical example of measurement failure as a capability failure. The system that should have surfaced the gap instead generated reports that hid it. The turnover among schedulers — the human capability operating the measurement instrument — meant the institution lost the knowledge to even notice it was lying to itself. The case stands as the strongest argument in this book for treating decision-grade evidence as a design requirement, not a reporting requirement.

LENS APPROACH

LENS treats the VA case in LEN 4 as the canonical evidence-gap case (the measurement system itself was the source of harm), in LEN 7 as a governance failure (multiple warnings unactioned over fifteen years), and in LEN 8 as a knowledge-loss case via turnover.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Set access targets against actual staffing capacity, so the metric cannot be met only by falsifying it.
- Design the scheduling instrument so correct data entry is the path of least resistance, not a discipline that churning staff must be retrained into.
- Build an independent read on real wait times — separate from the staff incentivized to report them — into the measurement architecture from the start.

IN OPERATION / AFTER

- Audit reported access against an out-of-band signal (direct veteran survey, third-party booking records) with authority to act when the two diverge.
- Treat scheduler turnover as a measurement risk: monitor it, and protect the knowledge of how to run the instrument honestly against constant churn.
- Track whether old data-reliability findings keep recurring — a repeat finding is the signal that the institution is relearning the same gap rather than closing it.

REFLECTION QUESTIONS

1. Identify a measurement system in your domain that is also operated by a high-turnover role. What is the institutional risk that the system stops measuring reality?
2. Design the evidence pipeline that would have surfaced the Phoenix VA gap without relying on the people who were gaming the metrics.
3. The 14-day target was unrealistic given staffing, so it was met by hiding reality. Identify a target in your organization that the people measured against it cannot honestly meet — and design the correction that surfaces the gap rather than burying it.

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LENS COURSES LEN 4 LEN 7 LEN 8

CASE 38 DEFENSE EDUCATION AT SCALE 2012 – PRESENT

GIFT and the Adoption Gap

K Knowledge & Institutional Memory G Governance & Trust N Normalization of Deviance

IMPACT Active open-source ITS framework with demonstrated learning effectiveness; ubiquitous fielded adoption in routine military training has not been achieved

IN BRIEF

The Generalized Intelligent Framework for Tutoring (GIFT) is an open-source framework, originated at the U.S. Army Research Laboratory, for authoring and delivering intelligent tutoring systems. Computer-based tutoring has been shown to be about as effective as expert human tutors in well-defined domains, and GIFT exists to lower the barrier to building it; the framework is actively developed, with regular releases and a peer-reviewed annual symposium. The puzzle is not that GIFT failed — it did not — but that ubiquitous fielded adoption across routine military training remains limited despite decades of supporting research. That gap is the canonical learning-engineering problem: the science is settled, the platform exists, the studies are positive — and the institutional pathway to scaled adoption is still being built.

THE SHIFT

Five decades of research established that computer-based tutoring can be about as effective as an expert human tutor in well-defined domains, and significantly better than traditional classroom instruction. The open question shifted from “does adaptive tutoring work?” to “why isn’t it everywhere?”[0] That shift matters because it moves the problem out of the laboratory: once the efficacy question is answered, every remaining obstacle to scaled use is institutional rather than scientific, and the field’s research strength stops being the binding constraint.

WHAT IS EMERGING

The Generalized Intelligent Framework for Tutoring (GIFT), originated under the U.S. Army Research Laboratory and now developed at DEVCOM, is an open-source framework for authoring, delivering, and evaluating intelligent tutoring systems — an effort to lower the authoring barrier that has historically made ITS expensive to build. It is actively maintained, with regular releases and a peer-reviewed annual symposium.[1] Open-sourcing the framework and sustaining a research community around it directly attacks the cost-to-build problem, since the expense of authoring a tutor from scratch had long been the practical reason adaptive tutoring stayed confined to well-funded demonstrations.

THE CAPABILITY QUESTION

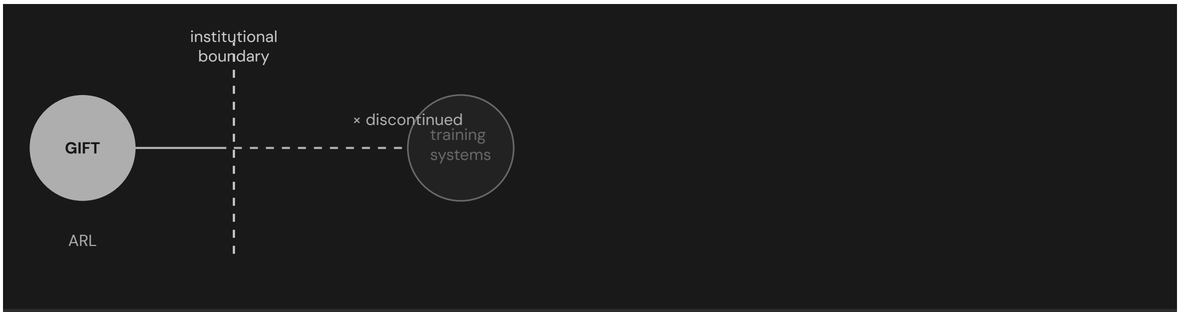
So the puzzle is not failure — GIFT was not discontinued. It is that ubiquitous fielded adoption across routine military training remains limited despite a working framework and decades of positive research. The science is settled and the platform exists; what is missing is the institutional pathway to scaled use.[2] This is the diagnostic feature of the case: the gap is not between idea and prototype but between a maintained, evidence-backed framework and the routine training pipelines that would have to adopt it, and that latter distance is the one no amount of further research closes.

EARLY EVIDENCE

The effectiveness evidence is strong — the tutoring-effectiveness literature is among the more robust in education — and GIFT-based studies continue to demonstrate learning gains. The bottleneck is not efficacy.[3] Because the supporting literature is among the more robust in the field, a decision-maker hesitating to field adaptive tutoring cannot honestly point to weak evidence as the reason; the hesitation traces instead to the missing pathway that would let the proven approach be bought, integrated, and made routine.

OPEN PROBLEMS

What has not been built is the adoption pathway: procurement that can buy adaptive tutoring, integration into existing training pipelines, instructor-workflow redesign, and the authority structure to make adaptive tutoring a default rather than an experiment.[4] GIFT is the case in this book closest to the LENS discipline itself — proof that a working technology and settled science do not adopt themselves, and that the adoption pathway is an engineering deliverable in its own right. Each of those pieces — a contracting vehicle, a pipeline integration, a redesigned instructor workflow, an owner with authority — is a concrete artifact someone must build, and their absence, not any technical shortfall, is what keeps the framework experimental.



ADAPTIVE INSTRUCTION WORKED; ADOPTION DID NOT CROSS THE BOUNDARY

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THE LEARNING ENGINEERING LENS

> *The technology works. The institutional pathway to ubiquitous fielded use does not yet.*
>
> EDITORS' SYNTHESIS OF THE GIFT ADOPTION RECORD

LE INSIGHT

GIFT is the most directly relevant case in this book to the learning-engineering discipline itself. The technology works, the learning science works, the framework is active and supported. What has not been built is the institutional adoption pathway — procurement, training-pipeline integration, instructor workflow redesign, the authority structure — that would make adaptive tutoring a default rather than an experiment.

LENS APPROACH

LENS treats GIFT in LEN 1 as the problem-framing case for the discipline, in LEN 8 as the foundational adoption-pathway case, and in LEN 10 (capstone) as a prompt for designing the institutional deliverables that would close an adoption gap of this shape.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat the adoption pathway — procurement, pipeline integration, instructor-workflow redesign, authority — as a named deliverable of the program, not a follow-on hope.
- Build the tutoring capability against an existing training pipeline so integration is designed in, rather than fielding a framework and expecting pipelines to bend to it.
- Specify who holds the authority to make adaptive tutoring a default, so the decision to scale is owned rather than left to volunteers.

IN OPERATION / AFTER

- Measure fielded routine use, not just study-level learning gains, so the institution can see whether adoption is actually happening.
- Sustain the open-source framework and its community so the authoring-cost barrier it lowered does not quietly rise again as releases age.
- Audit each stalled pipeline to find which adoption artifact is missing — contracting vehicle, integration, workflow, owner — and treat that as the engineering gap to close.

REFLECTION QUESTIONS

1. GIFT exists, is supported, and works. Adoption at scale does not. What is the equivalent in your domain — an effective intervention whose adoption pathway has not been engineered?

2. Design the institutional adoption deliverable that would move adaptive tutoring from “available framework” to “default routine practice” in one operational training pipeline.

3. GIFT’s bottleneck is procurement, pipeline integration, instructor-workflow redesign, and authority — not efficacy. For an effective tool in your domain, which of those four is the binding constraint, and who would have to own it for adoption to become the default?

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LENS COURSES LEN 1 LEN 10 LEN 8 LEN 6

CASE 40 EDUCATION AT SCALE DEFENSE ONGOING

xAPI / Total Learning Architecture — Interoperability Gap

K Knowledge & Institutional Memory G Governance & Trust

IMPACT Despite xAPI adoption, most implementations remain siloed; cross-organizational learning-data interoperability largely unrealized

IN BRIEF

The Experience API (xAPI) was created to track learning experiences across platforms, and the Advanced Distributed Learning Initiative's Total Learning Architecture envisioned learning records, competencies, and credentials flowing across organizational boundaries to support continuous capability development — the kind of evidence infrastructure LENS depends on. In practice, xAPI implementations remain largely siloed inside individual learning-management systems; the cross-organizational data sharing that matters most for high-consequence domains has not materialized at scale. The technical standard exists and reference implementations exist; what lags is governance — who owns the data, what consent applies, how quality is assured. xAPI mirrors inBloom (Case 8) in structure — technology ahead of governance — across the whole learning-technology ecosystem. It is the book's case for treating data governance as a capability deliverable.

THE SHIFT

Learning increasingly happens across many systems — courses, simulators, on-the-job tools — and the field recognized it needed a common way to record those experiences, so capability could be tracked over a career rather than a single course.[0] The career-long view is the point: a single course's records say little about whether a person can do the job, whereas experiences stitched across courses, simulators, and on-the-job tools are what let an institution reason about real capability rather than completed seat-time.

WHAT IS EMERGING

The Experience API (xAPI) was built to do exactly that, and the Advanced Distributed Learning Initiative's Total Learning Architecture envisioned learning records, competency frameworks, and credentials flowing across organizational boundaries — the evidence infrastructure a discipline like LENS depends on.[1] The vision was explicitly cross-boundary: records, competencies, and credentials that move with the learner between organizations are precisely the evidence base on which a capability discipline must stand, which is why the standard's promise mattered well beyond any single training shop.

THE CAPABILITY QUESTION

In practice, xAPI implementations remain largely siloed inside individual LMS platforms. The cross-organizational data sharing — the capability most relevant to high-consequence operational domains — has not materialized at scale. The standard exists; the ecosystem it promised does not.[2] The siloing is the diagnostic detail: the very cross-boundary flow that made the architecture worth building is the part that did not arrive, so the standard delivers tracking within each LMS while the career-long, cross-organizational record it envisioned stays out of reach.

EARLY EVIDENCE

The technical pieces work: the data model is sound and reference implementations exist. What has lagged is the governance — who owns the data, what consent frameworks apply, and how data quality is assured across organizations.[3] With the data model proven and reference implementations in hand, the remaining obstacles are not engineering questions an organization can solve alone but agreements between organizations — ownership, consent, and assured quality — that no technical specification can settle on their behalf.

OPEN PROBLEMS

xAPI mirrors inBloom (Case 8) in essential structure — technology in advance of governance — but across the whole learning-technology ecosystem rather than one initiative.[4] It is the book's clearest case that an interoperability standard is necessary but not sufficient: without the governance to make organizations willing and able to share, the data stays in its silos, and the evidence infrastructure remains a diagram rather than a system. The parallel to inBloom is instructive because it shows the pattern is not particular to one failed project: a sound standard arriving ahead of the ownership, consent, and quality-assurance arrangements will stall the same way at any scale.

STANDARD EXISTS; CROSS-ORGANIZATION DATA STILL SILO'D

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4. B. Saxberg, learning-engineering infrastructure essays; IEEE ICICLE LEBoK chapters on data and analytics.
5. Cf. inBloom (Case 8) — technology in advance of governance.

THE LEARNING ENGINEERING LENS

> *The standard exists. The governance does not.*
>
> EDITORS' SYNTHESIS OF ICICLE / ADL TLA DISCUSSION

LE INSIGHT

xAPI/TLA is the technical-standard analog of the implementation gap — the discipline has the data model, the spec, and the reference implementations. What it does not have is the governance and institutional architecture to make cross-organizational learning data flow. This is the canonical case in this book for treating data governance as a capability-engineering deliverable.

LENS APPROACH

LENS treats xAPI/TLA in LEN 4 as a data-governance and interoperability case and in LEN 8 as an example of organizational- learning infrastructure that has not scaled. The case is the technical underlay to the larger argument about evidence systems that decision-makers can trust.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify the governance — data ownership, consent, and cross-organization quality assurance — as a deliverable alongside the technical standard, not after it.
- Design for the cross-boundary flow from the start, so the standard is not merely implemented inside each LMS but engineered to move records between organizations.
- Secure the inter-organizational agreements that no specification can settle, treating willingness-to-share as a thing to be built rather than assumed.

IN OPERATION / AFTER

- Audit whether learning data is actually crossing organizational boundaries, not just whether xAPI is nominally adopted within each silo.
- Monitor data quality across organizations continuously, since shared records are only trustworthy evidence if their quality is assured at the seams.
- Watch for the inBloom pattern — a sound standard outrunning its governance — and treat any stall as a missing-governance signal, not a technical defect.

REFLECTION QUESTIONS

1. Why has the xAPI standard not produced cross-organizational interoperability at scale? What governance condition is missing?
2. Design the minimum governance architecture under which xAPI data could flow across two organizations in your domain.
3. The xAPI data model is sound; what stalled was ownership, consent, and quality assurance across organizations. For a data-sharing effort in your domain, which of those three is the unresolved question — and who would have to agree for the data to actually flow?

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LENS COURSES LEN 4 LEN 8 LEN 6

CASE 41 HEALTHCARE ONGOING

Implementation Science in Healthcare — The 17-Year Gap

K Knowledge & Institutional Memory G Governance & Trust N Normalization of Deviance

IMPACT Average time from research finding to clinical practice: 17 years; only ~14% of research findings ever reach practice

IN BRIEF

Implementation science has a canonical finding: it takes an average of about seventeen years for research evidence to reach clinical practice, and only roughly 14% of research findings ever make it at all. This is not a single incident but a systemic condition — effective interventions exist; the system to adopt, sustain, adapt, and measure them at scale does not. Frameworks like the Active Implementation Frameworks (Fixsen et al., 2005) and EPIS (Aarons et al., 2011) were built specifically to attack this gap, and LENS threads implementation science throughout its curriculum in direct response. The seventeen-year gap is the meta-case for the whole book: every success case is a closure of this gap in one domain, every failure case the gap left open. The discipline exists to make seventeen years shorter.

THE SHIFT

Medicine generates more validated knowledge than it can absorb. Implementation science arose to study a stubborn fact: knowing what works and having it practiced are different problems, separated by years.[0] Treating the two as one problem is the error the field formed to correct: a validated finding is not yet a changed practice, and the distance between them is itself a phenomenon to be studied, measured, and engineered rather than waited out.

WHAT IS EMERGING

The canonical figures are stark: it takes an average of about seventeen years for research evidence to be integrated into clinical practice, and only roughly 14% of research findings ever make it at all.[1] Read together, the two figures describe a pipeline that is both slow and leaky: most of what is learned never reaches the bedside at all, and the fraction that does arrives long after the patients who first needed it, so the delay is compounded by sheer attrition.

THE CAPABILITY QUESTION

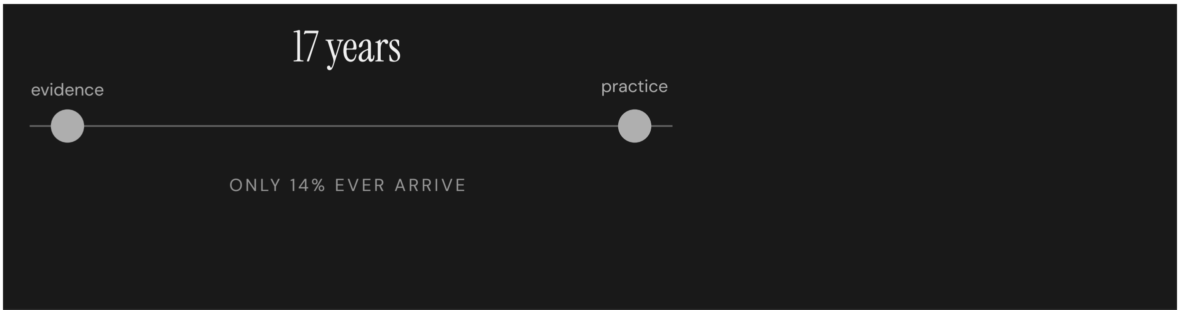
This is not a single case but a systemic condition — the same structural problem the medical-error data (Case 31) describes from the outcome side. Effective interventions exist; the institutional system to adopt, sustain, adapt, and measure them at scale does not.[2] Where the medical-error data counts the harm at the far end of the pipeline, the translation figures name the mechanism that produces it: the same missing adoption-and-measurement system shows up as a delay from one vantage and as a body count from the other.

EARLY EVIDENCE

Frameworks built to attack the gap — the Active Implementation Frameworks (Fixsen et al., 2005) and the EPIS framework (Aarons et al., 2011) — show that implementation can be engineered rather than left to chance, and they inform LENS's choice to thread implementation science through every course rather than isolate it in a module.[3] Threading the discipline through every course rather than confining it to a single module is itself a claim these frameworks support: if implementation is an engineerable property of any intervention, it cannot be quarantined as a specialty and must inform how every design is taught.

OPEN PROBLEMS

The seventeen-year gap is the meta-case for this book. Every success case in the chapters ahead is a closure of the gap in one domain; every failure case is the gap left open.[4] The open problem is general and unglamorous: building, funding, and owning the adoption-and-measurement pathway that turns a proven intervention into routine practice — which is, in one sentence, what the LENS discipline exists to do. Because the problem is general rather than domain-specific, no single clinical result closes it; what closes it is the repeatable, owned, and funded pathway that any proven finding can be run through, again and again.



RESEARCH TO PRACTICE — THE CANONICAL IMPLEMENTATION GAP

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THE LEARNING ENGINEERING LENS

> *The answer is 17 years. What is the question?*
>
> MORRIS, WOODING & GRANT, J ROYAL SOC MED, 2011

LE INSIGHT

The 17-year gap is the structural problem that LENS exists to address. It is the difference between knowing what works and having it deployed. Every case in this book is a sample from a distribution governed by that gap. The success cases shorten it; the failure cases let it run.

LENS APPROACH

LENS uses this case in LEN 1 as the foundational problem statement of the discipline, in LEN 10 as a studio constraint (designs must consider implementation, not just efficacy), and in LEN 8 as the central knowledge-transfer challenge. The case is referenced at least once in every required course.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design every intervention with its adoption-and-measurement pathway attached, so implementation is engineered in rather than left to chance after the evidence is published.
- Use an implementation framework (Active Implementation, EPIS) from the outset to plan adoption, sustainment, and adaptation as deliverables of the project.
- Name an owner and a funding line for the pathway, since a proven finding with no one accountable for fielding it is exactly what the gap is made of.

IN OPERATION / AFTER

- Measure both reach and speed — what fraction of practice has adopted the intervention and how long it took — to see the slow-and-leaky pipeline rather than assume publication equals uptake.
- Sustain and adapt fielded interventions, treating drift back to old practice as a measurable failure mode, not a one-time rollout that holds itself.
- Track the gap as a standing metric across the institution, so closing it in one domain becomes a repeatable pathway rather than a one-off success.

REFLECTION QUESTIONS

1. Pick an evidence-based intervention in your domain. Estimate the gap between when the evidence became conclusive and when the intervention reached majority of practice. What did the gap cost?

2. Design the deliverable that would shorten that gap by half in your domain. Be specific about who funds it, who owns it, and what evidence demonstrates the reduction.

3. The translation pipeline is both slow and leaky — most findings never reach practice, and those that do arrive late. For your domain, is the binding problem the delay or the attrition, and what would you measure to tell which one to attack first?

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LENS COURSES LEN 1 LEN 10 LEN 8 LEN 6

CASE 78 AVIATION 1972 – PRESENT

NASA Saturn V Documentation

K Knowledge & Institutional Memory

IMPACT Foundational U.S. aerospace case for the cost of losing the institutional capability to build a system

IN BRIEF

When NASA returned to heavy-lift rockets in the 2000s, it confronted an uncomfortable fact: the institutional capability to build a Saturn V — the rocket that sent Apollo to the Moon — had been lost. The drawings and documents survived; the practical knowledge of how the components were manufactured, tested, and assembled, and why each choice was made, had walked out with the workforce that retired by the 1990s. The vehicle could be redesigned but not reproduced. Apollo was documented to an unprecedented degree, yet the documentation was not the capability: when the engineers who built the system left, the system left with them. Saturn V is the book's strongest evidence that institutional capability lives in people and sustaining practices, not in the artifacts an institution leaves behind.

BACKGROUND

The Saturn V that launched Apollo to the Moon was, by any measure, one of the best-documented engineering programs in history — exhaustive drawings, specifications, and test records.[0] That thoroughness is what makes the case bite: the program left behind close to everything a successor could ask for on paper, so any later difficulty in rebuilding it cannot be blamed on a thin archive but must lie in what the archive could not hold.

WHAT HAPPENED

Yet by the early 2000s, as NASA worked the Constellation program, it had become apparent that the institutional capability to build a Saturn V had been lost. The drawings existed; the practical knowledge of how the components were manufactured, tested, and assembled — and why particular choices had been made — had walked out with the workforce that retired by the 1990s. The vehicle could be redesigned. It could not be reproduced.[1] The distinction is exact: redesign starts from requirements and rebuilds the reasoning afresh, while reproduction needs the original making-knowledge, and it was that knowledge — not the paperwork describing it — that had left with the people who held it.

THE INVESTIGATION

The case is canonical for the difference between documentation and institutional capability. Apollo's documentation was unprecedented, but it captured the **what**, not the tacit **how** — the judgment, the workarounds, the undocumented reasons — that lived in the people who did the work. When they left, that knowledge was in no archive to recover.[2] A drawing records the decision but not the deliberation behind it; the workaround that made a part manufacturable and the reason a tolerance was set where it was lived in the doing, and once the doers retired there was no document from which to reconstruct it.

THE CAPABILITY GAP

Saturn V is the strongest available evidence that institutional capability is not the same as the artifacts an institution produces. Capability lives in people and in the practices that sustain them; a library of drawings is a necessary record but not a transferable ability. An institution that treats documentation as preservation of capability is, quietly, letting the capability expire.[3] The danger is that the archive looks like insurance: its very completeness can reassure managers that the capability is safe, so the people and practices that actually carry it are allowed to disperse precisely because the paperwork seems to stand in for them.

AFTERMATH & REFORM

The lesson reshaped how serious programs think about knowledge retention — apprenticeship, continuity of teams, deliberate capture of tacit reasoning, and not letting a critical capability rest in a handful of soon-to-retire heads.[4] It pairs with the chapter's other memory cases: knowledge, unlike a document, has to be actively kept alive, or it is gone in a generation. Each of the retention practices addresses the same root cause the Saturn V exposed — that tacit making- knowledge transfers only person to person — so apprenticeship and team continuity are not nice-to-haves but the actual mechanism by which a capability outlives the people who first held it.

1969	2005
built it	couldn't rebuild

drawings persist; tacit institutional knowledge walks out with retirees

SATURN V — DOCUMENTATION IS NOT CAPABILITY

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THE LEARNING ENGINEERING LENS

> *The Saturn V drawings exist. The Saturn V does not.*
>
> PARAPHRASING THE NASA CONSTELLATION-ERA CAPABILITY DISCUSSION, C. 2005

LE INSIGHT

The Saturn V case is the canonical evidence that institutional knowledge is not equivalent to documentation. The documents persist; the capability does not. Capability engineering must treat the people who hold tacit knowledge as a primary institutional asset.

LENS APPROACH

LENS uses the Saturn V case in LEN 8 as the foundational organizational-memory case and in LEN 1 to teach the distinction between recorded knowledge and institutional capability.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Capture tacit reasoning as the work happens — the why behind a choice, the workaround that made a part buildable — not just the resulting drawing.
- Build capability into teams and apprenticeship, so the making-knowledge has a living carrier rather than resting in a handful of soon-to-retire heads.
- Treat documentation as a record, never as a substitute for the people and practices that hold the capability.

IN OPERATION / AFTER

- Audit critical capabilities for single points of human failure — knowledge held by a few near-retirement practitioners — and transfer it before they leave.
- Periodically test reproducibility, not just whether the archive is complete, since a full archive can mask a capability that has already expired.
- Sustain continuity of teams and practice deliberately, so a capability is kept alive across generations rather than rediscovered at need.

REFLECTION QUESTIONS

1. Identify a capability in your domain that is currently held in the tacit knowledge of a small number of senior practitioners. What is your institution's capacity to reproduce it after they retire?
2. Design the institutional practice that would preserve a capability against the retirement of its holders.
3. Saturn V was exhaustively documented yet could not be reproduced, because the archive captured the **what** and not the tacit **how**. Identify a capability in your institution whose documentation might be giving false reassurance — and describe what the paperwork is failing to capture.

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LENS COURSES LEN 1 LEN 8

CASE 79 SPACE / AEROSPACE 2019 – 2024

Boeing Starliner

K Knowledge & Institutional Memory D Designed Out

IMPACT Multiple delays; the 2024 crewed flight left two NASA astronauts at the ISS for months; contemporary case for capability erosion at a legacy contractor

IN BRIEF

Boeing's Starliner, meant to be the second U.S. commercial crew vehicle alongside SpaceX's Crew Dragon, accumulated failures across a decade: software errors on its 2019 uncrewed flight, valve corrosion scrubbing a 2021 launch, and propulsion-system trouble on its 2024 crewed test that left two NASA astronauts on the ISS rather than returning them as contracted — NASA brought them home on SpaceX months later. GAO and NASA reviews found capability erosion across multiple engineering disciplines at a contractor whose human-spaceflight record had once been definitive. The erosion looks partly generational (Apollo- and Shuttle-era engineers retired) and partly institutional (cost and schedule pressure, and the thinning of NASA's in-house depth to challenge the contractor). Starliner is the contemporary case for capability erosion at a legacy contractor.

BACKGROUND

Boeing was awarded a NASA Commercial Crew contract to build Starliner as the second U.S. vehicle to carry astronauts to the ISS, alongside SpaceX's Crew Dragon — drawing on a human-spaceflight heritage that had once been definitive in American aerospace.[0] The award rested in part on that heritage: a contractor with a definitive human-spaceflight record was a presumed safe choice, and that presumption is exactly what the program would go on to test, since reputation was standing in for a current measurement of capability.

WHAT HAPPENED

Instead Starliner stumbled across a decade: software errors marred its 2019 uncrewed test flight, valve corrosion scrubbed a 2021 launch, and propulsion-system problems on the 2024 crewed test flight left the two NASA astronauts aboard the ISS rather than returning on the contracted spacecraft — NASA brought them home on a SpaceX vehicle months later.[1] The three failures fell in different subsystems across three separate years, which is itself telling: a single bad part is bad luck, but software, valves, and propulsion failing in succession points to something broader than any one component — a decline running across the engineering organization rather than within one of its parts.

THE INVESTIGATION

GAO and NASA reviews identified the program as exhibiting capability erosion across multiple engineering disciplines — software, valves, propulsion, integration testing — at a contractor whose track record had previously been definitive.[2] The erosion looked partly generational, as Apollo- and Shuttle-era engineers retired, and partly institutional: cost pressure, schedule-driven decisions, and the loss of NASA's own in-house engineering depth to challenge the contractor.[3] The two causes compound: as the contractor's senior depth thinned through retirement, the buyer's own engineering depth had also eroded, so the very expertise NASA would have needed to catch the slipping contractor was the expertise it had let go of.

THE CAPABILITY GAP

Starliner is the contemporary case for capability erosion at scale at a legacy contractor. The decline happened over decades and was visible only in retrospect, because the institutional architecture for catching a supplier's slow capability decay — and updating the buyer's confidence to match — did not exist. Reputation outran reality, and no instrument was measuring the gap.[4] Slow decay is the hard case to catch: no single year's results look alarming, the brand keeps the buyer's confidence high, and without an instrument that tracks the supplier's actual current capability the divergence is only legible once a crewed flight forces the reckoning.

AFTERMATH & REFORM

NASA leaned harder on independent reviews and on SpaceX as the reliable alternative, and the episode sharpened questions about how to sustain — and verify — capability at sole-source and legacy suppliers.[5] It pairs with Saturn V (Case 78): where that case lost a capability to retirement, Starliner shows the same erosion in slow motion at a living institution still carrying the brand of the capability it had let thin. Having a second supplier to fall back on is what let NASA absorb the failure, which is itself the lesson: where a capability can erode unseen, a verified alternative is the difference between an embarrassment and a crew with no way home.

STARLINER — CAPABILITY DECAY AT A LEGACY CONTRACTOR

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THE LEARNING ENGINEERING LENS

> *Starliner has demonstrated significant readiness shortfalls across multiple engineering disciplines.*
>
> EDITORS' SYNTHESIS OF GAO AND NASA REVIEWS OF THE COMMERCIAL CREW PROGRAM

LE INSIGHT

Starliner is the case for sustained capability erosion at a legacy contractor whose track record had previously been definitive. The erosion happened over decades and was visible in retrospect; the institutional architecture for catching it did not exist.

LENS APPROACH

LENS uses Starliner in LEN 8 as a contemporary contractor- capability erosion case and in LEN 5 for the contractor-NASA interface capability that has thinned over decades.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Verify a supplier's current capability against present evidence rather than awarding on reputation and a once-definitive track record.
- Preserve enough in-house engineering depth to genuinely challenge a contractor, since a buyer who has let its own expertise erode cannot catch a slipping supplier.
- Maintain a verified second supplier where the stakes are crewed, so a capability that erodes unseen does not become a single point of failure.

IN OPERATION / AFTER

- Instrument supplier capability so slow, multi-year decay is visible before a high-stakes flight forces the reckoning.
- Watch for cross-subsystem patterns — failures in software, valves, and propulsion in succession — as a signal of organization-wide erosion, not isolated bad luck.
- Update institutional confidence to match measured reality, so a contractor's brand cannot keep outrunning its current performance.

REFLECTION QUESTIONS

1. Identify a legacy supplier in your domain whose capability track record may have eroded faster than your institutional confidence in them has updated. What is the signal?
2. Design the contractor-capability review that would have caught the Starliner gaps before the 2024 crewed flight.
3. Starliner's failures spanned software, valves, and propulsion across three separate years — a pattern, not a single bad part. In your domain, what would distinguish a one-off component failure from organization-wide capability erosion, and how soon could you tell them apart?

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LENS COURSES LEN 5 LEN 8 LEN 6

CASE 80 SPACE / AEROSPACE 1996

Ariane 5 Flight 501

D Designed Out K Knowledge & Institutional Memory H Human-System Interface

IMPACT Maiden flight destroyed itself 37 seconds after launch; ~$500M payloads lost; reused Ariane 4 code never re-verified for Ariane 5

IN BRIEF

On its 1996 maiden flight, the Ariane 5 rocket destroyed itself 37 seconds after launch, losing around half a billion dollars in payloads. The cause was reused software: the inertial reference system inherited code from Ariane 4, where a horizontal-velocity value never exceeded a 16-bit integer's range. Ariane 5's steeper, faster trajectory pushed it past that range; the integer overflow crashed both redundant reference systems almost simultaneously, the vehicle lost guidance and broke up. The inquiry found the offending code path had been irrelevant on Ariane 4 and was never removed or re-verified for Ariane 5 — software reuse had been treated as risk-free. Ariane 5 is the foundational safety-critical-software case for the hazard of reusing code without re-verifying it against the new system's operating envelope.

BACKGROUND

The Ariane 5 heavy-lift rocket reused proven components from its successful predecessor, Ariane 4 — including, in its inertial reference system, software that had flown reliably for years. Reuse of flight-proven code was regarded as a way to reduce risk and cost.[0] The reasoning was seductive precisely because the code's flight record was genuine: software that had worked on Ariane 4 looked like the safest possible choice, and that confidence is what allowed it to cross onto a different vehicle without being re-examined against the new vehicle's conditions.

WHAT HAPPENED

On 4 June 1996, Ariane 5 Flight 501 — its maiden flight — veered off course and broke up under aerodynamic stress 37 seconds after launch; the range-safety system destroyed it, and roughly half a billion dollars of payloads were lost.[1] The vehicle was destroyed in well under a minute of flight, before it had done anything but climb — the whole half-billion-dollar loss flowing from a fault that triggered almost immediately at liftoff, when the new trajectory first pushed the inherited code outside the range it had been built to handle.

THE INVESTIGATION

The inquiry board traced the cause to the reused code. A horizontal-velocity value that never exceeded a 16-bit signed integer on Ariane 4's trajectory did exceed it on Ariane 5's faster one; the resulting overflow shut down both redundant inertial reference systems almost simultaneously, and the vehicle lost guidance.[2] Redundancy gave no protection here because both units ran the identical inherited code and met the identical out-of-range value at the same instant, so the backup failed in lockstep with the primary — duplicating a system defends against a part breaking, not against a shared assumption being wrong. The offending code path had become irrelevant to Ariane 5′s flight after liftoff, yet had neither been removed nor re-verified against the new vehicle's envelope — reuse had been treated as risk-free.[3]

THE CAPABILITY GAP

Ariane 5 is the canonical software-engineering case for the fallacy of risk-free reuse. Code is fit for the envelope it was written and verified against; reusing it in a different envelope is a new design decision and must be re-verified as one. The institutional capability that was missing was the requirement that reused safety-critical code be re-verified, not merely trusted because it had worked before.[4] The flaw was never in the Ariane 4 code, which was correct for the trajectory it was written against; the flaw was the institutional assumption that a record of working in one envelope carried over to another, an assumption no test was required to challenge.

AFTERMATH & REFORM

The fix and the lesson reshaped safety-critical software practice: reuse became a verification event, with explicit checking of every inherited assumption against the new system's operating conditions.[5] The case rhymes with the Patriot (Case 19) and the Mars Climate Orbiter (Case 54) — small, unexamined assumptions inherited across a boundary, fatal when the boundary's conditions changed. What unites the three is that each inherited a quantity sound in its original setting and lethal in the new one, so the reform is the same in every case: make the boundary crossing the moment the assumption is re-checked, rather than the moment it is silently trusted.

37s

AFTER LAUNCH · 16-BIT INTEGER OVERFLOW

code path disabled by the previous vehicle's profile; re-enabled by the new one

ARIANE 5 — THE FALLACY OF RISK-FREE CODE REUSE

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6. Cf. Patriot (Case 19) and Mars Climate Orbiter (Case 54).

THE LEARNING ENGINEERING LENS

> *The internal SRI software exception was caused during execution of a data conversion from 64-bit floating point to 16-bit signed integer value.*
>
> ARIANE 5 FLIGHT 501 FAILURE INQUIRY BOARD, 1996

LE INSIGHT

Ariane 5 is the canonical software-engineering case for the fallacy of risk-free code reuse. Code is fit for its envelope of operation. Reusing it in a different envelope is a new design decision and must be re-verified as one. The institutional capability that was missing was the requirement to re-verify.

LENS APPROACH

LENS uses Ariane 5 in LEN 5 for software-engineering capability deliverables and in LEN 8 for the institutional discipline that treats reuse as a verification event rather than as a savings.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat every reuse of safety-critical code as a new design decision: re-verify each inherited assumption against the new system's operating envelope before flight.
- Document the original operating envelope a component was verified against, so a later reuse can be checked against it rather than blindly trusted.
- Design redundancy to fail independently — diverse implementations or inputs — so a shared inherited assumption cannot take out primary and backup at once.

IN OPERATION / AFTER

- Audit reused components for code paths that are irrelevant in the new system but still active, and remove or re-verify them rather than leaving them dormant.
- Make boundary crossings — code moving onto a new vehicle, a new envelope — a mandatory re-verification event in the institution's process, not a savings.
- Review past reuse decisions for the Ariane / Patriot / Mars Climate Orbiter pattern: a quantity sound in its original setting and unchecked in the new one.

REFLECTION QUESTIONS

1. Identify a piece of reused infrastructure in your domain whose original operating envelope is not documented. What is the silent assumption?
2. Design the verification deliverable that should accompany every reuse of safety-critical software.
3. Both redundant reference systems failed at the same instant because they shared the same inherited assumption. Identify a place in your domain where redundancy gives false comfort because the duplicated units share a common flaw rather than fail independently.

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LENS COURSES LEN 5 LEN 8

CASE 81 INDUSTRIAL 1940

Tacoma Narrows Bridge

D Designed Out K Knowledge & Institutional Memory

IMPACT Bridge collapsed in ~40 mph wind four months after opening; foundational case in aeroelasticity and structural-engineering pedagogy

IN BRIEF

The Tacoma Narrows Bridge over Puget Sound twisted itself apart on 7 November 1940, four months after opening, in a wind no stronger than a stiff breeze and well within its design speed. The piers and cables were sound; the failure mode was aeroelastic flutter — a wind-driven oscillation that the bridge's slender, shallow deck (chosen for elegance and economy over a deeper, stiffer truss) was prone to, and that the discipline had not yet learned to recognize in long-span suspension bridges. No one died, and the collapse was captured on film shown to nearly every civil-engineering student since. The cost of the lesson was a bridge; the payoff is that no comparable bridge has failed the same way. Tacoma Narrows is the book's clearest example of an entire discipline learning permanently from one failure.

BACKGROUND

The Tacoma Narrows Bridge, opened in July 1940 across Puget Sound, used a slender, shallow deck chosen for elegance and economy over an earlier proposal for a deeper, stiffer stiffening truss. It was sound by the design knowledge of its day.[0] The choice was a deliberate trade of a deeper, stiffer truss for a slimmer, cheaper, more elegant deck — and every analysis the discipline then had to bring to bear judged the result safe, which is what makes the collapse a failure of the available knowledge rather than of the calculation.

WHAT HAPPENED

On 7 November 1940, in a wind of only about 40 mph — well within what the bridge was designed for — the deck began to twist in growing oscillations until it tore itself apart and fell into the sound. The piers held and the cables held; the deck destroyed itself.[1] That the piers and cables survived is the diagnostic detail: the structure did not fail where engineers knew to look but in the slender deck, and not under an extreme load but in an ordinary wind the bridge was expressly built to withstand, which is why the cause lay outside what could then be calculated.

THE INVESTIGATION

The failure mode was aeroelastic flutter — a wind-driven, self-amplifying oscillation — that had not been recognized as a design hazard for long-span suspension bridges. The bridge had not been designed against the thing that destroyed it because that thing was not yet known to the discipline.[2] Because the oscillation fed on itself, a modest steady wind could pump ever more energy into the slender deck until it tore apart — a self-amplifying mechanism nothing in the era's static load analysis was equipped to anticipate, so no competent designer of the day would have thought to check for it.

THE CAPABILITY GAP

Tacoma Narrows is the canonical case of a failure that was not the engineer's fault but the discipline's not-yet-knowing. The missing capability lived above any individual: a body of knowledge that did not yet include flutter. It is the rarer, more hopeful kind of capability gap — the one a profession can close by learning, rather than one a single institution failed to manage.[3] The distinction matters for how the gap is closed: a failure of management is fixed by holding an institution to what was already known, whereas a failure of the discipline's knowledge can only be fixed by the whole profession learning something it did not previously have.

AFTERMATH & REFORM

And the profession did learn: wind-tunnel testing of deck sections became standard, aeroelasticity entered the curriculum, and the collapse film became a permanent teaching artifact. No comparable bridge has failed in the same mode since.[4] The case is the book's strongest evidence that discipline-level learning across an entire profession is possible — and, set against the recurrence in Challenger/Columbia (Case 6), a reminder of how rare it is to absorb a lesson that completely. The reform worked because it lodged the new knowledge in the field's permanent machinery — a required test, a curriculum topic, a teaching film every student sees — rather than in the memory of the engineers who happened to witness it.

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FROM OPENING TO COLLAPSE

aeroelastic flutter — a failure mode the discipline had not yet learned to recognize

TACOMA NARROWS — A DISCIPLINE-LEVEL LEARNING EVENT

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THE LEARNING ENGINEERING LENS

> *The bridge was not designed against the failure mode that destroyed it because that failure mode was not yet known to the discipline.*
>
> PARAPHRASING PETROSKI, TO ENGINEER IS HUMAN, 1985

LE INSIGHT

Tacoma Narrows is the canonical case for a discipline-level learning event. The failure was not the engineer's fault; it was the discipline's not yet knowing. The discipline absorbed the lesson permanently. The case is the strongest available evidence that institutional learning across an entire profession is possible — and rare.

LENS APPROACH

LENS uses Tacoma Narrows in LEN 1 as a problem-framing exemplar for discipline-level learning, in LEN 8 as the canonical case for successful institutional knowledge absorption, and in studio (LEN 10) as a worked example of what discipline-level reform requires.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Test designs against dynamic and self-amplifying behavior, not only static loads, so an unfamiliar failure mode has a chance to reveal itself before fielding.
- Treat a novel design choice — a slenderer deck, an untried configuration — as a reason to test beyond the standard envelope rather than to trust prior practice.
- Build physical or model testing into the design process for any structure pushing past the regime where the discipline's analysis is proven.

IN OPERATION / AFTER

- Lodge each hard-won lesson in the field's permanent machinery — required tests, curriculum, teaching artifacts — so it outlives the engineers who witnessed the failure.
- Distinguish discipline-level gaps from management failures when reviewing an incident, since only the former requires the whole profession to learn something new.
- Watch for novel structures operating where the discipline's knowledge is thin, and treat them as candidates for the next not-yet-recognized failure mode.

REFLECTION QUESTIONS

1. Identify a failure mode in your discipline that the field has not yet learned to recognize. What would be the equivalent Tacoma Narrows event?
2. What institutional architecture absorbed the Tacoma Narrows lesson into the structural-engineering curriculum so durably? Describe its equivalent in your discipline.
3. Tacoma Narrows was a failure of the discipline's knowledge, not of one engineer's management. For a recent failure in your field, decide which kind it was — and explain how the fix differs depending on the answer.

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LENS COURSES LEN 1 LEN 8

CASE 82 ENERGY 2015 – 2016

Aliso Canyon Methane Leak

G Governance & Trust N Normalization of Deviance K Knowledge & Institutional Memory

IMPACT Largest methane leak in U.S. history; ~8,000 households evacuated; ~100,000 tons of methane released

IN BRIEF

In October 2015 a well at SoCalGas's Aliso Canyon underground gas-storage field ruptured and could not be capped for nearly four months, releasing roughly 100,000 tons of methane — the largest such leak in U.S. history, equivalent to the annual emissions of half a million cars — and forcing the evacuation of about 8,000 households in Porter Ranch. The root-cause analysis found external corrosion of the well casing, a subsurface safety valve removed in the 1970s and never replaced, and a pattern of deferring integrity checks on the field's aging wells. The operator knew the wells were old and their integrity uncertain; so did the regulator; the inspection regime had not been engineered to catch the failure mode that occurred. Aliso Canyon is the book's case for legacy infrastructure aging past its oversight.

BACKGROUND

SoCalGas's Aliso Canyon facility stores natural gas underground in depleted oil wells, some dating to the 1950s. Over decades the field's wells aged while the regime for inspecting them did not keep pace — and a subsurface safety valve on the well that would fail had been removed in the 1970s and never replaced.[0] Storing high-pressure gas in wells first drilled for oil decades earlier meant the assets were being asked to do a job their age had not been qualified for, while the one piece of equipment that might have contained a downhole failure had been gone for forty years without replacement.

WHAT HAPPENED

In October 2015 well SS-25 ruptured and could not be capped for nearly four months. About 100,000 tons of methane vented to the atmosphere — the largest such leak in U.S. history, roughly the annual greenhouse emissions of half a million cars — and some 8,000 households were evacuated from the Porter Ranch neighborhood.[1] That it took nearly four months to cap a single well is its own measure of the gap: the failure was not only unprevented but, once underway, beyond the operator's ready means to stop, so the release ran on and the evacuation of the Porter Ranch households stretched across the whole period.

THE INVESTIGATION

The Blade Energy root-cause analysis identified external corrosion of the well casing as the failure — a known, inspectable mode the operator had not adequately inspected.[2] It also found missing safety equipment and an institutional pattern of deferring integrity checks on the older wells; SoCalGas knew the field's wells were aging and their integrity uncertain, and so did the regulator.[3] External corrosion was not an exotic or unforeseeable mode but a familiar one with an established inspection technique, so the failure turned on the deferral itself: the pattern of putting off integrity checks on the oldest wells meant the very inspections that existed to catch corrosion were the ones not being done.

THE CAPABILITY GAP

Aliso Canyon is the case for legacy infrastructure aging past the inspection regime designed for it. The wells were older than the regulations governing them; the operator knew, the regulator knew, and the capability deliverable to update the inspection regime to the assets' actual engineering condition did not exist. Knowledge of the risk was present everywhere except in the rules that would have acted on it.[4] The gap was therefore not one of awareness but of translation: everyone holding the knowledge that the wells were past their safe life lacked the mechanism to convert that knowledge into a mandatory, updated inspection, so the awareness sat inert until the well ruptured.

AFTERMATH & REFORM

The leak prompted tighter California rules for underground gas storage — required inspections, pressure limits, and risk-management plans — and a long fight over the field's future. The lesson pairs with the chapter's other memory failures: an institution can hold the knowledge that an asset is past its safe life and still not act, until the rules that translate knowledge into inspection are themselves rebuilt. Each of the new requirements addresses a specific part of the failure — inspections for the uninspected corrosion, pressure limits for the overtaxed old wells, risk plans for the deferral pattern — which is what it looks like to rebuild the rules so that knowledge of an aging asset is finally forced into action.

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METHANE RELEASED OVER ~4 MONTHS

legacy wells aging past the inspection regime designed for them

ALISO CANYON — INFRASTRUCTURE OUTPACING ITS OVERSIGHT

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THE LEARNING ENGINEERING LENS

> *The leak resulted from external corrosion of the well casing — a known and inspectable failure mode that the operator did not adequately inspect.*
>
> PARAPHRASING BLADE ENERGY PARTNERS, ALISO CANYON ROOT CAUSE ANALYSIS, 2019

LE INSIGHT

Aliso Canyon is the case for legacy infrastructure aging past the inspection regime designed for it. The wells were older than the inspection regulations. The operator knew. The regulator knew. The capability deliverable to update the inspection regime did not exist.

LENS APPROACH

LENS uses Aliso Canyon in LEN 7 for regulatory-architecture capability and in LEN 8 for the institutional pattern of legacy assets aging past the regime governing them.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Match the inspection regime to the asset's actual engineering condition and age, not to the rules written when the asset was new.
- Require, not just permit, the safety equipment a hazard calls for — a removed subsurface valve left unreplaced is a designed-in failure path.
- Build a mechanism that converts known risk into mandatory action, so awareness that an asset is aging cannot sit inert in the absence of a rule.

IN OPERATION / AFTER

- Audit for deferred integrity checks on the oldest assets, since a familiar, inspectable failure mode going uninspected is the warning the system is slipping.
- Re-qualify legacy infrastructure for the duty it now performs, especially where old assets have been repurposed beyond their original design intent.
- Maintain ready means to arrest a failure once begun, so a rupture is not compounded by months of inability to cap it.

REFLECTION QUESTIONS

1. Identify legacy infrastructure in your domain whose age exceeds the inspection regime designed for it. What is the capability deliverable to update the regime?
2. Design the regulator's deliverable that would force re-inspection of legacy infrastructure as the inspection regime is updated.
3. At Aliso Canyon the operator and the regulator both knew the wells were aging, yet the knowledge sat inert because nothing translated it into a mandatory inspection. Identify a known risk in your domain that everyone is aware of but no rule yet forces action on — and design the translation.

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LENS COURSES LEN 7 LEN 8

Chapter 7

The Evidence Gap

Measurement systems that cannot see their own failures.

People don't just die from heart attacks and bacteria, they die from system-wide failings and poorly coordinated care.

MARTIN MAKARY, BMJ, 2016

CASE 31 HEALTHCARE ONGOING

Medical Errors as Systemic Failure

T Training Gap H Human-System Interface N Normalization of Deviance
K Knowledge & Institutional Memory G Governance & Trust

IMPACT Makary & Daniel (2016) estimate of ~250,000 U.S. deaths/year from medical error — third leading cause; estimate is widely cited and substantively contested

IN BRIEF

Medical error in the United States is not a single incident but a systemic condition that the system's own measurement instruments cannot see. In 2016, Makary and Daniel of Johns Hopkins estimated, in the BMJ, that medical errors cause more than 250,000 deaths a year — which would make them the third leading cause of death behind heart disease and cancer. Death certificates do not record medical error as a cause, so the problem is structurally invisible to the systems meant to track it. The 250,000 figure has been substantively contested on methodological grounds — a dispute that is itself a worked example of the gap-attribution problem. The Institute of Medicine raised the alarm in 1999; seventeen years later the problem persisted at scale.

BACKGROUND

Modern medicine generates vast amounts of data, yet its core mortality-measurement instrument — the death certificate — records a proximate physiological cause and has no field for medical error. Because the certificate is keyed to a billing-and-classification taxonomy built for disease, a death set in motion by a care-process breakdown is recorded under whatever organ ultimately failed, and the causal role of the system disappears into the physiology. As a result, harm caused by the care system itself is largely absent from the statistics that are supposed to govern it.[0]

WHAT HAPPENED

In 2016, surgeon Martin Makary and Michael Daniel published an analysis in the BMJ estimating that medical errors cause more than 250,000 deaths a year in the United States — which would rank third behind heart disease and cancer. Their core claim was that "people don't just die from heart attacks and bacteria, they die from system-wide failings and poorly coordinated care." By relocating the cause from the individual clinician to the coordination of care, the framing recast a ledger of isolated mistakes as a single population-scale failure mode the existing statistics were never built to count.[1]

THE INVESTIGATION

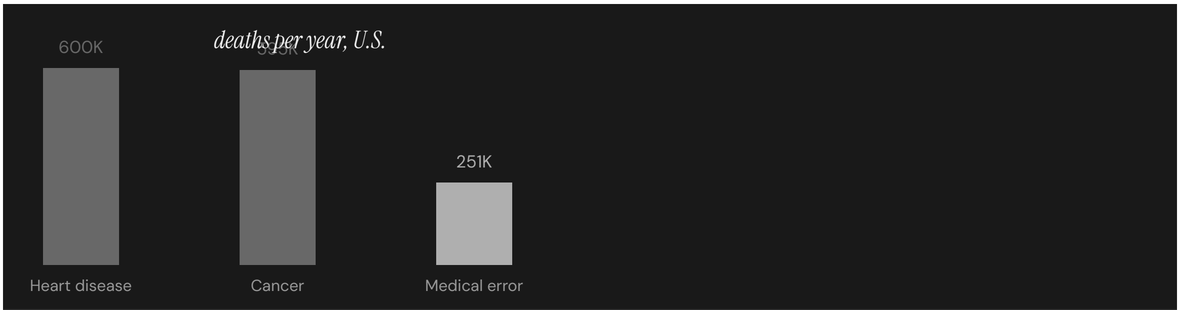
The 250,000 estimate was substantively contested. Shojania and Dixon-Woods, writing in BMJ Quality & Safety in 2017, challenged the extrapolation and the attribution method, arguing that counting deaths “due to” error is far harder than a single headline number implies. Their objection turned on counterfactual attribution — how confidently one can say a frail, already-dying patient would have survived but for the error — a judgment that resists the clean tallying a headline number demands. The dispute is itself a worked example of the gap-attribution problem: how much of a counted death is the learning system, how much the system design, how much the underlying disease.[2]

THE CAPABILITY GAP

The deeper failure is one of measurement: a system that cannot see its own failure modes cannot manage them. With no field on the certificate and no reliable count, every safety program competes for resources against a harm that the official record renders invisible, so even effective interventions struggle to prove their worth. The missing capability is an instrument that captures medical error as a tracked cause of harm — and an attribution method robust enough that the resulting number can guide intervention rather than fuel a methodological stalemate.[3]

AFTERMATH & REFORM

The Institute of Medicine’s To Err Is Human (1999) had earlier estimated 44,000–98,000 annual deaths and catalyzed the patient-safety movement — TeamSTEPPS (Case 27), the WHO surgical checklist (Case 13), the Keystone ICU project (Case 14). Each of those reforms targeted a specific process — teamwork, the operating room, the central line — leaving the system-wide count untouched, so the headline mortality kept escaping measurement even as bounded harms fell. Yet later work, including a 2023 NEJM study of inpatient harm, confirms the problem persists at scale: the interventions exist; the measurement and the implementation still lag.[4]



THIRD LEADING CAUSE OF DEATH — AND STRUCTURALLY INVISIBLE

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THE LEARNING ENGINEERING LENS

> *People don't just die from heart attacks and bacteria, they die from system-wide failings and poorly coordinated care.*
>
> MARTIN MAKARY, PRESS STATEMENTS ACCOMPANYING MAKARY & DANIEL, BMJ (2016)

LE INSIGHT

The Makary data is the anchor evidence for the LENS argument because of its scale, its provenance (Johns Hopkins), and its framing — system failure rather than individual error. The seventeen years between **To Err Is Human** and Makary's reassessment is also the implementation gap of Case 41 in another guise — and the cost of leaving it open is measured in lives at population scale.

LENS APPROACH

LENS treats this case as the central evidence anchor of the curriculum: LEN 1 as the foundational problem statement, LEN 4 as the canonical case for measurement systems blind to their own failures (death certificates do not capture the failure mode), and LEN 10 as the studio prompt for interventions at population scale.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Give the mortality-recording instrument an explicit field for care-process failure, so a system-caused death is captured, not absorbed into the proximate cause.
- Specify the attribution method up front — the counterfactual test for "due to" error — so the count guides action rather than collapsing into dispute.
- Pair each safety intervention with the population measure it should move, gating reforms on demonstrated effect.

IN OPERATION / AFTER

- Audit reported mortality against an independent count of care-related harm.
- Use active surveillance of inpatient harm rather than waiting for the death certificate.
- Hold the count and the intervention to the same measurement discipline, closing the implementation gap.

REFLECTION QUESTIONS

1. Identify a measurement instrument in your domain that systematically fails to capture the failure modes it should be designed to surface. What would it cost to fix?

2. Two hundred fifty thousand deaths a year is the third leading cause of death in the U.S. Design the measurement and intervention regime that would shift the curve over a five-year horizon. Estimate the deliverable and the evidence.

3. Makary and Shojania disagreed not on whether error kills but on how to attribute a death to it. Specify an attribution method robust enough to survive that dispute — and name who would hold the authority to act on the number it produces.

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LENS COURSES LEN 1 LEN 4 LEN 10 LEN 6

CASE 83 GOVERNMENT TECHNOLOGY 2003 – 2012

LIBOR Manipulation

G Governance & Trust N Normalization of Deviance

IMPACT Major banks fined more than $9B; the benchmark underlying ~$350T of contracts was manipulable; replaced by SOFR

IN BRIEF

LIBOR — the London Interbank Offered Rate — was the benchmark interest rate referenced in roughly $350 trillion of financial contracts, yet it was set each day not from observed transactions but from a small panel of banks' estimates of what they would pay to borrow. For about a decade, banks systematically shaded those estimates to favor their own derivatives positions; the coordination was later laid bare in subpoenaed trader-and-submitter messages. The capability gap was in the design of the benchmark itself: it asked for declarations, not observations, and the architecture invited gaming. Regulators levied more than $9 billion in fines, and LIBOR was eventually replaced by SOFR, a rate built from actual transactions. The reform changed the unit of measurement, not just the rules.

BACKGROUND

LIBOR was the benchmark interest rate underlying an estimated $350 trillion in loans, mortgages, and derivatives worldwide. It was set daily by a panel of large banks, each submitting an estimate of the rate at which it could borrow — a self-reported figure rather than a record of any transaction that had actually occurred. Because the same banks that set the rate held derivatives whose value moved with it, the instrument asked the parties with the most to gain to supply the very numbers that would decide their gains.[0]

WHAT HAPPENED

For roughly a decade from the early 2000s, traders at multiple banks asked their firms' rate submitters to shade the daily LIBOR figure up or down to benefit the banks' derivatives books. The requests were routine and documented in internal messages later subpoenaed by regulators — casual, repeated, and unconcealed, which is the signature of conduct the participants did not believe could be detected. During the 2008 crisis, some banks also lowballed submissions to appear healthier than they were, bending the rate to mask their own funding stress as well as to profit.[1]

THE INVESTIGATION

Investigations by the U.S. Department of Justice, UK regulators, and others produced settlements with Barclays, UBS, Deutsche Bank, RBS, and others totaling more than $9 billion. The UK Treasury commissioned the Wheatley Review, which concluded that the benchmark's reliance on subjective estimates rather than transactions was the structural flaw that made manipulation possible. The review located the fault in the instrument rather than only in the individuals who exploited it, treating the gaming as the predictable output of a design that left the door open.[2]

THE CAPABILITY GAP

The capability that was missing was the basic architectural choice to measure observations rather than declarations. A benchmark built on what banks said they would pay is gameable by anyone willing to misstate; a benchmark built on transactions that actually occurred is not. The gaming, once it began, was systematic rather than occasional — exactly what the design invited — and because the figure flowed into hundreds of trillions in contracts, a small shading of the submission moved enormous sums while leaving no trace in any executed trade.[3]

AFTERMATH & REFORM

LIBOR was phased out and replaced by SOFR, the Secured Overnight Financing Rate, which is constructed from observed overnight Treasury repo transactions. The transition took roughly a decade and required coordination across regulators in multiple jurisdictions; the reform re-engineered the measurement at the architecture level rather than merely policing the old one. By anchoring the benchmark to trades that actually settle, the redesign removed the discretion that had been the lever for manipulation, rather than asking submitters to exercise that discretion more honestly.[4]

LIBOR	SOFR
estimates	transactions

the reform changed the unit of measurement, not the regulator

LIBOR — AN INSTRUMENT WHOSE DESIGN INVITED THE GAMING

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THE LEARNING ENGINEERING LENS

> *The architecture of LIBOR invited the manipulation it experienced.*
>
> PARAPHRASING THE WHEATLEY REVIEW OF LIBOR, 2012

LE INSIGHT

LIBOR is the canonical financial-system case for an instrument whose design invites the gaming it then experiences. The capability that was missing was the basic architectural choice to measure observations rather than declarations. The reform — SOFR — re-engineered the measurement at the architecture level.

LENS APPROACH

LENS uses LIBOR in LEN 4 as the canonical measurement-architecture case at financial-system scale and in LEN 7 for the cross- jurisdictional governance reform that produced SOFR. The case pairs with Wells Fargo (Case 70) and Volkswagen (Case 71) as measurement-gaming cases.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Anchor the benchmark in observed, settled transactions from the outset, so the figure cannot be set by anyone's declaration of what they would pay.
- Separate the parties who supply the input from the parties who profit from its level, eliminating the conflict of interest the panel design built in.
- Stress-test the instrument against an adversary who wants to move it: if a small misstatement shifts large contract value undetectably, redesign before fielding.

IN OPERATION / AFTER

- Audit submissions against independent transaction data so a divergence between what is reported and what actually trades surfaces the gaming early.
- Monitor submitter-trader communications and position-aligned shading as a leading indicator, rather than waiting for a subpoena to reveal it.
- Sustain the reform at the architecture level — keep the benchmark transaction-based — rather than relying on policing discretion that can quietly return.

REFLECTION QUESTIONS

1. Identify a benchmark in your domain that is constructed from declarations rather than observations. What gaming pressure does it experience?
2. Design the observation-based replacement for that benchmark.
3. The same panel banks set LIBOR and held positions that moved with it. Identify a measurement in your domain where the party supplying the figure benefits from its level, and design the separation that would remove the conflict.

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LENS COURSES LEN 4 LEN 7

CASE 84 EDUCATION AT SCALE GOVERNMENT 2009 – 2015

Atlanta Public Schools Cheating Scandal

G Governance & Trust N Normalization of Deviance

IMPACT ~180 educators implicated; 35 indicted; 11 convicted under RICO statute; foundational U.S. education measurement-gaming case

IN BRIEF

Under a celebrated superintendent, Atlanta Public Schools was held up nationally as a high-performing urban district. The performance was substantially fabricated. A state special investigation, supported by the Georgia Bureau of Investigation, and reporting by the Atlanta Journal-Constitution found that for several years educators across dozens of schools had systematically erased and corrected students' answers on state standardized tests. The cheating was organized — principals pressured teachers, staff held "erasure parties" after testing days — and the incentive system rewarded the gaming with bonuses and promotions. Around 180 educators were implicated, 35 indicted, and 11 convicted under Georgia's racketeering statute. The capability gap lay in the measurement architecture: the institution being measured operated the instrument that measured it, with no independent audit.

BACKGROUND

During the 2000s, Atlanta Public Schools under superintendent Beverly Hall was celebrated nationally for rapid gains on Georgia's high-stakes standardized tests. Bonuses, public recognition, and job security were tied directly to those scores, and the district administered and reported the very tests on which it was being judged. With the reward attached to the number and the number produced in-house, the people under pressure to improve the score also controlled the answer sheets that determined it.[0]

WHAT HAPPENED

Over several years, educators at dozens of APS elementary and middle schools systematically changed students' answers after testing. The cheating was organized rather than incidental: principals pressured teachers to hit targets, and staff held after-hours "erasure parties" to correct answer sheets. Suspiciously high rates of wrong-to-right erasures flagged the pattern — a statistical fingerprint the gaming left behind precisely because correcting a wrong answer to a right one is far rarer, by chance, than the reverse.[1]

THE INVESTIGATION

A 2009 Atlanta Journal-Constitution analysis of erasure data, followed by the state's Office of Student Achievement and a Governor-ordered special investigation supported by the Georgia Bureau of Investigation, documented the scheme in a 2011 report. Roughly 180 educators were implicated; 35 were indicted and 11 convicted under Georgia's RICO statute, the report finding the administration had emphasized results and praise "to the exclusion of integrity and ethics." That an investigation came from outside the district — a newspaper and state investigators, not the schools — is itself the mark of the missing independent check.[2]

THE CAPABILITY GAP

The capability gap was at the measurement architecture: the institution being measured also operated the instrument that measured it. High-stakes incentives without an independent audit are a textbook setup for Campbell's Law — the more a measure is used for decision-making, the more it will be corrupted. The students were not learning; the system was rewarding the appearance of learning, so the score rose while the underlying capability it was meant to certify went unmeasured and, for the children, unmet.[3]

AFTERMATH & REFORM

The convictions made APS the most prominent U.S. case of high-stakes-testing fraud and fuelled a broader reassessment of accountability-testing regimes. The structural lesson — that a measurement used for consequential decisions needs an audit independent of the institution being measured — is direct, but is still rarely implemented at scale, because the independent audit costs money and political will that the high score, once reported, makes seem unnecessary.[4]

180

EDUCATORS IMPLICATED · DOZENS OF SCHOOLS

the institution being measured operated the instrument that measured it

ATLANTA PUBLIC SCHOOLS — MEASUREMENT GAMING UNDER HIGH-STAKES TESTING

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THE LEARNING ENGINEERING LENS

> *Dr. Hall's administration emphasized test results and public praise to the exclusion of integrity and ethics.*
>
> GEORGIA SPECIAL INVESTIGATORS' REPORT ON ATLANTA PUBLIC SCHOOLS, 2011

LE INSIGHT

The APS cheating scandal is the strongest available case for measurement-gaming under high-stakes incentives in education. The capability gap was at the measurement architecture: the institution being measured operated the instrument that measured it. The reform pattern is direct — independent measurement audit — but rarely implemented at scale.

LENS APPROACH

LENS uses APS in LEN 4 as the foundational U.S. education measurement-gaming case and in LEN 7 for the incentive- architecture failure. Studio projects examine how an independent audit would have caught the cheating.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Separate test administration and scoring from the institution being judged, so the party with the incentive to inflate the number never controls the instrument that produces it.
- Engineer a statistical integrity check — such as wrong-to-right erasure-rate monitoring — into the measurement system before scores carry consequences.
- Anticipate Campbell's Law when attaching bonuses and job security to a metric, and design the audit as part of the incentive, not as an afterthought.

IN OPERATION / AFTER

- Run independent, ongoing audits of the high-stakes measure rather than relying on a newspaper or state investigators to surface fraud after years.
- Monitor for the appearance of learning diverging from independent evidence of learning, so the gap is caught while the score still means something.
- Sustain the audit's funding and authority against the complacency a high reported score creates, since the cost of checking looks unnecessary exactly when it is most needed.

REFLECTION QUESTIONS

1. Identify a high-stakes measurement in your domain whose audit is operated by the institution being measured. What would change with independent audit?
2. Apply Campbell's Law to a current high-stakes measurement system. What distortion is predicted?
3. The erasure-rate analysis caught APS only after the fact. Design a continuous integrity signal — a statistical fingerprint of gaming — that would flag the pattern as it emerged rather than years later.

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LENS COURSES LEN 4 LEN 7

CASE 85 GOVERNMENT TECHNOLOGY 1992 – 2008

Bernard Madoff / SEC Failures

G Governance & Trust K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT ~$65B Ponzi scheme — the largest in history; SEC repeatedly investigated and cleared Madoff; foundational regulator-capability case

IN BRIEF

Bernard Madoff ran the largest Ponzi scheme in history — roughly $65 billion in fictitious account value — for years while the SEC repeatedly investigated and cleared him. Financial analyst Harry Markopolos delivered the agency a detailed memo in 2005, “The World’s Largest Hedge Fund is a Fraud,” showing that Madoff’s steady returns were mathematically impossible. The SEC opened an investigation and concluded no action was warranted. Madoff operated until his sons turned him in during the 2008 crisis, when redemptions became impossible to honor. The SEC Inspector General later found the staff assigned lacked the expertise to evaluate Markopolos’s arguments and had deferred to Madoff’s industry stature. The capability gap was at the regulator’s technical-evaluation pipeline, not at the evidence — which was specific and checkable.

BACKGROUND

Bernard Madoff was a former NASDAQ chairman and a respected Wall Street figure whose investment arm reported remarkably steady returns for years. Those returns were entirely fictitious: client funds were never invested, and earlier investors were paid with later investors’ money — a Ponzi scheme that eventually represented some $65 billion in fabricated account value. The very steadiness that reassured investors was the tell: real markets do not deliver an almost unbroken line of gains, and the absence of the normal volatility was itself evidence the numbers were manufactured.[0]

WHAT HAPPENED

Financial analyst Harry Markopolos concluded by analysis that Madoff’s returns were mathematically impossible and delivered the SEC a detailed memorandum in 2005 titled “The World’s Largest Hedge Fund is a Fraud.” The SEC opened an investigation and concluded that no enforcement action was warranted. The warning was not a vague suspicion but a quantitative case any competent reviewer could in principle retrace, which is what makes the dismissal so telling. Madoff continued operating until December 2008, when the financial crisis made redemptions impossible and his sons reported him.[1]

THE INVESTIGATION

The SEC Office of Inspector General’s 2009 report found the agency had received credible, specific complaints across more than a decade and “missed numerous opportunities” to uncover the fraud. The staff assigned to Madoff lacked the expertise to evaluate Markopolos’s technical arguments, and the institutional culture had defaulted to treating

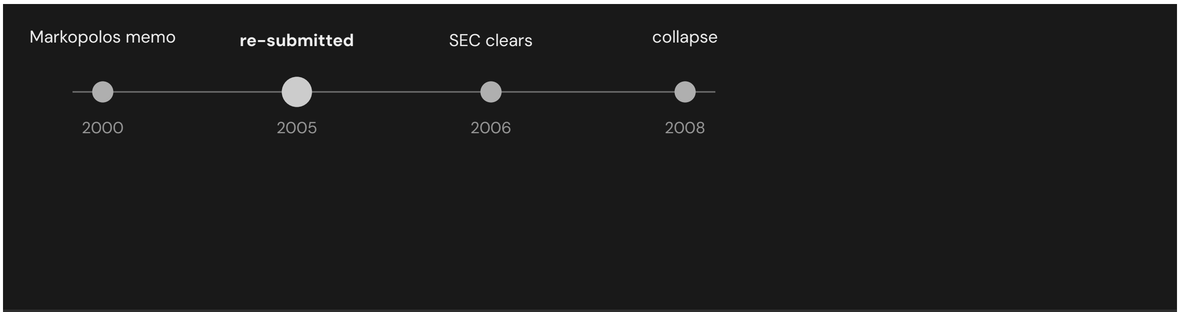
Madoff as a respected industry figure rather than following the evidence. Stature stood in for analysis: the reviewers weighed who Madoff was instead of whether the math could be true, and the reputational halo did the work that scrutiny should have.[2]

THE CAPABILITY GAP

The complaint was specific; the math was checkable; the institution simply did not have the people to check it. The capability gap was at the regulator's technical-evaluation pipeline, not at the evidence. A regulator whose technical depth lags the entities it oversees cannot act on even a correct and well-documented warning — and because the gap is in the evaluator rather than the tip, more tips would not have helped; the agency could not have used them.[3]

AFTERMATH & REFORM

The collapse wiped out tens of thousands of investors and prompted SEC reforms to its handling of tips and referrals, its examination procedures, and its recruitment of staff with quantitative and trading expertise, including the creation of an Office of Market Intelligence. Each reform addressed a different part of the same pipeline — getting the tip in, getting it to someone who could read it, and having someone who could — so that a future Markopolos memo would meet an evaluator able to test it. Madoff is paired with Theranos (Case 69) as a case in which a regulator lacked the technical capability to challenge the evidence in front of it.[4]



MADOFF — THE REGULATOR RECEIVED THE EVIDENCE; THE REGULATOR LACKED THE STAFF TO EVALUATE IT

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THE LEARNING ENGINEERING LENS

> *The agency missed numerous opportunities to discover the fraud.*
>
> SEC OFFICE OF INSPECTOR GENERAL REPORT OIG-509, 2009

LE INSIGHT

The Madoff case is the canonical example of a regulator without the technical capability to evaluate the evidence it received. The complaint was specific. The math was checkable. The institution did not have the people to check it. The capability gap was at the regulator's expertise pipeline, not at the evidence.

LENS APPROACH

LENS uses Madoff in LEN 4 for technical-evidence evaluation and in LEN 7 for regulator-capability deliverables. The case pairs with Theranos (Case 69) as cases where regulators lacked the technical depth to challenge the evidence presented to them.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Staff the evaluation pipeline with quantitative and trading expertise matched to the entities being overseen, so a checkable claim meets someone able to check it.
- Build tip-and-referral triage that routes a specific, technical complaint to a qualified evaluator rather than letting the subject's reputation decide its fate.
- Require warnings to be assessed on their merits first, structurally separating the analysis of the math from any weighing of who the subject is.

IN OPERATION / AFTER

- Audit closed investigations for cases dismissed despite specific, checkable evidence, surfacing where the evaluator — not the evidence — was the failure point.
- Monitor for returns or results too smooth to be real as a leading indicator, treating the absence of expected volatility as a signal to investigate.
- Sustain the regulator's technical depth against the regulated industry's growing sophistication, since a depth gap quietly reopens the same blind spot over time.

REFLECTION QUESTIONS

1. Identify a regulator in your domain whose technical evaluation capability has not kept pace with the entities it regulates. What is the resulting blind spot?
2. Design the regulator-side technical-capability deliverable that should have allowed the SEC to evaluate the Markopolos memo on its merits.
3. The SEC weighed Madoff's stature over Markopolos's math. Design a triage process that forces a tip to be evaluated on its technical merits before the subject's reputation is allowed to enter the decision.

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LENS COURSES LEN 4 LEN 7

CASE 86 DEFENSE GOVERNMENT 1996 – 2001

9/11 Intelligence Sharing Failures

G Governance & Trust K Knowledge & Institutional Memory

IMPACT 2,977 killed; the 9/11 Commission found systemic intelligence-sharing failures across the U.S. government

IN BRIEF

The September 11, 2001 attacks, which killed 2,977 people, were enabled in part by intelligence the U.S. government already held but never integrated. The CIA knew, from a January 2000 meeting in Kuala Lumpur, that two future hijackers had entered the country; it did not tell the FBI. The FBI separately flagged suspicious flight-training activity in Phoenix and Minneapolis in 2001, but that information was never aggregated. Visa issuance, immigration tracking, and watch-listing were each run by a different agency, and the handoffs between them depended on individual initiative that no institution required. The 9/11 Commission called it a "failure of imagination" — a framing critics say understates the structural gap. The reform that followed built the cross-agency architecture, the ODNI and the NCTC, that had not existed.

BACKGROUND

By 2001, U.S. counterterrorism depended on many agencies — CIA, FBI, NSA, State, INS, FAA — each holding a piece of the picture. No institution was responsible for integrating those pieces; cross-agency information sharing depended on individuals choosing to pass information along rather than on any architecture that required it. Each agency's incentives, classification rules, and turf reinforced the boundary, so the natural tendency was to hold information rather than to push it across a line no one was charged with bridging.[0]

WHAT HAPPENED

On 11 September 2001, nineteen hijackers seized four aircraft and killed 2,977 people. In the months and years before, the warning signs had been distributed across the government: the CIA tracked two future hijackers from a January 2000 meeting in Kuala Lumpur but did not watch-list them or notify the FBI; FBI field offices flagged suspicious flight-training activity in Phoenix and Minneapolis; none of it was aggregated into a single picture. Any one fragment looked minor in isolation; only assembled would they have shown the shape of the plot, and assembly was exactly the function no one performed.[1]

THE INVESTIGATION

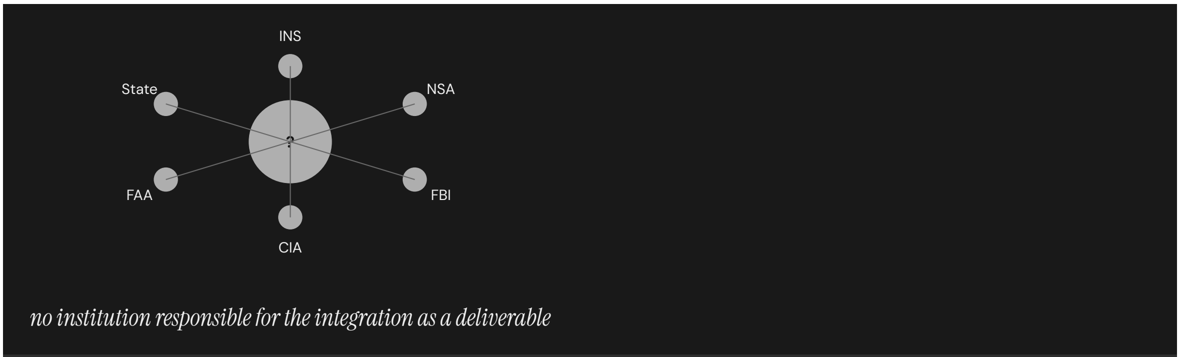
The 9/11 Commission and the earlier Congressional Joint Inquiry documented specific failures of information sharing across the FBI, CIA, and NSA. The Commission famously concluded that "the most important failure was one of imagination" — a framing later criticized as understating the structural nature of the gap, which was less a lack of imagination than an absence of any institution responsible for integration. The distinction matters for the remedy: a failure of imagination invites exhortation to think harder, while a structural gap demands an institution be built to close it.[2]

THE CAPABILITY GAP

The intelligence sharing did not happen because no institution owned it as a deliverable. Each agency was competent inside its own boundary; the integration across boundaries existed nowhere as a required function. The missing capability was an architecture — shared watch-lists, mandated handoffs, a body responsible for fusing the picture — rather than more raw collection. The government did not lack data; it lacked the connective tissue to turn data held in many places into a single picture anyone was accountable for assembling.[3]

AFTERMATH & REFORM

The Intelligence Reform and Terrorism Prevention Act of 2004 created the Office of the Director of National Intelligence and the National Counterterrorism Center, and fusion centers followed — the institutional architecture for cross-agency information sharing that had not previously existed. By creating bodies whose explicit mandate was integration, the reform converted information sharing from an act of individual initiative into a required function someone owned. The case is foundational in U.S. national-security policy for treating cross-agency capability as an engineerable institutional deliverable.[4]



9/11 — CROSS-AGENCY INFORMATION SHARING AS MISSING ARCHITECTURE

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THE LEARNING ENGINEERING LENS

> *The most important failure was one of imagination.*
>
> THE 9/11 COMMISSION REPORT, 2004

LE INSIGHT

9/11 is the foundational U.S. case for cross-agency information- sharing as an engineering deliverable. The architecture had not been built. The reform built it. The cost of the missing architecture was paid in 2001. The discipline LENS represents is the kind of work that, applied across the U.S. intelligence community in the 1990s, would have produced the architecture earlier.

LENS APPROACH

LENS uses 9/11 in LEN 8 as the foundational case for cross- organizational capability and in LEN 7 for the governance architecture of multi-agency systems. Studio projects compare 9/11 with Eagle Claw (Case 46) as institutional-architecture failures of different kinds.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Assign integration as an explicit deliverable owned by a named body, so the function does not depend on individuals choosing to pass information across a boundary.
- Engineer mandated handoffs and shared watch-lists into the architecture, making cross-agency sharing a required function rather than an act of initiative.
- Design for fusion of fragments that look minor in isolation, so the system is built to assemble the picture no single holder can see.

IN OPERATION / AFTER

- Audit cross-boundary information flows for reliance on individual initiative, and treat any flow that depends on goodwill as an unowned failure mode.
- Monitor whether the integrating body actually receives and fuses what the component agencies hold, rather than assuming the mandate guarantees the practice.
- Sustain the integration architecture against the agency incentives, classification rules, and turf that pull information back behind boundaries over time.

REFLECTION QUESTIONS

1. Identify a cross-organizational information flow in your domain that depends on individual initiative rather than institutional architecture. What is the foreseeable failure mode?
2. Design the institutional deliverable that would have produced ODNI-level information sharing across U.S. intelligence agencies in the 1990s.
3. The Commission called it a “failure of imagination”; critics called it a structural gap. Take a near-miss in your domain and argue which framing fits — and show how the remedy differs depending on which you choose.

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LENS COURSES LEN 7 LEN 8 LEN 3

CASE 87 HEALTHCARE 1999 – 2004

Vioxx Withdrawal

G Governance & Trust D Designed Out

IMPACT Tens of thousands of excess cardiovascular events estimated; Merck withdrew Vioxx in 2004; ~$4.85B settlement

IN BRIEF

Merck's painkiller Vioxx (rofecoxib) was approved by the FDA in 1999 and prescribed to millions. A 2000 trial, VIGOR, found a roughly five-fold higher rate of heart attacks in the Vioxx arm than in the comparison drug — a signal many read as a protective effect of the comparator rather than a cardiovascular risk of Vioxx. Not until the placebo-controlled APPROVe trial was halted in 2004 was the risk established and the drug withdrawn. For nearly four years, the disclosure architecture between Merck's internal data, FDA reviewers, and prescribers failed to surface the magnitude of the danger to a decision point; an FDA scientist estimated tens of thousands of excess cardiovascular events. The reforms that followed — REMS and the Sentinel Initiative — built post-market surveillance that had not existed.

BACKGROUND

Vioxx, a COX-2 inhibitor for arthritis pain, was approved by the FDA in 1999 and became one of the most widely prescribed drugs of its era. Detecting rare or delayed harms in a drug at that scale depends on post-market surveillance — aggregating adverse-event data after approval — a function that, at the time, was thin relative to the size of the exposed population. A risk too small to surface in a pre-approval trial becomes a large absolute toll once a drug reaches millions, which is precisely the regime that demands strong after-market monitoring.[0]

WHAT HAPPENED

The VIGOR trial, published in 2000, reported about a five-fold higher rate of myocardial infarction in patients taking Vioxx than in those taking naproxen. Merck and many readers interpreted the gap as naproxen being protective rather than Vioxx being harmful. That reading was not absurd — it was the more comfortable of two explanations for the same numbers — but it required the absent placebo arm to be true, and the data could not adjudicate between them. The placebo-controlled APPROVe trial, terminated in September 2004, established the cardiovascular risk directly, and Merck withdrew the drug.[1]

THE INVESTIGATION

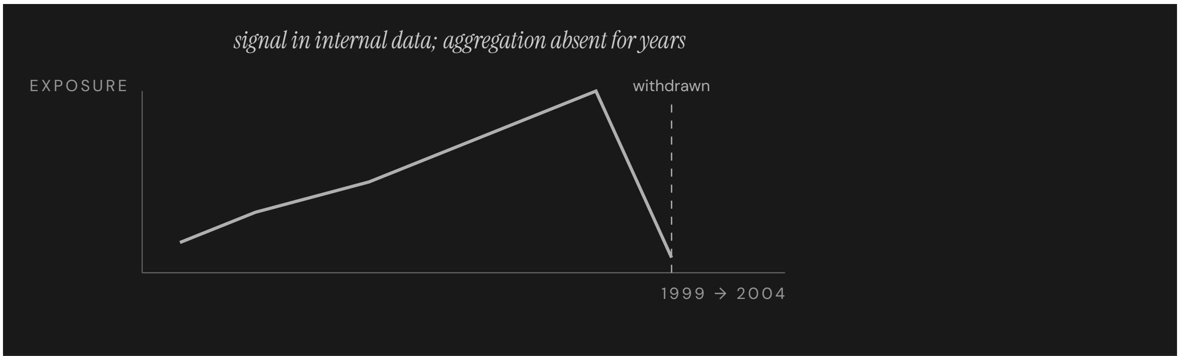
Congressional hearings and subsequent analysis found that signals of cardiovascular harm had been present in the trial record for years before withdrawal. FDA scientist David Graham testified that the risk had been visible well before 2004 and estimated tens of thousands of excess heart attacks and sudden cardiac deaths; a 2005 Lancet analysis quantified the population-level harm. The harm was not hidden in some unmeasured corner — it sat in the trial record the whole time, waiting for an architecture that would carry it to a decision rather than leave it to interpretation.[2]

THE CAPABILITY GAP

The capability gap was in post-market surveillance: the disclosure architecture between manufacturer-held data, FDA reviewers, and prescribers did not aggregate the signal to a decision boundary for nearly four years. The FDA's adverse-event reporting system relied on voluntary submissions structured to minimize signal, and was not adequate to the volume of the drug's distribution. A system that waits for a clinician to choose to file a report will always lag a harm spread thinly across millions of prescriptions, because no single prescriber sees enough of the pattern to recognize it.[3]

AFTERMATH & REFORM

Merck eventually settled litigation for about $4.85 billion. The case drove the creation of Risk Evaluation and Mitigation Strategies (REMS) and, in 2008, the FDA's Sentinel Initiative — an active post-market surveillance system that queries health-data partners rather than waiting for voluntary reports. By going out to the data instead of waiting for it to arrive, the reform inverted the logic that had let the signal sit unaggregated for years. The reform built the surveillance infrastructure that had not previously existed.[4]



VIOXX — POST-MARKET SURVEILLANCE AS A MISSING DELIVERABLE

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THE LEARNING ENGINEERING LENS

> *The cardiovascular risk was visible in Merck's internal data years before the drug was withdrawn.*
>
> PARAPHRASING DAVID GRAHAM (FDA OFFICE OF DRUG SAFETY), SENATE FINANCE COMMITTEE TESTIMONY, NOVEMBER 2004

LE INSIGHT

Vioxx is the canonical pharmaceutical case for post-market surveillance as a capability deliverable. The signal existed; the institutional architecture to aggregate it did not. The reform pattern — Sentinel — built the architecture. The case is the drug-industry analog of the EHR case (Case 25): a measurement architecture too thin for the system that depended on it.

LENS APPROACH

LENS uses Vioxx in LEN 4 for post-market surveillance as a measurement deliverable and in LEN 7 for the corporate-regulator dynamics that allowed signal aggregation to be deferred.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Field an active post-market surveillance architecture sized to the exposed population before scale-up, so a harm spread thinly across millions can still aggregate to a decision boundary.
- Design trials and data collection to adjudicate between competing readings of a signal, rather than leaving an ambiguous result to the more comfortable interpretation.
- Treat signal-aggregation as a defined deliverable carrying evidence from manufacturer data through reviewers to prescribers, not a byproduct of voluntary reporting.

IN OPERATION / AFTER

- Audit the trial record and adverse-event data for signals already present but never carried to a decision, closing the lag between visibility and action.
- Monitor by querying health-data partners actively rather than waiting for voluntary reports that no single prescriber sees enough of the pattern to file.
- Sustain post-market surveillance funding against the commercial pressure to defer aggregation, since the cost of waiting compounds across the exposed population.

REFLECTION QUESTIONS

1. Identify a post-deployment surveillance architecture in your domain that is too thin for the scale of the system it monitors. What is the missing deliverable?
2. Design the Sentinel-equivalent post-market surveillance system for a new domain.
3. The VIGOR signal admitted two readings — harmful drug or protective comparator — and the data could not decide. Identify a measurement in your domain whose interpretation is underdetermined, and design the study or instrument that would force the question to a decision.

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LENS COURSES LEN 4 LEN 7

Chapter 8

The Paired Intervention

Technical change with cultural change.

The checklist alone did not work. The checklist plus the nurse's authority to enforce it did.

KEYSTONE ICU PROJECT, MICHIGAN, 2004

CASE 12 AVIATION 1981 – PRESENT

Crew Resource Management & CAST

T Training Gap H Human-System Interface N Normalization of Deviance

IMPACT 83% reduction in U.S. commercial aviation fatality risk (1998–2008); 95% reduction in fatalities per 100M passengers over 20 years

IN BRIEF

In March 1977, two 747s collided in fog at Tenerife and 583 people died — in part because a KLM flight engineer's twice-voiced doubt about the runway was overridden by his captain. The failure was not of skill but of the system by which skill in one seat reached another. Crew Resource Management, formalized by United Airlines in 1981, re-engineered the cockpit as a coordinated team: explicit communication protocols, named authority gradients, structured briefings. Twenty years later the Commercial Aviation Safety Team (CAST) added closed-loop analysis of operational data to find and fix hazards before they caused accidents. The pairing — cultural redesign plus continuous evidence — drove an 83% reduction in U.S. commercial-aviation fatality risk between 1998 and 2008, earning the 2008 Collier Trophy.

BACKGROUND

By the 1970s, accident investigations were repeatedly finding that crashes stemmed not from a lack of individual flying skill but from breakdowns in how a crew worked together. The 1977 Tenerife disaster — 583 dead — was the starkest example: a KLM flight engineer questioned, twice and indirectly, whether the runway was clear, and the senior captain dismissed him and continued the takeoff. The pattern that emerged was consistent across investigations — skilled crews failing not because anyone lacked competence, but because the crew's most senior voice could close off information held by a more junior one before it reached the decision.[0]

THE INTERVENTION

Crew Resource Management, formalized by United Airlines in 1981, treated crew coordination as an engineerable property of the system rather than a matter of personality. It introduced explicit communication protocols, named and trained against authority gradients, and instituted structured briefings and debriefings — rebuilding the cockpit as a team in which information from any seat could reach the decision. Rather than exhorting captains to listen better, it built coordination into the standard procedure itself, so the behavior that Tenerife had lacked became the trained default rather than a matter of individual temperament.[1]

HOW IT WORKED

CRM did not teach individual airmanship; it engineered the system of interaction in which airmanship operates. By making it legitimate and expected for a junior officer to challenge a captain, and by giving crews a shared protocol for doing so, it closed the path by which

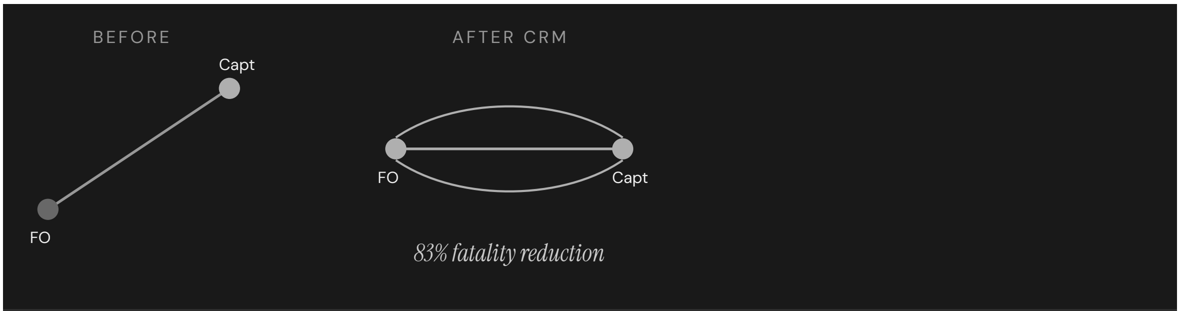
Tenerife-style deference had absorbed safety-critical information instead of transmitting it. The change was structural rather than attitudinal: a challenge that had once depended on a junior officer's nerve now had a named procedure behind it, so the same doubt that went unheard at Tenerife had an authorized route to the decision.[2]

THE EVIDENCE

Twenty years on, the Commercial Aviation Safety Team added the missing second half: closed-loop hazard identification on operational data, prioritized enhancements, tracked implementation, and measured outcomes. CAST set an 80% fatality-reduction target and exceeded it, reaching 83% by 2007 — work recognized with the 2008 Collier Trophy. Over twenty years, fatalities per 100 million passengers fell roughly 95%. The loop closed on itself: data surfaced the next hazard, enhancements were prioritized against it, and the measured outcome fed back into the priorities, so improvement continued rather than plateauing once the cultural change had taken hold.[3]

WHAT TRANSFERRED

CRM and CAST together define what a mature capability-engineering apparatus looks like: a cultural redesign paired with a continuous-evidence loop, where neither half works alone. The model has been exported to surgery, firefighting, and other high-consequence domains, and is now the template for redesigning human roles in AI-augmented systems. What transferred was not the specific protocols but the design logic itself — that coordination is an engineerable system property and that the cultural change needs a measurement loop to keep it honest over time.[4]



AUTHORITY GRADIENT FLATTENED; COMMUNICATION CLOSES THE LOOP

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THE LEARNING ENGINEERING LENS

> *CRM succeeded because it treated crew coordination as an engineerable property of the system.*
>
> PARAPHRASING HELMREICH & FOUSHEE, IN KANKI ET AL., CREW RESOURCE MANAGEMENT (2010)

LE INSIGHT

CRM is the canonical evidence that capability is engineerable at the system level, not just the individual. Tenerife was not solvable by hiring better pilots — only by changing the authority structure inside the cockpit. The intervention worked because it was **paired**: a procedural change plus a cultural change in how the procedure was authorized. Either alone fails.

LENS APPROACH

LENS treats CRM/CAST as the foundational success case across the curriculum. LEN 1 uses it as the problem-framing exemplar; LEN 4 uses CAST as the model closed-loop evidence system; LEN 2 uses CRM as the template for redesigning human roles in automated environments — the logic now being applied to AI-augmented systems.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Build the coordination behavior into standard procedure — explicit communication protocols, named authority gradients, and structured briefings — so it is the trained default, not a matter of temperament.
- Authorize the junior-to-senior challenge explicitly and drill it, so safety-critical information has a procedural route to the decision rather than depending on individual nerve.
- Stand up the evidence half from the start: instrument operational data so hazards can be found and prioritized before they cause an accident.

IN OPERATION / AFTER

- Run the closed loop continuously — surface hazards from data, prioritize enhancements, track implementation, and measure outcomes — so improvement does not plateau once the culture shifts.
- Set and exceed an explicit reduction target (the CAST model) so the intervention is judged against a measured outcome, not against its own activity.
- Export the design logic, not just the protocols, when transferring to new domains, adapting the paired cultural-plus-evidence structure to each setting.

REFLECTION QUESTIONS

1. Identify a high-consequence domain whose authority gradient absorbs information rather than transmitting it. What would the CRM equivalent intervention look like there?

2. CRM is paired with CAST. What is the closed-loop evidence system in your domain — and if there is not one, what would it cost to build?

3. CRM made a junior officer's challenge a named procedure rather than an act of nerve. Identify a moment in your domain where the right information depends on someone's courage, and design the protocol that would make raising it the expected default instead.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs · incident analysis · safety dashboards

LENS COURSES LEN 1 LEN 4 LEN 2 LEN 3

CASE 14 HEALTHCARE 2004 – PRESENT

Keystone ICU / Pronovost Checklist

T Training Gap N Normalization of Deviance

IMPACT Central-line-associated bloodstream infections (CLABSI) reduced to near zero across 103 Michigan ICUs; ~1,500 lives saved in 18 months; ~$100M saved; sustained at ten years

> IN BRIEF
>
> Peter Pronovost's central-line checklist has five items — wash hands, clean the skin with chlorhexidine, drape the patient, use full barrier precautions, apply a sterile dressing — and not one of them was unknown to the physicians it governed. The question was never what to do, but whether it would be done every time. The Keystone project, launched across 103 Michigan ICUs in 2004, paired the checklist with a cultural change: nurses were not merely permitted but required to stop the procedure if a step was skipped. That authorization was the load-bearing element. Central-line-associated bloodstream infections fell to near zero, an estimated 1,500 lives and $100 million were saved in eighteen months, and the effect was sustained at ten years.

BACKGROUND

Central-line-associated bloodstream infections were a common, often fatal complication of intensive care, and the steps to prevent them were well established and uncontroversial. The problem was reliability: in the existing culture, a nurse who saw a physician skip a sterile step had no procedural path to intervene without crossing the hospital's authority gradient. The knowledge was universal and the steps cheap; what was missing was any mechanism that made the right action happen every time rather than most of the time, which is precisely where a fatal infection finds its opening.[0]

THE INTERVENTION

In 2004, Peter Pronovost's team launched the Keystone ICU project across 103 Michigan units. It combined a simple five-item central-line checklist — hand hygiene, chlorhexidine skin prep, full-barrier draping, sterile gown-mask-gloves, and a sterile dressing — with an explicit authorization: nurses were required, not merely permitted, to stop any procedure in which a step was missed. The distinction between permitted and required was deliberate — a permission a nurse might decline to exercise against a senior physician became an obligation the institution stood behind, removing the personal risk of intervening.[1]

HOW IT WORKED

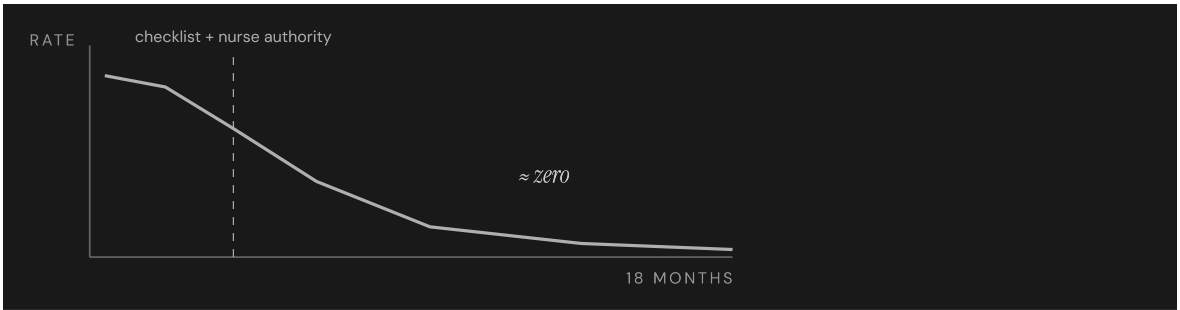
The checklist was the technical half; the nurses' enforcement authority was the cultural half, and it was the load-bearing one. Before Keystone, the path to intervene did not exist; after it, the path was institutional and expected. The pairing converted knowledge everyone already had into behavior that happened every time. The checklist gave the nurse an objective, impersonal basis for the stop — a missed item, not a judgment about the physician — which is what made the authority usable in practice rather than merely on paper.[2]

THE EVIDENCE

Across the Michigan ICUs, the median quarterly CLABSI rate fell to zero, and the state's units outperformed 90% of ICUs nationwide. The program was estimated to have saved roughly 1,500 lives and $100 million within eighteen months. Results were published in the New England Journal of Medicine in 2006, and follow-up showed the effect sustained at ten years. The durability mattered as much as the size of the drop: an improvement that survives a decade is evidence the change was built into the system's structure rather than riding on the initial enthusiasm of a single project.[3]

WHAT TRANSFERRED

Keystone became the clearest evidence in healthcare that a technical intervention without an authority change produces no durable improvement — and vice versa. The model was packaged as the AHRQ CUSP toolkit, adopted in more than forty states, and replicated internationally, establishing the design principle of intervening in matched technical-and-cultural pairs. Packaging the approach as a reusable toolkit was itself part of what transferred — it turned a single project's success into something other institutions could adopt without rediscovering the load-bearing role of the authority change.[4]



BLOODSTREAM-INFECTION RATE ACROSS 103 MICHIGAN ICUS

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THE LEARNING ENGINEERING LENS

> *The checklist was the technical intervention. The nurses' authority to enforce it was the cultural intervention. Neither worked without the other.*
>
> PARAPHRASING PRONOVOST & VOHR, SAFE PATIENTS, SMART HOSPITALS, 2010

LE INSIGHT

Keystone is the clearest evidence in healthcare that a technical intervention without authority intervention produces no durable change, and authority intervention without a technical artifact produces no measurable change. Both are necessary. The empirical record of Keystone is the strongest available argument for designing interventions as **pairs** — and treating the cultural half as engineering, not aspiration.

LENS APPROACH

LENS uses Keystone in LEN 4 and LEN 10 as the canonical worked example of measurement linked to intervention. Studio projects require students to specify both halves of the pair, name the authority that authorizes the cultural half, and identify the measurement signal that will confirm or refute the effect. The course treats "is the cultural change actually authorized?" as falsifiable, not stated.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pair the technical artifact with the authority change from the outset — a checklist plus the explicit, institution-backed right of a junior member to stop the procedure when a step is missed.
- Make the intervention an obligation rather than a permission, so the corrective action does not depend on an individual's willingness to challenge a senior colleague.
- Anchor the stop to an objective, impersonal trigger (a missed checklist item) so the authority is usable without it reading as a judgment of the more senior operator.

IN OPERATION / AFTER

- Measure the outcome directly (the CLABSI rate) and publish it, so the cultural half can be confirmed as authorized in practice rather than merely declared.
- Track the effect over years, not months, to confirm the change is built into the system's structure rather than riding on a project's initial enthusiasm.
- Package the paired design as a reusable toolkit so other institutions can adopt it without rediscovering the load-bearing role of the authority change.

REFLECTION QUESTIONS

1. Identify an evidence-based protocol in your domain that is **known** to work but is not consistently performed. What is the authority change required to pair with it?
2. Design the measurement signal that would confirm the cultural half of a Keystone-style intervention is actually being authorized — not merely declared.
3. Keystone made the nurse's stop an obligation the institution stood behind, not a personal risk. Identify a corrective action in your domain that currently costs the person who takes it, and design the institutional backing that would remove that cost.

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · incident analysis · safety dashboards

LENS COURSES LEN 4 LEN 10 LEN 5

CASE 16 ENERGY 1979 – PRESENT

INPO and the Nuclear Academy

T Training Gap K Knowledge & Institutional Memory G Governance & Trust

IMPACT No INES-level event at U.S. commercial reactors post-INPO; sustained improvement in INPO/WANO performance indicators across the industry

IN BRIEF

Three Mile Island did not produce a reactor accident at the next plant over — it produced an institution. Within months of the 1979 Kemeny Commission report, the U.S. commercial nuclear industry founded the Institute of Nuclear Power Operations on a stark premise: an accident at any single plant would threaten every operator's license, and no utility could engineer its safety capability alone. Funded by the utilities it evaluated and operating without statutory authority, INPO set training and certification standards, accredited every plant's programs through the National Academy for Nuclear Training, and ran peer evaluations in which operators from one utility scrutinized another's control rooms and records. The pre-TMI culture of complacency gave way to mandated vigilance. No U.S. commercial reactor has had a significant INES-level event since.

BACKGROUND

The 1979 partial meltdown at Three Mile Island exposed not just a plant-level failure but an industry with no shared mechanism for learning. The Kemeny Commission traced the accident in part to a pervasive "mindset" of complacency, in which each utility operated alone and no institution carried lessons from one plant to the rest. The structural problem sat above any single control room: a lesson learned at one plant had no path to the others, so the same latent failure could surface repeatedly across an industry that never compared notes.[0]

THE INTERVENTION

Within months of the Kemeny report, the utilities founded the Institute of Nuclear Power Operations. Its premise was that an accident anywhere threatened everyone's license to operate. INPO set training and certification standards for operators and supervisors, and in 1985 the National Academy for Nuclear Training began accrediting each facility's programs. The shared-exposure premise was what gave a body with no statutory power its teeth — every utility had a direct stake in every other utility's competence, because one failure could end the whole industry's license to operate.[1]

HOW IT WORKED

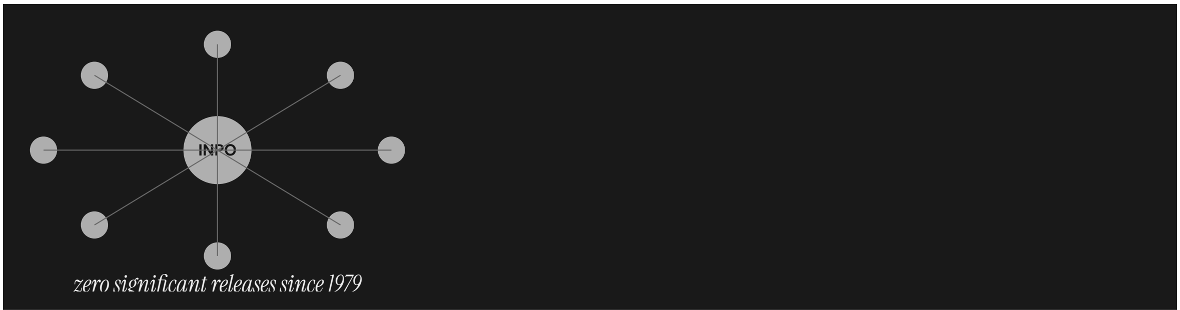
INPO's load-bearing mechanism was honest peer review: teams of operators from one utility examined another's procedures, control rooms, and incident records, reporting candidly because every utility was, in the title of one history, a hostage of the others. Funded by the utilities it evaluated and holding no statutory authority, INPO depended on shared catastrophic exposure to make its findings stick. Peer review by working operators rather than distant regulators also meant the examiners knew what to look for and the findings carried the weight of professional judgment, not just rule compliance.[2]

THE EVIDENCE

The post-TMI culture shifted from smugness to mandated vigilance, and U.S. commercial reactors have recorded no significant INES-level event since INPO's founding. Industry performance indicators tracked by INPO and its international counterpart WANO improved steadily and broadly across the fleet. The broad, steady improvement across the whole fleet — not just the strongest plants — is the signature of a learning mechanism working as designed: the laggards were pulled up by the same peer-review architecture that held the leaders to standard.[3]

WHAT TRANSFERRED

INPO is the strongest evidence in any domain that capability engineering can be undertaken at the level of an entire industry rather than a single organization. Its enabling conditions — shared catastrophic exposure, regulatory legitimacy, and an honest peer-review architecture — recur wherever one operator's failure can damage every operator, and it informed the founding of WANO after Chernobyl. That the model crossed national borders to WANO is itself evidence that the design is portable: the enabling conditions, not the particular American institution, are what make the mechanism work.[4]



INDUSTRY-WIDE PEER REVIEW: A LEARNING RING AFTER TMI

REFERENCES

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THE LEARNING ENGINEERING LENS

> *Every utility recognized that an accident at any single plant would affect every operator's license to operate.*
>
> PARAPHRASING THE INSTITUTIONAL ANALYSIS IN REES, HOSTAGES OF EACH OTHER, 1994

LE INSIGHT

INPO is the strongest evidence in any domain that capability engineering can be undertaken at the level of an **industry**, not just an organization. The conditions that made it possible — shared catastrophic exposure, regulatory legitimacy, an honest peer-review architecture — appear wherever a single failure can damage every operator.

LENS APPROACH

LENS uses INPO in LEN 8 as the canonical example of industry-level learning: students identify the structural conditions in their own domain that would permit (or block) an INPO-equivalent and design the peer-review architecture required. LEN 1 uses the founding moment — nine months after TMI — to discuss the **speed** a credible response to catastrophe demands.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Found the body on shared catastrophic exposure — make explicit that one operator's failure threatens every operator's license — so members have a direct stake in each other's competence.
- Set common training, certification, and accreditation standards across the industry rather than leaving each organization to learn alone.
- Staff peer review with working operators, not distant regulators, so examiners know what to look for and findings carry professional weight beyond rule compliance.

IN OPERATION / AFTER

- Track fleet-wide performance indicators and confirm the laggards are being pulled up, not just the leaders held to standard — the signature of a learning mechanism working.
- Sustain candid peer review by keeping the body funded by and accountable to its members, so the honest examination that makes it effective does not erode into formality.
- Export the enabling conditions rather than the institution when scaling (as INPO informed WANO), adapting the shared-exposure-plus-peer-review design to each new context.

REFLECTION QUESTIONS

1. What is the equivalent of "an accident at any single plant affects every operator" in your domain? If the answer is "nothing," what does that tell you?
2. INPO was stood up in nine months. Pick a current cross-organizational capability problem and write the nine-month deliverable that would constitute a credible response.
3. INPO held no statutory authority yet made its findings stick through shared catastrophic exposure and peer review. Design the non-statutory mechanism that could enforce a standard in your domain, and name the shared stake that would give it teeth.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · governance frameworks · bias & evidence audits

LENS COURSES LEN 1 LEN 8 LEN 3

CASE 13 HEALTHCARE 2008 – PRESENT

WHO Surgical Safety Checklist

T Training Gap N Normalization of Deviance

IMPACT Death rate 1.5% → 0.8%; complications down >33%; adopted by the majority of surgical providers worldwide

IN BRIEF

In 2008, Atul Gawande's team and the WHO introduced a single-page, nineteen-item surgical checklist applied at three junctures — before anesthesia, before incision, and before the patient leaves the operating room. Piloted across eight hospitals in eight countries, from Toronto to Tanzania, it nearly halved the surgical death rate (1.5% to 0.8%) and cut serious complications by more than a third — results published in the NEJM in 2009. The artifact was the checklist; the intervention was the system of forced pauses it created, three moments when a moving team had to stop and confirm shared state. A later Ontario study found no mortality benefit after a mandated rollout, surfacing the fidelity question: the artifact works only when the institution authorizes its honest use.

BACKGROUND

Surgical harm — wrong-site operations, retained instruments, missed allergies, post-operative infection — was widespread across health systems rich and poor, and much of it stemmed not from a lack of skill but from teams that never paused to confirm shared understanding before acting. The knowledge to prevent these events existed; the reliable practice did not. A surgical team in motion rarely stopped to verify that everyone held the same picture of the patient and procedure, so a mismatch any member could have caught passed silently into the operation.[0]

THE INTERVENTION

In 2008, the WHO and a Harvard team led by Atul Gawande introduced a single-page, nineteen-item Surgical Safety Checklist applied at three critical junctures: before anesthesia, before skin incision, and before the patient leaves the operating room. The team piloted it across eight hospitals in eight countries spanning very different economic and operational conditions. Testing the same artifact from Toronto to Tanzania was deliberate — it had to work in settings with vastly different resources to demonstrate that the gain came from the forced pause itself, not from any one system's wealth.[1]

HOW IT WORKED

The checklist was the artifact; the intervention was the system of pauses it imposed. At each juncture a team that would otherwise keep moving had to stop, look at one another, and confirm names, the procedure, allergies, and equipment aloud. The pauses were the load-bearing element — the requirement to halt and speak mattered more than the specific list of items. Saying the confirmations aloud to one another, rather than each member assuming them privately, is what surfaced the mismatches a moving team would otherwise have carried into the incision.[2]

THE EVIDENCE

The 2009 NEJM study reported the surgical death rate falling from 1.5% to 0.8% and major complications dropping by more than a third across all eight sites — confirming Gawande's framing that "gaps in teamwork and safety practices in surgery are substantial in countries both rich and poor." The checklist was subsequently adopted by the majority of surgical providers worldwide. That the death rate roughly halved across all eight sites, despite their differences in wealth and practice, is what made the result so persuasive — the effect tracked the intervention rather than the setting.[3]

WHAT TRANSFERRED

A 2014 Ontario population study found no statistically significant mortality reduction after province-wide mandated adoption — a result that did not refute the checklist so much as illustrate its dependence on implementation fidelity. Where a mandate replaced genuine authorization of the pause, the measured effect attenuated, making the checklist a paired lesson in both minimal-artifact design and the limits of mere compliance. The contrast between the pilot and the mandated rollout is the lesson: when the pause was genuinely used it worked, and when it became a box to check before proceeding the same paper produced nothing.[4]

THREE PAUSE POINTS ACROSS THE SURGICAL TIMELINE

REFERENCES

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THE LEARNING ENGINEERING LENS

> *Gaps in teamwork and safety practices in surgery are substantial in countries both rich and poor.*
>
> ATUL GAWANDE, HARVARD GAZETTE, JANUARY 2009

LE INSIGHT

The Surgical Safety Checklist is the canonical evidence that a tiny artifact — one page, nineteen items — can produce population-scale effects when paired with the structural change of **requiring a pause.** The checklist alone is paper. The pause alone is anxiety. Together they constitute the smallest effective capability intervention in the dataset. The Ontario follow-up underscores the secondary requirement: the artifact carries the effect only when the institution actually enforces the pause. Where mandate replaced authorization, the measured effect attenuated.

LENS APPROACH

LENS uses the WHO checklist in LEN 10 (capstone) as the canonical end-to-end design exemplar: a problem identified, a minimal artifact prototyped, a multi-site pilot, a measurement regime, and global scale-out. LEN 4 examines the measurement architecture that made the effect provable.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Design the minimal artifact — a single-page checklist — around a small number of forced pauses at the highest-risk junctures, keeping the list short enough to be used every time.
- Make the confirmations spoken aloud and shared across the team, so the pause surfaces mismatches rather than letting each member assume them privately.
- Pilot across deliberately varied sites so the measured gain can be shown to track the intervention rather than any one setting's resources.

IN OPERATION / AFTER

- Measure outcomes (death and complication rates) directly so the effect can be confirmed and the artifact is not assumed to work merely because it is in use.
- Guard implementation fidelity as the scale grows — a mandate that turns the pause into a box to check reproduces the artifact without the effect.
- Build a governance signal that distinguishes genuine, authorized use from compliance check-off, so the fidelity gap is visible before a rollout is judged a failure.

REFLECTION QUESTIONS

1. What is the smallest possible capability artifact in your domain that, paired with a required pause, would shift outcomes?
2. The WHO checklist was studied across eight countries. Design the multi-site evaluation that would establish whether your candidate intervention generalizes.
3. The Ontario mandated rollout produced no measurable mortality reduction. What governance signal would have surfaced the fidelity gap between authorized use and compliance check-off before the trial was declared a failure?

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TOOLS scenario simulation · competency models · incident analysis · safety dashboards

LENS COURSES LEN 4 LEN 10

CASE 15 DEFENSE 2018 – PRESENT

Navy Surface Warfare Readiness Reform

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Tripled ship-driving training hours; 10 pass-or-fail career assessments; Ready-for-Sea Assessments — 3 of 18 forward-deployed ships immediately sidelined

IN BRIEF

After the fatal 2017 Fitzgerald and McCain collisions (Case 1) exposed gutted seamanship training, the U.S. Navy overhauled how it builds surface-warfare competence. It restored the Surface Warfare Officers School from CD-ROM self-study to classroom and simulator instruction, stood up Maritime Skills Training Centers on both coasts, created ten pass-or-fail career assessments — three of them no-go gates — and adopted aviation-style debriefing. New Ready-for-Sea Assessments evaluated forward-deployed ships against a deliverable standard; three of the first eighteen were immediately sidelined. The structural change is real and the investment substantial. What is missing is the third half: the GAO has noted the Navy still lacks systematic evaluation of whether the reforms work — a live success in structure, with evidence of effect still outstanding.

BACKGROUND

In 2017, the destroyers USS Fitzgerald and USS John S. McCain suffered fatal collisions (Case 1) that investigations traced in part to seamanship and navigation training degraded to CD-ROM self-study. The Navy faced a clear capability gap: officers were going to sea without the hands-on ship-driving competence the job required. The diagnosis was unusually unambiguous — two avoidable collisions in one year pointing at the same eroded fundamentals — which is what converted a long-tolerated shortfall into a mandate for structural change.[0]

THE INTERVENTION

Beginning in 2018, the Navy restored the Surface Warfare Officers School from self-study to classroom-plus-simulator instruction, established Maritime Skills Training Centers on both coasts, roughly tripled ship-driving training hours, and created ten pass-or-fail assessments across an officer's career path — three of them no-go gates that can halt advancement. The no-go gates were the structural teeth: by tying advancement to demonstrated competence rather than time served, they made the qualification something the system would stop on, not merely something it recorded.[1]

HOW IT WORKED

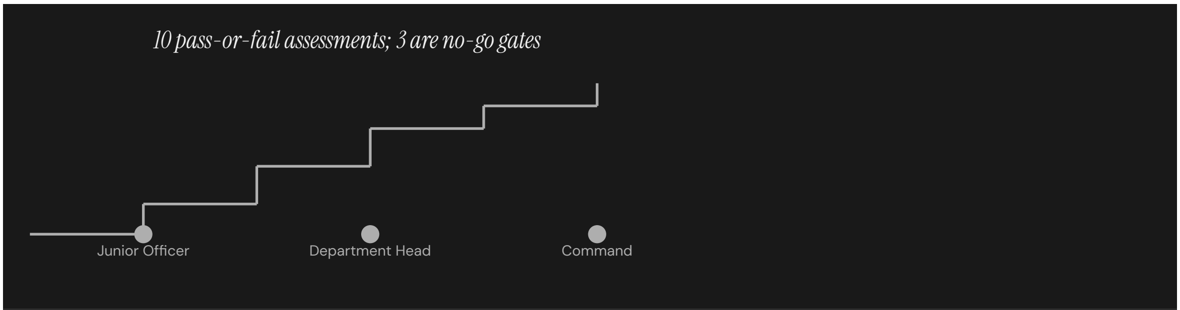
The reform paired technical investment — simulators, restored curricula, qualification gates — with a cultural change borrowed from aviation: structured debriefing and explicit gate ownership. New Ready-for-Sea Assessments evaluated forward-deployed ships against a deliverable standard rather than a paper one; three of the first eighteen ships assessed were immediately sidelined as not ready. That a sixth of the first cohort failed against the new standard showed the assessment had teeth — it measured demonstrated readiness rather than accepting the paper certifications that had masked the pre-collision gap.[2]

THE EVIDENCE

Here the case turns instructive. The Government Accountability Office has noted that the Navy still lacks systematic evaluation of whether the reforms actually improve readiness. The structural intervention happened and the investment was large, but decision-grade evidence on outcomes is incomplete — the measurement to confirm the effect has lagged the change itself. Without an outcome time-series, the Navy cannot yet distinguish a real capability gain from activity that looks like one, which is the same blindness, in milder form, that let the pre-collision gap go unseen.[3]

WHAT TRANSFERRED

The reform is a live, in-progress success: structural change real, evidence of effect outstanding. As a teaching case it argues that mature capability engineering must build the measurement infrastructure from the start — the time-series that lets an institution know whether the capability it bought is materializing — rather than treating evaluation as an afterthought. The two halves the reform did deliver, technical investment and a cultural change, are necessary but not sufficient; the missing third half is the evidence regime that would let the Navy prove, not assume, the gap has closed.[4]



POST-FITZGERALD REFORM: TRIPLED TRAINING; PASS-OR-FAIL GATES

REFERENCES

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THE LEARNING ENGINEERING LENS

> *The Navy still lacks systematic evaluation of whether the reforms work.*
>
> PARAPHRASING GAO-21-168 ON NAVY READINESS REFORM, 2021

LE INSIGHT

The Navy reform is a paired intervention in progress: technical (training restored, simulators procured, assessments created) plus cultural (debriefing, gate ownership). What it lacks is the third half: evidence that the intervention has worked. LENS treats this as a teachable case for what mature capability engineering should include from the start — the measurement infrastructure that lets the institution know whether its investment is producing the capability it bought.

LENS APPROACH

LENS uses Navy SWO reform in LEN 10 (capstone) as a worked exercise in capability intervention at scale, and in LEN 4 as a case where the measurement infrastructure has lagged the intervention. Students design the evidence regime that should accompany the reform.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pair the technical investment — restored curricula, simulators, training hours — with a cultural change such as structured debriefing and explicit gate ownership, rather than buying tools alone.
- Install no-go gates that tie advancement to demonstrated competence, so qualification is something the system will stop on, not merely something it records.
- Assess units against a deliverable standard from the first cohort, accepting that a real test will sideline some — that failure rate is the evidence the gate has teeth.

IN OPERATION / AFTER

- Build the measurement infrastructure from the start — the outcome time-series that lets the institution distinguish a real capability gain from activity that merely looks like one.
- Give the senior commander a readiness dashboard that reports demonstrated capability over time, so the same blindness that masked the pre-collision gap cannot recur.
- Treat evidence of effect as the third half of the intervention, not an afterthought, so the reform can be proven rather than assumed to have closed the gap.

REFLECTION QUESTIONS

1. Navy SWO reform happened without an evidence regime to confirm it worked. Identify a current reform in your domain and the evidence regime that should accompany it.
2. Design the dashboard the Chief of Naval Operations should have to know whether SWO capability is improving in time-series.
3. The Ready-for-Sea Assessment sidelined three of the first eighteen ships because it measured readiness against a deliverable standard. Identify a certification in your domain that ratifies rather than tests, and design the gate that would be willing to fail a sixth of its first cohort.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

LENS COURSES LEN 4 LEN 10 LEN 5

CASE 23 AVIATION 2000 – PRESENT

Korean Air Safety Transformation

T Training Gap N Normalization of Deviance

IMPACT From industry pariah (16 aircraft written off, 700+ lives lost, 1970–1999) to spotless passenger safety record since 1999

IN BRIEF

Between 1970 and 1999, Korean Air had one of the worst safety records in commercial aviation — a loss rate roughly seventeen times United's, more than 700 lives lost — and after the 1997 Guam crash of Flight 801 killed 228, the president of South Korea called it "an embarrassment to the nation." The NTSB traced root causes to cockpit authority gradients: junior officers were culturally unable to challenge a captain's errors. In 2000, Korean Air brought in Delta's David Greenberg to rebuild flight operations — mandating English as the cockpit language, adapting CRM for a high-power-distance culture, and bringing in outside consulting from Boeing and Delta. The airline has had no fatal passenger accident since, winning the 2006 Phoenix Award. Cultural legacy is not destiny when it is deliberately redesigned.

BACKGROUND

From 1970 to 1999, Korean Air suffered repeated fatal crashes — a loss rate roughly seventeen times United Airlines', with over 700 lives lost. The NTSB's investigation of the 1997 Guam crash of Flight 801, which killed 228, identified steep cockpit authority gradients as a root cause: junior officers were culturally unable to challenge a captain's erroneous decisions in time. The pattern echoed Tenerife's lesson but ran deeper, rooted in a national hierarchy the crew carried into the cockpit, so the silence in the right seat was not a lapse of training but an artifact of how rank was expressed in speech.[0]

THE INTERVENTION

After foreign carriers suspended code-shares and the FAA downgraded South Korea's safety rating, Korean Air in 2000 hired David Greenberg from Delta to overhaul flight operations. The interventions were deliberate and cultural: mandated English fluency for all pilots, CRM training adapted for a high-power-distance setting, external consulting from Boeing and Delta, and fleet modernization. The external commercial pressure — suspended code-shares and a downgraded rating — supplied the leverage to act, turning a cultural problem the airline had long lived with into one it could no longer afford to ignore.[1]

HOW IT WORKED

The load-bearing move operated on language. Making English the cockpit language stripped out the Korean honorific hierarchy that had silenced first officers, because English has no honorifics to enforce rank. The CRM adaptation then gave crews an explicit, culturally workable protocol for raising concerns — converting deference into communication. The choice was elegant precisely because it changed the medium rather than asking pilots to override their own culture: with no honorifics to encode, the language itself flattened the gradient that had absorbed the first officer's warnings.[2]

THE EVIDENCE

Korean Air has recorded no fatal passenger accident since the reforms took hold. Air Transport World recognized the turnaround with its 2006 Phoenix Award, and the carrier moved from international pariah to a safety record indistinguishable from the best global operators. The categorical nature of the shift — from a loss rate many times the industry's to none — is what makes the case persuasive: the same crews and the same national culture produced opposite outcomes once the cockpit's communication structure was redesigned.[3]

WHAT TRANSFERRED

Korean Air is the strongest aviation evidence that cultural legacy is not destiny: a specific cultural feature — high power distance in the cockpit — was the binding constraint, and once it was redesigned, the safety record changed categorically. The case also shows that the interface for cultural redesign can itself be engineered, in this instance through language. The transferable lesson is to locate the single cultural feature that actually binds the capability rather than attempting to remake a culture wholesale, then change the medium through which that feature operates.[4]

CULTURAL HIERARCHY REMOVED FROM THE COCKPIT

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THE LEARNING ENGINEERING LENS

> *Korean Air's record before 2000 was changed by an external intervention into cockpit culture, not by criticism of it.*
>
> EDITORS' SYNTHESIS OF NTSB FINDINGS ON KAL 801 AND THE KOREAN AIR TRANSFORMATION

LE INSIGHT

Korean Air is the strongest aviation evidence that cultural legacy is not destiny. A specific cultural feature — high power distance in the cockpit — was the binding capability constraint. Once it was redesigned, the safety record changed categorically. The intervention is also methodologically interesting because it operated on language: English became the cockpit language because English has no honorifics to enforce. The interface for cultural redesign was a linguistic one.

LENS APPROACH

LENS uses Korean Air in LEN 8 as the canonical organizational- learning case under cultural constraint and in LEN 2 as a CRM- extension case for high-power-distance contexts. The case is paired in this book with the Toyota Andon Cord (Case 24) as cultural intervention success stories.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Locate the single cultural feature that actually binds the capability — here, high power distance expressed through honorifics — rather than attempting to remake the whole culture.
- Change the medium through which that feature operates (making English the cockpit language) so the constraint is removed structurally rather than by asking people to override their own culture.
- Pair the structural change with an adapted protocol — CRM tuned for a high-power-distance setting — that gives crews a culturally workable way to raise concerns.

IN OPERATION / AFTER

- Use external pressure (suspended code-shares, a downgraded rating) as durable leverage to hold the reform in place against any drift back to the prior norm.
- Sustain the change with continued outside benchmarking and consulting so the new communication structure is reinforced rather than quietly eroding.
- Track the safety record over years to confirm the categorical shift holds, treating a maintained accident-free record as the evidence the redesign took.

REFLECTION QUESTIONS

1. Identify a cultural feature in an institution you work with that constrains capability. Is it engineerable? What would the redesign look like?
2. Korean Air's intervention operated on language. What surface of culture is engineerable in your domain that you have not yet considered?
3. Korean Air acted only once suspended code-shares and a downgraded rating made the status quo unaffordable. What external pressure exists in your domain that could supply the leverage to redesign a long-tolerated cultural constraint — and how would you use it?

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · incident analysis · safety dashboards

LENS COURSES LEN 2 LEN 8

CASE 24 INDUSTRIAL 1950S – PRESENT

Toyota Production System / Andon Cord

N Normalization of Deviance G Governance & Trust

IMPACT Front-line authority to stop the line resolves most defects quickly at the source; defect-propagation cost minimized; the system adopted globally

IN BRIEF

The andon cord lets any assembly-line worker stop Toyota's entire production line on detecting a defect — handing the lowest-ranking person on the floor the power to halt operations worth millions per hour. The cord is trivially cheap; the authority it confers is the design. The case is decisive for capability engineering because when American automakers copied the cord in the 1980s and 1990s, workers were too afraid to pull it: the tool was present, the empowerment was not. Toyota's system works because it pairs the mechanism with a culture of psychological safety, fast supervisor response, no-blame problem-solving, and the codified "Five Whys" method. When the line stops at Toyota, the team treats it as a learning opportunity. The Andon Cord is the manufacturing twin of the Keystone nurse-authority intervention (Case 14).

BACKGROUND

In high-volume manufacturing, a defect that passes undetected propagates downstream, multiplying the cost of every later correction. Catching problems at the source requires the person who sees them — usually the lowest-ranking worker on the line — to be able to act, in an environment where stopping a line running at millions of dollars per hour is otherwise unthinkable. The economics and the authority structure point in opposite directions: the cheapest moment to fix a defect is the one at which the person who sees it has the least standing to halt production.[0]

THE INTERVENTION

As part of the Toyota Production System, Toyota installed the andon cord: a physical pull-cord that lets any worker signal a problem and, if unresolved, stop the entire line. The inversion of authority is the entire point — the cord itself costs almost nothing, while the protected authority it confers on a front-line worker is the actual design. Handing the lowest-ranking person the power to halt operations worth millions per hour deliberately resolves the contradiction the background poses: it puts the authority to stop exactly where the defect is first visible.[1]

HOW IT WORKED

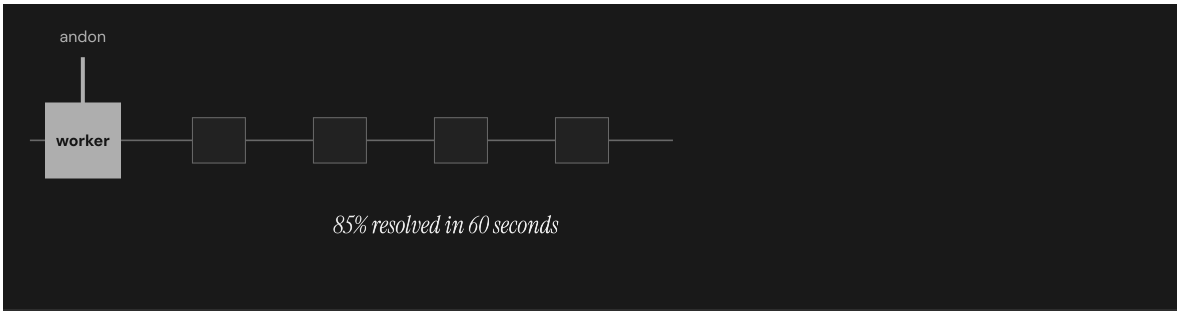
The cord works because Toyota pairs the mechanism with a cultural system: psychological safety, a rapid supervisor-response protocol when the cord is pulled, no-blame root-cause analysis, and the codified "Five Whys" method. A stop is treated as a learning opportunity rather than a failure, so workers actually use it — the technical artifact and the protected authority are inseparable. The rapid supervisor response is what makes the protection credible in practice: a worker who pulls the cord sees help arrive rather than blame, so the no-blame norm is demonstrated each time, not just asserted.[2]

THE EVIDENCE

The proof of the pairing is negative as well as positive. When American manufacturers copied the andon cord in the 1980s and 1990s, workers were too afraid to pull it; the artifact without the authority produced nothing. At Toyota, where the authority is protected, the great majority of activations are resolved within a minute and the system has been sustained and exported for decades. The natural experiment is unusually clean — the same physical cord produced opposite results across two settings, isolating the protected authority, not the hardware, as the variable that mattered.[3]

WHAT TRANSFERRED

The Andon Cord is the foundational evidence that authority interventions and technical artifacts are inseparable — the cord means nothing without the protected authority to pull it, and vice versa. It is the manufacturing counterpart of the Keystone nurse-stop authority (Case 14): same logic, different industry, same load-bearing element, and the same failure mode when only the artifact is copied. That the identical pattern recurs across manufacturing and medicine is what elevates it from a Toyota practice to a design principle: wherever the person who sees the problem lacks the standing to stop, copying the tool alone reproduces the failure.[4]



LOWEST-RANKING WORKER AUTHORIZED TO STOP THE LINE

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THE LEARNING ENGINEERING LENS

> *When American manufacturers copied the andon cord, workers were too afraid to pull it.*
>
> PARAPHRASING LIKER, THE TOYOTA WAY (2ND ED., 2020)

LE INSIGHT

The Andon Cord is the foundational evidence that authority interventions and technical artifacts are inseparable. The cord means nothing without the protected authority to pull it. The protected authority means nothing without the cord to act through. The pair is irreducible — and that irreducibility is the LENS co-optimization commitment in physical form.

LENS APPROACH

LENS uses the Andon Cord in LEN 2 as the foundational example of paired technical-cultural intervention, in LEN 8 to discuss cross-domain transfer (and why imitation without the cultural half fails), and in LEN 10 as a studio exemplar of minimal-artifact design.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Place the authority to stop exactly where the problem is first visible — with the front-line worker — resolving the contradiction that the cheapest moment to fix a defect is when the seer has the least standing.
- Pair the cheap artifact with the protected authority deliberately, recognizing the cord costs almost nothing while the authority it confers is the actual design.
- Stand up a rapid supervisor-response protocol so that pulling the cord brings help, not blame, making the no-blame norm credible from the first activation.

IN OPERATION / AFTER

- Sustain the pairing with no-blame root-cause analysis and a codified method (the Five Whys) so each stop becomes embedded learning rather than a one-off interruption.
- Demonstrate the protected authority repeatedly — help arriving within a minute — so the norm is shown each time rather than merely asserted in policy.
- When transferring the model, export the protected authority and response culture, not just the artifact, since copying the tool alone reproduces the failure mode.

REFLECTION QUESTIONS

1. Identify a technical artifact in your domain whose effectiveness depends entirely on a protected authority. Is the authority protected, or only declared?
2. American manufacturers copied the cord without the authority. What is the equivalent surface-level imitation you have observed in your domain, and what was missing?
3. Toyota's no-blame norm is demonstrated each time help arrives within a minute of a pull rather than blame. Design the visible, repeated response that would prove a protected authority is real in your domain rather than merely written into policy.

WHO BUILDS THIS safety science & organizational analysis · governance, ethics & measurement — plus domain experts and a learning engineer to integrate.

TOOLS incident analysis · safety dashboards · governance frameworks · bias & evidence audits

LENS COURSES LEN 10 LEN 2 LEN 8

CASE 27 HEALTHCARE 2006 – PRESENT

TeamSTEPPS

T Training Gap N Normalization of Deviance

IMPACT Improved teamwork, communication, and patient-safety culture across diverse settings; OR on-time first start +21%; adopted by thousands of healthcare organizations

IN BRIEF

TeamSTEPPS — developed jointly by the Agency for Healthcare Research and Quality and the Department of Defense and released in 2006 — is the healthcare analog of Crew Resource Management (Case 12): an evidence-based team-training framework distilled from fifty years of aviation, military, and nuclear research and adapted for clinical settings. It trains four core competencies: communication, leadership, situation monitoring, and mutual support. It is explicitly the translation pathway from high-reliability research into bedside practice — the cross-domain capability transfer LENS is built to teach. Because its implementation infrastructure was funded as part of the program, TeamSTEPPS moved from research to scaled deployment in years rather than decades, and has been adopted by thousands of healthcare organizations with measurable gains in teamwork and safety culture.

BACKGROUND

Decades of research in aviation, the military, and nuclear power had shown that teamwork — not just individual expertise — drives safety in high-consequence work, and the IOM's To Err Is Human had identified poor communication as a leading cause of medical harm. What clinical settings lacked was a structured, evidence-based way to build those team skills. The evidence existed and the diagnosis was clear; what was missing was the translation — a route from the high-reliability research base into a curriculum a hospital could actually run at the bedside.[0]

THE INTERVENTION

In 2006, the Agency for Healthcare Research and Quality and the Department of Defense jointly released TeamSTEPPS — Team Strategies and Tools to Enhance Performance and Patient Safety — the healthcare analog of Crew Resource Management. It trains four core competencies: communication, leadership, situation monitoring, and mutual support, with a structured curriculum and ready-made implementation materials. The joint AHRQ-DoD authorship mattered: it paired a healthcare research agency's clinical reach with the military's deep team-training experience, so the framework arrived already grounded in both the evidence base and the practicalities of running it.[1]

HOW IT WORKED

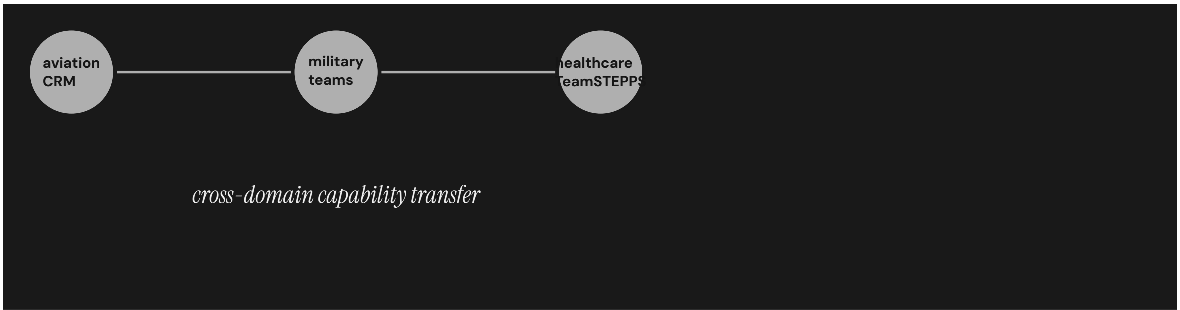
TeamSTEPPS is explicitly the translation pathway from high-reliability-organization research into clinical practice — fifty years of cross-domain evidence adapted for the bedside. Crucially, its implementation infrastructure (master-trainer programs, toolkits, an institutional support center) was funded as part of the intervention, so adopting organizations had a route from training to sustained practice rather than a binder on a shelf. Funding the implementation alongside the curriculum is precisely what compressed the usual decades-long gap between a proven method and its scaled use, because the path from adoption to sustained practice was paid for in advance.[2]

THE EVIDENCE

Studies across diverse settings report improved teamwork, communication, and patient-safety culture, with concrete operational gains such as a 21% improvement in on-time first surgical starts. Thousands of healthcare organizations have adopted the framework, and AHRQ has continued to develop it, releasing TeamSTEPPS 3.0 in 2023. The continued development into a third version is itself evidence the transfer took hold — a framework that is still being maintained and revised nearly two decades on is one institutions kept using, not one that was issued and abandoned.[3]

WHAT TRANSFERRED

TeamSTEPPS is the canonical evidence that cross-domain capability transfer is engineerable — and that the long implementation gap can be dramatically shortened when the transfer is funded as part of the intervention rather than as an afterthought. Its four competencies map directly onto the argument that capability engineering is itself a teachable discipline. The deeper lesson is that the bottleneck in cross-domain transfer is rarely the knowledge, which already existed, but the funded path that carries it from research into routine practice — the part programs most often leave unbudgeted.[4]



AVIATION → MILITARY → HEALTHCARE; THE TRANSLATION PATHWAY

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THE LEARNING ENGINEERING LENS

> *TeamSTEPPS represents the translation pathway from high-reliability research into clinical practice.*
>
> EDITORS' SYNTHESIS DRAWING ON AHRQ TEAMSTEPPS 3.0 (2023) AND SALAS ET AL.

LE INSIGHT

TeamSTEPPS is the canonical evidence that cross-domain capability transfer is engineerable and that it can shorten the implementation gap dramatically when the transfer is funded as part of the intervention rather than as an afterthought. The four competencies map directly to the LENS curriculum's argument for what capability engineering competence looks like as a deliverable.

LENS APPROACH

LENS treats TeamSTEPPS in LEN 8 as the canonical cross-domain transfer case and in LEN 1 as evidence that the program's core proposition — that learning engineering is a discipline — has institutional precedent. Studio projects (LEN 10) reference TeamSTEPPS as the worked example of cross-domain adaptation methodology.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Distill the cross-domain evidence base into a structured curriculum of a few teachable competencies rather than leaving adopters to translate the research themselves.
- Pair authorship across the source and target domains (as AHRQ and DoD did) so the framework arrives grounded in both the evidence and the practicalities of running it.
- Fund the implementation infrastructure — master-trainer programs, toolkits, a support center — as part of the intervention, not as an afterthought.

IN OPERATION / AFTER

- Maintain and revise the framework over time (as with successive TeamSTEPPS versions) so it stays in use rather than being issued and abandoned.
- Track concrete operational gains (such as on-time first surgical starts) alongside culture measures, so the transfer's effect is shown, not assumed.
- Identify and budget the specific path from adoption to sustained practice when scaling, since that unbudgeted step is the usual bottleneck in cross-domain transfer.

REFLECTION QUESTIONS

1. What is the cross-domain transfer your domain has not yet executed? What evidence base in another industry should it draw on?
2. TeamSTEPPS funded its implementation infrastructure. Design the implementation-infrastructure budget for an equivalent transfer in your domain.
3. TeamSTEPPS shortened the usual decades-long gap because the path from adoption to sustained practice was paid for in advance. Identify a proven method in your domain that has not scaled, and name the specific unbudgeted step that is blocking it.

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · incident analysis · safety dashboards

LENS COURSES LEN 1 LEN 10 LEN 8 LEN 3

CASE 28 DEFENSE 1954 – PRESENT

U.S. Nuclear Navy / Rickover Training Model

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT Zero reactor accidents in 60+ years of U.S. Naval nuclear operations; the most demanding nuclear operator training program in the world

IN BRIEF

Admiral Hyman Rickover built a training and qualification culture for the Naval Nuclear Propulsion Program that has produced zero reactor accidents across more than 60 years and thousands of reactor-years of operation, often in extreme conditions. Every nuclear-trained sailor must pass rigorous qualification to zero-defect standards and demonstrate competence in oral examination by senior nuclear-qualified officers; the culture demands personal accountability, deep technical mastery, and a questioning attitude that obliges operators to challenge assumptions, including superiors'. The sharpest contrast in this book is internal: the same Navy that ran the nuclear program to this standard let surface-warfare training decay to CD-ROMs and paid for it at Fitzgerald and McCain (Case 1). Same institution, two philosophies — and the choice shows up in the casualty columns.

BACKGROUND

In the early 1950s the U.S. Navy set out to put nuclear reactors aboard ships and submarines — machines whose failure could be catastrophic and irreversible, operated by young sailors far from any help. The capability problem was absolute: there was no acceptable accident rate, so the human operating system had to be engineered to an extraordinary standard from inception. Because a reactor at sea could not be evacuated or handed to outside experts, the operators themselves had to be the entire margin of safety — a constraint that forced training to a level no ordinary program would justify.[0]

THE INTERVENTION

Admiral Hyman Rickover established a training and qualification regime requiring every nuclear-trained officer and enlisted sailor to pass demanding programs held to zero-defect standards. Operators must demonstrate competence through oral examination by senior nuclear-qualified officers, and the program embeds continuous re-qualification rather than one-time certification. The oral board tested understanding rather than recall, and the continuous re-qualification meant competence had to be sustained rather than banked once — so the standard governed the operator's whole career, not just entry to it.[1]

HOW IT WORKED

The culture pairs technical mastery with a deliberately engineered attitude: personal accountability and a mandatory questioning posture in which operators are trained to challenge assumptions, including those of superiors. Rickover's premise — that people, not organizations or management systems, get things done — made the qualification ladder, not paperwork, the load-bearing element of safety. The questioning posture is the cultural half of the pair: deep technical mastery alone could still defer to a mistaken superior, so the obligation to challenge assumptions is what keeps competence from being silenced by rank.[2]

THE EVIDENCE

The result is the longest-running continuous capability-engineering record in any high-consequence domain: zero reactor accidents across more than six decades and thousands of reactor-years. The cost — the qualification ladder, the zero-defect oral boards, the continuous re-qualification — is the visible budget-line price of that record. The duration is the key evidence: a record sustained across six decades and many generations of sailors shows the safety came from the engineered system rather than from any one cohort's talent or luck.[3]

WHAT TRANSFERRED

The cleanest test of the model is internal. The same Navy that engineered the nuclear program to this standard let surface-warfare training decay to CD-ROM self-study and paid the price at Fitzgerald and McCain (Cases 1 and 15). Same institution, same era, opposite philosophies, radically different outcomes — the strongest available demonstration of capability treated as a system parameter versus deferred as a cost. The internal comparison controls for nearly everything that usually confounds such claims — one service, one manpower system, one budget process — leaving the training philosophy itself as the variable that diverged.[4]

NUCLEAR
0
reactor accidents
60+ years

SURFACE (PRE-2017)
17
sailors killed
two collisions

SAME NAVY, OPPOSITE PHILOSOPHIES — RADICALLY DIFFERENT OUTCOMES

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THE LEARNING ENGINEERING LENS

> *Human experience shows that people, not organizations or management systems, get things done.*
>
> ADMIRAL HYMAN G. RICKOVER, "DOING A JOB," COLUMBIA UNIVERSITY, 1982

LE INSIGHT

The Nuclear Navy is the longest-running continuous capability- engineering program in any high-consequence domain. The choice to treat training as a system parameter rather than as a cost center has produced sixty-plus years of zero reactor accidents. The contrast with the Surface Navy is the cleanest available test of what happens when capability is engineered versus when it is deferred — and the price of that engineering (the qualification ladder, the zero-defect oral boards, the continuous re-qualification) is visible on the budget line.

LENS APPROACH

LENS treats the Rickover model in LEN 8 as the foundational organizational-learning case and in LEN 5 as a worked example of capability requirements traceable from operational analysis through qualification standards. The internal Navy comparison anchors the program's core argument about capability as a system parameter.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Engineer the human operating system to the standard the consequences demand from inception — where there is no acceptable failure rate, make the operators themselves the margin of safety.
- Gate qualification on demonstrated understanding through oral examination by senior qualified people, testing comprehension rather than recall.
- Pair technical mastery with a mandatory questioning posture so competence is obliged to challenge assumptions, including superiors', rather than be silenced by rank.

IN OPERATION / AFTER

- Embed continuous re-qualification rather than one-time certification, so competence must be sustained across a career and cannot be banked once and assumed.
- Protect the training as a durable, visible budget line, since the qualification ladder and oral boards are the price of the safety record and the first thing tempo will erode.
- Sustain the standard across generations of operators, treating a multi-decade record as the evidence the safety comes from the engineered system rather than any one cohort.

REFLECTION QUESTIONS

1. Identify an institution that operates two divisions under different capability philosophies. What does the comparison reveal that neither division could see alone?
2. Rickover's standard was zero-defect oral examination. What is the equivalent in your domain — and would you survive it?
3. The Nuclear Navy's safety record is paid for by a durable, visible training budget. What is the equivalent line-item investment in your domain that a comparable safety claim would require — and is it being made?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards

LENS COURSES LEN 5 LEN 8 LEN 3

CASE 39 EDUCATION AT SCALE 2012 – PRESENT

Georgia State University Predictive Analytics

T Training Gap K Knowledge & Institutional Memory

IMPACT Six-year graduation rate 32% → 54%; equity gaps in graduation eliminated; 2,000+ more graduates per year

IN BRIEF

Georgia State University built a predictive-analytics advising system that tracks some 800 risk factors per student and triggers proactive outreach when early warning signs appear. Crucially, it was designed with equity as a primary constraint — the explicit goal was to eliminate, not reproduce, graduation gaps — and the alerts prompt human advisors rather than making automated decisions. The six-year graduation rate rose from 32% to 54%; Black and Pell-eligible students now graduate at the same rate as their peers, and GSU produces roughly 2,000 more graduates a year. The difference from the algorithmic-bias cases (35–37) is design: GSU built equity in from the start, used predictions to trigger more human support rather than gatekeeping, and tracked outcomes by demographic group as a primary metric.

BACKGROUND

Georgia State was a regional commuter university where only about a third of students finished in six years, with large gaps by race and income. Like many institutions it had predictive data but such systems, where they existed elsewhere, were typically used for triage or gatekeeping — risking the reproduction of existing inequities rather than their repair. A model that flags at-risk students can just as easily steer them away as toward help; the same prediction serves opposite ends depending on what the institution decides to do with it.[0]

THE INTERVENTION

Beginning in 2012, GSU deployed a predictive-analytics advising system that monitors roughly 800 behavioral and academic risk factors per student daily and fires an alert to an advisor when warning signs — a missed assignment, a poor grade in a gateway course — appear. The system was built with equity as a primary design constraint, with the explicit aim of closing graduation gaps. Daily monitoring meant the alert fired while there was still time to act — a slipping student was caught at the missed assignment rather than at the failed semester, when intervention could still change the outcome.[1]

HOW IT WORKED

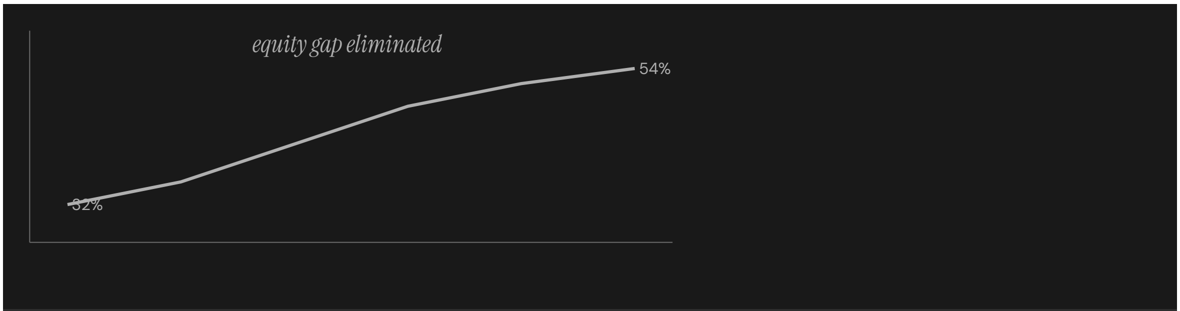
The load-bearing design choice was the human-loop architecture: alerts trigger proactive advising — a phone call, a meeting, a financial-aid check — rather than automated decisions. Predictions are used to deliver more support to at-risk students, not to gatekeep them out. Human judgment stays in the loop, and the model functions as decision support rather than decision-maker. Routing every alert through an advisor rather than an automated action is what kept the prediction in service of the student: the model identified who needed attention, and a person decided what that attention should be.[2]

THE EVIDENCE

GSU's six-year graduation rate rose from 32% to 54%, and the institution now produces some 2,000 additional graduates a year. The graduation rate for Black students rose to match the overall rate, and Pell-eligible students graduate at the same rate as non-Pell students — the equity gap was eliminated rather than merely narrowed. Eliminating the gap rather than narrowing it is the decisive result: the overall rate rose while the disparities by race and income closed, so the gain did not come at the expense of the students the system was most at risk of leaving behind.[3]

WHAT TRANSFERRED

GSU is the positive counterpart to the algorithmic-harm cases of Chapter 5 (A-Level, Robodebt, educational bias). The same technical capability — a predictive model — produced an equity gain rather than an equity harm because of how it was framed and governed. The case is the strongest evidence that construct definition and human-loop architecture, not the model itself, determine whether prediction helps or harms. Holding the technology constant and varying only the design constraint and governance is what isolates the lesson: the same predictive capability that harmed in Chapter 5 helped here, so the framing and the human loop, not the model, are where intent lives.[4]



GRADUATION RATE, WITH EQUITY AS A PRIMARY CONSTRAINT

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THE LEARNING ENGINEERING LENS

> *Predictions trigger support, not gatekeeping.*
>
> EDITORS' SYNTHESIS OF THE GSU ADVISING MODEL, DRAWN FROM RENICK & STROM (2020) AND NEW YORK TIMES COVERAGE (2018)

LE INSIGHT

GSU is the positive counterpart to A-Level (35), Robodebt (36), and educational algorithmic bias (37). The same technical capability — a predictive model — was deployed under a different design constraint and produced an equity outcome rather than an equity harm. The case is the strongest available evidence that the **construct definition** and the **human-loop architecture** determine whether a predictive model produces good or harm, not the model itself.

LENS APPROACH

LENS treats GSU as the canonical positive case for predictive analytics in education. LEN 4 examines the measurement architecture that made equity a primary outcome. LEN 7 examines the governance structure that kept the system as decision support rather than automated decision. LEN 1 uses it as a problem-framing exemplar.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Set equity as a primary design constraint from the start — the explicit aim of closing gaps — rather than discovering disparities after deployment.
- Build a human-loop architecture so alerts trigger proactive support routed through an advisor, with the model as decision support and a person deciding the action.
- Tune the monitoring to fire early — at the missed assignment, not the failed semester — so the intervention reaches the student while it can still change the outcome.

IN OPERATION / AFTER

- Track outcomes by demographic group as a primary metric, so the system is judged on whether it closes gaps rather than merely raises the average.
- Confirm the overall gain does not come at the expense of the most at-risk students — eliminating the gap, not just narrowing it, is the test that the design held.
- Keep human judgment in the loop as the system scales, so prediction continues to deliver more support rather than drifting into automated gatekeeping.

REFLECTION QUESTIONS

1. What is the difference between GSU's predictive analytics and the algorithmic-bias cases of Chapter 5? Be specific about what makes the GSU implementation work.
2. Design the equity-as-primary-constraint version of a predictive system in your domain. What would you measure first?
3. GSU used predictions to deliver more support rather than to gatekeep, with an advisor between the alert and the action. Identify a predictive system in your domain and specify the human-loop architecture that would keep it serving the people it flags rather than screening them out.

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines

LENS COURSES LEN 1 LEN 4 LEN 7

CASE 42 EDUCATION AT SCALE 1990S – PRESENT

Cognitive Tutor / Carnegie Learning

T Training Gap

IMPACT Randomized controlled trials showed significant learning gains; RAND study found positive effects on Algebra I achievement; adopted across 3,000+ schools

IN BRIEF

Carnegie Learning's Cognitive Tutor, built from John Anderson's ACT-R cognitive architecture at Carnegie Mellon, is the most rigorously evaluated intelligent tutoring system in education. It uses Bayesian knowledge tracing to model each student's mastery and adapts instruction accordingly, and a RAND Corporation evaluation found statistically significant positive effects on Algebra I achievement. The system is a learning-engineering success in the discipline's own terms — learning science to engineered software to randomized-trial evidence to deployment across 3,000-plus schools. Its limitations are instructive: it works best in well-defined domains like algebra and less well in ill-structured ones, making it the canonical evidence that the pipeline delivers for problems that fit it — leaving open whether the same discipline can deliver where problems do not.

BACKGROUND

For decades, intelligent tutoring systems promised individualized instruction at scale, but few were grounded in a validated theory of how people learn or rigorously tested for effect. The opportunity was to build a tutor from a real cognitive model and prove its impact with the methods of experimental science. The field's recurring weakness was that promising systems rested on intuition about learning rather than a validated theory, and were rarely subjected to the kind of controlled trial that could separate genuine effect from novelty.[0]

THE INTERVENTION

Carnegie Learning's Cognitive Tutor, developed from John Anderson's ACT-R cognitive architecture at Carnegie Mellon, models the specific skills underlying a subject and tracks each student's mastery using Bayesian knowledge tracing. It adapts problem selection to the individual learner and provides step-level feedback, embodying a full learning-science theory in software. Grounding the tutor in ACT-R rather than designer intuition is what made the system testable: a theory that decomposes a subject into specific skills can be turned into a measurement of mastery and an instrument that responds to it.[1]

HOW IT WORKED

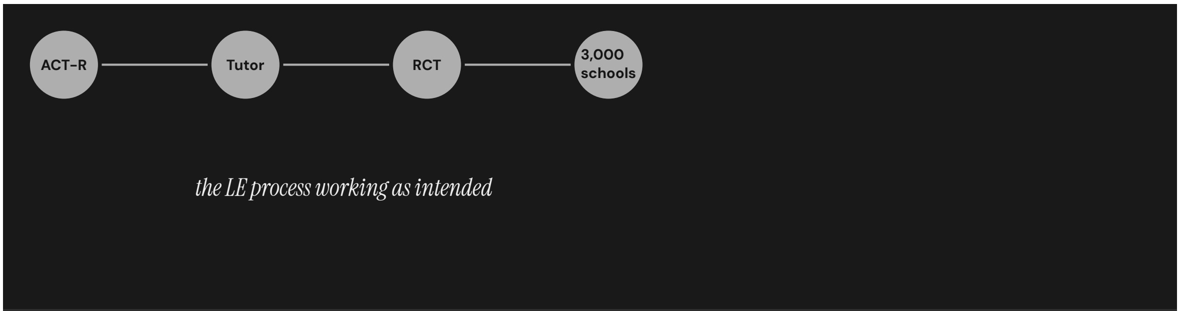
The tutor's effectiveness rests on the chain from theory to instrument: a decomposable skill model, a measurement method (knowledge tracing) that estimates mastery from student actions, and an instrumentable interface that adapts in response. Instruction is targeted where the model detects weakness, so practice concentrates on skills not yet mastered rather than on a fixed sequence. Each link in that chain has to hold for the next to work — the skill model makes mastery measurable, the measurement makes adaptation possible, and the adaptation is what concentrates practice where it pays off.[2]

THE EVIDENCE

The RAND Corporation's multi-site evaluation found statistically significant positive effects on Algebra I achievement, and the program scaled to more than 3,000 schools. The case demonstrates the learning-engineering process working end to end: learning science, to engineered software, to randomized-controlled-trial evidence, to scaled implementation. A multi-site randomized evaluation is the strong form of the claim — it shows the effect survived contact with many real classrooms rather than a single favorable setting, which is what the field's earlier, untested systems had lacked.[3]

WHAT TRANSFERRED

The limitations are as instructive as the success. Cognitive Tutor performs best in well-defined domains like algebra and less well in ill-structured ones, making it the canonical evidence that the learning-engineering pipeline works for problems that fit the pipeline. The frontier question — whether the same discipline can deliver in operational, ill-structured domains where capability matters most — remains open. The dependence on a decomposable skill model is the boundary condition: where a subject cannot be cleanly broken into trackable skills, the very chain that made algebra tractable has nothing to attach to.[4]



THEORY → ENGINEERING → EVIDENCE → SCALE

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THE LEARNING ENGINEERING LENS

> *Cognitive Tutors demonstrate the LE process working: theory → engineering → evidence → scale.*
>
> EDITORS' SYNTHESIS OF ANDERSON ET AL. (1995) AND KOEDINGER & CORBETT (2006)

LE INSIGHT

Cognitive Tutor is the canonical evidence that the learning- engineering process exists, works, and produces measurable benefits at scale — when the problem is tractable. It is also the canonical case for what tractability looks like: well-defined domain, decomposable skill model, instrumentable interface, rigorous evaluation. The frontier for the discipline is whether the same pipeline can deliver in the operational, ill-structured domains where capability matters most.

LENS APPROACH

LENS uses Cognitive Tutor in LEN 1 as the foundational LE-process exemplar, in LEN 4 as the canonical case for Bayesian knowledge tracing as a measurement instrument, and in LEN 9 as a technical case for model-based adaptive instruction.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Ground the intervention in a validated theory that decomposes the subject into specific skills, rather than building on designer intuition about how people learn.
- Build the full chain — a skill model, a measurement method that estimates mastery from learner actions, and an interface that adapts in response — so each link supports the next.
- Target instruction where the model detects weakness, concentrating practice on unmastered skills rather than marching through a fixed sequence.

IN OPERATION / AFTER

- Prove the effect with a multi-site randomized evaluation, so the gain is shown to survive many real settings rather than one favorable classroom.
- Scale only where the problem fits the pipeline — well-defined, decomposable domains — and treat the decomposability of a skill model as the boundary condition for applicability.
- Keep the frontier question explicit when extending the method, testing whether the same theory-to-instrument chain can be built in less-structured domains before assuming it transfers.

REFLECTION QUESTIONS

1. Cognitive Tutor works best in well-defined domains. Identify a problem in your domain that is currently ill-structured. What would have to be true to make the LE pipeline applicable?

2. The case is the strongest evidence the LE process works. What is the equivalent piece of evidence required to demonstrate the same pipeline in a non-cognitive domain — for example, surgical skill, or operational watchstanding?

3. Cognitive Tutor's chain runs from a decomposable skill model through measurement to adaptation. Take a capability in your domain and attempt the decomposition — where it resists being broken into trackable skills, what does that tell you about whether the pipeline can apply?

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LENS COURSES LEN 1 LEN 4 LEN 9

CASE 88 HEALTHCARE INDUSTRIAL 1982

Tylenol Recall

G Governance & Trust N Normalization of Deviance

IMPACT Foundational U.S. corporate crisis-management case; produced tamper-evident packaging regulation and modern recall practice

IN BRIEF

In 1982, seven people in the Chicago area died after taking Tylenol capsules laced with potassium cyanide. Not knowing who was responsible or how widespread the tampering was, Johnson & Johnson recalled every Tylenol product nationwide — 31 million bottles, at a cost of roughly $100 million — suspended advertising, and engaged openly with the FBI and FDA. The response was unprecedented in U.S. corporate practice, and it was a direct application of the J&J Credo, written in 1943, which had pre-specified that the company's first responsibility was to its customers. The reform that followed reshaped consumer packaging worldwide — tamper-evident seals and blister packs — and the FDA issued tamper-resistant-packaging rules within months. Tylenol recovered its market share within a year.

BACKGROUND

Johnson & Johnson's corporate Credo, written in 1943, pre-specified that the company's first responsibility was to the patients and consumers who used its products, ahead of shareholders. For four decades it was a statement of values; in 1982 it became an operational decision rule under extreme pressure. The ordering was explicit — customers ahead of shareholders — which is precisely the ranking that crisis pressure inverts, so committing to it in advance pre-decided the hardest trade-off before it had to be faced.[0]

THE INTERVENTION

After seven people in the Chicago area died from Tylenol capsules laced with potassium cyanide, and with the source and scope of the tampering unknown, Johnson & Johnson recalled every Tylenol product nationwide — about 31 million bottles, at a cost near $100 million — suspended all advertising, and engaged publicly with the FBI and FDA rather than minimizing exposure. Recalling nationwide despite the tampering being known only in Chicago was the decisive choice — it treated the unknown scope as a reason to protect every customer rather than as room to limit the company's own exposure.[1]

HOW IT WORKED

The load-bearing element was a commitment pre-committed in writing. Because the Credo had already decided, decades earlier, that the customer came first, the 1982 leadership did not have to improvise an ethical calculus under crisis — it executed a pre-made decision. CEO James Burke later credited the Credo with making clear "exactly what we were all about" the moment the deaths occurred. Pre-commitment worked because it moved the decision out of the moment of maximum pressure — when fear and legal caution push hardest toward minimization — and into a calmer time when the right ordering could be set down without that distortion.[2]

THE EVIDENCE

The response, unprecedented in U.S. corporate practice, preserved public trust: Tylenol recovered its market share within a year despite the enormous short-term cost. The case became the canonical positive example in business education of crisis response driven by capability commitment rather than legal minimization. The market recovery is what makes the case persuasive rather than merely admirable — the $100 million spent protecting customers was repaid in the trust that brought them back, so the pre-committed choice proved sound on its own terms.[3]

WHAT TRANSFERRED

The reform reshaped consumer-product packaging worldwide — tamper-evident seals, blister packs, and caplet forms became standard — and the FDA promulgated tamper-resistant-packaging regulations within months. The deeper transfer is the principle that values must be pre-committed in writing to be operational under stress, not invented in the moment. The packaging reform and the decision-rule principle are the two layers of the transfer — one a physical safeguard against the specific threat, the other an institutional safeguard against the improvisation that crisis invites.[4]

31M

BOTTLES RECALLED · ~$100M COST

the pre-committed institutional credo became operational under stress

TYLENOL — VALUES PRE-COMMITTED IN WRITING, EXECUTED UNDER CRISIS

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THE LEARNING ENGINEERING LENS

> *The Credo is all about the consumer. When those seven deaths occurred, the Credo made it very clear at that point exactly what we were all about.*
>
> JAMES BURKE (JOHNSON & JOHNSON CEO), QUOTED IN LASTING LEADERSHIP (WHARTON)

LE INSIGHT

Tylenol is the canonical positive case for institutional response to crisis. The capability that was load-bearing was the pre-specified institutional commitment in the Credo. The crisis decision had been made decades earlier; in 1982 it was executed. The case is the strongest evidence in the business-ethics dataset that values must be pre-committed in writing to be operational under stress.

LENS APPROACH

LENS uses Tylenol in LEN 7 as the foundational positive case for institutional governance under crisis and in LEN 10 (capstone) as a worked example of pre-committed capability that executed under operational pressure.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pre-commit the hardest trade-off in writing before the crisis — rank customer safety ahead of shareholder exposure — so leadership executes a pre-made decision rather than improvising under pressure.
- Set the rule in a calm period when fear and legal caution cannot distort the ordering, since those forces push hardest exactly when the decision must be made.
- Make the commitment concrete enough to act on — a nationwide recall, open engagement with regulators — so unknown scope becomes a reason to protect everyone rather than room to limit exposure.

IN OPERATION / AFTER

- Pair the institutional decision rule with a physical safeguard against the specific threat (tamper-evident packaging) so the response addresses both the improvisation problem and the vulnerability.
- Treat the preserved trust and market recovery as the measure that the pre-committed choice was sound, not merely admirable, and document it to defend the principle internally.
- Embed the commitment durably enough that it survives leadership turnover, so the next crisis meets the same pre-decided rule rather than a fresh improvisation.

REFLECTION QUESTIONS

1. What is your institution’s equivalent of the J&J Credo, and is it operational under crisis or aspirational?
2. Pre-commitment is hard to enforce later. Design the institutional architecture that makes a Tylenol-style response the only available option in the worst case.
3. J&J recalled nationwide while the tampering was known only in Chicago, treating unknown scope as a reason to protect everyone. Identify a decision in your domain where uncertainty currently licenses minimizing exposure, and write the pre-committed rule that would flip it toward protection instead.

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TOOLS governance frameworks · bias & evidence audits · incident analysis · safety dashboards

LENS COURSES LEN 10 LEN 7 LEN 6

CASE 89 AVIATION 1976 – PRESENT

Aviation Safety Reporting System (ASRS)

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance

IMPACT NASA-administered confidential reporting system; more than 2M reports received; foundational architecture for evidence-driven aviation safety

IN BRIEF

The Aviation Safety Reporting System, run by NASA on behalf of the FAA since 1976, accepts confidential reports from pilots, controllers, mechanics, and cabin crew about incidents, near-misses, and safety concerns. Its load-bearing feature is institutional, not technical: reporting an event to ASRS confers immunity from FAA enforcement for the conduct reported, within specified limits, making honest reporting the rational choice. Over nearly fifty years and more than two million reports, ASRS has become the world's largest repository of aviation-safety information, surfacing patterns — automation surprise, runway incursions, fatigue effects — before they reached formal investigation thresholds. The architecture has been emulated across domains, and is the canonical success case for an evidence system paired with a credible commitment to non-punitive use.

BACKGROUND

The most valuable safety information in any high-consequence domain lives with front-line operators — the near-misses and quiet errors that never reach an accident report. But operators will not surrender that information to an institution that can punish them for it, so the data that could prevent the next accident stays locked in the people who hold it. The incentives run backward: the person best placed to report a near-miss is the same person a punitive system gives the strongest reason to stay silent, so the data that matters most is the data least likely to surface.[0]

THE INTERVENTION

In 1976 the FAA and NASA created the Aviation Safety Reporting System, a confidential channel for pilots, controllers, mechanics, and cabin crew to report incidents, near-misses, and concerns. NASA — not the regulator — administers it, and reporting an event confers immunity from FAA enforcement for the conduct reported, subject to specified limits. Putting a neutral party between the reporter and the enforcer directly addressed the backward incentive — an operator now had a positive reason to report, because doing so converted potential jeopardy into protection.[1]

HOW IT WORKED

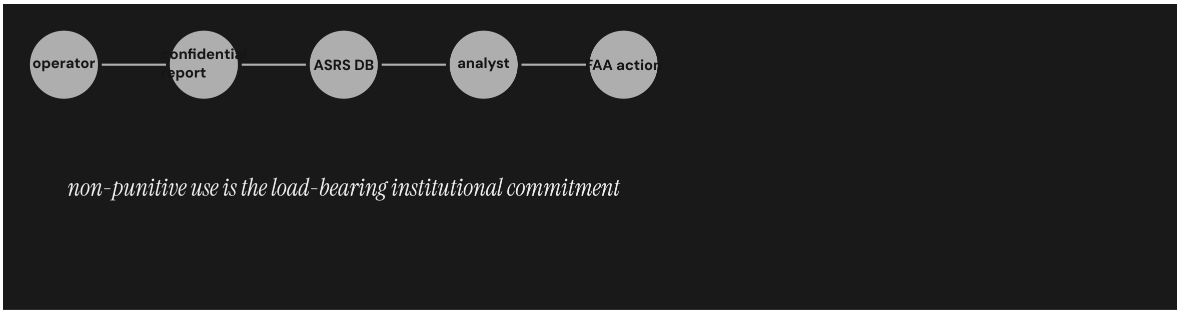
The system pairs a technical artifact (the reporting form and a searchable database) with a cultural commitment (protected, non-punitive use). The immunity provision makes reporting the rational choice for the operator, and NASA's role as a neutral third party makes the protection credible. Either half alone fails; together they make the data flow to the institution that can act on it. The credibility of the protection is what does the work — a promise of non-punishment from the regulator itself would be doubted, so routing it through NASA is what makes operators trust it enough to report.[2]

THE EVIDENCE

Over nearly fifty years ASRS has accumulated more than two million reports — the largest single repository of aviation-safety information in the world. Patterns such as automation surprise, runway incursions, and fatigue effects were first identified at scale through ASRS data before they crossed formal investigation thresholds. Surfacing a pattern before it reaches an accident is the whole point — the value of the system is the hazards it lets the industry act on while they are still near-misses, not the reports it collects after the fact.[3]

WHAT TRANSFERRED

ASRS has been studied and emulated across domains — patient-safety reporting systems, the maritime and aviation CHIRP scheme, and similar systems in rail and nuclear power. It is the canonical positive case for evidence architecture paired with an institutional commitment to non-punitive learning, the defining design pattern of a "just culture." The breadth of emulation underscores that the load-bearing element travels — wherever the most valuable safety data sits with operators who fear punishment, the same protected-reporting design recurs as the way to unlock it.[4]



ASRS — PAIRED TECHNICAL ARTIFACT AND PROTECTED REPORTER

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THE LEARNING ENGINEERING LENS

> *ASRS is the model for confidential, voluntary, non-punitive incident reporting in any high-consequence domain.*
>
> PARAPHRASING JAMES REASON, MANAGING THE RISKS OF ORGANIZATIONAL ACCIDENTS, 1997

LE INSIGHT

ASRS is the canonical positive case for paired-intervention evidence architecture. The technical artifact (the reporting form and the database) is paired with the cultural commitment to non-punitive use. Either alone fails. Together they have produced the most comprehensive operational-safety dataset in any domain.

LENS APPROACH

LENS uses ASRS in LEN 4 as the foundational positive case for evidence architecture and in LEN 8 for institutional commitment to non-punitive learning. Studio projects design ASRS-equivalents for new domains.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pair a simple reporting artifact (a form and searchable database) with a credible commitment to non-punitive use, since the channel without the protection collects nothing of value.
- Confer immunity for reported conduct so that reporting becomes the rational choice, directly reversing the incentive that otherwise keeps the most valuable data hidden.
- Route the protection through a neutral third party rather than the enforcer, so operators trust the non-punitive promise enough to report against their own interest.

IN OPERATION / AFTER

- Analyze the accumulated reports to surface patterns — automation surprise, runway incursions, fatigue — and act on them while they are still near-misses rather than accidents.
- Protect the immunity provision over time, since a single high-profile punishment of a reporter would collapse the trust the whole system depends on.
- Export the protected-reporting design, not just the database, to new domains, adapting the neutral-party structure wherever valuable safety data sits with operators who fear punishment.

REFLECTION QUESTIONS

1. Identify a domain in your institution that would benefit from an ASRS-equivalent. What cultural commitment would be required for it to function?
2. Design the institutional commitment that makes an ASRS-equivalent operational rather than merely declared.
3. ASRS made the protection credible by routing it through NASA rather than the regulator. Identify a reporting channel in your domain that operators distrust, and specify the neutral party or structural separation that would make its non-punitive promise believable.

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LENS COURSES LEN 4 LEN 8

CASE 90 HEALTHCARE 1984 – PRESENT

Bristol Heart Babies Reform

G Governance & Trust N Normalization of Deviance

IMPACT Foundational UK case in clinical outcomes transparency; produced specialty-level performance reporting in UK cardiac surgery

IN BRIEF

Through whistleblowing and a public inquiry, the Bristol Royal Infirmary was found to have had substantially worse pediatric cardiac-surgery outcomes than other UK centers for years. The Kennedy Inquiry — one of the most influential UK healthcare inquiries in modern times — located the capability gap in the absence of routine specialty-level outcomes reporting: surgeons did not know how their results compared with peers, patients did not know, and referrals did not respond to actual outcome data. The reform built national specialty-level outcomes registries, first in cardiac surgery and then across other specialties, making the UK one of the few countries that routinely publishes surgeon-level results — a paired intervention of technical registry plus institutional commitment to publish that ended outcomes data as the private property of clinicians.

BACKGROUND

In pediatric cardiac surgery, small differences in a unit's performance can mean the difference between a child living and dying — yet in the UK of the 1980s and early 1990s, no system routinely compared outcomes between centers. A surgeon, a hospital, or a referring physician had no reliable way to know whether a given unit's results were normal or dangerously poor. Without comparison across centers, a dangerously poor unit and an ordinary one looked alike from inside, so the very gap that cost children's lives was the one no one was positioned to see.[0]

THE INTERVENTION

After whistleblowing and a public inquiry into deaths at the Bristol Royal Infirmary between 1984 and 1995, the Kennedy Inquiry recommended routine, risk-adjusted, specialty-level outcomes reporting. The reform built national registries — beginning with cardiac surgery and extending to other specialties — together with a commitment to publish results, including at the level of individual surgeons. Starting in cardiac surgery and then extending outward was deliberate sequencing — the specialty where the harm had been exposed proved the model, and the registry then spread to fields that had not had their own Bristol.[1]

HOW IT WORKED

The intervention was explicitly paired. The technical half — registry design, statistical risk adjustment so that surgeons taking hard cases are not penalized, and a publication architecture — was necessary but not sufficient. The cultural half — surgeons accepting that their comparative results would be visible — was equally load-bearing, and was the harder of the two to secure. The risk adjustment was what made the cultural half securable: without it, surgeons who took the sickest patients would have been punished by the raw numbers, giving them every reason to resist publication or avoid hard cases.[2]

THE EVIDENCE

The UK became one of the few countries that routinely publishes surgeon- and unit-level outcomes, and the cardiac-surgery registry is among the most mature specialty-outcomes regimes in any country. Outcomes data ceased to be the private property of clinicians and became a shared resource for patients, referrers, and surgeons themselves. That surgeons themselves became users of the data, not just subjects of it, is part of why the regime endured — comparison that had once felt like exposure became a tool the profession relied on to know where it stood.[3]

WHAT TRANSFERRED

Bristol is the foundational UK case for outcomes transparency as a paired-intervention deliverable, and its registry model has been extended across NHS specialties and influenced later national-quality reforms. It pairs with Keystone ICU (Case 14) as healthcare-outcomes interventions operating at different layers — the bedside protocol and the system-level measurement regime. The two layers are complementary rather than competing: Keystone changes what happens at the point of care, while Bristol changes what the system can see about results across centers, and a mature regime needs both.[4]

BEFORE	AFTER
private	published

outcomes data ceased to be the private property of clinicians

BRISTOL — OUTCOMES TRANSPARENCY AS PAIRED INTERVENTION

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THE LEARNING ENGINEERING LENS

> *Outcomes data must cease to be the private property of clinicians and become a shared institutional resource.*
>
> PARAPHRASING THE KENNEDY INQUIRY FINAL REPORT (LEARNING FROM BRISTOL), 2001

LE INSIGHT

Bristol is the canonical UK case for outcomes-transparency as a paired intervention. The technical capability — registry design, statistical risk adjustment, publication architecture — was necessary. The cultural capability — surgeons accepting that their results would be visible and comparable — was equally necessary. The pair has produced one of the most mature specialty-outcomes regimes in any country.

LENS APPROACH

LENS uses Bristol in LEN 4 for outcomes-transparency as a paired- intervention deliverable and in LEN 7 for the institutional reform that made surgeon-level publication acceptable. The case pairs with Keystone ICU (Case 14) as healthcare-outcomes interventions at different layers.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pair the technical instrument — a registry with statistical risk adjustment and a publication architecture — with the cultural change of practitioners accepting that comparative results will be visible.
- Build risk adjustment in from the start so those who take the hardest cases are not punished by raw numbers, which is what makes the cultural half securable.
- Sequence the rollout to begin where the harm was exposed, proving the model in one specialty before extending it to fields that have not had their own crisis.

IN OPERATION / AFTER

- Make practitioners users of the data, not just subjects of it, so comparison becomes a tool the profession relies on rather than an exposure it resents — which is what sustains the regime.
- Extend the registry model across specialties over time, turning a single reform into a system-wide measurement regime.
- Pair the system-level transparency layer with point-of-care interventions (as with Keystone), since a mature outcomes regime needs both what the system can see and what happens at the bedside.

REFLECTION QUESTIONS

1. What is the equivalent of surgeon-level outcomes transparency in your domain? What cultural change would have to accompany the technical instrument?
2. Design the registry and publication architecture for a specialty in your domain currently operating without outcomes transparency.
3. Bristol's risk adjustment was what let surgeons accept publication, by ensuring those who took the hardest cases were not punished by raw numbers. Identify a transparency measure in your domain that practitioners resist, and design the adjustment that would make the comparison fair enough to accept.

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LENS COURSES LEN 4 LEN 7 LEN 3

CASE 91 AVIATION 1980S – PRESENT

Singapore Airlines Safety Transformation

T Training Gap N Normalization of Deviance

IMPACT Sustained safety record over decades despite challenging operating conditions; among the most safety-invested carriers in commercial aviation

IN BRIEF

Singapore Airlines has invested in safety capability across decades in a way that sets it apart from carriers operating under comparable conditions — early adoption of CRM, heavy simulator investment, a young-fleet policy, and a strong reporting culture, sustained even through rapid growth. The 2000 crash of Flight SQ006, which attempted takeoff from a closed, partly-constructed runway at Taipei during a typhoon and killed 83, prompted a sustained institutional self-examination — operational changes, training updates, and public transparency about what had happened — that became a model in the literature on post-accident learning. The airline is the operational successor to Korean Air (Case 23): an Asian carrier that engineered its safety capability deliberately and sustained the investment as a competitive differentiator, not only in response to crisis.

BACKGROUND

Commercial aviation runs on thin margins, and safety investment — simulators, training hours, fleet renewal — is a cost that competitive pressure constantly pushes downward. The question for any carrier is whether to treat capability as a floor set by regulation or as a deliberate, sustained investment ahead of the minimum. The pressure is structural rather than occasional — every budget cycle invites trimming the margin between regulatory minimum and actual capability, so sustaining the investment requires deciding the question deliberately rather than by default.[0]

THE INTERVENTION

Singapore Airlines chose sustained investment. From the 1980s it was an early adopter of Crew Resource Management and CRM-style culture work tuned to its operating context, and it committed to heavy simulator investment, a deliberately young fleet, and a strong internal reporting culture — maintaining these even during periods of rapid expansion. Holding the investment through rapid growth is the hard test — expansion is precisely when the temptation to let capability lag the fleet is strongest, and maintaining it then is what separates a sustained commitment from a fair-weather one.[1]

HOW IT WORKED

The carrier treated safety capability as a competitive differentiator rather than a regulatory burden, pairing technical investment — training systems, modern aircraft — with a culture of transparency and reporting. Investing ahead of regulatory minimums made the capability a managed system parameter, not a residual of cost-cutting decisions made elsewhere.

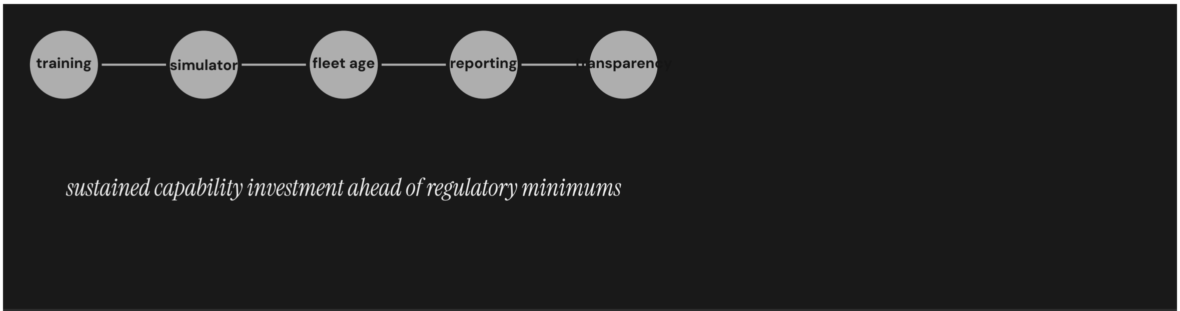
Framing safety as a differentiator rather than a burden is what made the investment defensible against cost pressure — it tied capability to the brand and the premium the airline charged, giving the spend a commercial rationale, not just a safety one.[2]

THE EVIDENCE

The 2000 crash of Flight SQ006 — an attempted takeoff from a closed, partly-constructed runway at Taipei during Typhoon Xangsane, killing 83 — tested the institution. Its response, documented in the Taiwan investigation, became a reference case in post-accident institutional learning: operational changes, training updates, and public transparency about what had happened and why, rather than defensiveness. The test of a safety culture is how it behaves after its own accident, and choosing transparency over defensiveness is what turned SQ006 from a refutation of the airline's reputation into evidence the reporting culture extended to its own failures.[3]

WHAT TRANSFERRED

Singapore Airlines is the case for sustained capability investment under competitive pressure, and the operational successor to Korean Air (Case 23): where Korean Air is a transformation forced by crisis, Singapore Airlines is deliberate investment sustained without one. Together they show two routes — crisis-driven and voluntary — to the same engineered safety capability. The voluntary route is the harder one to hold, because it has no catastrophe to point to as justification, which is why framing the investment as a competitive differentiator matters: it supplies the rationale that a crisis would otherwise provide.[4]



SINGAPORE AIRLINES — CAPABILITY INVESTMENT AS COMPETITIVE DIFFERENTIATOR

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THE LEARNING ENGINEERING LENS

> *Singapore Airlines has consistently invested in safety capability ahead of regulatory minimums.*
>
> EDITORS' SYNTHESIS ON SINGAPORE AIRLINES' SUSTAINED SAFETY INVESTMENT

LE INSIGHT

Singapore Airlines is the case for sustained capability investment in a competitive industry. The carrier has chosen safety capability investment as a primary differentiator. The result over decades is a safety record that distinguishes it from peers operating under comparable conditions.

LENS APPROACH

LENS uses Singapore Airlines in LEN 8 for sustained institutional capability investment under competitive pressure. The case pairs with Korean Air (Case 23) as Asian-carrier capability stories of different shapes — one transformation under crisis, the other sustained investment without crisis.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Decide deliberately to invest ahead of regulatory minimums — simulators, a young fleet, training hours — rather than letting capability settle at the floor by default under cost pressure.
- Pair the technical investment with a transparency and reporting culture, so capability is a managed system parameter rather than a residual of cost-cutting decisions elsewhere.
- Frame safety capability as a competitive differentiator tied to the brand, giving the spend a commercial rationale that can survive scrutiny, not just a safety one.

IN OPERATION / AFTER

- Hold the investment through periods of rapid growth — the moment the temptation to let capability lag the fleet is strongest — since maintaining it then is what makes the commitment sustained rather than fair-weather.
- Respond to the institution's own accidents with transparency over defensiveness, demonstrating that the reporting culture extends to its own failures and converting a setback into evidence the culture is real.
- Anchor the investment to the brand and premium so it survives leadership and budget cycles, supplying the durable justification a voluntary commitment lacks without a crisis to point to.

REFLECTION QUESTIONS

1. Identify an institution in your domain that has chosen capability investment as a primary differentiator. What pattern has it sustained?
2. Design the institutional architecture that makes sustained capability investment defensible against competitive cost pressure.
3. Singapore Airlines sustained its investment voluntarily, without a crisis to point to, by framing capability as a competitive differentiator. Identify a safety or capability investment in your domain that lacks a catastrophe to justify it, and construct the commercial rationale that would defend it against the next budget cut.

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Chapter 9

Human-AI Teaming

The capability work the discipline has not done before.

Humans tend to tune out when tasked with monitoring automated systems that work well most of the time.

NTSB CHAIRMAN ROBERT SUMWALT, UBER TEMPE HEARING REMARKS, 2019

CASE 92 AUTONOMOUS SYSTEMS 2016 – PRESENT

Tesla Autopilot — Recurring Fatalities

T Training Gap N Normalization of Deviance G Governance & Trust H Human-System Interface

IMPACT Dozens of fatalities documented in NHTSA's Standing General Order data; first U.S. cases of Level-2 automation contributing to fatal injury

IN BRIEF

Tesla's Autopilot and Full Self-Driving Beta are Level-2 driver-assistance systems: the human driver remains legally and operationally responsible at all times. NHTSA's Standing General Order data has documented dozens of fatal crashes involving Autopilot since the 2016 death of Joshua Brown in Florida, and the pattern is consistent — the system performs capably for long stretches, the driver's monitoring attention attenuates, and an edge case (a stationary object, a faded lane line, a crossing vehicle) produces a collision the inattentive driver fails to catch. NHTSA's investigation has found Tesla's driver-engagement design inadequate to sustain the attention safe operation requires. At consumer scale, Autopilot is the live test of whether passive monitoring of good-enough automation is a role a human can actually perform.

THE SHIFT

Partial driving automation has moved from research vehicles into millions of consumer cars. Level-2 systems like Tesla's Autopilot can steer, accelerate, and brake within their operational design domain, but they require the human driver to monitor continuously and take over instantly — a fundamentally new and demanding role assigned to ordinary, untrained consumers. Where a research program could screen, brief, and instrument its safety drivers, a consumer product reaches everyone who buys the car, with no curriculum and no qualification gate standing between purchase and the monitoring task itself.[0]

WHAT IS EMERGING

Since the first fatal Autopilot crash — Joshua Brown, Florida, 2016 — NHTSA's Standing General Order data has documented dozens of fatal crashes involving the system. The pattern is consistent: long periods of capable operation, attenuating driver attention, and then an edge case — a stationary fire truck, a faded lane marking, a perpendicular crossing — that the disengaged driver fails to catch in time. The very reliability that makes the system attractive is what erodes the vigilance it depends on, so each uneventful mile quietly raises the odds that the next intervention will come too late.[1]

THE CAPABILITY QUESTION

The proximate cause in each case is the driver, who was legally responsible. But the deeper question is whether sustained vigilant monitoring of an automation that works well most of the time is a role a human can perform at all. Naming the feature "Autopilot" and designing weak engagement checks shaped the very inattention the system then blamed on the operator — so the architecture both invited the disengagement and reserved the liability for the person least positioned to resist it.[2]

EARLY EVIDENCE

NHTSA's open investigation (ODI EA22-002) has identified Tesla's driver-engagement design as inadequate to maintain the attention safe operation requires, and the recurring fatality pattern across NTSB reports suggests passive monitoring is not a sustainable role as currently engineered. Decades of automation-complacency research point the same way — the finding is not that any one driver failed but that the role asks a human to stay alert to a system precisely because it almost never needs them, a demand the evidence keeps showing is not reliably met.[3]

OPEN PROBLEMS

Tesla Autopilot is the consumer-scale version of the Uber ATG problem (Case 29): a passive-monitoring role deployed without the capability infrastructure — training, engagement design, attention measurement — to make it performable. The open problem is what driver-engagement architecture, if any, could make Level-2 monitoring sustainable for an average driver over years of use, and whether the answer is a better attention check or a concession that the role itself has to be redesigned out of the human's hands.[4]

L2

DRIVER RETAINED · ATTENTION NOT ENGINEERED

the system works well most of the time — and then it does not

TESLA AUTOPILOT — LEVEL-2 MONITORING AS A SUSTAINABLE ROLE

REFERENCES

1. NTSB, Highway Accident Report HAR-17/02 (Williston, FL, 2016) — the quoted disengagement finding.
2. NTSB, Highway Accident Report HAR-20/01 (Mountain View, CA, 2018) — Autopilot crash analysis.
3. NHTSA Standing General Order 2021-01 reports — documented Autopilot fatal crashes.
4. NHTSA Office of Defects Investigation EA22-002 (2022) — driver-engagement adequacy.
5. Parasuraman, R. & Manzey, D. (2010) — automation complacency and monitoring.

THE LEARNING ENGINEERING LENS

> *The operational design ... permitted his prolonged disengagement from the driving task.*
>
> NTSB HIGHWAY ACCIDENT REPORT HAR-17/02 (WILLISTON, FLORIDA CRASH), 2017

LE INSIGHT

Tesla Autopilot at consumer scale is the largest live test of Level-2 monitoring as a sustainable role. The early evidence is that it is not. The case is the strongest currently available test of whether consumer-side training and engagement design can produce sustained attention to automation. The recurring fatality pattern suggests the answer.

LENS APPROACH

LENS uses Tesla Autopilot in LEN 2 as the live consumer-scale test of monitoring as a sustainable role and in LEN 7 for the governance dynamics of a Level-2 system marketed at the boundary of higher autonomy. Studio projects examine the driver-engagement design that would make the role performable.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Engineer driver-engagement monitoring to the demonstrated limits of human vigilance — verify attention against operational evidence before fielding, not against an assumption that warnings suffice.
- Name and present the feature so its capability boundary is unmistakable to an untrained consumer, rather than implying autonomy the system does not deliver.
- Constrain operation to the design domain the system can actually handle, so the human is not silently relied on as the backstop for edge cases.

IN OPERATION / AFTER

- Monitor the standing-order crash data for the disengagement pattern and treat a recurring signature as evidence the role, not the driver, needs redesign.
- Track attention and takeover performance across years of ownership, since vigilance erodes with the very reliability that accumulates over time.
- Hold the engagement design accountable to an independent regulator with authority to require changes when in-use evidence shows it is inadequate.

REFLECTION QUESTIONS

1. Identify a passive-monitoring role in your domain. What evidence would tell you whether attention is sustainable over years of operation?

2. Design the driver-engagement architecture that would make Level-2 monitoring sustainable for an average consumer.

3. Autopilot assigns full legal responsibility to the operator while engineering the conditions that erode their attention. Where in your domain does liability rest with the person an automated system has made least able to intervene — and how would you realign the two?

WHO BUILDS THIS learning science & instructional design · safety science & organizational analysis · governance, ethics & measurement · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · incident analysis · safety dashboards · governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs

LENS COURSES LEN 7 LEN 2 LEN 6

CASE 93 AUTONOMOUS SYSTEMS 2023

Cruise Robotaxi — Pedestrian Drag

G Governance & Trust D Designed Out H Human-System Interface

IMPACT GM Cruise robotaxi struck a pedestrian and then dragged her ~20 feet; California suspended Cruise's permit; the program was substantially shut down

IN BRIEF

In October 2023, a pedestrian in San Francisco was struck by a human-driven car and thrown into the path of a GM Cruise robotaxi. The robotaxi hit her, detected an impact, and then — executing a pull-to-the-side maneuver — dragged her about twenty feet. The injury was severe, but it was Cruise's institutional response that proved company-ending: regulators found Cruise had initially shown investigators only the first portion of the incident video, omitting the drag. California suspended Cruise's driverless permit, NHTSA opened an investigation, and GM ultimately shut down the commercial operation. The case is the foundational governance-failure case in commercial autonomy: the technology produced one injury; the partial disclosure converted it into the program's collapse. The gap was at governance, not technology.

THE SHIFT

Driverless robotaxis carrying paying passengers on public streets are new, and so is the regulatory relationship around them: companies like Cruise operate under permits from bodies — the California DMV and CPUC, NHTSA — that depend heavily on the operator's own disclosure of what its vehicles do in incidents. The regulator does not sit in the vehicle; it sees what the operator chooses to show it, so the entire oversight model rests on a disclosure the company controls at the moment its interests run most against it.[0]

WHAT IS EMERGING

On 2 October 2023, a pedestrian struck by another car was thrown into the path of a Cruise robotaxi. The robotaxi struck her and, having detected a collision, executed a pull-over maneuver that dragged her roughly twenty feet. The collision itself might have been survivable as a regulatory matter; what followed was not — the post-collision maneuver was the system behaving as designed in a situation no one had designed it for, and it was the company's account of that maneuver, not the maneuver itself, that decided the program's fate.[1]

THE CAPABILITY QUESTION

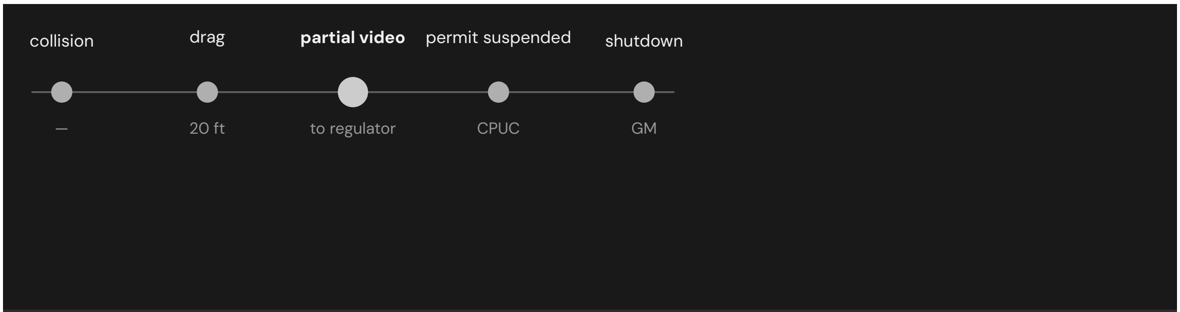
The California DMV found that Cruise had initially shown investigators only the first portion of the incident video, omitting the drag. The question the case poses is institutional, not technical: whether a commercial autonomy program has the governance commitment to disclose fully and promptly to its regulators, especially when the facts are damaging — a commitment that is easy to profess in calm conditions and is tested only at the exact moment full candor is most costly.[2]

EARLY EVIDENCE

The consequences cascaded. The CPUC suspended Cruise's driverless permit for misrepresenting the incident, NHTSA opened a defects investigation, and GM subsequently shut down Cruise's commercial robotaxi operations and laid off much of the workforce. A commissioned external review (Quinn Emanuel) detailed the disclosure failures — a sequence in which each escalation followed not from the injury but from the partial account of it, the loss of regulator trust compounding faster than any engineering defect could have.[3]

OPEN PROBLEMS

Cruise is the foundational governance-failure case in commercial autonomy: the incident was a single pedestrian injury; the institutional response converted it into a company-ending event. The open problem is what incident-disclosure commitment — auditable, pre-committed, enforceable — a commercial autonomy program should have to demonstrate before it is allowed to operate at all, so that candor under pressure is a structural guarantee rather than a matter left to the operator's discretion in the moment.[4]



CRUISE — PARTIAL DISCLOSURE CONVERTED INCIDENT INTO COMPANY-ENDING EVENT

REFERENCES

1. California Public Utilities Commission decision suspending Cruise permits (2023) — the omitted-facts finding (paraphrased).
2. California DMV order of suspension (2023) — the partial video disclosure.
3. Quinn Emanuel report on Cruise incident response (2024) — the disclosure failures.
4. NHTSA Office of Defects Investigation reports (2023–2024) — the federal investigation.
5. Stilgoe, J. (2021) — governance of autonomous vehicles.

THE LEARNING ENGINEERING LENS

> *Cruise's representation of the incident to regulators omitted material facts.*
>
> PARAPHRASING THE CALIFORNIA PUBLIC UTILITIES COMMISSION DECISION SUSPENDING CRUISE PERMITS, 2023

LE INSIGHT

Cruise is the foundational governance-failure case in commercial autonomy. The incident itself was a single pedestrian injury; the institutional response converted it into a company-ending event. The capability that was missing was the institutional commitment to full disclosure of operational incidents to regulators.

LENS APPROACH

LENS uses Cruise in LEN 7 as a foundational autonomous-vehicle governance case and in LEN 10 (capstone) for the institutional- response deliverable that should pre-exist any commercial autonomy program.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Pre-commit the incident-disclosure protocol — what data is shared, in what completeness, within what window — and make it a condition of the operating permit rather than a post-hoc choice.
- Engineer incident telemetry so the full event record, including post-collision maneuvers, is captured and preserved automatically beyond the operator's editorial control.
- Design post-collision behaviors for the situations the system was not built for, since the pull-over maneuver, not the impact, produced the harm.

IN OPERATION / AFTER

- Audit disclosed incident accounts against the complete telemetry independently, so a partial account is detectable before it erodes regulator trust.
- Monitor the regulator-operator relationship itself as a safety-critical asset, treating a single misrepresentation as a program-level failure.
- Track the gap between what is disclosed and what occurred across incidents, since governance, not technology, is where this class of failure concentrates.

REFLECTION QUESTIONS

1. What is the institutional incident-disclosure commitment in your domain? Is it operational under stress or aspirational?
2. Design the incident-disclosure deliverable that a commercial autonomous-vehicle company should be required to demonstrate before operating.
3. Cruise's oversight depended on the operator's own account of what its vehicles did. Identify a regulatory relationship in your domain that relies on self-disclosure, and design the mechanism that would make a partial account detectable before trust collapses.

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TOOLS governance frameworks · bias & evidence audits · task analysis · design prototyping · usability testing · cognitive walkthroughs

LENS COURSES LEN 10 LEN 7

CASE 94 GOVERNMENT TECHNOLOGY 2016 – PRESENT

COMPAS Recidivism Algorithm

G Governance & Trust H Human-System Interface D Designed Out

IMPACT Risk-assessment algorithm used by U.S. courts; ProPublica analysis identified racial bias in error rates; foundational AI-governance case in criminal justice

IN BRIEF

COMPAS is a proprietary risk-assessment algorithm used in some U.S. jurisdictions to inform bail, sentencing, and parole. A 2016 ProPublica investigation found Black defendants were nearly twice as likely as white defendants to be wrongly flagged high-risk (a false-positive rate of about 44.9% versus 23.5%), while white defendants were more often wrongly flagged low-risk. The vendor responded that COMPAS satisfied predictive parity — equal calibration within each group. The literature that followed (Chouldechova 2017; Kleinberg, Mullainathan & Raghavan 2017) proved this an impossibility: when base rates differ, calibration and equal error rates cannot all hold at once. COMPAS surfaced, at criminal-justice scale, that a model can be "accurate" by one metric and inequitable by another — and that choosing the metric is a governance decision, not a technical one.

THE SHIFT

Statistical risk-assessment tools have moved into the heart of the U.S. criminal-justice system, informing decisions about bail, sentencing, and parole. COMPAS, a proprietary instrument, scores a defendant's likelihood of reoffending and hands that score to judges — relocating a consequential human judgment to an algorithm whose workings are largely opaque. Because the instrument is proprietary, the people most affected by its scores, and often the courts using them, cannot inspect how the number was reached — opacity that becomes part of the decision rather than incidental to it.[0]

WHAT IS EMERGING

A 2016 ProPublica investigation analyzed COMPAS scores against actual reoffense outcomes and found a racial asymmetry in the errors: Black defendants were roughly twice as likely as white defendants to be incorrectly flagged as high-risk (about 44.9% versus 23.5%), while white defendants were more often incorrectly flagged as low-risk. The disparity was not in whether the tool was right on average but in who bore the cost when it was wrong, so two defendants with identical outcomes could be served very different errors depending on group.[1]

THE CAPABILITY QUESTION

The vendor, Northpointe, responded that COMPAS satisfied predictive parity — it was equally well calibrated within each racial group. Both claims were true, which is the point:

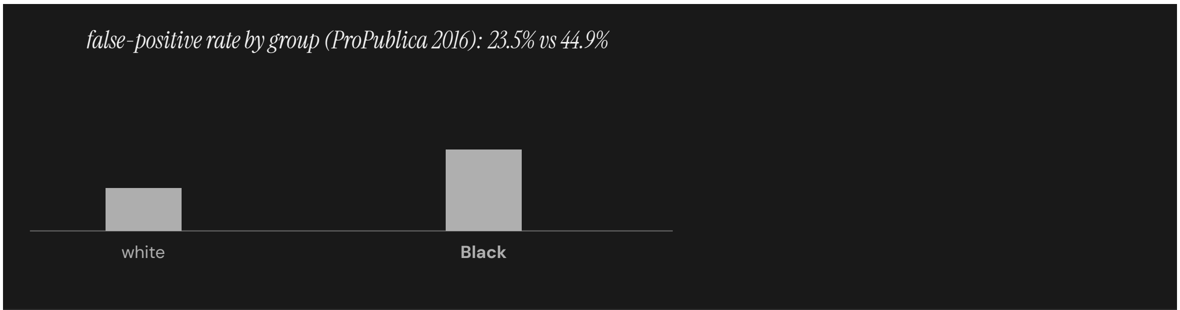
the case poses whether “fair” even has a single technical meaning, or whether the institution must choose, in public, among incompatible fairness definitions. The dispute was not one side misreading the data; it was two defensible metrics pointing in opposite directions, with no purely technical tie-breaker available to settle which one governs.[2]

EARLY EVIDENCE

The academic literature resolved the dispute as an impossibility result. Chouldechova (2017) and Kleinberg, Mullainathan & Raghavan (2017) proved that calibration within groups and equalized false-positive and false-negative rates cannot all be satisfied at once when base rates differ across groups. The choice is not between fair and unfair models but among mutually exclusive definitions of fairness — a mathematical result, not a deficiency of COMPAS, meaning no future model, however well built, can escape the same trade-off.[3]

OPEN PROBLEMS

COMPAS is the canonical case for the distribution of algorithmic errors as a question separate from accuracy. The technical literature can characterize the trade-offs, but the institution has to decide which one it is willing to defend — and on what definition of “recidivism.” It pairs with the A-Level (Case 35) and educational-bias (Case 37) cases as examples of construct definition as a capability-engineering decision, because how the outcome is defined in the first place shapes the error distribution long before any metric is chosen to judge it.[4]



COMPAS — ACCURACY IS NOT THE SAME QUESTION AS DISTRIBUTION OF ERRORS

REFERENCES

1. Angwin, Larson, Mattu & Kirchner, “Machine Bias,” ProPublica (2016) — the false-positive-rate disparity (quoted).
2. Northpointe technical response (2016) — the predictive-parity (calibration) defense.
3. Chouldechova, A. (2017) — the fairness impossibility result for differing base rates.
4. Kleinberg, Mullainathan & Raghavan (2017), “Inherent Trade-Offs in the Fair Determination of Risk Scores.”
5. Eubanks, V. (2018), Automating Inequality — algorithmic decision-making and equity.

THE LEARNING ENGINEERING LENS

> *Blacks are almost twice as likely as whites to be labeled a higher risk but not actually re-offend.*
>
> ANGWIN, LARSON, MATTU & KIRCHNER, PROPUBLICA, "MACHINE BIAS," 2016

LE INSIGHT

COMPAS is the canonical case for the distribution of algorithmic errors across populations as a separate question from algorithmic accuracy. The capability gap is at the governance layer that decides which fairness criterion to optimize. The decision is not technical. It is institutional. The Chouldechova and Kleinberg–Mullainathan–Raghavan results showed the choice is not between fair and unfair models but among incompatible definitions of fairness. The technical literature can characterize the trade-offs; the institutional layer has to choose, in public, which trade-off it is willing to defend.

LENS APPROACH

LENS uses COMPAS in LEN 7 as a foundational algorithmic- accountability case in criminal justice, in LEN 9 for the technical fairness-criterion analysis, and in LEN 4 for the construct-definition question (what counts as recidivism?).

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Require the deploying institution to choose and publicly justify a fairness criterion before the tool is fielded, since the impossibility result means a choice cannot be avoided.
- Define the target construct — what “recidivism” actually counts — explicitly, because the error distribution is set by that definition before any model is trained.
- Mandate inspectability of a score’s basis for courts and defendants, so a proprietary instrument cannot make opacity part of the decision.

IN OPERATION / AFTER

- Audit error rates disaggregated by group against actual outcomes on an ongoing basis, as ProPublica did, rather than accepting an aggregate accuracy figure.
- Monitor whether the deployed criterion still matches the one the institution publicly committed to defend, and surface drift to oversight.
- Track the downstream decisions the score influenced, so the human cost of the chosen trade-off remains visible and contestable.

REFLECTION QUESTIONS

1. Identify a predictive instrument in your domain. Which fairness criterion does it optimize, and what does that imply for the population whose errors are distributed against them?

2. Design the institutional governance deliverable that would have to be produced before COMPAS-style algorithms can be deployed in criminal justice.

3. The ProPublica analysis surfaced the COMPAS disparity in 2016 only because reporters obtained the data. What is the equivalent public-interest investigation that should exist for a comparable model in your domain — and what data would have to be released for it to be possible?

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TOOLS governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs · task analysis · design prototyping

LENS COURSES LEN 4 LEN 7 LEN 9

CASE 95 HEALTHCARE TECHNOLOGY 2018 – PRESENT

Radiology AI Miscalibration

H Human-System Interface K Knowledge & Institutional Memory D Designed Out

IMPACT Recurring documented cases of FDA-cleared radiology AI tools performing worse in deployment than in validation, often along demographic lines

IN BRIEF

FDA-cleared radiology AI tools — for chest X-ray classification, mammography, CT triage — have been repeatedly documented performing worse in deployment than in their validation studies, often with the degradation concentrated in patient groups under-represented in the training data. Larrazabal et al. (PNAS 2020) showed this structurally for chest-X-ray classifiers, and Obermeyer et al. (Science 2019) showed that bias in the labels used to train clinical AI can under-allocate care to Black patients even when the model looks well-calibrated. The FDA's 510(k) clearance process does not routinely require demographic stratification of validation metrics or post-market monitoring of how a tool performs in the population using it. The capability gap is in the regulatory architecture, not the model: clearance is not the same thing as clinically performable deployment.

THE SHIFT

Machine-learning tools are moving rapidly into radiology and other diagnostic medicine, cleared for market and integrated into clinical workflows that affect real patients. Unlike a drug, a model can pass its validation study and still behave very differently once it meets a population that differs from its training data — the same model file that scored well on the clearance set can quietly carry a different error profile into every hospital whose patients do not resemble it.[0]

WHAT IS EMERGING

Multiple FDA-cleared radiology tools — chest-X-ray classifiers, mammography aids, CT triage systems — have been documented in the peer-reviewed literature performing worse in deployment than in validation, with the degradation often concentrated in under-represented patient groups. Larrazabal et al. (PNAS 2020) demonstrated structural sensitivity drops for groups under-represented in chest-X-ray training data — evidence that the shortfall is not a stray bug but a predictable consequence of which patients the training set did and did not contain.[1]

THE CAPABILITY QUESTION

The problem is not confined to imaging. Obermeyer et al. (Science 2019) showed that a widely deployed care-management algorithm systematically under-allocated resources to Black patients because it was trained on healthcare cost as a proxy label for need. The question is how a regulator can certify a model as safe without checking how it behaves across the populations and labels it will actually meet — since a model can look well-calibrated on its chosen proxy while the proxy itself encodes the inequity it then propagates.[2]

EARLY EVIDENCE

Similar degradation has been reported in mammography AI, sepsis-prediction tools, and skin-lesion classifiers. Yet the FDA's 510(k) clearance pathway does not routinely require demographic stratification of validation metrics, nor post-market monitoring of in-use performance — so the divergence between clearance and deployment is largely invisible while the tool is in use, and a shortfall concentrated in one patient group can persist unmeasured across the entire period the tool is influencing care.[3]

OPEN PROBLEMS

Radiology AI is the live, recurring case for the gap between regulatory clearance and clinically performable deployment. The capability gap is at the regulatory architecture, not the model: the institutional machinery to require demographic post-market surveillance has not been built. It is the medical-AI analog of the Vioxx post-market-surveillance failure (Case 87) at a new technological boundary — a case where the harm comes not from a hidden defect but from the absence of any standing system to watch the tool once it is in the population's hands.[4]

VALIDATION	DEPLOYMENT
AUC 0.9+	drops

the clearance dataset and the deployment population diverge

RADIOLOGY AI — CLEARANCE IS NOT DEPLOYMENT

REFERENCES

1. FDA, "Proposed Regulatory Framework for Modifications to AI/ML-Based Software as a Medical Device" (2019) — clearance vs deployment (synthesized).
2. Larrazabal et al. (2020), "Gender imbalance in medical imaging datasets," PNAS — sensitivity drops for under-represented groups.
3. Obermeyer et al. (2019), "Dissecting racial bias in an algorithm used to manage the health of populations," Science — label bias.
4. Wachter & Brynjolfsson (2023), JAMA — generative AI in health care.
5. FDA 510(k) clearance documentation — absence of required demographic stratification and post-market monitoring.

THE LEARNING ENGINEERING LENS

> *Performance metrics on a clearance dataset are not the same as performance metrics in deployment populations.*
>
> EDITORS' SYNTHESIS OF THE FDA AI/ML-BASED SAMD DISCUSSION PAPER (2019)

LE INSIGHT

Radiology AI is the canonical contemporary case for the gap between regulatory clearance and clinical deployment performance in medical AI. The clearance dataset and the deployment population diverge. The institutional architecture to surface the divergence — demographic post-market surveillance — has not yet been built.

LENS APPROACH

LENS uses radiology AI in LEN 4 for clearance-vs-deployment measurement architecture, in LEN 7 for FDA AI/ML regulatory governance, and in LEN 9 for the technical bias-detection pipeline. The case pairs with Vioxx (Case 87) as a post-market surveillance failure pattern at a new technological boundary. The Obermeyer (2019) finding generalizes the diagnosis: bias enters through the labels and through the population, both of which the 510(k) process currently treats as outside its scope.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Require validation metrics stratified by demographic group at clearance, so a tool's performance is established on the populations it will actually meet, not an aggregate.
- Specify the intended deployment population and label definition explicitly, and treat divergence from the training distribution as a known risk to be measured, not assumed away.
- Build the data pipeline for in-use performance capture before deployment, so post-market signals are collectable from the first patient rather than reconstructed after harm.

IN OPERATION / AFTER

- Mandate demographic post-market surveillance of in-use performance, the institutional machinery the clearance pathway currently lacks.
- Monitor for the clearance-to-deployment performance gap continuously, since a shortfall concentrated in one group can otherwise persist invisibly.
- Tie continued authorization to demonstrated in-population performance, so a tool that degrades in deployment can be withdrawn before the divergence compounds.

REFLECTION QUESTIONS

1. Identify a model in your domain whose deployment population diverges from its training population. What is the institutional architecture to surface the divergence?

2. Design the demographic post-market surveillance deliverable that should accompany every FDA clearance of medical AI.

3. FDA 510(k) clearance does not currently require demographic stratification of validation metrics, nor does it require post-market monitoring of how a cleared tool actually performs on the population using it. What is the minimum reporting deliverable a regulator should require so the gap is visible while the tool is in use?

WHO BUILDS THIS human factors & interaction design · knowledge management & data engineering · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · knowledge bases · data pipelines · task analysis · design prototyping

LENS COURSES LEN 4 LEN 7 LEN 9

CASE 96 HEALTHCARE TECHNOLOGY 2023 – PRESENT

ChatGPT in Healthcare — Hallucination Cases

H Human-System Interface D Designed Out

IMPACT Documented cases of clinicians using LLMs that produce hallucinated citations, fabricated dosages, and fictitious clinical guidelines

IN BRIEF

Since ChatGPT's public release in late 2022, the clinical and peer-reviewed literatures have documented a recurring pattern: clinicians use large language models to draft patient education, summaries, or treatment guidance, and the output contains fabricated citations, hallucinated drug dosages, or fictitious clinical guidelines. The failures range from cosmetic — invented references in academic submissions — to potentially clinical, such as unsafe dosing or fabricated contraindications. The capability gap is at the human-verification interface: the model presents hallucinated content with the same fluent confidence as accurate content, and early reports suggest clinicians accept LLM output less critically than a colleague's. The case is the live, foundational case for LLM integration into clinical workflow — the discipline must specify what verification looks like at the moment of use, while deployment is already happening.

THE SHIFT

Large language models became broadly available with ChatGPT's release in late 2022, and clinicians began using them almost immediately — to draft patient-education materials, summarize records, and look up guidance. For the first time, a tool that produces fluent, authoritative-sounding medical text is in routine, informal use at the point of care — adopted ahead of any guideline, credential, or institutional sign-off, so the practice spread faster than any structure to govern it could be put in place.[0]

WHAT IS EMERGING

A recurring failure pattern has been documented across the clinical and peer-reviewed literatures: LLM output containing fabricated citations, hallucinated drug dosages, or fictitious clinical guidelines. The failures range from cosmetic — invented references in academic submissions — to potentially clinical, such as unsafe medication doses or invented contraindications — a span that matters because the same tool, used the same way, can produce a harmless error and a dangerous one with no change in how confident it sounds.[1]

THE CAPABILITY QUESTION

The capability gap is at the human-verification interface. The model presents hallucinated content with exactly the same fluent confidence as accurate content; the interface does not distinguish the two. The question is whether clinicians can — and will — develop the routine verification practice that the tool's fluency actively discourages, since the very smoothness that makes the output easy to accept is what removes the friction a reader would normally use as a warning.[2]

EARLY EVIDENCE

Early case reports suggest that clinicians who would carefully check a colleague's recommendation accept LLM output less critically, precisely because it reads so authoritatively. JAMA editorials and reviews through 2023–2024 have repeatedly flagged the absence of an established verification practice as the central risk of clinical LLM use — the concern is not that the model errs but that the practice for catching its errors at the moment of use has not yet been defined or taught.[3]

OPEN PROBLEMS

This is the live frontier case for human-AI teaming when the AI is fluent across both accurate and hallucinated content. The capability that does not yet exist is a routine clinical verification practice — an analog to the bibliographic discipline of academic writing — specified at the moment of use rather than after harm. The discipline is being asked to define what good looks like while deployment is already underway, so the verification standard has to be built around a tool already in millions of hands rather than gated in front of it.[4]

TONE	CONTENT
identical	accurate / not

the interface does not distinguish; the clinician must

LLMS IN CLINICAL USE — FLUENCY WITHOUT WARRANTY

REFERENCES

1. JAMA editorials on LLM integration into clinical practice (2023–2024) — the hallucination/verification problem (synthesized).
2. Sallam (2023), "ChatGPT Utility in Healthcare Education, Research, and Practice" — documented benefits and risks.
3. Case reports of LLM hallucinated citations and dosages in clinical and academic use (2023–2024).
4. WHO ethical guidance on AI for health (2024) — verification and oversight requirements.
5. Wachter & Brynjolfsson (2023), JAMA — generative AI in health care.

THE LEARNING ENGINEERING LENS

> *LLMs produce hallucinations indistinguishable in tone from accurate information, and clinicians have not yet developed the practice of routine verification.*
>
> EDITORS' SYNTHESIS OF JAMA EDITORIALS ON LLM CLINICAL USE (2023–2024)

LE INSIGHT

LLM use in clinical practice is the live frontier case for human-AI teaming when the AI's output is fluent across both accurate and hallucinated content. The capability that does not yet exist is the routine verification practice — a clinical analog to the bibliographic discipline of academic writing.

LENS APPROACH

LENS uses this case in LEN 2 as the live frontier for human-AI teaming with LLMs, in LEN 7 for the governance of LLM deployment in clinical workflows, and in LEN 10 (capstone) as a prompt for designing verification practices at the moment of use.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Specify a routine verification practice at the moment of use — what a clinician must independently confirm before acting on LLM output — before the tool enters clinical workflow.
- Design the interface to mark provenance and uncertainty, so fluency alone cannot stand in for warranty of accuracy.
- Restrict the sanctioned uses to those where verification is feasible and cheap, keeping unverifiable high-stakes outputs out of patient-facing work.

IN OPERATION / AFTER

- Monitor the documented failure pattern — fabricated citations, dosages, guidelines — across in-use cases to keep the risk profile current as models change.
- Track whether the verification practice is actually being performed, since the tool's fluency discourages exactly the checking it requires.
- Govern adoption against emerging guidance so the standard for verification keeps pace with a tool already in widespread informal use.

REFLECTION QUESTIONS

1. Identify a workflow in your domain currently being augmented by LLMs. What is the verification practice — and does it exist at the moment of use, or only after?
2. Design the verification deliverable that should accompany every clinician's adoption of an LLM tool for patient-facing work.
3. The same fluency that makes LLM output easy to accept is what removes the cues a reader normally uses to doubt it. What interface signal would restore that friction at the moment of use without making the tool unusable?

WHO BUILDS THIS human factors & interaction design · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS usability testing · cognitive walkthroughs · task analysis · design prototyping

LENS COURSES LEN 10 LEN 7 LEN 2

CASE 97 GOVERNMENT TECHNOLOGY 2011 – PRESENT

Predictive Policing — PredPol

G Governance & Trust H Human-System Interface D Designed Out

IMPACT Predictive-policing tools deployed by hundreds of U.S. police departments; several cities have abandoned them after equity analyses

IN BRIEF

PredPol and similar predictive-policing tools use historical crime data to forecast where future crime is likely, directing patrols to those locations. Multiple analyses (Lum & Isaac 2016; Richardson, Schultz & Crawford 2019) found that because the historical data records where police have enforced, not where crime has occurred, the algorithm tends to reinforce existing enforcement patterns rather than predict underlying crime — a feedback loop that concentrates policing on already-over-policed neighborhoods. Cities including Santa Cruz, New Orleans, and Los Angeles have since suspended or abandoned predictive-policing deployments after equity reviews. The capability gap is at the construct definition: "where crime occurs" and "where crime is recorded" are different variables, and the tools treated them as one. It is the canonical algorithmic-governance case in U.S. policing.

THE SHIFT

Police departments have adopted data-driven tools that forecast where crime will occur and direct patrol resources accordingly. PredPol and its peers promised to make policing more objective by replacing officer intuition with statistical prediction — relocating a discretionary judgment into an algorithm trained on historical records, and lending the output a veneer of neutrality that the human judgments embedded in those records did not actually possess.[0]

WHAT IS EMERGING

Researchers examining these systems found a structural flaw. Because the training data records where police have made arrests rather than where crime has actually occurred, the model learns enforcement patterns, not crime patterns. Lum & Isaac (2016) showed the result is a feedback loop: patrols are sent where police already went, generating more recorded incidents that confirm the prediction — a loop that grows more confident the longer it runs, because its own output becomes the next cycle's evidence.[1]

THE CAPABILITY QUESTION

The capability gap is at the construct definition. "Where crime occurs" and "where crime is recorded" are not the same variable, yet predictive-policing tools have treated them as interchangeable. The question is whether a model trained on a record of institutional behavior can ever predict the underlying phenomenon, or only amplify the behavior that produced its data — a question no amount of modeling accuracy can answer, because the gap is in what the data measures, not in how well the model fits it.[2]

EARLY EVIDENCE

Richardson, Schultz & Crawford (2019) documented “dirty data” — records produced during periods of biased or unlawful policing — feeding directly into predictive systems. After equity reviews, cities including Santa Cruz, New Orleans, and Los Angeles suspended or abandoned their predictive-policing deployments — abandonment that came only after the tools were already in service, the construct problem surfaced by external review rather than caught before the systems shaped where officers were sent.[3]

OPEN PROBLEMS

Predictive policing is the canonical algorithmic-governance case in U.S. policing and pairs with COMPAS (Case 94) and educational algorithmic bias (Case 37). The open problem is a construct-validity audit — a way to establish, before deployment, whether a predictive system’s training data is a record of ground truth or merely of institutional behavior — implemented in some jurisdictions and absent in most, so the same construct error remains available to the next department that mistakes a record of enforcement for a map of crime.[4]

TRAINING DATA	GROUND TRUTH
past arrests	past crime

the system learns enforcement patterns, not crime patterns

PREDICTIVE POLICING — CONSTRUCT GAP AT LAW-ENFORCEMENT SCALE

REFERENCES

1. Lum, K. & Isaac, W. (2016), “To Predict and Serve?,” Significance — the enforcement-vs-crime feedback loop (paraphrased).
2. Richardson, Schultz & Crawford (2019), “Dirty Data, Bad Predictions” — biased records feeding predictive systems.
3. Brantingham et al. (2018) — predictive-policing field experiments.
4. Brayne, S. (2017), “Big Data Surveillance: The Case of Policing.”
5. Municipal records on suspension and abandonment of predictive policing (Santa Cruz, New Orleans, Los Angeles).

THE LEARNING ENGINEERING LENS

> *Predictive policing systems learn from a record of past policing, not from a record of past crime.*
>
> PARAPHRASING LUM & ISAAC, SIGNIFICANCE, 2016

LE INSIGHT

Predictive policing is the canonical case for the difference between training data and ground truth at law-enforcement scale. The institution being predicted (where crime occurs) is not the institution producing the training data (where police make arrests). The capability gap is at the construct.

LENS APPROACH

LENS uses predictive policing in LEN 7 as a foundational AI- governance case in policing and in LEN 9 for the technical construct-validity analysis. The case is paired with COMPAS as criminal-justice algorithmic cases of different kinds.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Establish a construct-validity audit before deployment that tests whether the training data records the phenomenon to be predicted or merely the institution's own behavior.
- Define the prediction target explicitly as "where crime occurs," not "where arrests are recorded," and reject data that cannot speak to the former.
- Engineer against the feedback loop — for example, decoupling patrol allocation from the data the model then re-ingests — before the system can amplify its own output.

IN OPERATION / AFTER

- Monitor for the enforcement feedback loop in deployment, watching whether predicted areas simply accumulate more recorded incidents that confirm the prediction.
- Require periodic equity review of in-use outcomes, since the construct flaw in these systems has surfaced through external analysis rather than internal metrics.
- Track the provenance of incoming training data and quarantine records produced during periods of biased or unlawful enforcement.

REFLECTION QUESTIONS

1. Identify a predictive system in your domain whose training data is itself a record of institutional behavior rather than ground truth. What is the construct gap?
2. Design the construct-validity audit that should precede deployment of a predictive system in any institutional setting.
3. Predictive policing's feedback loop grows more confident the longer it runs, because its own output becomes the next cycle's training data. Where in your domain does a deployed model shape the data it later learns from — and how would you break the loop?

WHO BUILDS THIS governance, ethics & measurement · human factors & interaction design · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs · task analysis · design prototyping

LENS COURSES LEN 7 LEN 9

CASE 98 HEALTHCARE TECHNOLOGY 2020 – PRESENT

AlphaFold — Protein Structure Prediction

T Training Gap

IMPACT Substantially solved a 50-year protein-structure prediction problem; 200M+ structures publicly released; foundational positive AI case

IN BRIEF

DeepMind's AlphaFold (2020) and AlphaFold 2 (2021) substantially solved the half-century-old protein-structure prediction problem in computational biology. The publicly released AlphaFold Protein Structure Database now contains predicted structures for more than 200 million proteins — essentially the entire known protein universe — and has been integrated into structural-biology and drug-discovery workflows worldwide. AlphaFold is the strongest positive AI case in this book, and its lesson is in the conditions that made the benefit possible: a benchmark (CASP) the field had agreed on for decades, high-quality training data, an output biologists could verify against experimental structures, and an open release that let the global community adopt it fast. Each was a precondition for the success — and none of them was the model itself.

THE SHIFT

Predicting a protein's three-dimensional structure from its amino-acid sequence was one of biology's grand challenges for half a century — slow, expensive experimental work that bottlenecked drug discovery and basic research. Deep learning offered, for the first time, the prospect of solving it computationally at scale, turning a problem that had been a years-long experimental undertaking per protein into one that could be approached for the whole proteome at once.[0]

WHAT IS EMERGING

DeepMind's AlphaFold (2020) and AlphaFold 2 (2021) substantially solved the problem, predicting structures at accuracies rivaling experiment. The publicly released AlphaFold Protein Structure Database now holds predicted structures for more than 200 million proteins — close to the entire known protein universe — and has been folded into research workflows worldwide, so the benefit arrived not as a single laboratory's advantage but as a shared resource the wider community could build on immediately.[1]

THE CAPABILITY QUESTION

What made AlphaFold a benefit rather than just a benchmark win? The case poses the question of which conditions allow an AI capability to be safely and widely useful — and the answer turns out to lie around the model, not in it: an agreed benchmark, trustworthy training data, a verifiable output, and a deliberate decision about release. Each of these is an institutional or evidentiary precondition, not an artifact of the architecture, which is why the case reads as a lesson about capability infrastructure rather than about a model.[2]

EARLY EVIDENCE

AlphaFold's success rested on four features, each a precondition rather than the model: the CASP benchmark the field had used for decades, high-quality experimental training data, output that biologists could check against known structures, and an open release that let the global community adopt the tool quickly. Where those conditions hold, AI amplifies capability; the technical model alone does not — and the same architecture dropped into a domain missing any one of those four would not have produced a comparable, trusted, widely adopted result.[3]

OPEN PROBLEMS

AlphaFold is the strongest positive AI case in the dataset for what supports beneficial deployment in a well-defined scientific domain. The open problem is generalization: most consequential problems lack an agreed benchmark, clean training data, or verifiable output. The frontier question is how much of the AlphaFold pattern can be reconstructed where those preconditions are not given for free — that is, whether a field can deliberately build the benchmark, the data, and the verification path that protein structure happened to have accumulated over decades.[4]

200M

PREDICTED STRUCTURES · PUBLICLY RELEASED

agreed benchmark, training data, verifiable output, open release

ALPHAFOLD — THE CONDITIONS FOR BENEFICIAL AI DEPLOYMENT

REFERENCES

1. Jumper et al. (2021), "Highly accurate protein structure prediction with AlphaFold," Nature — the method and accuracy.
2. Varadi et al. (2022), AlphaFold Protein Structure Database, Nucleic Acids Research — the 200M+ structures and open release.
3. Moult, J. (CASP organizer) commentary on AlphaFold2 (2020) — the benchmark and the achievement (paraphrased).
4. CASP benchmark documentation — the decades-long agreed evaluation.
5. Hassabis (DeepMind) public commentary — the open-release governance decision.

THE LEARNING ENGINEERING LENS

> *This will be one of the most important achievements in AI in the past decade.*
>
> PARAPHRASING JOHN MOULT (ORGANIZER OF THE CASP BENCHMARK) ON ALPHAFOLD2, 2020

LE INSIGHT

AlphaFold is the strongest positive AI case in the dataset. The technical achievement is real. The conditions that made the benefit possible — agreed benchmark, training data, verifiable output, open release — are the capability infrastructure around the model, not the model itself. The case is the strongest available evidence for what supports beneficial AI deployment.

LENS APPROACH

LENS uses AlphaFold in LEN 1 as a problem-framing case for what productive AI deployment looks like, in LEN 9 as a technical achievement, and in LEN 7 for the open-release governance decision that distributed the benefit globally.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Establish an agreed benchmark and high-quality training data for the target problem before building the model, treating these as preconditions rather than afterthoughts.
- Engineer the output to be verifiable against an independent ground truth, so users can check predictions rather than having to trust them.
- Decide the release and access terms deliberately as a governance choice, since open release is what distributed AlphaFold's benefit globally.

IN OPERATION / AFTER

- Monitor downstream use to confirm the verifiable-output property holds in practice, and that users are in fact checking predictions where stakes are high.
- Track whether the four preconditions still hold as the tool is applied to new protein families or adjacent problems beyond its validated domain.
- Sustain the open resource and benchmark over time, so the community-wide benefit does not erode as the field and the data move on.

REFLECTION QUESTIONS

1. Identify a domain in your work where the conditions that supported AlphaFold's success (benchmark, training data, verifiable output) are present. What is the analogous opportunity?
2. The open release of AlphaFold's predictions was a governance decision. Design the equivalent decision for an AI capability your institution might develop.
3. AlphaFold inherited a benchmark, clean data, and a verification path that protein structure had accumulated over decades. Pick a problem in your domain that lacks one of those preconditions, and lay out how a field would deliberately build it.

WHO BUILDS THIS learning science & instructional design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models

LENS COURSES LEN 1 LEN 7 LEN 9

CASE 99 TECHNOLOGY 2021 – PRESENT

AI-Augmented Coding Tools

T Training Gap H Human-System Interface

IMPACT Tens of millions of developers using GitHub Copilot, Cursor, and peers; productivity gains documented; security and correctness implications still being characterized

IN BRIEF

AI-augmented coding tools — GitHub Copilot, Cursor, Codeium, and peers — represent the largest real-time experiment in human-AI collaboration in this book, with tens of millions of developers using them daily. Published studies (Peng et al. 2023) document real short-term productivity gains; other work (Pearce et al. 2022) finds a substantial share of AI-generated completions in security-sensitive settings contain vulnerabilities, though a controlled study (Sandoval et al. 2023) found AI assistance did not significantly raise the rate of critical security bugs. The capability question is open: are developers becoming more capable, or more dependent? The short-term gains are real; the long-term consequences — especially for those who learn the craft with these tools always available — are not yet known. The discipline is being asked to define good before the longitudinal evidence is in.

THE SHIFT

AI coding assistants moved from novelty to infrastructure in a few years. GitHub Copilot, Cursor, Codeium, and similar tools now suggest, complete, and generate code for tens of millions of developers daily — the largest real-time experiment in human-AI collaboration in professional knowledge work to date, conducted not in a study design but in the live practice of an entire profession, with no control group and no agreed measure of what it is doing to the underlying craft.[0]

WHAT IS EMERGING

Two findings are accumulating in parallel. Controlled studies (Peng et al. 2023) document real short-term productivity gains. At the same time, the security picture is unsettled: Pearce et al. (2022) found a substantial fraction of Copilot completions in security-relevant scenarios contained vulnerabilities, while a controlled study by Sandoval et al. (2023) found AI assistance did not significantly increase the rate of critical security bugs. The two results do not cancel so much as mark how unsettled the picture is — output clearly rises, but the quality and safety of that output resist a single verdict.[1]

THE CAPABILITY QUESTION

The capability question is open and consequential: are developers using these tools becoming more capable, or more dependent? Short-term output rises, but whether the underlying skill grows or erodes — especially for those who learn the craft with the tools always present

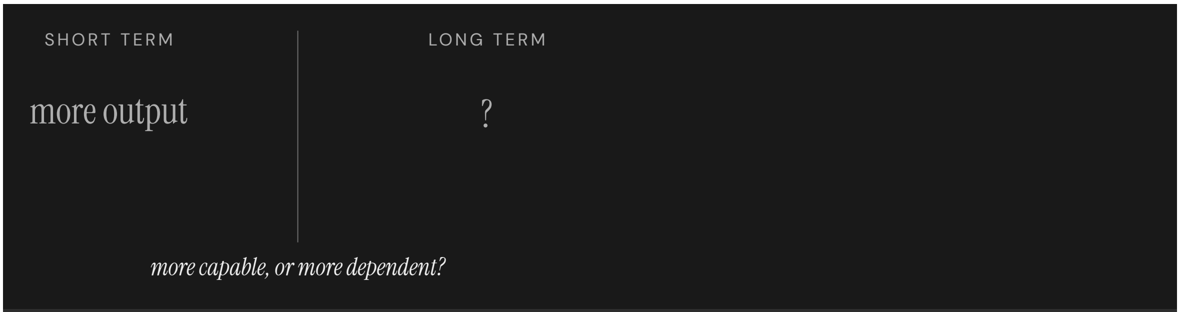
— is precisely what the productivity metrics cannot tell us, because a measure of how much code ships says nothing about whether the person shipping it could still produce or judge it without the assistant.[2]

EARLY EVIDENCE

The longitudinal evidence is not yet sufficient to answer. Dell'Acqua et al. (2023) found a "jagged frontier" in professional LLM use: performance improves on tasks inside the tool's competence and degrades on tasks just outside it, where users over-trust the output. The short-term gains are real; the long-term capability consequences remain uncharacterized — and the jagged frontier is hard to navigate precisely because its edge is invisible from inside the task, so the user cannot tell when they have crossed from where the tool helps to where it misleads.[3]

OPEN PROBLEMS

AI-augmented coding is the live frontier for human-AI teaming in professional knowledge work, and the discipline LENS represents is being asked to specify what good looks like before the long-term evidence is in. The open problem is the longitudinal study that could distinguish capability growth from capability erosion — and a training practice that keeps the human's skill on the growing side, built and adopted while a generation is already learning the craft with the tools always within reach.[4]



AI-AUGMENTED CODING — THE CAPABILITY QUESTION IS OPEN

REFERENCES

1. Peng et al. (2023), "The Impact of AI on Developer Productivity" — short-term productivity gains.
2. Pearce et al. (2022), "Asleep at the Keyboard? Assessing the Security of GitHub Copilot's Code Contributions."
3. Sandoval et al. (2023), "Lost at C," USENIX Security — AI assistance did not significantly increase critical security-bug rates.
4. Dell'Acqua et al. (2023), "Navigating the Jagged Technological Frontier" (HBS / BCG) — professional LLM use.
5. L. Bainbridge, "Ironies of Automation," Automatica 19(6) (1983) — the classic deskilling and over-reliance problem, applied to AI-augmented work.

THE LEARNING ENGINEERING LENS

> *AI assistance changes what developers can do; it may also change what they need to know.*
>
> EDITORS' SYNTHESIS

LE INSIGHT

AI-augmented coding is the live foundational case for human-AI teaming in professional knowledge work. The short-term gains are real. The long-term capability question — does the tool make the operator more capable, or more dependent — is the question the discipline must learn to ask and answer.

LENS APPROACH

LENS uses AI-augmented coding in LEN 2 as the largest live human- AI teaming case in the dataset, in LEN 8 for the long-term capability-development question, and in LEN 10 (capstone) as a prompt for evaluating AI augmentation in professional practice.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Define what competence the human must retain independently of the tool, and design the workflow so that skill is exercised rather than quietly handed off.
- Engineer the assistant to surface the jagged-frontier edge — flagging where a task sits outside its reliable competence — so users do not over-trust output just beyond it.
- Keep verification of generated code, especially in security-relevant settings, a required step rather than an optional one, given the unsettled quality picture.

IN OPERATION / AFTER

- Run the longitudinal study that productivity metrics cannot substitute for, measuring whether underlying skill is growing or eroding over years of use.
- Monitor for over-reliance at the competence boundary, where the evidence shows performance degrading as users trust the tool past its reliable range.
- Track outcomes for practitioners who learned the craft with the tools always present, the cohort whose long-term capability is most uncertain.

REFLECTION QUESTIONS

1. In your domain, identify a class of practitioners whose work is currently being augmented by AI tools. What evidence would tell you whether their capability is growing or eroding?
2. Design the longitudinal study that would distinguish capability growth from capability erosion in an AI-augmented professional practice.
3. The “jagged frontier” is hard to navigate because its edge is invisible from inside the task. Design a signal or practice that would tell a practitioner in your domain when they have crossed from where the tool helps to where it misleads.

WHO BUILDS THIS learning science & instructional design · human factors & interaction design — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · usability testing · cognitive walkthroughs

LENS COURSES LEN 10 LEN 2 LEN 8

CASE 100 EDUCATION AT SCALE HEALTHCARE DEFENSE ONGOING

The Discipline We Build Next

T Training Gap K Knowledge & Institutional Memory N Normalization of Deviance G Governance & Trust H Human-System Interface D Designed Out

IMPACT **Open: the case that has not yet been written. The discipline that this casebook documents is the one practitioners now in training must build.**

IN BRIEF

The ninety-nine cases that precede this one are closed: the investigations are complete, the failures diagnosed, the interventions evaluated. This one is open. It belongs to the practitioners now in training — the students of the LDT program and the LENS specialization, the graduates who will take positions in healthcare, defense, education at scale, and the human-AI frontier, and the institutions they will build. The strongest evidence for the discipline is the cumulative record of what its absence has cost; the strongest argument for its possibility is the record of what its presence has produced. Capability engineering in 2026 sits roughly where INPO sat in 1979 or CRM in 1981 — evidence accumulating, practitioners in training, institutional architecture not yet complete. What the reader does next is what fills in this case.

THE SHIFT

Every preceding case in this book is closed: the investigation is complete, the failure diagnosed or the intervention evaluated. This one is open. It belongs to the practitioners now in training — the students of the LDT program and the LENS specialization, and the institutions they will build across healthcare, defense, education, and the human-AI frontier. What distinguishes this case is not a verdict already reached but a verdict still to be earned, written by the choices its readers make once they leave the page.[0]

WHAT IS EMERGING

What is emerging is a discipline. The argument of this volume is that capability is engineerable — that training, interfaces, evidence systems, and institutions can be designed rather than left to chance — and that the work of designing them is a teachable practice with its own methods and growing body of knowledge. If that argument holds, then the gaps the preceding cases document are not the price of progress but the absence of work that could have been done.[1]

THE CAPABILITY QUESTION

The capability question this case poses is the largest one: can the discipline be built fast enough, and at enough scale, to close the gaps the preceding ninety-nine cases document? The strongest evidence for the discipline is the cumulative record of what its absence has cost; the strongest argument for its possibility is the record of what its presence has produced — and the case stays open precisely because that question is answered not by argument but by what the next generation of practitioners actually builds.[2]

EARLY EVIDENCE

The early evidence is the success cases themselves. Capability engineering in 2026 sits roughly where INPO sat in 1979 or CRM in 1981 — a discipline whose evidence is accumulating, whose practitioners are being trained, and whose institutional architecture is not yet complete. Those precedents became mature institutions within a generation, which is the most that can yet be claimed here: a comparable trajectory is plausible, but it is not given, and the comparison is an invitation rather than a guarantee.[3]

OPEN PROBLEMS

No specific incident sits behind this case because the work is not done. The institutions of capability engineering that exist today are insufficient to the volume of capability work the world now requires. What the practitioners reading this volume do next — a failure prevented, a success engineered, an institution built — is what fills in this case, and the page is left deliberately unfinished so that the record of the discipline's possibility can still be added to.[4]

100th

CASE · OPEN

what the practitioners reading this volume do next is what fills it in

THE CASE THAT HAS NOT YET BEEN WRITTEN

REFERENCES

1. LDT / LENS program statement, Johns Hopkins School of Education — the program and its commitments.
2. Goodell & Kolodner (2022), Learning Engineering Toolkit — the discipline's methods.
3. The preceding ninety-nine cases of this volume — the cumulative record of cost and of possibility.
4. INPO (Case 16) and CRM/CAST (Case 12) — institution-building precedents (1979, 1981).
5. IEEE ICICLE Learning Engineering Body of Knowledge (LEBoK); the introduction to this volume.

THE LEARNING ENGINEERING LENS

> *Capability is engineerable. The institution to engineer it has not yet been built. That is the work.*
>
> LDT / LENS PROGRAM STATEMENT, JOHNS HOPKINS SCHOOL OF EDUCATION

LE INSIGHT

Every case in this book is closed. This one is open. The discipline LENS represents is the discipline that closes cases that are now open. The institutions of capability engineering that exist in 2026 are insufficient to the volume of capability work the contemporary world requires. The graduates of this program are the people who will build the institutions that are insufficient now.

LENS APPROACH

LENS is organized to produce the practitioners who can do this work. The curriculum's commitments — capability as a system parameter, evidence as a deliverable, ethics as design, implementation as a thread — are the operational form of the argument the ninety-nine preceding cases evidence. Every studio project (LEN 10) is a small attempt at this case.

APPROACHES TO CONSIDER

DURING DEVELOPMENT

- Treat capability as a system parameter to be specified and engineered from the outset of any deployment, not left to chance and diagnosed only after failure.
- Build the evidence system as a deliverable alongside the capability, so the discipline can show what its presence produces rather than only argue for it.
- Design the institution that will carry the practice forward, taking INPO and CRM as precedents for how a discipline matures within a generation.

IN OPERATION / AFTER

- Monitor whether the discipline is in fact closing the gaps the preceding cases document, treating the open case as a standing measure of progress.
- Sustain the institutional architecture once built, since the precedents show maturity is reached only by carrying the work across a generation.
- Keep the cumulative record of cost and of possibility current, so each failure prevented or success engineered is added to the evidence the discipline rests on.

REFLECTION QUESTIONS

1. What is the case you are positioned to add to this dataset in the next ten years — a failure prevented, a success engineered, an institution built?
2. If this book is read by the practitioners who will write its hundred-and-first case, what should that case look like?
3. Capability engineering in 2026 is compared here to where INPO stood in 1979 — a plausible trajectory, not a guaranteed one. What would have to be true a generation from now for that comparison to hold, and what is the first institution you would build to make it so?

WHO BUILDS THIS learning science & instructional design · knowledge management & data engineering · safety science & organizational analysis · governance, ethics & measurement · human factors & interaction design · systems & human-factors engineering — plus domain experts and a learning engineer to integrate.

TOOLS scenario simulation · competency models · knowledge bases · data pipelines · incident analysis · safety dashboards · governance frameworks · bias & evidence audits · usability testing · cognitive walkthroughs · task analysis · design prototyping

LENS COURSES LEN 1 LEN 10 LEN 8 LEN 3

DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganises the hundred cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Each section opens with a callout naming the dataset's standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE	Case 46	**Operation Eagle Claw (1980).** Cross-organizational capability that existed on no service's deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater-Nichols.
STANDOUT SUCCESS	Case 28	**U.S. Nuclear Navy / Rickover Training Model (1954–).** The longest continuous capability-engineering program in any high-consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.

1	USS Fitzgerald & USS John S. McCain	2017	T·K·N
9	Marine Corps Training in the INDOPACOM AOR	ongoing	T·K
11	V-22 Osprey	1991–	T·H·N
15	Navy Surface Warfare Readiness Reform	2018–	T·K·N
18	USS Vincennes & Iran Air Flight 655	1988	H·T
19	Patriot Missile / Dhahran	1991	D·H·K
26	F-35 Sustainment & Maintainer Shortage	ongoing	T·K·D
28	U.S. Nuclear Navy / Rickover Training Model	1954–	T·K·N
33	Military Fratricide — Desert Storm to Afghanistan	1991–2014	T·H·K

AVIATION

STANDOUT FAILURE — Case 2 — **Boeing 737 MAX / MCAS (2018–19).** A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS — Case 12 — **Crew Resource Management & CAST (1981–).** First-officer authority engineered into the cockpit, a continuous learning architecture across the industry, and the safety record that followed.

78	NASA Saturn V Documentation 1972–	K
89	Aviation Safety Reporting System (ASRS) 1976–	T·K·N
91	Singapore Airlines Safety Transformation 1980s–	T·N

HEALTHCARE

STANDOUT FAILURE — **Case 25** — **EHR / CPOE Implementation (2005–).** Roughly $30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.

STANDOUT SUCCESS — **Case 13** — **WHO Surgical Safety Checklist (2008–).** One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

7	Therac-25 1985–87	H·D·G
13	WHO Surgical Safety Checklist 2008–	T·N
14	Keystone ICU / Pronovost Checklist 2004–	T·N
25	EHR / CPOE Implementation 2005–	H·D·G
27	TeamSTEPPS 2006–	T·N
31	Medical Errors as Systemic Failure ongoing	T·H·N·K·G
34	ACGME 80-Hour Resident Duty-Hour Reform 2003–17	T·K·N
41	Implementation Science — The 17-Year Gap ongoing	K·G·N
58	Mid Staffordshire NHS Foundation Trust 2005–09	G·N·K
69	Theranos 2003–18	G·D
87	Vioxx Withdrawal 1999–2004	G·D
88	Tylenol Recall 1982	G·N
90	Bristol Heart Babies Reform 1984–	G·N
95	Radiology AI Miscalibration 2018–	H·K·D

ENERGY

STANDOUT FAILURE — **Case 5** — **Three Mile Island (1979).** Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.

STANDOUT SUCCESS — **Case 16** — **INPO and the Nuclear Academy (1979–).** The industry built the institution TMI revealed it needed. No INES-level event at U.S. commercial reactors since.

INDUSTRIAL

STANDOUT FAILURE	Case 17	**Bhopal (1984).** Training, knowledge, normalisation of deviance, and governance all collapsed in the same plant. Roughly 4,000 immediate deaths; chronic harms continue. The cautionary case for outsourced operational responsibility.
STANDOUT SUCCESS	Case 24	**Toyota Production System / Andon Cord (1950s–).** Any operator can stop the line. The smallest authority-architecture intervention in the dataset, and the most durably copied.

17	Bhopal 1984	T·K·N·G
20	Grenfell Tower 2017	G·T·K·N
24	Toyota Production System / Andon Cord 1950s–	N·G
51	Ford Pinto Fuel Tank 1971–78	D·G
52	Takata Airbag Inflators 2008–23	D·G
53	GM Ignition Switch 2002–14	D·G
71	Volkswagen Dieselgate 2015	D·G
74	Hyatt Regency Walkway Collapse 1981	D·G
75	FIU Pedestrian Bridge Collapse 2018	D·G·N
81	Tacoma Narrows Bridge 1940	D·K

SPACE & AEROSPACE

STANDOUT FAILURE	Case 6	**Challenger & Columbia (1986 / 2003).** Normalisation of deviance across two decades and two crews, captured in the Rogers and CAIB reports as a culture that accepted flying with flaws when problems were defined as normal and routine.
NOTE	Case 28	No paired-intervention success in the aerospace dataset. See Defense (Rickover, Case 28) for the safety-engineering counterpart that aerospace did not build.

6	Challenger & Columbia 1986/2003	N·K·G

54	Mars Climate Orbiter — Unit Mismatch	1999	D·K
79	Boeing Starliner	2019–24	K·D
80	Ariane 5 Flight 501	1996	D·K·H

GOVERNMENT

STANDOUT FAILURE	Case 36	**Australia Robodebt (2016–20).** An income-averaging algorithm that reversed the burden of proof onto welfare recipients. The Royal Commission found "venality, incompetence and cowardice"; the scheme was linked to deaths, including suicide.
NOTE	Case 89	No paired-intervention success in the government dataset. The reforms that follow these cases — Royal Commissions, statutory restructures — are documented in the narratives but did not produce a LENS-style success case in time for this volume.

10	Healthcare.gov Launch	2013	K·T·G
32	VA Wait-Time Scandal	2014	G·K·N
36	Australia Robodebt	2016–20	G·D·H
83	LIBOR Manipulation	2003–12	G·N
85	Bernard Madoff / SEC Failures	1992–2008	G·K·N
94	COMPAS Recidivism Algorithm	2016–	G·H·D
97	Predictive Policing — PredPol	2011–	G·H·D

EDUCATION AT SCALE

STANDOUT FAILURE	Case 21	**Summit Learning / Personalized Learning Rollout (2014–19).** A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.
STANDOUT SUCCESS	Case 39	**Georgia State University Predictive Analytics (2012–).** 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

8	inBloom	2014	G
21	Summit Learning / Personalized Learning Rollout	2014–19	G·T·K
22	Tennessee Voluntary Pre-K Study	2009–18	G·D
35	UK A-Level Algorithm / Ofqual	2020	G·H·D
37	Algorithmic Bias in Educational Predictive Analytics	ongoing	G·H·D
39	**Georgia State University Predictive Analytics**	2012–	T·K
40	xAPI / Total Learning Architecture	ongoing	K·G
42	**Cognitive Tutor / Carnegie Learning**	1990s–	T
84	Atlanta Public Schools Cheating Scandal	2009–15	G·N
100	**The Discipline We Build Next**	ongoing	all

TECHNOLOGY

STANDOUT FAILURE	Case 68	**UK Post Office Horizon Scandal (1999–2015).** Institutional refusal to believe sub-postmasters' reports of computer error, sustained over fifteen years. Hundreds of wrongful convictions; ruined lives and at least four suicides — the longest-running operator-feedback failure in the dataset.
NOTE	Case 98	The technology dataset's success story is filed under Healthcare: AlphaFold (Case 98), the contemporary worked example of computational capability-engineering done as a discipline.

55	Knight Capital Trading Loss	2012	D·K
65	CrowdStrike Falcon Outage	2024	D·K·G
67	TSB Bank IT Migration	2018	H·G
68	UK Post Office Horizon Scandal	1999–2015	G·H·K
70	Wells Fargo Fake Accounts	2011–16	G·N
72	Cambridge Analytica / Facebook	2014–18	G
73	Equifax Data Breach	2017	G·K

AUTONOMOUS SYSTEMS

STANDOUT FAILURE	**Case 29**	**Uber ATG / Tempe Fatality (2018).** The safety driver was nominally responsible for what the system was designed to do without her attention — the classic vigilance contradiction at the heart of partial-autonomy operator roles.
NOTE	**Case 100**	No paired-intervention success in the autonomous-systems dataset. The capability engineering this category needs is still being built — see Case 100 (**The Discipline We Build Next**) for the open question this volume closes on.

About this index. Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

LENS COURSE INDEX

The cases by LENS course.

Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the specialization, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

LEN 1

21 cases

LEN 2

23 cases

14 Keystone ICU / Pronovost Checklist

15 Navy Surface Warfare Readiness Reform

22 Tennessee Voluntary Pre-K Study

31 Medical Errors as Systemic Failure

32 VA Wait-Time Scandal

34 ACGME 80-Hour Resident Duty-Hour Reform

35 UK A-Level Algorithm / Ofqual

37 Algorithmic Bias in Educational Predictive Analytics

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LEN 5

36 cases

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13 cases

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49 cases

LEN 8

45 cases

LEN 9

10 cases

LEN 10

19 cases

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Each case spread in the body of the book includes its own primary sources. Sources cited only within case spreads are not repeated here; this list collects the works the discipline as a whole draws on, together with the references used in the introductory essay.

ABOUT THE EDITORS

The editors.

The casebook's central argument — that capability engineering is the applied form of the educational sciences, taught from the School of Education and informed by operational practice — is embodied in the two editors. Each has spent a career working in one half of that pairing and a substantial portion of it in the other. Each contributes to the five pillars on which LENS rests: Mission Literacy · JHU Ecosystem · Intersectional Expertise · Capability Focus · Flywheel Iteration.

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